

ANALYZERS OF CHEMICAL CURRENT SOURCES

ASK75.10.20.2

ASK75.10.20.12



OPERATING MANUAL 022үүүү.0000-0 үү

023YARST.0000-0 RE

Moscow, 2023.

<https://battery-analyzers.ru/>

Dear Customer, Thank you for choosing our equipment. This Manual contains information about the operating principle, design and characteristics of the Chemical Current Source Analyzers ASK75.10.20.2 and ASK75.10.20.12, as well as instructions necessary for correct and safe operation, compliance with which will ensure uninterrupted, long-term and trouble-free operation of the devices.

The HIT ASK75.10.20.2 and ASK75.10.20.12 analyzers are complex electronic devices that operate from an alternating current network with a voltage of 220 V. Only persons with an electrical safety group of at least II and who have carefully studied this Manual are allowed to work with the Analyzers.

The figures in this Manual are provided for ease of description. They may differ slightly from the upgraded versions of the devices.

For all questions related to the operation and maintenance of devices, please contact us by e-mail: Yarst@mail.ru

Also, please contact us at the specified e-mail address with your wishes and comments regarding the operation of the devices and software.

Software updates, additional information, product catalogs of YAROSTANMASH LLC and documentation for it can be found on our website on the Internet: <https://battery-analyzers.ru/>.

CONTENT

Section	p.
List of accepted abbreviations and designations 1 Purpose of the	4
device 2 Completeness of	5
delivery	7
3 Safety precautions 4	9
Operating conditions 5 Getting to	11
know the device 6 Design and	12
operating principle 7 Technical and	15
metrological characteristics 8 Preparation for the first switch-on 9	19
Software installation and configuration 10	21
Connecting the tested CCS and device control 11 CCS testing program	23
11.1 General instructions 11.2 CCS testing parameter signs 11.3 Constant current	24
charge (CHCC) 11.4 Constant power charge	26
(CHCP) 11.5 Constant voltage	26
recharge (CHCV) 11.6 Constant current discharge	28
(DCHCC) 11.7 Constant power discharge (DCHCP)	29
11.8 Constant resistance discharge (DCHCR) 11.9 Constant	30
voltage recharge (DCHCV) 11.10 Relaxation 11.11 Voltage sweep	32
11.12 Current sweep 11.13 Power sweep 11.14	33
Resistance sweep 11.15 Voltage pulse mode 11.16 Current	34
pulse mode 11.17 Power pulse mode 11.18 Resistance pulse mode	36
11.19 Working with tables 11.19.1 General instructions 11.19.2 Working	37
with voltage table	38
11.19.3 Working with current table	39
11.19.4 Working with power	40
table 11.19.5 Working with resistance	41
table 11.20 Voltage recorder 11.21 Pause	42
11.22 MPPT mode	43
	44
	46
	48
	50
	50
	53
	54
	55
	57
	59
	60
	60

CONTENTS. CONTINUED

Section	p.
12 Determining the parameters of the HIT 13	62
Measuring the ESR of the HIT	67
13.1 General instructions 13.2	67
Measuring the ESR of the HIT using the periodic current interruption method 13.3 Measuring the	67
ESR of the HIT in pulse modes 13.4 Measuring the ESR of the HIT	70
when changing program steps	71
14 Determining the electrical capacitance of the HIT 15	73
Measurement result files 15.1 General instructions	74
15.2 Primary data files 15.3	74
Summary measurement result files 16	76
Thinning measurement result files 17 Time format in measurement	78
result files 18 Working with result files in MS Excel 19 Computer Interface	82
19.1 General instructions 19.2 "Tasks" page 19.3 "Tables" page 19.4 "Graphs"	84
page 19.5 "Results" page 19.6 "Analysis" page 19.7 "Settings" page 19.8	86
"Status" page 19.9 "Postprocessing" page	88
20 Viewing previously recorded	88
measurement results 21 Operating	92
the device in autonomous mode 22	117
Device protection and emergency	121
modes 23 Firmware update 24	133
Information on the content of	134
precious metals 25 Maintenance and	140
repair 26 Transportation 27 Storage	142
28 Disposal 29 Certification information	144
	145
	152
	154
	162
	164
	164
	164
	164
	164
	165

LIST OF ACCEPTED ABBREVIATIONS AND DESIGNATIONS

CHCC – CHarge Constant Current – charging the heat source with a given constant current;
CHCP – CHarge Constant Power – charging the heat source with a given constant power;
CHCV – CHarge Constant Voltage – additional charge of the HIT at a given constant voltage;
DCHCC – DisCHarge Constant Current – discharge of the chemical energy source with a given constant current;
DCHCP – DisCHarge Constant Power – discharge of a chemical energy source with a given constant power;
DCHCR – DisCHarge Constant Resistance – discharge of the HIT to a given constant resistance;
DCHCV – DisCHarge Constant Voltage – additional discharge of the HIT at a given constant voltage;
ESR – Equivalent Series Resistance – equivalent series resistance of the heat sink;
MPPT – Maximum Power Point Tracking – maximum power point search mode;
L – Low – low, junior (sub-range of reproduction and measurement of voltage, current, or the level of voltage, current);
H – High – high, senior (sub-range of reproduction and measurement of voltage, current, or the level of voltage, current);
U – voltage, V;
I – current, A;
P – power, W;
R – resistance, Ohm;
C – electrical capacity of the heat source, F;
Q – charge received (given) by the CIT, or the total capacity of the CIT, A h;
E – energy received (given) by the heat source, or the total energy capacity of the heat source, W h;
t – time, sec, min, h or day;
T – temperature, °C;
N – the number of the current cycle of the HIT testing program, or the total specified number of cycles of the HIT testing program;
dU – voltage jump (drop, change, increment);
ADC – Analog-to-Digital Converter;
HIT – Chemical Current Source (battery, supercapacitor, or other);
Efficiency – Coefficient of Performance;
PO – Software (device control program);
SB – Solar Battery;
see – look;
Wi-Fi is a general term for hardware and software methods of wireless transmission of data used, in particular, in personal computers for connection with various devices and the Internet;
MS Excel is a program used on personal computers to work with spreadsheets;
NEDC, EPA, JC08, WLTC, NRTC – various international standards and methods testing of cars and self-propelled equipment;
Computer Interface – specialized application software installed on the User's computer, with the help of which the User controls the Analyzer and receives measurement results;
point – a set of values of the HIT testing parameters (charge-discharge cycle number, step marker, voltage on the HIT, current through the HIT, HIT temperature and HIT ESR) at a specific point in time.

1 PURPOSE OF THE DEVICE

Analyzers ASK75.10.20.2 and ASK75.10.20.12 are designed for testing:

- Batteries of various electrochemical systems (Li-ION, Li-Po, LiFePO₄, Ni-Cd, Ni-MH, Pb-Acid and others);
- Supercapacitors (Ionistors);
- Galvanic cells;
- Solar panels;
- Fuel cells;
- Redox batteries;
- Electrochemical cells;
- Any other sources and storage devices of electrical energy.

The ASK75.10.20.2 analyzers have two channels, and the ASK75.10.20.12 analyzers have twelve channels that are completely independent of each other. Each channel can have an individual program for testing the chemical supply devices, independent of other channels. Different channels of the device can simultaneously test a wide variety of chemical supply devices, of different types, with different operating voltages, having different capacities and different natures. Moreover, the channels of the Analyzer are electrically isolated from each other, which allows one and the same device to simultaneously test several chemical supply devices, connected in series into a common battery, without the need to disassemble the battery into individual elements.

Also, the ASK75.10.20.2 and ASK75.10.20.12 Analyzers can be used as multi-channel programmable power sources, the output parameters of which (voltage, current, power) can be configured within wide limits and automatically changed over time according to a program specified by the User, which is relevant for testing not only HIT, but any energy consumers, including third-party electronic equipment, communications equipment, etc. In this case, the Analyzers will calculate the amount of charge and energy consumed.

In addition, the ASK75.10.20.2 and ASK75.10.20.12 Analyzers can be used as multi-channel programmable electronic loads, the input parameters of which (voltage, current, power, resistance) can be adjusted within wide limits and automatically changed over time according to a program specified by the User, which is relevant for testing not only HIT, but any other energy sources, including various power supplies and energy converters. In this case, the Analyzers will calculate the amount of charge and energy received.

The tests of the HIT are carried out according to a program prepared in advance by the User, containing:

- up to 3 test preparation steps, performed once at the beginning of the test;
- up to 42 steps of the main cyclic part of the program, executed a specified number of times (cycles);
- up to 3 test completion steps, performed once at the end of the test.

The total number of configurable steps of the testing program is 48.

At each step, the devices can perform:

- Charging the HIT with a given constant current (CHCC);
- Charging the heat source with a given constant power (CHCP);
- Additional charging of the CHIT at a given constant voltage (CHCV);
- Discharge of the CHIT with a given constant current (DCHCC);
- Discharge of a given constant power CHIT (DCHCP);
- Discharge of the chemical energy source to a given constant resistance (DCHCR);

- Additional discharge of the chemical battery at a given constant voltage (DCHCV);
- Relaxation;
- Voltage sweep (U) on the heat exchanger over time;
- Current sweep (I) through the HIT in time;
- Power sweep (P) on the heat source over time;
- Time sweep of the discharge resistance of the chemical energy source (R);
- Voltage pulse mode;
- Current pulse mode;
- Power pulse mode;
- Resistance pulse mode;
- Work according to a given voltage table (given voltage profile);
- Work according to a given current table (given current profile);
- Work according to a given power table (given power profile);
- Work according to a given resistance table (given resistance profile);
- Recording the voltage at the terminals of the HIT (U recorder);
- Pause;
- Maximum Power Point Search (MPPS) mode.

The testing modes for each step of the program can be set by the User in any desired (arbitrary) order, the testing parameters for each step can be selected by the User of any desired values from the permissible operating range of the device.

If necessary, adjacent channels of one Analyzer can operate in parallel to increase current and power, as well as reduce discharge resistance.

Analyzers ASK75.10.20.2 and ASK75.10.20.12 determine:

- Voltage on the CCS, V, current through the CCS, A, CCS temperature, °C, instantaneous value of the equivalent series resistance of the CCS (ESR), Ohm, current delivered (received) capacity, A h, current delivered (received) energy, W h – as a function of the discharge (charge) time;
- Capacity of the HIT by charge, A h;
- Capacity of the heat storage device by energy (energy capacity), W h;
- Electrical capacity (supercapacitors), F;
- Charge storage efficiency (by A h), %;
- Energy storage efficiency (Wh), %;
- Equivalent series resistance of the heat sink (ESR), mOhm;
- Average leakage current through the HIT for each charge-discharge cycle, A.

To expand the operating range of charge and discharge current, the devices have four sub-ranges for reproducing and measuring current through the CIT.

The analyzers HIT ASK75.10.20.2 and ASK75.10.20.12 are designed to operate under control from the User's computer. The device communicates with the computer via a local computer network using the Ethernet interface.

The application software, the Ym128 Computer Interface, is installed on the User's computer, which is necessary for setting the parameters for testing the HIT, starting and stopping the test, and monitoring the progress of testing. obtaining and displaying measurement results.

The computer can be switched off during testing. In this case, the Analyzer will continue to operate in autonomous mode, and the current measurement results will be recorded in the internal memory of the device. The next time the computer is switched on and a connection with the device is established, the accumulated data will be automatically transferred to the computer, processed and saved in the measurement results files.

2 DELIVERY CONTENTS

Table 1 - Completeness of delivery of the ASK75.10.20.2 Analyzer

№	Name	Quantity	Note
1	HIT analyzer 	1 pc.	
2	Connecting wires with crocodile clips 	2 set	Length 0.6 m.
3	Temperature sensor HIT 	2 pcs.	Length 0.6 m.
4	220V power cord with grounding 	1 pc.	Length 1.8 m.
5	Ethernet interface cable 	1 pc.	Length 1.8 m
6	Ym128 software on media	1 copy.	
7	Form 022yyy.0000-0 FO	1 copy.	
8	Operation manual 022yyy.0000-0 yy	1 copy.	
9	Packaging containers	1 pc.	

The completeness of the delivery and the appearance of the device may be changed by the manufacturer.
telescope and are not reflected in this document.

Table 2 - Completeness of delivery of the ASK75.10.20.12 Analyzer

№	Name	Quantity	Note
1	HIT analyzer 	1 pc.	
2	Connecting wires with crocodile clips 	12 set	Length 0.6 m.
3	Temperature sensor HIT 	12 pcs.	Length 0.6 m.
4	220V power cord with grounding 	1 pc.	Length 1.8 m.
5	Ethernet interface cable 	1 pc.	Length 1.8 m
6	Ym128 software on media	1 copy.	
7	Form 023yyy.0000-0 FO 8 Operation manual 023yyy.0000-0 yy	1 copy.	
9	Packaging containers	1 copy.	
		1 pc.	

The completeness of the delivery and the appearance of the device may be changed by the manufacturer.
and are not reflected in this document.

3 SAFETY MEASURES

The HIT ASK75.10.20.2 and ASK75.10.20.12 analyzers are complex electronic devices that operate from an alternating current network with a voltage of 220 V. Only persons with an electrical safety group of at least II and who have carefully read this document are allowed to work with the Analyzers.

Incorrect use of the Analyzer and failure to observe safety precautions may result in failure of the device, electric shock, burns, fire, serious injury or death.

In this section, safety instructions are divided into "Danger" and "Warning", so pay special attention to the symbols.

 (Danger) and  (Warning) and the corresponding content of the text.

Symbol  means that incorrect use of the appliance may result in electric shock, serious injury or death.

Symbol  means that improper use of the device may lead to malfunctions of the device, damage to its internal electronic circuits or to other negative consequences.

1.  Connect the device to a power grid with guaranteed reliable grounding. If there is no socket with a working grounding conductor connected to the grounding circuit of the building or power supply network, use a copper grounding wire with a cross-section of at least 1 mm² (not included in the delivery set) connected between the grounding contact on the front panel of the device and the working grounding circuit in the room. Operating the device without grounding may cause the device to fail, electric shock, serious

injury or death.

2.  Always connect the connecting wires first to the device, and only then to the tested HIT. Disconnect in the reverse order – first from the HIT, then from the device. Free uninsulated ends of the connecting wires connected to the HIT can cause a short circuit of the HIT, fire, electric shock and death.

3.  Do not perform any actions with the HIT, as well as with the wires connecting the HIT to the device, in the Analyzer operating mode other than **"IDLE"**. This may result in electric shock, fire, or failure of the device.

4.  When connecting the tested HIT to the device, use the connecting wires included in the delivery set, or, when replacing the standard ones, with a cross-section of at least 1 mm²
Wires with a smaller cross-section can become very hot and cause a fire.

5.  Always check the electrical connections first before turning on the unit and running a test. Poor contact can cause heating and fires.



6. Never short-circuit the terminals of the device when the HIT is connected to it and/or the test program is running. This may result in electric shock, fire, or failure of the device.



7. Do not disassemble the appliance or remove the cover while it is plugged in or for 10 minutes after it is unplugged. This may result in electric shock.
electric shock and death.



8. When the tested CCS is discharged, the energy taken from it is dissipated as heat and released into the environment using fans on the rear panel of the device. In maximum power modes, the air flow coming out through the rear panel of the device may have a temperature of over 70 °C. To avoid burns, do not touch the rear panel of the device during operation. Do not place flammable objects and liquids in the path of the air flow coming out of the rear panel of the device, at a distance closer than 1 m from the device body - this may cause ignition and fire.



9. Do not operate the appliance with wires and cables with damaged insulation. This may result in electric shock, fire or appliance failure.



10. Do not operate the appliance in conditions that impede air flow to the ventilation openings on the front and bottom panels of the appliance or the air flow from the fans on the rear panel. This may cause overheating and damage to the appliance.



11. Do not install any switching equipment in the circuits between the tested CIT and the Analyzer. The exception is fusible links. Control of such switching equipment in the device operating modes other than "IDLE" may cause the device to fail.



12. The device is intended for indoor use in normal, non-chemically aggressive atmospheres. Do not operate the device outdoors or in conditions where the device body and ventilation openings may be exposed to snow, splashes of moisture, conductive dust (including carbon dust), conductive fibers (including carbon fibers), liquids of any type,

insects, foreign mechanical objects, corrosive gases, vapors, etc. This will cause the device to fail.



13. When bringing the device into a warm room during the cold season, it is necessary to wait at least three hours before turning on the device. This time is necessary for the evaporation of moisture condensate, which may appear on the external and internal elements of the Analyzer. This will prevent the device from failing.



14. Do not apply **more than 20 V** to the device terminals. This may cause the device to fail.

15.  Do not apply voltage to the device channel terminals relative to the device body (ground circuit) with an absolute value of **more than 100 V**. This may cause the device to fail.

16.  Do not operate the device near objects and installations that are sources of strong electromagnetic radiation and interference, including pulsed ones, to which the device may be exposed. This may lead to distortion of measurement results and failure of the device.

17.  Do not connect other electrical circuits that are not related to the operation of the Analyzer to the tested HIT and to the terminals of the device. This may lead to distortion of the measurement results and failure of the device.

18.  Do not immerse the HIT temperature sensors in liquid. The temperature sensors are not hermetically sealed. Liquid penetration into the sensor will cause distortion in temperature readings and will lead to failure of the sensor.

19.  Do not operate the HIT temperature sensors with damaged insulation. Accidental contact of the temperature sensor circuits with the HIT circuits may result in distortion of the measurement results and failure of the electronic circuits of the device.

4 OPERATING CONDITIONS

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are intended for use inside closed ventilated rooms with a normal, chemically inert atmosphere, away from electric heating and heat-generating devices and installations, in places where the devices will not be exposed to direct sunlight, away from installations and devices that are sources of strong electromagnetic radiation and interference,

under the following conditions:

- Ambient air temperature: from 18 to 28 °C;
- Relative air humidity: from 20 to 80% (without moisture condensation);
- Atmospheric pressure: from 84 to 107 kPa (from 630 to 800 mmHg).

5 GETTING TO KNOW THE DEVICE

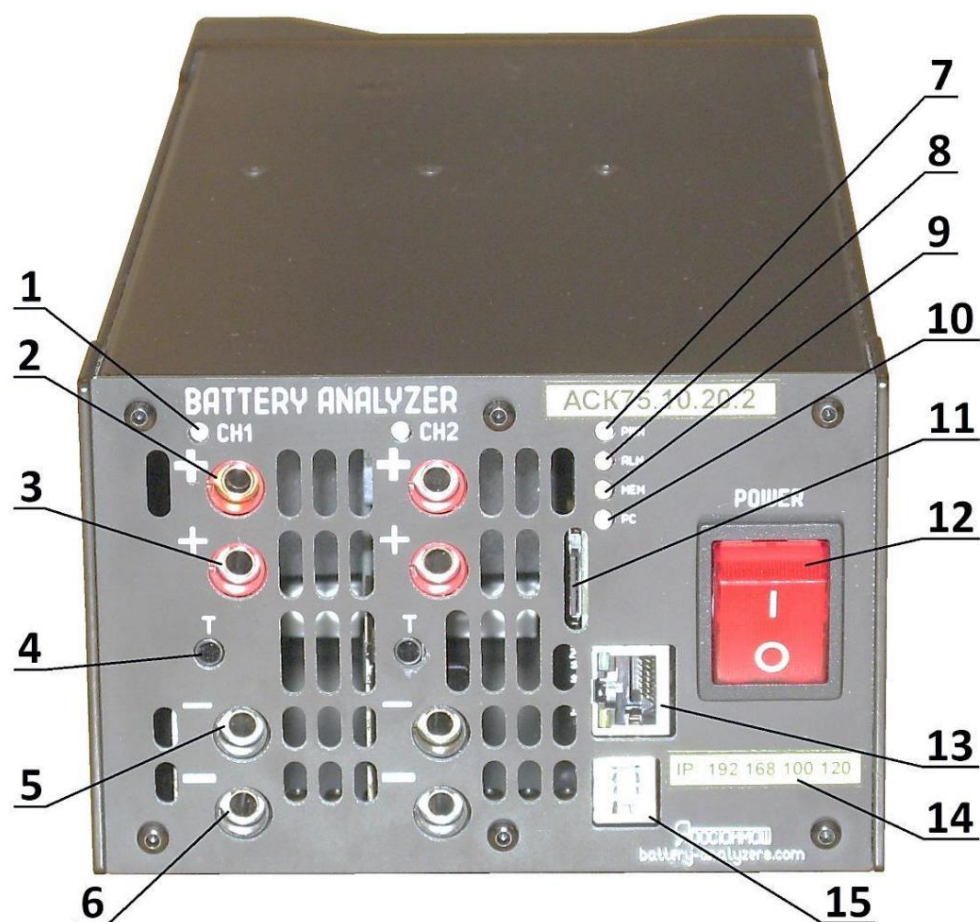


Figure 1 - Front panel of the ASK75.10.20.2 Analyzer



Figure 2 - Rear panel of the ASK75.10.20.2 Analyzer



Figure 3 - Sticker with the serial number of the ASK75.10.20.2 Analyzer (located on the bottom panel of the device)



Figure 4 - Front panel of the ASK75.10.20.2 Analyzer



Figure 5 - Rear panel of the ASK75.10.20.12 Analyzer



Figure 6 - Sticker with the serial number of the ASK75.10.20.12 Analyzer (located on the bottom panel of the device)

Table 3 - Legend to Figures 1, 2, 4 and 5

Position	Description	Note
1	Channel status indicator of the device Power terminal	see section 6
2	"+"	see section 10
3	Connector of potential wire "+" Connector of	see section 10
4	temperature sensor HIT	see section 10
5	Potential wire connector Power terminal "-" "-"	see section 10
6	LED indicator "PWR"	see section 10
7	(POWER) LED indicator "ALM" (ALARM) LED indicator	see. sections 6
8	"MEM" (MEMORY) LED indicator "PC" (PC) Micro SD	see. sections 6, 22
9	memory card	see. sections 6, 21
10		see. sections 6, 8
11		see section 21
12	Power switch	see. sections 6, 8
13	Ethernet RJ45 communication connector Sticker with the	see section 8
14	device's IP address USB communication	see. sections 9, 19.7
15	connector	
16	Cooling fan Grounding contact.	see. sections 3, 8
17	Thread M5 x 0.8.	see. sections 3, 8
18	220V power socket with grounding	see. sections 3, 8

6 DEVICE AND OPERATING PRINCIPLE

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are technically complex electronic devices consisting of two (for ASK75.10.20.2) or twelve (for ASK75.10.20.12) identical charge-discharge modules **MZR (MZR_x)** and a common control and management module **MKU**. Data exchange between the modules is carried out via a galvanically isolated data bus **SD**.

Each charge-discharge module contains a power source **PS**, a charge regulator, a discharge regulator, a current sensor A, a voltmeter V, a temperature sensor XIT T, a channel status indicator (**CHANNEL_x**) and a microcontroller:

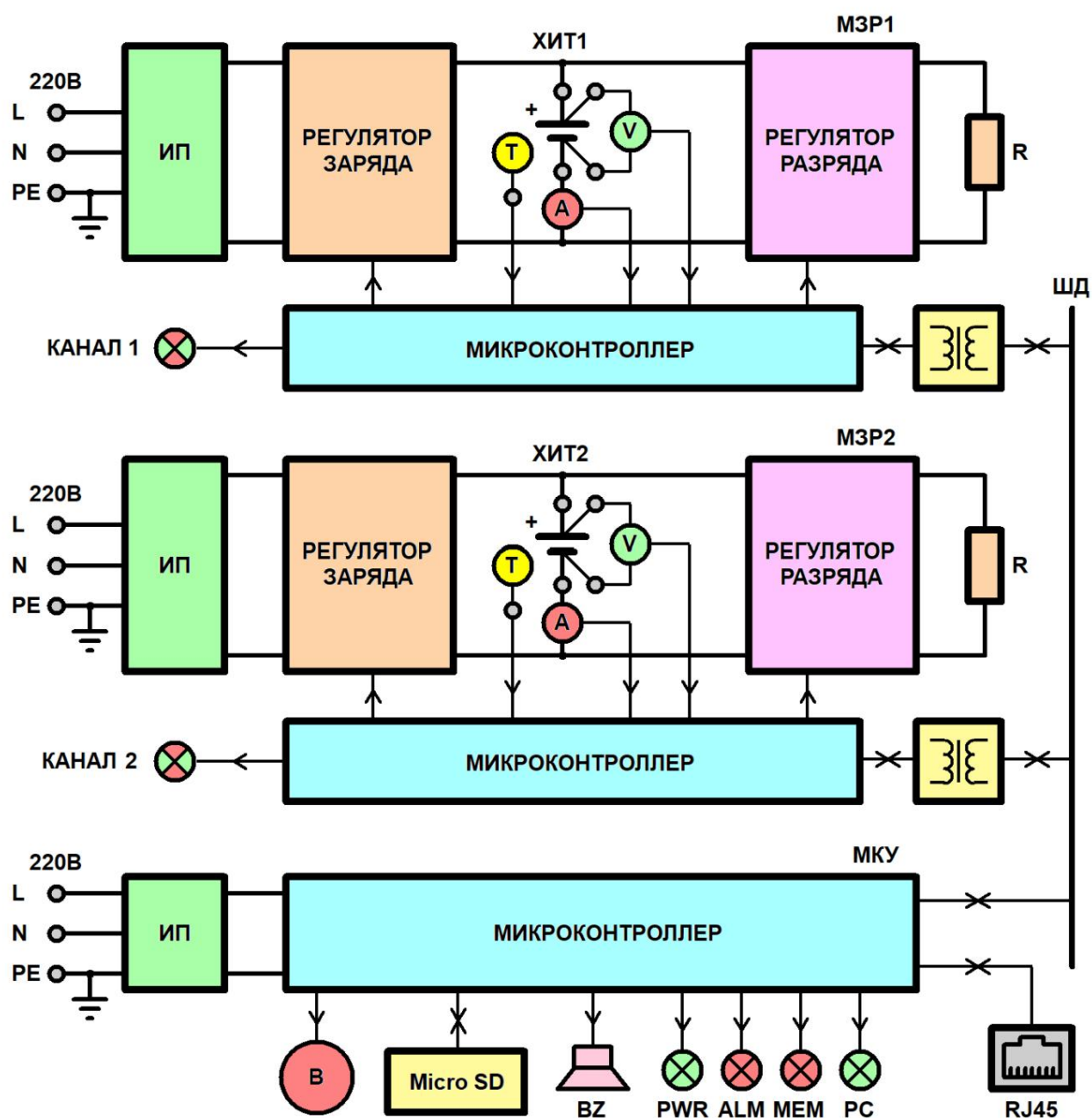


Figure 7 - Structural diagram of the ASK75.10.20.2 Analyzer

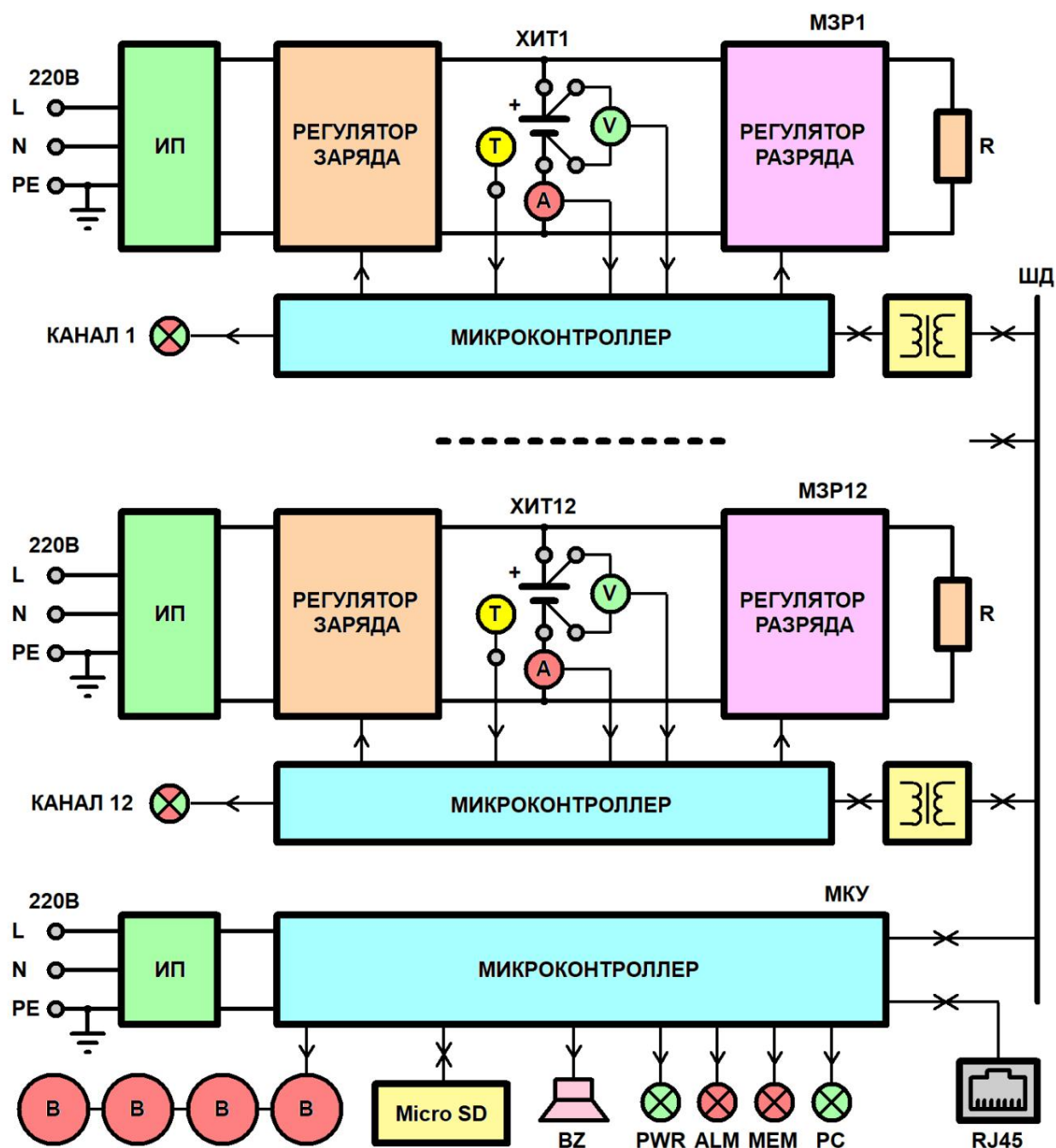


Figure 8 - Structural diagram of the ASC75.10.20.12 Analyzer

To charge the tested chemical power supply on the selected channel of the Analyzer, the energy from the 220 V network is converted by the power supply of the specific charge-discharge module of the MZR into a constant voltage and is fed to the chemical power supply through the charge regulator. The charge regulator, controlled by the microcontroller, provides all the necessary charging modes, namely:

- charging the heat exchanger with a given constant current (including the current pulse mode at positive current values);
- charging the heat source with a given constant power (including the power pulse mode at positive power values);
- recharging of the HIT at a given constant voltage (including voltage pulse mode at positive current values);
- HIT charge current sweep;
- power scan of the HIT charge;
- voltage sweep of the heat exchanger (at positive current values).

The discharge of the CIT is carried out by a discharge regulator, which takes energy from the CIT, converts it as necessary and dumps it onto the resistive load R. The discharge regulator operates under the control of a microcontroller and provides:

- discharge of the HIT with a given direct current
(including current pulse mode at negative current values);
- discharge of a high-voltage power source with a given constant power
(including power pulse mode at negative power values);
- discharge of the chemical energy source to a given constant resistance
(including the mode of pulses of resistance of the discharge of the chemical energy source);
- additional discharge of the HIT at a given constant voltage
(including voltage pulse mode at negative current values);
- scanning of the discharge current of the chemical energy source;
- power sweep of the discharge of the chemical energy source;
- scanning of the discharge resistance of the chemical energy source;
- voltage sweep of the discharge of the chemical energy source.

Current measurement through the HIT is carried out by current sensor A, voltage measurement on the HIT is carried out by voltmeter V. Current sensor **A** has four hardware subranges measurements.

The state of the charge-discharge module **MZR** is indicated by a three-color display indicator **CHANNELx** (position 1, figures 1 and 4).

The indicator operates in the following modes:

Table 4 - Channel status indicator operating modes

Color	Mode	Description
Green	Constant	The channel is ready for work. Testing is not running.
Red		The channel is in operation. The HIT testing process is underway.
Yellow		Pause for ESR measurement HIT
Green	Flashing	The testing process of HIT is complete
Red		HIT or device channel failure

The control and management module contains a power supply unit (**PS**), which supplies power to the module and cooling fan(s) B, a memory card for storing measurement results (**Micro SD**), a sound alarm (**BZ**), and **PWR** indicators .

(**POWER**), **ALM** (ALARM), **MEM** (MEMORY), **PC** (PC), Ethernet **RJ45** connector and a microcontroller.

The microcontroller of the control and management module polls the charge-discharge modules **of the MZR**, monitors the specified conditions for the end of the testing of the HIT on all channels of the device, accumulates and transmits the measurement results to the User's computer.

The Analyzer and the User's computer communicate via the Ethernet network, which is connected via the **RJ45** connector on the front panel of the device. The device can be connected to the User's computer directly or via a local computer network.

The universal application software Ym128 is installed on the User's computer, with the help of which the device is controlled, the program for testing the chemical and information technology is compiled and launched (see Section 11), monitoring the progress of testing, as well as receiving, processing and displaying measurement results.

After the required HIT testing program has been compiled and launched (see Section 11) The User's computer can be switched off, after which the Analyzer will switch to autonomous operation mode (see Section 21). During autonomous operation of the device, the current measurement results are saved to the **Micro SD memory card**. The next time the User's computer is switched on and a connection with the device is established, the accumulated measurement results will be read from the memory card, transferred to the computer and saved in measurement results files.

To indicate the operating modes of the ASK75.10.20.2 and ASK75.10.20.12 Analyzers LED indicators are used:

- **PWR** (POWER SUPPLY) – lights up if the device is supplied with 220 V power and is turned on "POWER" switch (position 12, figures 1 and 5);
- **ALM** (ALARM) – flashes in an alarm condition on any of the device channels;
- **MEM** (MEMORY) – flashes if the measurement results memory is close to full. If the measurement results memory is full, it lights continuously. In addition, this indicator lights continuously if the Micro SD card is not installed or is faulty (see Section 21);
- **PC** – lights up if a connection is established with the User's computer on which the device control program is running.

The **BZ** sound alarm is designed to give sound signals in various operating modes of the Analyzer, as well as an emergency signal if one occurs.

7 TECHNICAL AND METROLOGICAL CHARACTERISTICS

Table 5 - Technical characteristics of the ASK75.10.20.2 and ASK75.10.20.12 analyzers

Characteristic	Meaning for Analyzer	
	ASK75.10.20.2	ASK75.10.20.12
Number of independent channels	2	12
Possible device operations at every step of the HIT1 testing program	CHCC; CHCP; CHCV; DCHCC; DCHCP; DCHCR; DCHCV; Relaxation; Reamer U; Scan I; Scan P; Scan R; Pulses U, Pulses I, Pulses P, Pulses R, Table U, Table I, Table P, Table R, Recorder U	
Operating range of voltage on the heat exchanger,	from 0.12 to 20	
V Range of reproduction of voltage on the heat exchanger, V	from 0.12 to 20	
Resolution of the ADC for measuring voltage on the heat	24 bits	
exchanger Input resistance of the voltage meter at the terminals of the HIT,	not less than 1,000,000	
Ohm3 Range of reproduction of the charge current through the HIT,	from 0.01 to 10.00	
A Range of reproduction of the discharge current through the HIT, A	from minus 10.00 to minus 0.01	
Number of subranges of measuring the current through the HIT	4	
Current measurement subranges through the heat exchanger, A	from minus 10.0 to plus 10.0; from minus 2.0 to plus 2.0; from minus 0.5 to plus 0.5; from minus 0.1 to plus 0.1;	
ADC bit depth for measuring current through the HIT	24 bits	
Selecting the subrange of measuring current through the HIT	Automatically, in accordance with specified parameters of the HIT testing program, or manually	
Reproduction range of the HIT charge power, W	from 0.001 to 50,000	
Reproduction range of the HIT discharge power, W	from minus 75,000 to minus 0.001	
Maximum HIT charge power, W	50	
Maximum HIT discharge power, W	75	
range of the HIT discharge resistance, Ohm	Operating	
range of voltage on the HIT for the MPPT mode, V	Rate of change of	
conductivity in the MPPT mode	Reproduction range of the step time	
Measurement range of the step time	Measurement range of the	
HIT temperature, °C	Mode of the end of the HIT charge	
step according to the "-dU" condition	Absolute	
value of the negative increment	from minus 40 to plus 100	
	There is	
voltage on the heat exchanger, leading to termination	from 0.001 to 10,000	
charge of the heat source according to the condition "-dU", V		
Maximum frequency of parameter registration	Minimum	
frequency of parameter registration	100 points1 / s	
	1 point1 / h	
Memory capacity of measurement results for autonomous operation of the device without connection to a computer	Determined by the volume of the installed Micro SD card. 1 GByte y 450,000,000 points1	

Notes:

1. For more details, see the list of accepted abbreviations and designations, as well as section 11;
2. the maximum discharge current at low voltages on the HIT is limited by the minimum input resistance of the device channel. The maximum discharge current of 10 A is provided at a voltage on the CIT of more than 1.2 V. At lower voltages on the CIT, the maximum discharge current decreases proportionally to the decrease in voltage on the CIT;
3. under proper operating conditions (see section 4).

Table 5 - Technical characteristics of the Analyzers. Continued

Characteristic	Meaning for Analyzer	
	ASK75.10.20.2	ASK75.10.20.12
Maximum number of consecutively executed charge-discharge cycles of the HIT during one run of the HIT testing program	9 999 998	
Interface for connecting to a computer	Ethernet, 10 Mbps	
Minimum requirements for computer and monitor	P600, 128RAM, Win 2000, XP, 7, 8, 8.1, 10, 11, Ethernet	
	Monitor with a resolution of at least 1280 x 1024 pixels	
Supply voltage Power	220 V $\pm$ 10 %, 50/60 Hz	
consumption from the network, W Overall	no more than 170	no more than 920
dimensions, mm, no more than		
- height	88	88
- width	135	482
- length (depth)	310	312
Weight without packaging, kg	2,8	8,8

Table 6 - Metrological characteristics of Analyzers
ASK75.10.20.2 and ASK75.10.20.12

Characteristic	Meaning
Limits of permissible absolute error of reproduction DC voltage at terminals HIT2	$\pm 0,050$
Limits of permissible absolute error of reproduction of DC voltage at the terminals of HIT2	$\pm 0,050$
Limits of permissible absolute error in reproducing the direct current through the HIT1 A - in the range from minus 0.1 to plus 0.1 - in the range from minus 0.5 to plus 0.5 - in the range from minus 2.0 to plus 2.0 - in the range from minus 10.0 to plus 10.0	$\pm 0,0005$ $\pm 0,0025$ $\pm 0,01$ $\pm 0,05$
Limits of permissible absolute error in measuring direct current through HIT1 A - in the range from minus 0.1 to plus 0.1 - in the range from minus 0.5 to plus 0.5 - in the range from minus 2.0 to plus 2.0 - in the range from minus 10.0 to plus 10.0	$\pm 0,0005$ $\pm 0,0025$ $\pm 0,01$ $\pm 0,05$
Limits of permissible absolute error of reproduction units of time ² , Limits of	± 0.001 t _{zad} ± 0.2
permissible absolute error in measuring time intervals ³ ,	± 0.001 · t _{ism} ± 0.2

Notes:

1. under proper operating conditions (see section 4);
2. t_{zad} – specified duration of the testing program step, s;
3. t_{ism} – measured duration of the time interval, s.

8 PREPARATION FOR THE FIRST USE

Before turning on the Analyzer for the first time, you must make sure that:

- The device has no external mechanical damage after transportation, there are no loose parts inside the device;
- The wires used have undamaged insulation;
- The socket for connecting the device complies with the "Euro" standard and has a working (not fictitious) grounding conductor connected to the building's grounding circuit or to the grounding circuit of the power supply network, or there is a working (not fictitious) grounding circuit in the room;
- The voltage in the socket for connecting the device is in the range from 198 to 242 V ($220\text{ V} \pm 10\%$), and has a frequency of 47 to 63 Hz;
- The User's computer to which the device will be connected has a free Ethernet port, or this computer is part of a local computer network that has a switch with a free port to which the Analyzer will be connected;
- The computer monitor has a resolution of at least 1280 pixels horizontally and at least 1024 pixels vertically.

Procedure for preparing the device for operation:

1. Install the device on an open, flat, stable surface, or secure it in a standard 19" rack (only for ASK75.10.20.12). In the latter case, first remove the four instrument feet located at the bottom of the device body;
2. Make sure that there are no foreign objects around the device that could interfere with normal cooling of the Analyzer;
3. Make sure that there is a fire extinguisher behind the appliance and at least 1 m away from the rear there are no flammable materials or liquids on the instrument panel;
4. Make sure that the power switch (position 12, Figure 1 and Figure 5) is in position "0" ("Off");
5. If the socket for connecting the device complies with the "Euro" standard and has a working (not dummy) grounding conductor, using the 220 V power cord from the delivery set (see item 4 of Table 1 and Table 2) or similar, connect the device to a 220 V power supply with grounding;
6. If the power supply network does not have a working grounding conductor, however there is a working (not fictitious) grounding circuit in the room, using the grounding contact on the rear panel of the device (position 17, Figure 2 and Figure 5, thread M5 x 0.8) and a copper conductor with a cross-section of at least 1 mm² (not included in the delivery set), securely connect the device body to the grounding circuit of the room. Then, using the 220 V power cord from the delivery set (see item 4 of Table 1 and Table 2) or similar, connect the device to the 220 V power supply network;
7. If the 220 V power supply network does not have a working grounding conductor and there is no working grounding circuit in the room, operation of the device is not permitted (see section 3);
8. Turn on the device using the power switch on the front panel (position 12, Figure 1 and Figure 5);
9. Make sure that the device is turned on and the "**PWR**" indicator is lit. (position 7, figure 1 and figure 4), the "**ALM**" indicator (position 8, figure 1 and figure 4) goes out two seconds after switching on and does not blink, the indicator

"MEM" (MEM) (position 9, Figure 1 and Figure 4) also went out two seconds after switching on and does not light up;

10. If the device is connected to the User's computer directly, connect the Ethernet interface cable from the delivery set (see item 5 of Table 1 and Table 2) between the connector on the front panel of the device (position 13, Figure 1 and Figure 4) and the corresponding connector on the User's computer;
11. If the device is connected to a computer network via switch, connect the Ethernet interface cable from the delivery set (see item 5 of Table 1 and Table 2) between the connector on the front panel of the device (position 13, Figure 1 and Figure 4) and the switch port that is part of the local computer network to which the User's computer is connected;
12. Install and configure the software (Computer Interface) on the User's computer (see section 9), launch the Computer Interface;
13. On the "Settings" page of the Computer Interface in the area "Instrument(s) Type" select the type of instrument you are using – ASK75.10.20.2 or ASK75.10.20.12;
14. Establish a connection between the Computer Interface and the Analyzer by clicking the "Connect" button in the "TCP" area on the "Settings" page or by checking the "Connect automatically" box (see subsection 19.7);
15. Make sure that the connection between the Computer Interface and the device, which will be indicated by the continuously changing value of "Analyzer clock" in the "Additionally" area (see subsection 19.7), as well as the continuous glow of the **"PC"** indicator (position 10, figure 1 and figure 4).

The procedure for preparing, performing and completing HIT testing

1. Make sure that the selected channel of the device is in the **"IDLE"** mode. (IDLE) (ready for operation) (see subsection 19.1);
2. Connect the tested chemical analyzer to the selected channel of the device (see section 10);
3. Set up the HIT testing program (see subsections 19.2, 19.3);
4. Start the HIT testing process (see subsection 19.2);
5. Wait until the measurement results are received (see subsections 19.4, 19.5, 19.6);
6. After the testing of the HIT is completed and the device channel switches to the mode **"TST CMPLT"** (TEST COMPLETE), press the "Stop" button for the corresponding channel of the instrument to return the channel to the **"IDLE"** mode of the instrument (see subsection 19.1);
7. If necessary, you can interrupt the testing process of the HIT manually by pressing the "Stop" button for the corresponding channel of the device (see subsection 19.1).

9 INSTALLING AND SETTING UP THE SOFTWARE

To control the device, create and launch the HIT testing program (see section 11), to monitor the progress of testing, and to receive, process and display the measurement results, application software is used, which is installed on the User's computer – the **Ym128 Computer Interface**.

The Ym128 Computer Interface is supplied with the Analyzer, or can be taken from the website <https://battery-analyzers.ru/> on the Internet.

The Computer Interface does not require special installation, but must be copied to the User's computer in any location on the hard drive.

The Computer Interface folder contains three files:

- *Ym128.exe* – executable program file (Computer Interface);
- *Settings.bin* – a settings file that stores the selected operating modes, values entered into the input fields of the Computer Interface, etc. This file must be available for rewriting during the operation of the Computer Interface;
- *Settings.bi2* – a backup copy of the settings file, necessary in case of damage settings file *Settings.bin*.

ATTENTION! If it is necessary to operate several devices under control from one User computer, the software for each device must be copied to a separate folder on the computer's hard drive, including the settings files *Settings.bin* and *Settings.bi2*!

When you first launch the Computer Interface, on the "Settings" page (see subsection 19.7) in the "Instrument(s) Type" area, you must select the type of the instrument being used - ASK75.10.20.2 or ASK75.10.20.12, then in the "TCP" area you must specify the correct IP address of the connected Analyzer, which is indicated on the front panel of the instrument housing (position 14, Figure 1 and Figure 4). This address usually has the value 192.168.100.xxx, where xxx is a random number specific to a specific instance of the device.

In order to establish a connection with the Analyzer, it is generally necessary that the first three digits of the IP address of the User's computer and the same digits of the IP address of the device match. If they do not match, establishing a connection between the device and the computer becomes impossible. In this case, there are several options for solving the problem:

- Change (set manually) the IP address of the User's computer to 192.168.100.yyy, where yyy is a random value from 0 to 255, different from the value xxx in the IP address of the device;
- Add to the computer's local network connection settings
The user has an additional subnet with IP address 192.168.100.yyy;
- When ordering the device, please specify the desired IP address of the device or the IP address of the computer
The User with whom the device is to work. In this case, the device will be supplied with the required IP address or an IP address whose first three digits will match the IP address of the User's computer.

Even with the correct settings of the IP address of the User's computer, some antivirus programs can block the data flow between the device and the Computer Interface running on the User's computer. In this case, in the antivirus program settings, it is necessary to add the *Ym128.exe* program to the list of programs that are allowed to exchange data with external devices.

To configure the connection of the device to the computer, it is recommended Contact your system administrator.

More detailed information on setting up the device connection to the computer You can find it on our website <https://battery-analyzers.ru/> on the Internet.

10 CONNECTING THE TESTED HIT AND CONTROLLING THE DEVICE

Each channel of the Analyzer has four terminals for connecting the HIT - two power and two potential. The charge and discharge current through the HIT is supplied through the power terminals, while the voltage on the HIT is controlled using potential terminals (see Figure 1 and Figure 4, as well as Table 3). This allows us to eliminate calculations of the voltage drop on the connecting wires, and to perform measurements of the parameters of the chemical energy source being studied.

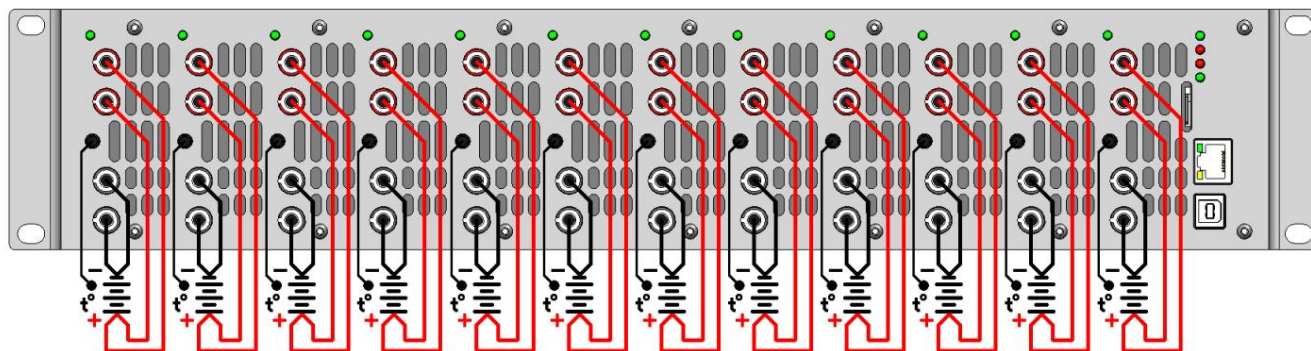


Figure 9 - Connection diagram of the tested HIT to the device

The connecting wires must have a cross-section of at least 1 mm². . All contacts must be securely fastened and checked before switching on.

To control the temperature of each chemical source, the Analyzers are equipped with remote chemical source temperature sensors (see item 3 of Table 1 and Table 2). The temperature sensors are connected to the corresponding connectors on the front panel of the Analyzer (position 4, Figure 1 and Figure 4). For the most accurate control of the chemical source temperature, reliable thermal contact should be ensured between the sensor and the chemical source being studied.

Without connecting the HIT temperature sensor, the HIT temperature measurement result will always be 0 °C, and the protective functions of the device when the HIT temperature goes beyond the specified limits (see subsection 19.2 and section 22) will not work.

To increase the testing current of the CIT beyond the capabilities of a single channel of the device, the channels of a specific device can be connected in parallel into groups. To do this, connect the connecting wires from several channels of the Analyzer to one CIT at once. In addition to the mechanical connection of the wires, for all slave channels it is necessary to enable the mode "|| to the previous" (Parallel to the previous channel) on the "Tasks" page of the Computer Interface in the "Channel mode" area (see subsection 19.2).

For the ASK75.10.20.2 Analyzer, the only option for parallel operation of channels is when both channels of the device work together. In this case, the device will be able to test only one HIT at a time, but the maximum HIT testing current increases to 20 A.

For the ASK75.10.20.12 Analyzer, the options for combining channels are more diverse. Thus, when combining every two channels of the device, the number of simultaneously tested chemical and electrical devices is reduced to six ($2 + 2 + 2 + 2 + 2 + 2$), but the maximum charge (discharge) current of each tested chemical and electrical device increases to 20 A. When combining channels into groups of four, the number of simultaneously tested chemical and electrical devices is reduced to three ($4 + 4 + 4$), but the maximum charge (discharge) current of each chemical and electrical device increases to 40 A. Combining all twelve channels of the ASK75.10.20.12 Analyzer allows testing only one HIT at a time, however, the maximum charge (discharge) current of the HIT in this case increases to 120 A, the maximum charge power – to 600 W, and the maximum discharge power – to 900 W. Other options for combining the channels of the ASK75.10.20.12 Analyzer are also possible, for example, $2 + 4 + 3 + 3$.

To combine several channels of the device, it may be necessary to use aluminum or copper buses, fixed to the terminals of the tested HIT (buses are not included in the delivery set).

An example of combining the channels of the ASK75.10.20.12 Analyzer into six channels for increasing the charge and discharge current to 60 A is shown in Figure 10.

The first channel in the group is configured as the master, the rest as slaves. The voltage on the CIT is measured and controlled only by the leading (first) channel in the group, therefore the potential wires from the leading (first) channel should be connected as close as possible to the terminals of the tested CIT. The potential wires from the other channels in the group should also be connected to the CIT circuits, but the place their connections are not so important.

Also, it is the leading (first) channel in the group that measures the temperature of the heat exchanger. Sensors. There is no need to connect the temperature sensor to the slave channels.

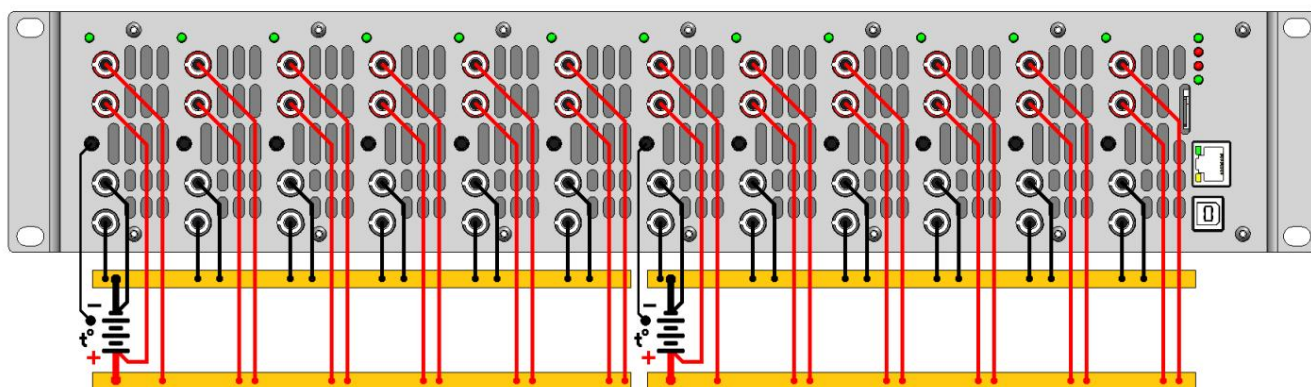


Figure 10 - Connecting the tested HITs to the ASK75 Analyzer.10.20.12
with parallel operation of channels according to the scheme (6 + 6)

When combining the Analyzer channels into a group, the HIT testing program is configured only for the leading (first) channel in the group, while the channel's capabilities for current, power, and HIT testing resistance are expanded in proportion to the number of channels in the group (see subsection 19.2).

The Analyzer channels are electrically isolated from each other (see Section 6). This allows for the simultaneous testing of individual elements of the CCS assembled into a battery on one device without disassembling the battery. An example of connection for such cases is shown in Figure 11.

ATTENTION! Do not apply voltage to the terminals of the device channels relative to the device body (ground circuit) with an absolute value of more than 100 V. This may cause the device to fail! (see Section 3)

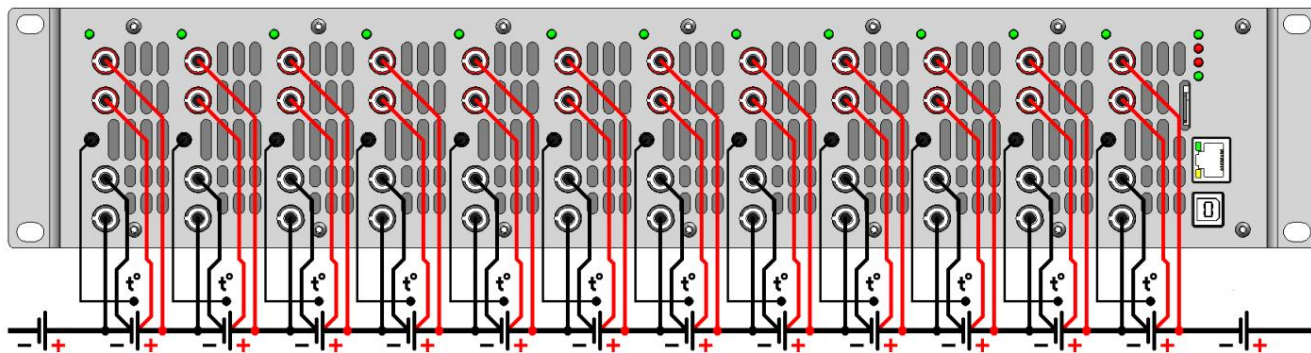


Figure 11 - Testing of the HIT connected in a series battery

11 HIT TESTING PROGRAM

11.1 General instructions

The HIT testing program generally consists of a sequence of identical cycles performed a specified number of times.

In turn, each cycle consists of a set of individual modes – program steps executed sequentially one after another. In general, the cycle contains at least one charging step of the tested CCS (for example, CHCC*) and at least one discharge step (for example, DCHCC). The maximum number of steps in a cycle is 42.

Before the cyclic part of the program is executed, one, two or three steps of preparation for testing can be executed once. These steps are executed at the beginning of the testing of the CCT and can be used, for example, for preliminary charging of the tested battery before the main cyclic part of the test program, so that its full capacity can be determined from the first cycle.

After the specified number of cycles have been completed, one, two or three steps of the HIT test program completion may be performed once immediately before the end of the test. These steps may be used, for example, for the final charge of the battery under test, so that immediately after testing the battery is fully charged and ready for its intended use.

The total number of configurable steps in the HIT testing program is $3 + 42 + 3 = 48$.

For each step of the HIT testing program (with some exceptions, see subsections 11.15, 11.16, 11.17 and 11.18), the following device operating modes can be selected:

- Charging the HIT with a given constant current (CHCC);
- Charging the heat source with a given constant power (CHCP);
- Additional charging of the CHIT at a given constant voltage (CHCV);
- Discharge of the CHIT with a given constant current (DCHCC);
- Discharge of a given constant power CHIT (DCHCP);
- Discharge of the chemical energy source to a given constant resistance (DCHCR);
- Additional discharge of the chemical battery at a given constant voltage (DCHCV);
- Relaxation;
- Voltage sweep (U) on the heat exchanger over time;
- Current sweep (I) through the HIT in time;
- Power sweep (P) on the heat source over time;
- Time sweep of the discharge resistance of the chemical energy source (R);
- Voltage pulse mode (U);
- Current pulse mode (I);
- Power pulse mode (P);
- Discharge resistance pulse mode of the chemical energy source (R);
- Work according to the voltage table (U);
- Work on the current table (I);
- Work according to the power table (P);
- Work according to the resistance table (R);
- Recording the voltage at the terminals of the device (U recorder);
- Pause;
- Maximum Power Point Search (MPPS) mode.

The block diagram of the HIT testing program is shown in Figure 12.

A more detailed description of each possible mode, a description of the permissible values of the specified parameters and the accompanying criteria for the end of the step, as well as the necessary recommendations for selecting modes, their sequence and values of operating parameters, are given in the following subsections, as well as in subsection 19.2.

* see the list of accepted abbreviations and designations.

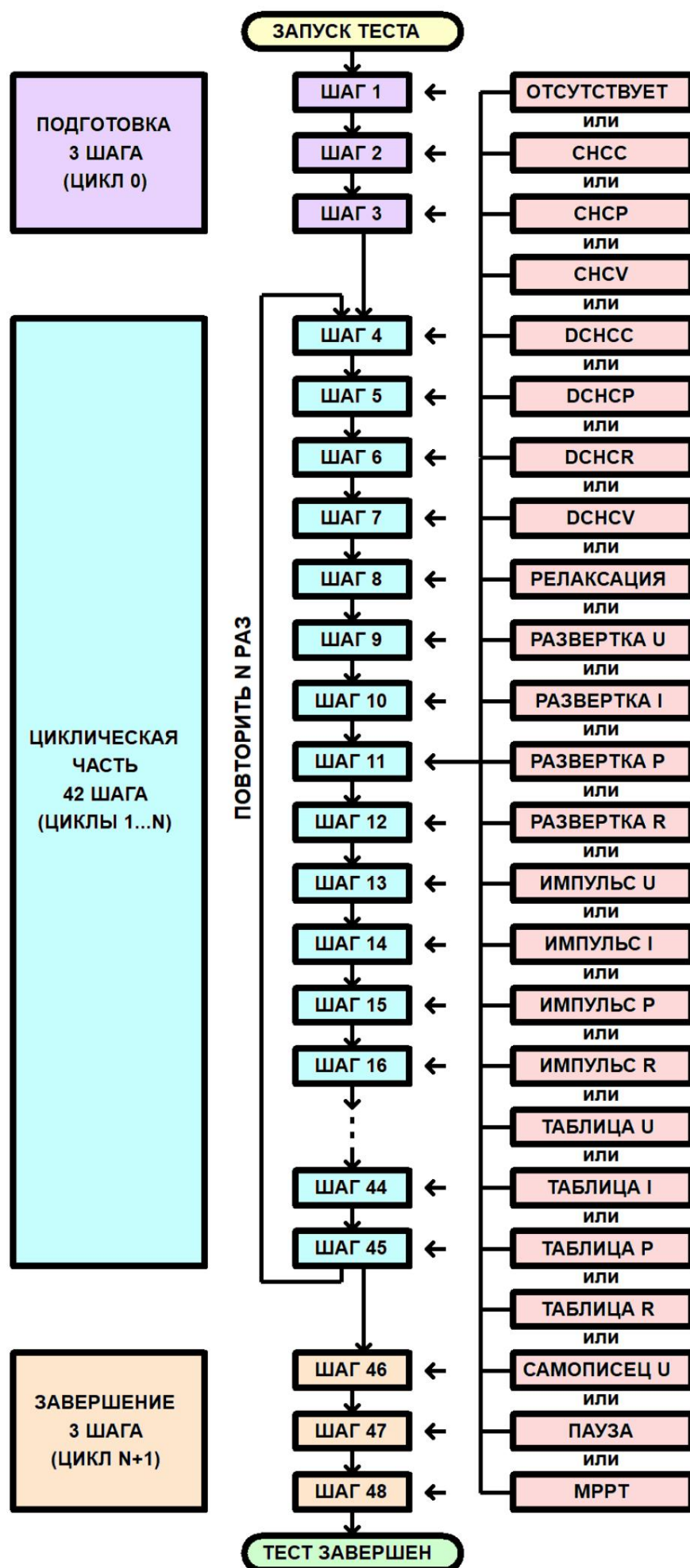


Figure 12 - HIT testing program

11.2 Signs of the parameters of testing of the HIT

This subsection describes the rules used in the work of Analyzers to determine the signs of the parameters of the HIT testing. The rules used are introduced to simplify the understanding of the processes taking place and to reduce the probability of errors when setting the testing program and analyzing the measurement results.

The voltage on the tested CHIT, V , is considered positive if the potential of the "+" terminal of the tested chemical energy source exceeds the potential of the "-" terminal.

The voltage on the tested chemical current source is considered negative if the potential of the "+" terminal of the tested chemical current source is less than the potential of the "-" terminal.

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are designed to operate in the field only positive voltages on the heat exchanger.

The current through the tested CCS, A , is considered positive if it flows out of the "+" terminal of the Analyzer channel, passes through the tested CCS and returns back to the device through the channel terminal, generally leading to an increase in voltage on the CCS or to a charge of the CCS. With a positive current, the Analyzer is a current source, and the tested CCS is a receiver or consumer.

The current through the tested CCS is considered negative if it flows out of the tested CCS, flows into the "+" terminal of the Analyzer channel, passes through the device and returns back to the tested CCS through the "-" terminal of the Analyzer channel, generally leading to a decrease in voltage on the CCS or to the discharge of the CCS. With a negative current, the tested CCS is a current source, and the Analyzer is a receiver or consumer.

The power on the tested CCS, W , is considered positive if a positive current passes through the tested CCS. Positive power generally leads to an increase in voltage on the CCS, or to a charge of the CCS.

The power on the tested CCS is considered negative if a negative current passes through the tested CCS. Negative power generally leads to a decrease in voltage on the CCS, or to the discharge of the CCS.

The discharge resistance of the chemical power supply, Ω , is always considered positive, despite the fact that the chemical power supply discharge is carried out by a negative current. This is convenient when setting the testing program and analyzing the obtained measurement results.

The charge, $A\ h$, is considered positive if it is transferred from the Analyzer to the tested CCS. In this case, the device is the source of the charge, and the tested CCS is the receiver or consumer.

The charge is considered negative if it is transferred from the tested CHIT to the Analyzer. In this case, the tested CHIT is the source of the charge, and the device is the receiver or consumer.

Energy, $W\ h$, is considered positive if it is transferred from the Analyzer to the tested chemical measuring device. In this case, the device is the energy source, and the tested chemical measuring device is the receiver or consumer.

Energy is considered negative if it is transferred from the tested CIT to the Analyzer. In this case, the tested CIT is the energy source, and the Analyzer is the receiver or consumer.

11.3 Constant Current Charging (CHCC)

At the constant current charging step, the Analyzer ensures that a stabilized positive current (charging current) of the specified value I_{SET} flows through the tested CIT (see Figure 13).

The current value is maintained by the Analyzer at the level specified by the User and does not depend on the voltage on the CHIT (within the operating range of the voltage and power of the Analyzer channel, see Table 5).

The inclusion of the DC charging step in the HIT testing program is performed by selecting the "**Charge CC**" mode from the drop-down list of operating modes for a specific step for a specific channel of the device (see subsection 19.2).

The charging current for the CHSS step can be set by the User at any level from the working range of the Analyzer (see Table 5), while the selection of the hardware current sub-range is carried out by the device automatically, in accordance with the specified current value.

When the device channels operate in parallel (see Section 10), the operating current range increases proportionally to the number of channels operating in parallel in the group. In this case, the specified current is divided equally between all channels in the group.

The DC charging step has three configurable step end criteria (see Figure 13):

- The voltage on the heat sink reaches the specified step end voltage U_{END} ;
- Expiration of the allotted time of the **TEND** step;
- Registration of the voltage drop on the power supply by a specified value (criterion "-dU").

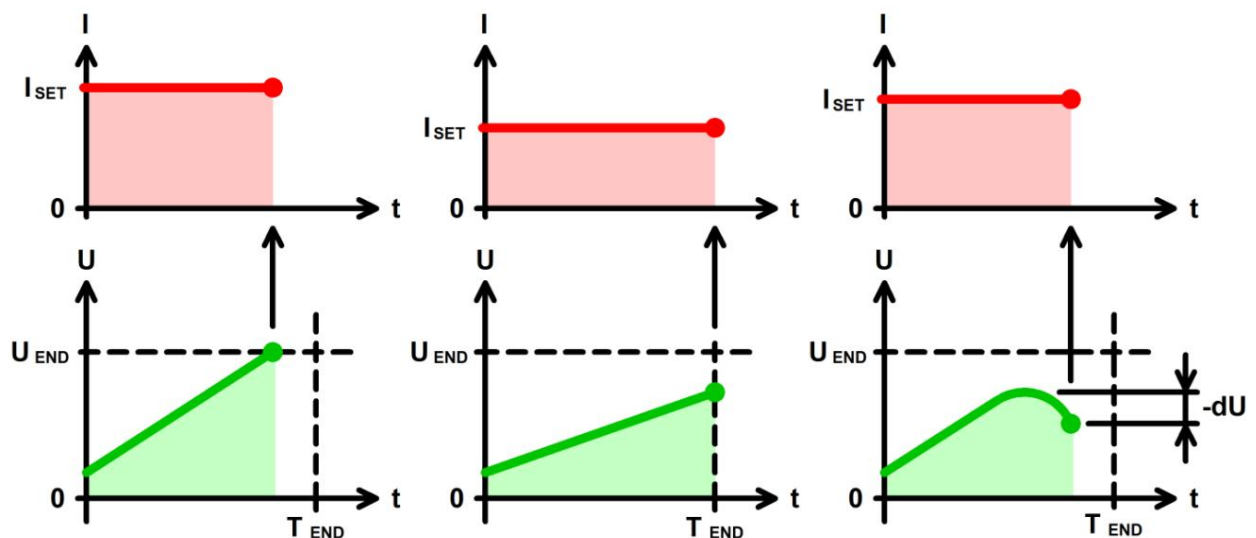


Figure 13 - Charging the CHIT with constant current (CHCC)

The specified step end voltage U_{END} is selected by the User when forming the testing program from the operating range of the device (see Table 5). The criterion for the end of the step upon reaching the specified voltage on the CIT can be

disabled, leaving the end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The maximum step duration (allocated step time) **TEND** is selected by the User from the working range of the Analyzer (see Table 5). The criterion for ending a step after a specified time cannot be disabled, but it is possible to set a deliberately larger value for the maximum step time than expected when

testing of the HIT so that this criterion never works.

The end-of-charge criterion "-dU" is typical for Ni-Cd and Ni-MH batteries. batteries of these types, as well as batteries of other types, are characterized by

increase in voltage during the charging process. However, after Ni-Cd or Ni-MH the battery is fully charged, the incoming charge energy can no longer be stored by the battery, and begins to be released in the battery structure as heat. The battery temperature begins to rise, while the voltage on the battery decreases slightly (instead of the expected increase, there is a decrease in the voltage on the battery, or "negative increment", "-dU").

Thus, the beginning of the voltage decrease on the Ni-Cd or Ni-MH battery during the charging process, it indicates that the battery has reached full charge and that the charging step needs to be stopped.

The "-dU" criterion can be enabled or disabled when specifying the test program. The magnitude of the negative voltage increment at which the device stops the charging step, is set simultaneously for all charging steps and can be selected from the operating range of the device (see Table 5).

The end of the CHCC step occurs when any (at least one) is triggered from the specified termination criteria.

During the CHCC step, the Analyzer can measure the ESR of the tested CHC by the method periodic interruption of current (see subsection 13.2).

In the measurement results files (see section 15) the DC charging step has the step marker "**CCC**".

11.4 Constant Charge Power (CHCP)

At the constant power charging step, the Analyzer ensures that a positive current (charging current) of such a value flows through the tested CIT that the value of the electric power on the CIT corresponds to the specified power value **PSET** (see figure 14).

The magnitude of the charging power is maintained by the Analyzer at the level specified by the User and does not depend on the voltage on the HIT (within the operating voltage range and the selected hardware subrange of the Analyzer channel current, see Table 5).

The inclusion of the constant power charging step in the HIT testing program is performed by selecting the "**Charge CP**" mode from the drop-down list of operating modes for a specific step for a specific channel of the device (see subsection 19.2).

The charging power for the CHCP step can be set by the User at any level from the working range of the Analyzer used (see Table 5).

The maximum current that the Analyzer can provide at the constant power charging step corresponds to the maximum current of the device (see Table 5). Thus, if the specified charging power value is 50 W, and the current voltage on the CCS is less than 5 V, the device will limit the CCS charging current at 10 A until the voltage on the CCS increases above 5 V.

When the Analyzer channels operate in parallel (see Section 10), the working range of the charging power increases proportionally to the number of parallel-operating device channels. In this case, the specified power is divided equally between all channels in group.

The constant power charging step has three configurable step end criteria (see Figure 14):

- The voltage on the heat sink reaches the specified step end voltage **UEND**;
- Expiration of the allotted time of **the TEND step**;
- Registration of the voltage drop on the power supply by a specified value (criterion "-dU").

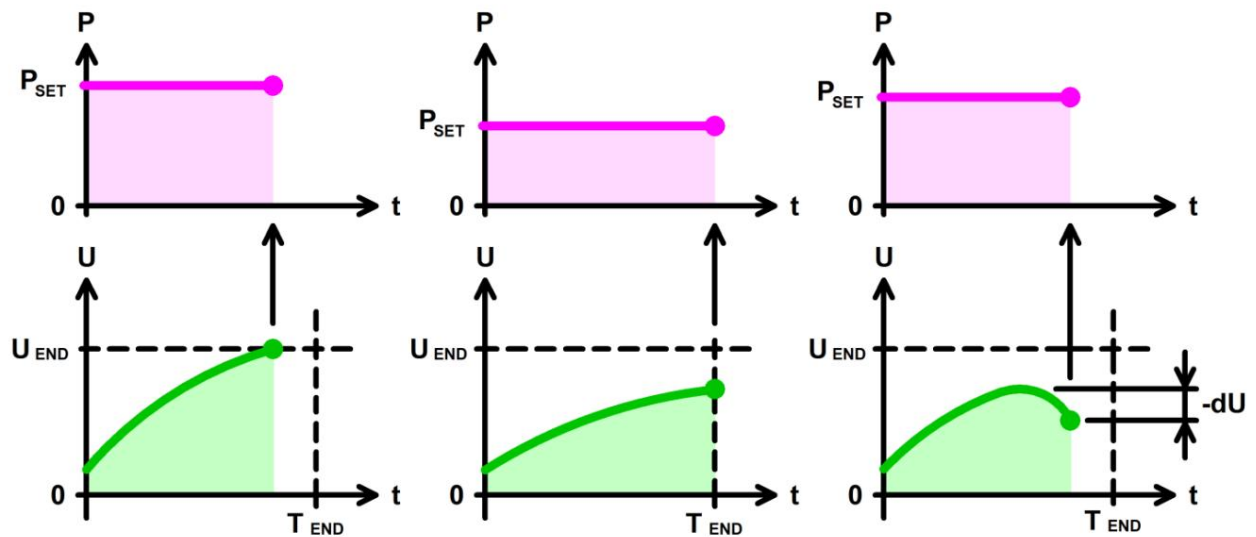


Figure 14 - Constant Power Charge (CHCP)

The specified step end voltage **UEND** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for ending the step upon reaching the specified voltage on the CIT

can be disabled by leaving the end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The selection of the hardware current subrange is performed by the device automatically, in accordance with the specified value of the charging power, as well as the specified minimum voltage limitation on the CHCP "Umin:" (see Figure 104). For example, with a specified charging power of 10 W and a specified minimum voltage limitation on the CHCP of 8 V, the current sub-range of 2 A will be automatically selected, since during the step execution the CHCP charging current will not exceed $10 \text{ W} / 8 \text{ V} = 1.25 \text{ A}$. If the minimum voltage limitation on the CHCP is not used (the minimum voltage limitation input field is left empty when specifying the CHCP testing program, see subsection 19.2), the minimum voltage value from the Analyzer operating range is used to calculate the maximum current during the CHCP step execution and the selection of the hardware current sub-range - 0.1 V, which will almost always lead to the selection of the highest current sub-range - 10 A.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umin:" (see Figure 104)!

The maximum step duration (allocated step time) **TEND** is selected by the User from the operating range of the device (see Table 5). The criterion for ending a step after a specified time cannot be disabled, but it is possible to set a deliberately larger value for the maximum step time than expected during testing of the CIT,

so that this criterion never works.

The "-dU" charge end criterion is similar to the constant current charge step (CHCC) (see subsection 11.3).

The end of the CHCP step occurs when any (at least one) of the specified end criteria is triggered.

During the CHCP step, the Analyzer can measure the ESR of the tested CHIT using the method periodic interruption of current (see subsection 13.2).

In the measurement results files (see section 15) the constant power charging step has a step marker "**CCP**".

11.5 Constant Voltage Charge (CHCV)

At the constant voltage recharging step, the Analyzer ensures stabilization of the voltage at the terminals of the tested power supply unit at a given **USET** level (see Figure 15).

The capabilities of the Analyzer channel to maintain a given voltage on the CIT are limited by the maximum current for the selected hardware current subrange and the maximum power of the device channel (see Table 5).

The inclusion of the constant voltage recharging step in the HIT testing program is performed by selecting the "**Charge CV**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The recharge voltage of the HIT can be set by the User at any level from the operating range of the device (see Table 5), however, this voltage is usually set at the voltage level of the end of the previous step.

The constant voltage recharge step has two adjustable criteria end of step (see Figure 15):

- Reducing the charging current through the HIT to the specified step end current **IEND**;
- Expiration of the allotted time of **the TEND step**.

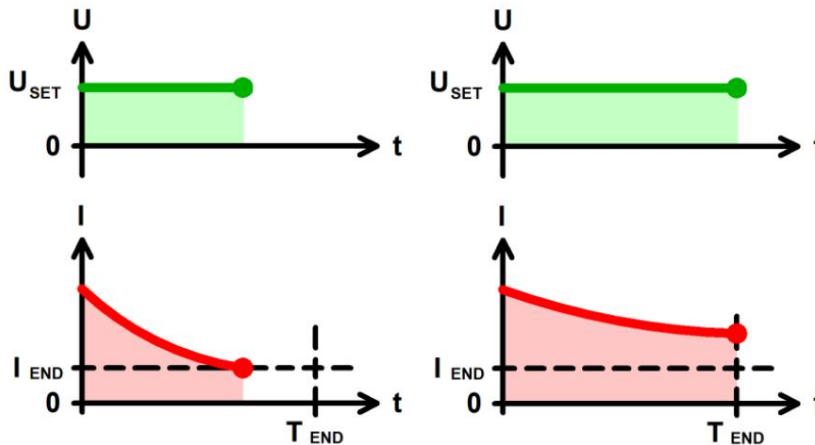


Figure 15 - Recharging of the CHC at constant voltage (CHCV)

The specified step end current **IEND** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for the end of the step by reducing the current through the CIT to the specified step end current can be disabled, leaving the end current input field empty when specifying the HIT testing program (see subsection 19.2).

When the Analyzer channels operate in parallel (see Section 10), the range of the end current setting increases proportionally to the number of parallel-operating device channels. Current between parallel-operating channels in CHCV mode distributed randomly within the capabilities of the Analyzer channels for current and power.

The hardware current subrange for the CHCV step is kept the same as it was selected in the previous step of the CHCV test program. If the CHCV step is the first of those specified in the preparation part of the test program, or the first of those specified in the cyclic part of the CHCV test program, the current subrange is selected based on the larger in absolute value of the specified constraints

maximum charge current of the HIT "**ICmax**:" and the maximum discharge current of the HIT "**IDmax**:" (see Figure 104). If at least one of the restrictions of the maximum charge current or maximum discharge current of the HIT is not used (the restriction input field

maximum charge current or the maximum discharge current input field is left empty when specifying the HIT testing program), the senior hardware current sub-range is automatically selected.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "ICmax:" and "IDmax:" (see Figure 104)!

The maximum step duration (allocated step time) **TEND** is selected by the User from the operating range of the device (see Table 5). Step end criterion after the specified time has elapsed, it is impossible to switch it off, however, it is possible to set a deliberately greater value of the maximum step time than expected during testing of the HIT, so that this criterion never works.

The end of the CHCV step occurs when any (at least one) is triggered from the specified termination criteria.

In the measurement results files (see section 15), the recharging step at constant voltage has a step marker "**CCV**".

11.6 Direct Current Discharge (DCHCC)

At the DC discharge step, the Analyzer ensures that a stabilized negative current (discharge current) of the specified value **ISSET** flows through the tested CIT (see Figure 16).

The current value is maintained by the Analyzer channel at a level specified by the User and does not depend on the voltage on the CIT (within the operating range of the voltage and power of the device channel, see Table 5).

The DC charging step is included in the HIT testing program by selecting the "**Discharge CC**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The discharge current for the DCHCC step can be set by the User at any level from the working range of the Analyzer (see Table 5), while the selection of the hardware current sub-range is carried out by the device automatically, in accordance with the specified current value.

When the Analyzer channels operate in parallel (see Section 10), the operating range of the discharge current increases proportionally to the number of channels operating in parallel. In this case, the specified current is divided equally between all channels in the group.

The DC discharge step has two adjustable step end criteria (see Figure 16):

- The voltage on the heat sink reaches the specified step end voltage **UEND**;
- Expiration of the allotted time of the **TEND** step.

The specified step end voltage **UEND** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for ending the step upon reaching the specified voltage on the CIT

can be disabled by leaving the end voltage input field empty when specifying the CCS testing program (see subsection 19.2). This allows the tested CCS to be discharged down to 0 V (for example, when testing supercapacitors).

The maximum step duration (allocated step time) **TEND** is selected by the User from the operating range of the device (see Table 5). Step end criterion after the specified time has elapsed, it is impossible to switch it off, however, it is possible to set a deliberately greater value of the maximum step time than expected during testing of the HIT, so that this criterion never works.

The end of the DCHCC step occurs when any (at least one) is triggered from the specified termination criteria.

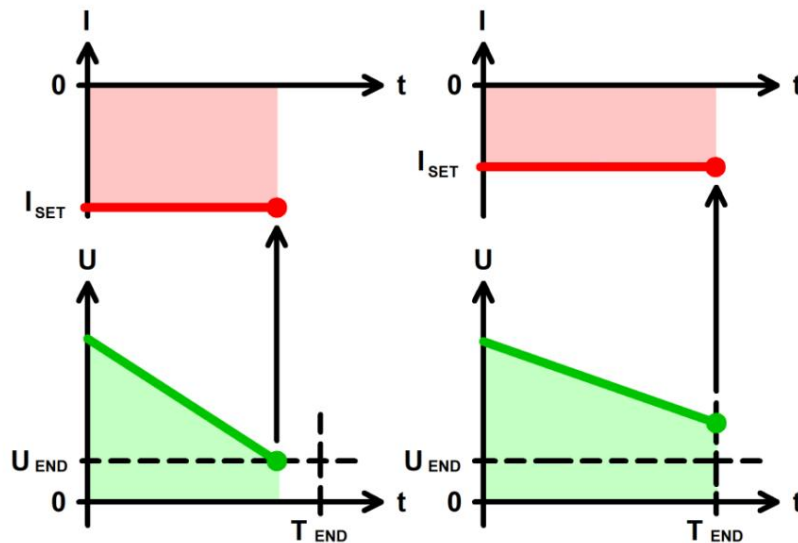


Figure 16 - Discharge of the CHIT with direct current (DCHCC)

During the DCHCC step, the Analyzer can measure the ESR of the tested CHIT by the method periodic interruption of current (see subsection 13.2).

In the measurement results files (see section 15) the DC discharge step has the step marker **"DCC"**.

11.7 Constant Power Discharge (DCHC $\ddot{y}$)

At the constant power discharge step, the Analyzer ensures that a negative current (discharge current) of such a value flows through the tested CCS so that the value of the electrical power on the CCS corresponds to the specified power value **PSET** (see Figure 17).

The discharge power value is maintained by the Analyzer at a level specified by the User and does not depend on the voltage on the HIT (within the operating voltage range and the selected hardware subrange of the Analyzer channel current, see Table 5).

The inclusion of the constant power discharge step in the HIT testing program is performed by selecting the **"Discharge CP"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The discharge power for the D $\ddot{y}$ $\ddot{y}$ $\ddot{y}$ $\ddot{y}$ P step can be set by the User at any level from the working range of the Analyzer (see Table 5).

The maximum current that the Analyzer can provide at the constant power discharge step corresponds to the maximum current of the device. Thus, if the specified discharge power value is 40 W, and the current voltage on the CCS is less than 4 V, the device channel will limit the CCS discharge current at 10 A.

When the Analyzer channels operate in parallel (see Section 10), the working range of the discharge power increases proportionally to the number of channels operating in parallel. In this case, the specified power is divided equally between all channels in the group.

The constant power discharge step has two adjustable step end criteria (see Figure 17):

- The voltage on the heat sink reaches the specified step end voltage **UEND**;
- Expiration of the allotted time of **the TEND step**.

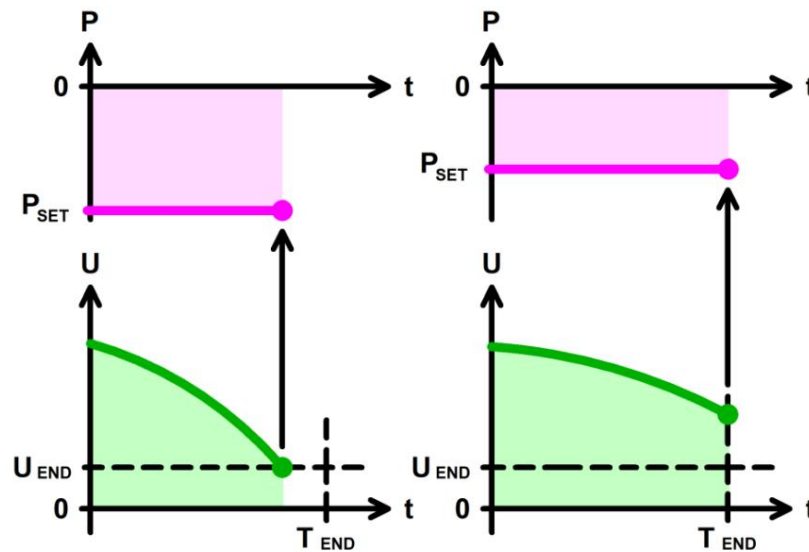


Figure 17 - Discharge of the constant power CHIT (DCHCP)

The specified step end voltage **U_{END}** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for ending the step upon reaching the specified voltage on the CIT

can be disabled by leaving the end voltage input field empty when specifying the CCS testing program (see subsection 19.2). This allows the tested CCS to be discharged down to 0 V (for example, when testing supercapacitors).

The selection of the hardware current subrange is performed by the device automatically based on the specified discharge power and the greater of the specified step end voltage **U_{END}** and the specified minimum voltage limitation on the HIT "U_{min}:" (see Figure 104). For example, with a specified discharge power of 16 W, a specified step end voltage of 10 V and a specified minimum voltage limitation on the CCS of 8 V, the current subrange of 2 A will be automatically selected, since during the step execution the CCS discharge current will not exceed $16 \text{ W} / 10 \text{ V} = 1.6 \text{ A}$. If the criterion for ending the step upon reaching the specified voltage on the CCS is not used (the input field for the end voltage value is left empty when specifying the CCS testing program) and the minimum voltage limitation on the CCS is also not used (the input field for the minimum voltage limitation on the CCS is also left empty), the minimum voltage value from the Analyzer operating range of 0.1 V is used to calculate the maximum current through the CCS during the DCHCP step execution and the selection of the hardware current subrange, which will almost always lead to the selection of the highest current subrange - 10 A.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "U_{max}:" (see Figure 104)!

The maximum step duration (allocated step time) **T_{END}** is selected by the User from the operating range of the device (see Table 5). Step end criterion after the specified time has elapsed, it is impossible to switch it off, however, it is possible to set a deliberately greater value of the maximum step time than expected during testing of the HIT, so that this criterion never works.

The end of the DCHC_y step occurs when any (at least one) is triggered from the specified termination criteria.

During the DCHC_y step, the Analyzer can measure the ESR of the tested HIT by the method periodic interruption of current (see subsection 13.2).

In the measurement results files (see section 15) the constant power discharge step has a step marker "**DCP**".

11.8 Discharge to Constant Resistance (DCHCR)

At the discharge step to a given constant resistance, the Analyzer simulates resistive load with a given electrical resistance value, ensuring the flow of a negative current (discharge current) through the tested CIT of such a value that the input resistance of the device corresponds to the given resistance value **RSET** (see Figure 18).

The value of the discharge resistance is maintained by the Analyzer at a level specified by the User and does not depend on the voltage on the CIT (within the operating range of voltage, current and power, see Table 5).

The inclusion of the constant resistance discharge step in the HIT testing program is performed by selecting the **"Discharge CR"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The discharge resistance for the DCNR step can be set by the User at any level from the working range of the Analyzer (see Table 5).

The maximum current that the Analyzer can provide at the constant resistance discharge step corresponds to the maximum current of the device. Thus, if the specified discharge resistance value is 0.5 Ohm, and the current voltage on the HIT is more than 5 V, the device will limit the HIT discharge current at 10 A until the voltage on the HIT drops to less than 5 V.

When the Analyzer channels operate in parallel (see Section 10), the working range of the discharge resistance decreases proportionally to the number of channels operating in parallel. In this case, the specified resistance is multiplied equally between all channels in the group.

The constant resistance discharge step has two adjustable criteria end of step (see Figure 18):

- Reducing the voltage on the heat sink to the specified step end voltage **UEND**;
- Expiration of the allotted time of **the TEND step**.

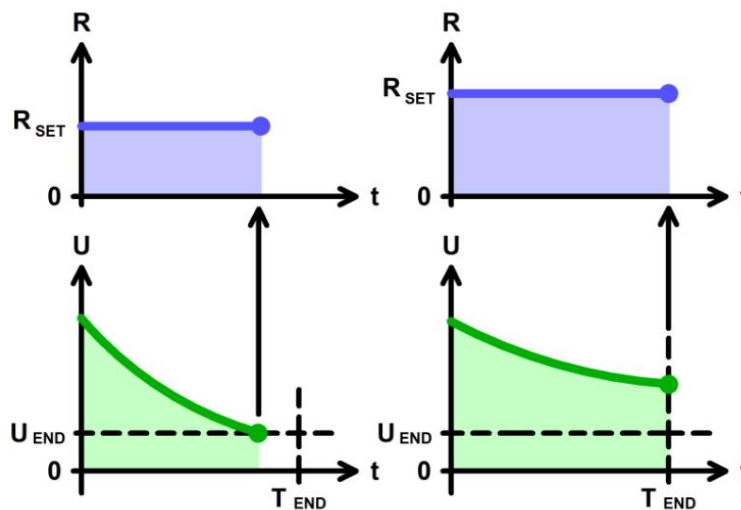


Figure 18 - Discharge of the CHIT to constant resistance (DCHCR)

The specified step end voltage **UEND** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for ending the step upon reaching the specified voltage on the CIT

can be disabled by leaving the end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The selection of the hardware current subrange is performed by the device automatically based on a given discharge resistance and a given maximum limitation voltage on the HIT "Umax:" (see Figure 104). If the maximum voltage limitation

voltage on the CIT is not used (the input field for the maximum voltage limit on the CIT is left empty when specifying the CIT testing program), the maximum operating voltage of the device is used to calculate the maximum current through the CIT during the execution of the DCHCR step and selection of the hardware current subrange (see Table 5).

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umax:" (see Figure 104)!

The maximum step duration (allocated step time) **TEND** is selected by the User from the operating range of the device (see Table 5). Step end criterion after the specified time has elapsed, it is impossible to switch it off, however, it is possible to set a deliberately greater value of the maximum step time than expected during testing of the HIT, so that this criterion never works.

The end of the DCHCR step occurs when any (at least one) is triggered from the specified termination criteria.

During the DCHCR step, the Analyzer can measure the ESR of the tested CIT by the method periodic interruption of current (see subsection 13.2).

In the measurement results files (see section 15) the discharge step is constant The resistance has a step marker "**DCR**".

11.9 Constant Voltage High Voltage (DCHCV)

At the stage of additional discharge at constant voltage, the Analyzer ensures stabilization of the voltage at the terminals of the tested chemical energy source at a given **USET** level (see Figure 19).

The Analyzer's capabilities to maintain a given voltage on the CIT are limited by the maximum current for the selected hardware current subrange and the maximum power of the device channel (see Table 5).

The inclusion of the step of additional discharge at constant voltage in the testing program of the HIT is carried out by selecting the "**Discharge CV**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The pre-discharge voltage of the HIT can be set by the User at any level from the working range of the Analyzer (see Table 5), however, usually this voltage is set at the level of the end voltage of the previous step.

The constant voltage recharge step has two adjustable criteria end of step (see Figure 19):

- Reduction of the absolute value of the discharge current through the HIT to a specified current end of step **IEND**;
- Expiration of the allotted time of the **TEND** step.

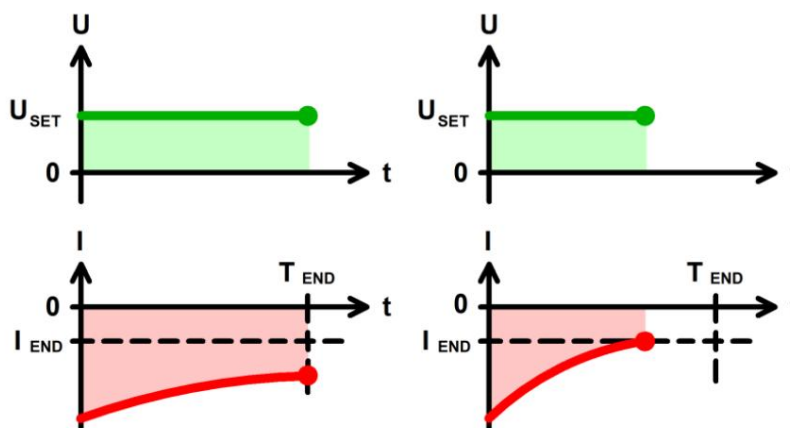


Figure 19 - DCHCV recharge of the HIT

The specified step end current **IEND** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The step end criterion for decreasing the current through the CCS to the specified step end current can be disabled by leaving the end current input field empty when specifying the CCS testing program (see subsection 19.2).

When the Analyzer channels operate in parallel (see Section 10), the range of the end current setting increases proportionally to the number of channels operating in parallel. Current between parallel channels in DCHCV mode

distributed randomly within the capabilities of the Analyzer channels for current and power.

The hardware current subrange for the DCHCV step remains the same as it was selected in the previous step of the HIT test program. If the DCHCV step is the first of those specified in the preparation part of the testing program, or the first of those specified in the cyclic part of the testing program of the chemical equipment, the current subrange is selected based on the greater absolute value of the specified maximum charge current limit of the HIT "ICmax:" and maximum discharge current limit of the HIT "IDmax:" (see Figure 104). If at least one of the maximum charge current limits

or the maximum discharge current of the HIT is not used (restriction input field maximum charge current or maximum discharge current input field left empty when specifying the testing program), the highest hardware current subrange is automatically selected.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "ICmax:" and "IDmax:" (see Figure 104)!

The maximum step duration (allocated step time) **TEND** is selected by the User from the operating range of the device (see Table 5). The criterion for ending a step after a specified time has elapsed cannot be disabled, but it is possible to set a deliberately larger value for the maximum step time than expected during testing of the CIT, so that this criterion never triggers.

The end of the DCHCV step occurs when any (at least one) is triggered from the specified termination criteria.

In the measurement results files (see section 15), the step of additional discharge at a constant voltage has a step marker "**DCV**".

11.10 Relaxation

At the relaxation step (rest), the Analyzer interrupts the current supply through the CHIT for a specified time, continuing to measure and record the voltage at the terminals of the heat exchanger.

The change in voltage on the CIT at the relaxation step is caused by internal processes occurring in the HIT itself.

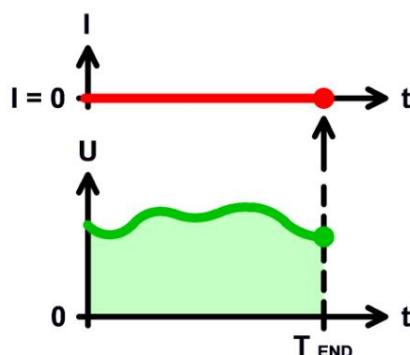


Figure 20 - Relaxation Step

Unlike the "Recorder U" mode (see subsection 11.20), during the relaxation step the power circuits of the device channel remain on, and the HIT will pass through additional leakage current associated with ensuring the operation of the internal power electronic circuits of the device (no more than 100 μ A).

For short relaxation times, the contribution of this additional current will be negligible, however, if the capacitance of the CCS is very small or the specified relaxation time is large enough, this leakage current may make a certain contribution to the additional discharge of the CCS during the relaxation step. For such experiments, it is recommended to use the "U recorder" mode instead of the relaxation mode (see subsection 11.20).

The relaxation step is included in the HIT testing program by selecting the **"Rest"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The relaxation step has a single criterion for the end of the step – the expiration of the specified relaxation time. The specified relaxation time is selected by the User from the working range of the device (see Table 5).

In the measurement result files (see section 15) the relaxation step has the marker **"RLX"**.

11.11 Voltage sweep

At the voltage sweep step, the Analyzer will linearly change (sweep) the voltage on the CIT from the initial sweep voltage **U1** to the final voltage with a specified rate of change (sweep) of the sweep voltage **U2** (see Figure 21).

In this case, the device channel operates as a source or as a consumer of energy with a stabilized voltage at the output (input) equal to the current required instantaneous sweep voltage. If the current voltage on the CCS differs from the instantaneous required sweep voltage, the device will give the CCS a charge current or take from the CCS a discharge current of such a level that the voltage on the CCS becomes equal to the instantaneous required one.

The capabilities of the Analyzer channel to maintain the required instantaneous voltage on the HIT during the voltage sweep are limited by the maximum current for the selected hardware current subrange and the maximum channel power (see Table 5).

The voltage sweep step is included in the HIT testing program by selecting the **"Scanning U"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The initial sweep voltage **U1** can be set by the User at any level from the operating range of the device (see Table 5), but usually this voltage is set at the voltage level of the end of the previous step.

The final sweep voltage **U2** can also be set by the User at any level from the working range of the device.

The specified voltage sweep rate must be such that the duration of the sweep step fell within the operating range of the device (see Table 5).

The hardware subrange of the current in the voltage sweep mode cannot be pre-determined by the device and must be set manually by the User when creating the HIT testing program (see subsection 19.2).

When the Analyzer channels operate in parallel (see section 10), the current between the parallel operating channels in a group is distributed randomly within the current and power capabilities of the Analyzer channels.

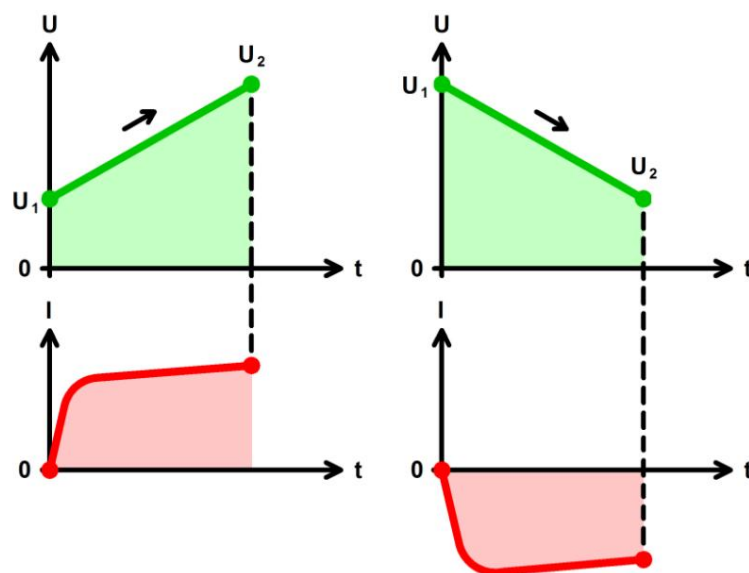


Figure 21 - Voltage sweep on the heat sink

A sequential combination of two voltage sweep steps with corresponding initial and final sweep voltages in the cyclic part of the testing program (see the example in subsection 19.2) allows the CCS to be investigated using the Cyclic VoltAmperometry (CVA) method (see Figure 22).

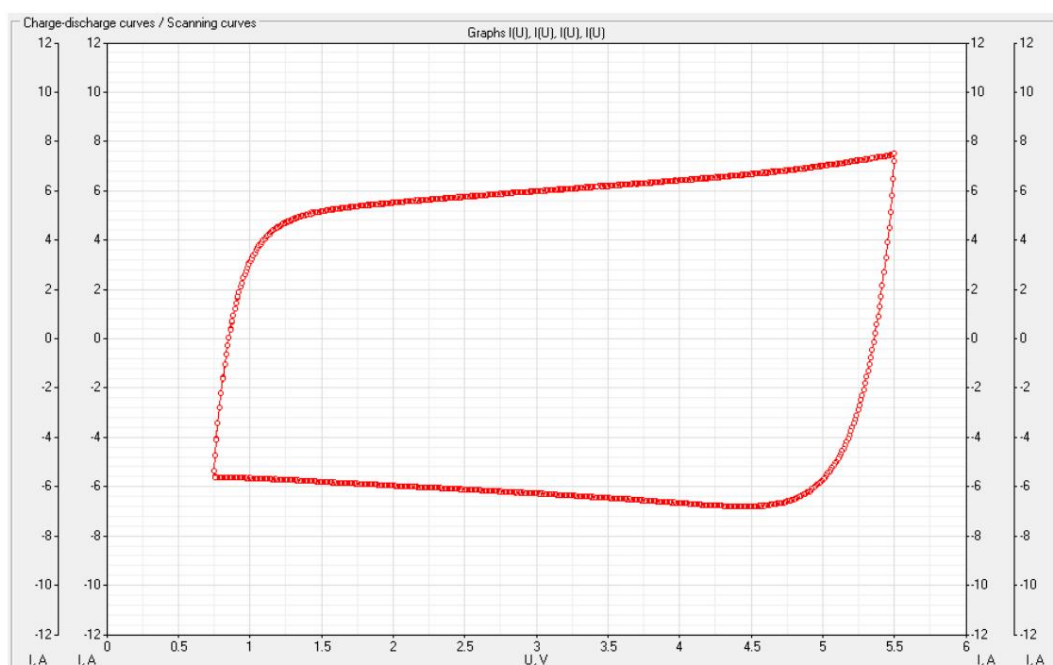


Figure 22 - Testing of the HIT using the CVA method

In the measurement results files (see section 15) the voltage sweep step has step marker **"SCU"**.

11.12 Current sweep

At the current sweep step, the Analyzer will linearly change (sweep) the current through the current source from the initial current **I1** to the final current **I2**, with a specified rate of change (sweep) of the current, regardless of the voltage on the current source (within the operating range of voltage and power of the Analyzer channel) (see Figure 23).

The inclusion of the current sweep step in the HIT testing program is performed by selecting the **"Scanning I"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The initial sweep current **I1**, as well as the final sweep current **I2**, can be selected to any value from the operating current range of the device (see Table 5), and can be either positive (charge current) or negative (discharge current), or equal to zero.

When the device channels operate in parallel (see Section 10), the operating current range increases proportionally to the number of channels operating in parallel in the group. In this case, the instantaneous sweep current is divided equally between all channels in the group.

The specified current sweep rate must be such that the current sweep step duration falls within the operating range of the device (see Section 5).

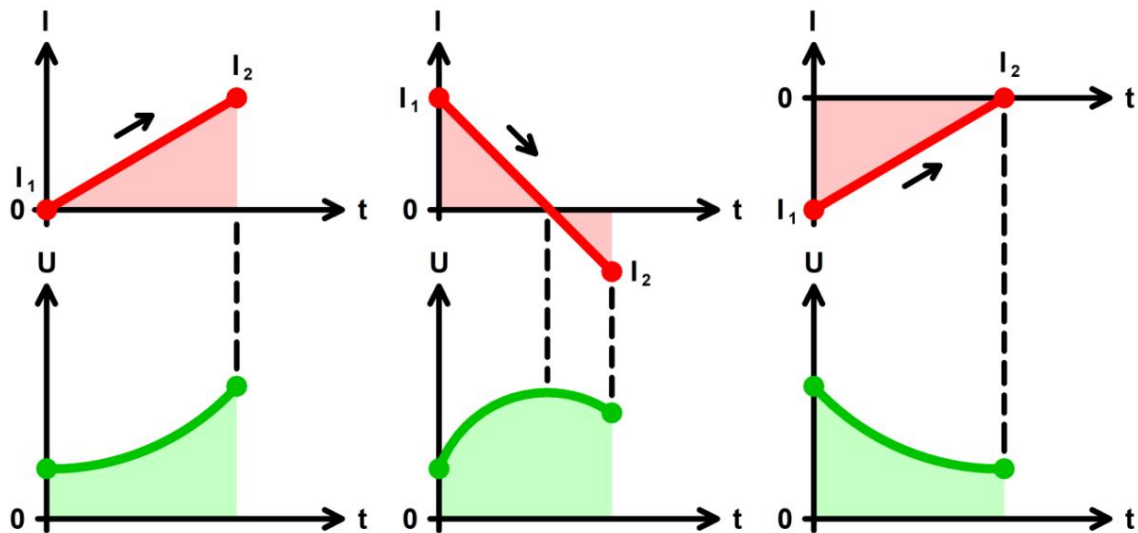


Figure 23 - Current sweep through the HIT

The hardware current subrange is selected automatically based on the larger of the current values **I1** and **I2** (in absolute value).

During the current sweep step, the Analyzer can measure the ESR of the tested CCT using the periodic current interruption method (see subsection 13.2).

In the measurement results files (see section 15), the current sweep step through the HIT has the step marker "SCI".

11.13 Power sweep

At the power sweep step, the Analyzer will linearly change (sweep) the power on the power supply from the initial power **P1** to the final power **P2**, with a specified rate of change (sweep), regardless of the voltage on the power supply (within the operating voltage range and the selected hardware current subrange of the Analyzer channel, see Table 5) (see Figure 24).

The power sweep step is included in the HIT testing program by selecting the "**Scanning P**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The initial sweep power **P1**, as well as the final sweep power **P2**, can be selected to any value from the operating range of the device (see Table 5), and can be either positive (charge power) or negative (discharge power), or equal to zero.

When the device channels operate in parallel (see Section 10), the operating power range increases proportionally to the number of channels operating in parallel in the group. In this case, the instantaneous sweep power is divided equally between all channels in the group.

The specified current sweep rate must be such that the step duration the power sweep fell within the operating range of the device (see Table 5).

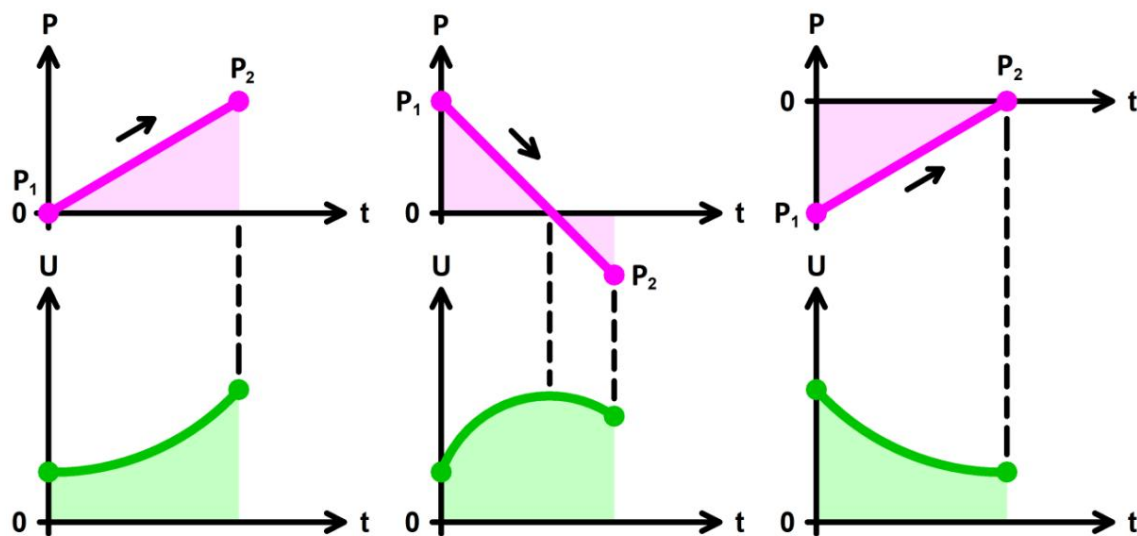


Figure 24 - Power sweep on the HIT

The selection of the hardware current sub-range is performed by the device automatically based on the larger of the power values **P1** and **P2** (in absolute value) and the specified minimum voltage limit on the power supply "Umin:" (see Figure 104). For example, with a specified initial sweep power of 18 W, a specified final sweep power minus 10 W and a specified minimum voltage limitation on the HCS of 10 V, the current subrange of -2 A will be automatically selected, since during the step execution the current through the HCS will not exceed $18 \text{ W} / 10 \text{ V} = 1.8 \text{ A}$. If the minimum voltage limitation on the HCS is not used (the minimum voltage limitation input field on the HCS is left empty when specifying the HCS testing program), the minimum voltage value from the Analyzer operating range of 0.1 V is used to calculate the maximum current through the HCS during the power sweep step and selection of the hardware current sub-range, which will almost always lead to the selection of the highest current sub-range - 10 A.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umin:" (see Figure 104)!

During the power sweep step, the Analyzer can measure the ESR of the tested power supply using the periodic current interruption method (see subsection 13.2).

In the measurement results files (see section 15), the power sweep step on the power analyzer has the step marker "**SyP**".

11.14 Resistance sweep

At the resistance sweep step, the Analyzer will linearly change (sweep) the resistance of the discharge of the chemical current source from the initial resistance **R1** to the final resistance **R2**, with a specified rate of change (sweep), regardless of the voltage on the chemical current source (within the operating range of voltage, current and power for the Analyzer channel) (see Figure 25).

The inclusion of the resistance scanning step in the HIT testing program is performed by selecting the "**Scanning R**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The initial sweep resistance **R1**, as well as the final sweep resistance **R2**, can be selected to any value from the operating range of the device (see Table 5).

When the device channels operate in parallel (see section 10), the operating range of the sweep resistance decreases proportionally to the number of channels operating in parallel.

channels of the device in the group. In this case, the instantaneous sweep resistance is multiplied equally between all channels in the group.

The specified current sweep rate must be such that the step duration the resistance sweep fell within the operating range of the device (see Table 5).

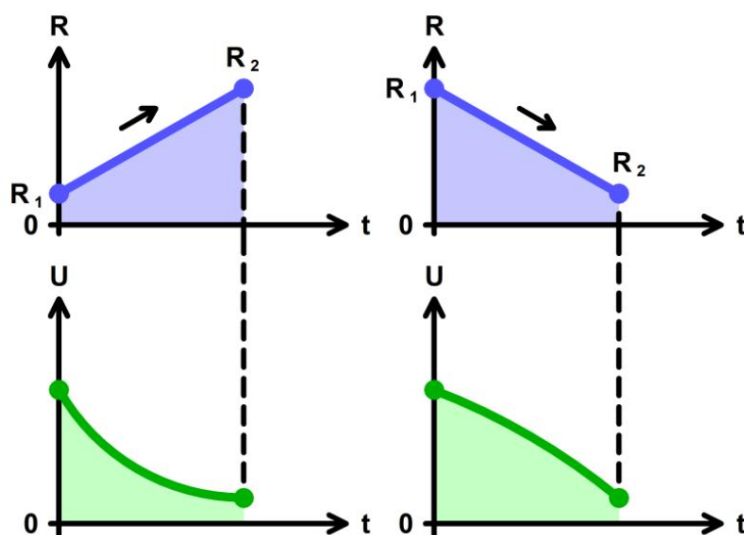


Figure 25 - Sweep of the discharge resistance of the HIT

The hardware voltage subrange for the resistance sweep step is selectable by the device automatically based on the smaller of the specified resistances **R1** and **R2** and the specified limitation of the maximum voltage on the CCS "Umax:" (see Figure 104). If the limitation of the maximum voltage on the CCS is not used (the input field for the limitation of the maximum voltage on the CCS is left empty when specifying the CCS testing program), the maximum operating voltage of the device is used to calculate the maximum current through the CCS during the resistance sweep step and selection of the hardware current subrange (see Table 5).

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umax:" (see Figure 104)!

During the resistance sweep step, the Analyzer can measure ESR tested CIT by the periodic current interruption method (see subsection 13.2).

In the measurement results files (see section 15), the resistance sweep step The HIT discharge has a step marker "**SCR**".

11.15 Voltage pulse mode

At the voltage pulse step, the Analyzer supplies alternating voltage pulses of specified values **U1** and **U2** to the tested CIT, where voltage **U1** is supplied for a specified time **T1**, and voltage **U2** is supplied for a specified time **T2** (see Figure 26).

In this case, the device channel operates as a source or as a consumer of energy with a stabilized voltage at the output (input) equal to the required voltage for a specific pulse. If the current voltage on the CCS differs from the required voltage of a specific pulse, the device will give the CCS a charge current or take from the CCS a discharge current of such a level that the voltage on the CCS becomes equal to the required one.

The Analyzer's ability to maintain a given voltage on the CIT for each pulse is limited by the maximum current for the selected hardware current subrange and the maximum power of the device channel (see Table 5).

The inclusion of the voltage pulse step in the testing program of the power supply system is performed by selecting the "**Impulses U**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

Voltage **U1**, as well as voltage **U2**, can be selected from any of the device's operating range (see Table 5).

The specified duration **T1** and **T2** of current pulses can be selected from any of the device's operating range (see Table 5).

The hardware subrange of the current in the voltage pulse mode cannot be predetermined by the device and must be set manually by the User when forming the HIT testing program (see subsection 19.2). The specified hardware subrange of the current (the same) is used for both voltage pulses **U1** and voltage pulses **U2**.

When the Analyzer channels operate in parallel (see section 10), the current in each pulse between the parallel operating channels in a group is distributed randomly within the current and power capabilities of the Analyzer channels.

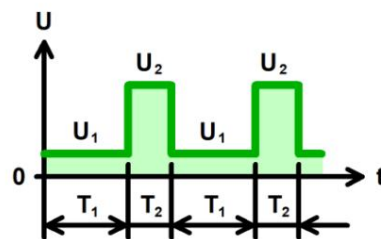


Figure 26 - Voltage pulse mode

The voltage pulse step has a single configurable step end criterion - after the allotted step time **TEND has expired**. Allotted step time **TEND** is selected by the User from the permissible range of the device (see Table 5).

During the execution of the voltage pulse step, the Analyzer measures and records ESR of the tested CIT at each voltage change (from **U1** to **U2** and back, from **U2** to **U1**). For more details, see subsection 13.3.

When setting the test program (see subsection 19.2), the voltage pulse step setting takes up two fields (two lines). For this reason, the voltage pulse mode cannot be selected for the last preparation step (step #3), the last step of the cyclic part (step #45), and the last test completion step (step #48).

In the measurement result files (see section 15) the voltage pulse step has the step marker "**IPU**".

11.16 Current pulse mode

At the current pulse step, the Analyzer supplies alternating current pulses of specified values **I1** and **I2** through the tested CIT, whereby current **I1** is supplied for a specified time **T1**, and current **I2** is supplied for a specified time **T2** (see Figure 27).

The magnitude of the current through the CIT during the supply of each current pulse is maintained by the Analyzer at a level specified by the User and does not depend on the voltage on the CIT (within the operating range of voltage and power of the Analyzer channel).

The inclusion of the current pulse step in the testing program of the heat-transfer device is performed by selecting the "**Impulses I**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

Current **I1**, as well as current **I2**, can be selected to any value from the operating range of the device (see Table 5), and can be either positive (charge current) or negative (discharge current).

When the device channels operate in parallel (see section 10), the operating current range increases proportionally to the number of channels operating in parallel in the group.

In this case, the current of each pulse is divided equally between all channels in the group.

The specified duration **T1** and **T2** of current pulses can be selected from any of the device's operating range (see Table 5).

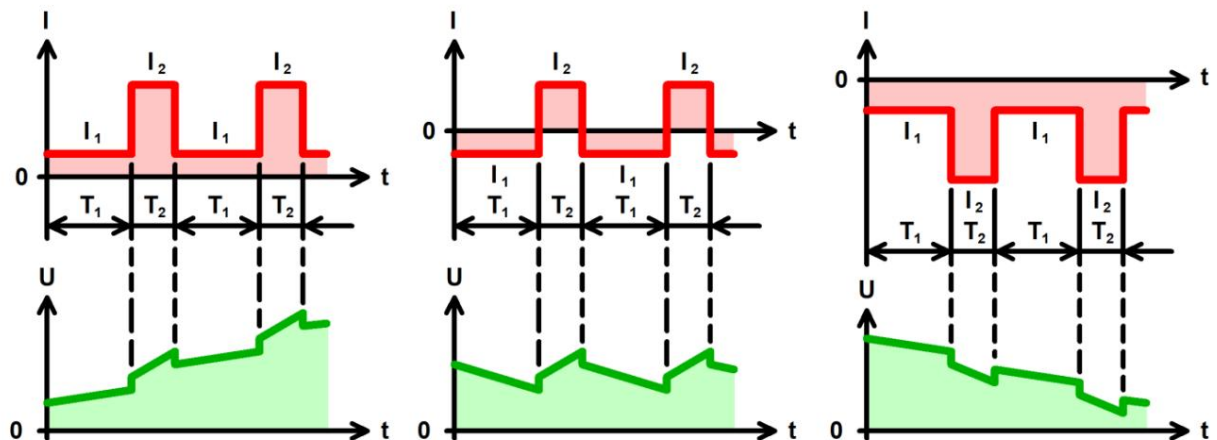


Figure 27 - Current pulse mode

The hardware current subrange for each current pulse is selectable automatically, based on the value of the specified current (absolute value).

The current pulse step has two adjustable step end criteria (see Figure 28):

- The voltage on the heat sink reaches the specified step end voltage **UEND**;
- Expiration of the allotted time of the **TEND** step.

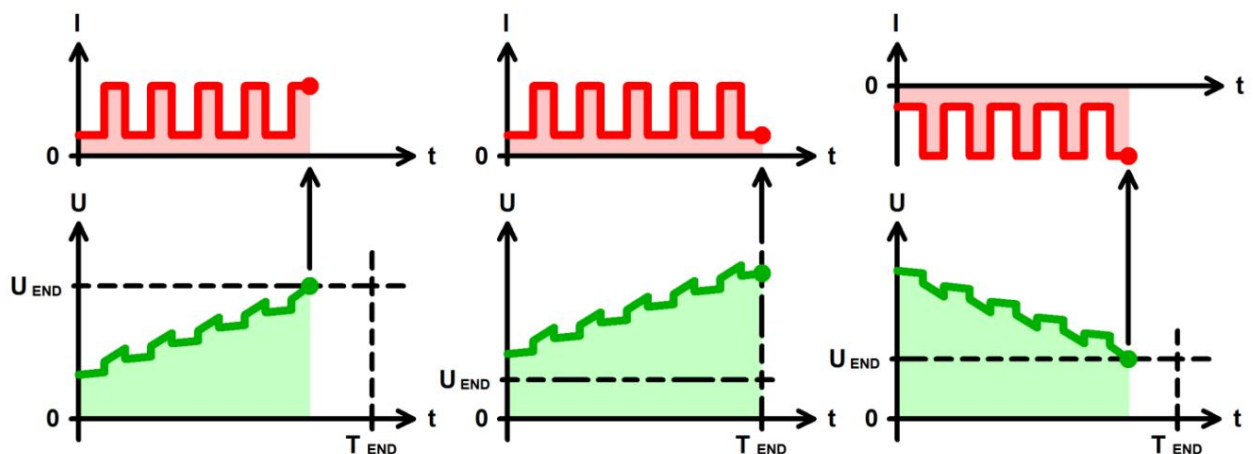


Figure 28 - Current pulse step termination criteria

The specified step end voltage **UEND** is selected by the User when forming the testing program for the HIT from the working range of the Analyzer (See Table 5). The end of the current pulse step occurs as soon as the current voltage on the current measuring device touches the specified level of step end voltage **UEND**. The criterion for the end of the step upon reaching the specified voltage on the current measuring device can be disabled by leaving the end voltage input field empty when specifying the testing program (see subsection 19.2).

The maximum step duration (allocated step time) **TEND** is selected by the User from the permissible range of the device (see Table 5). End Criterion

It is not possible to disable the step after a specified time, but it is possible to set

a deliberately larger value of the limiting step time than expected at testing of the HIT so that this criterion never works.

The end of the current pulse step occurs when any (at least one) of the specified termination criteria.

During the execution of the current pulse step, the Analyzer measures and records the ESR of the tested CIT at each change of current (from **I1** to **I2** and back, from **I2** to **I1**). For more details, see subsection 13.3.

When setting the test program (see subsection 19.2), the current pulse step setting takes up two fields (two lines). For this reason, the current pulse mode cannot be selected for the last preparation step (step #3), the last step of the cyclic part (step #45), and the last test completion step (step #48).

In the measurement result files (see section 15) the current pulse step has the step marker "**IPI**".

11.17 Power Pulse Mode

At the power pulse step, the Analyzer supplies alternating power pulses of specified values **P1** and **P2** to the tested CIT, whereby power **P1** is supplied for a specified time **T1**, and power **P2** is supplied for a specified time **T2** (see Figure 29).

The power value on the power supply unit during the supply of each power pulse is maintained by the Analyzer at the level specified by the User and does not depend on the voltage on the power supply unit (within the operating voltage range and the selected hardware subrange of the Analyzer channel current, see Table 5).

The inclusion of the power pulse step in the testing program of the power supply system is performed by selecting the "**Impulses P**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

Power **P1**, as well as power **P2**, can be selected to any value from the operating range of the device channel, and can be either positive (charge power) or negative (discharge power).

When the device channels operate in parallel (see Section 10), the operating power range increases proportionally to the number of parallel channels in the group. In this case, the power of each pulse is divided equally between all channels.

in a group.

The specified duration **T1** and **T2** of power pulses can be selected from any of the device's operating range (see Table 5).

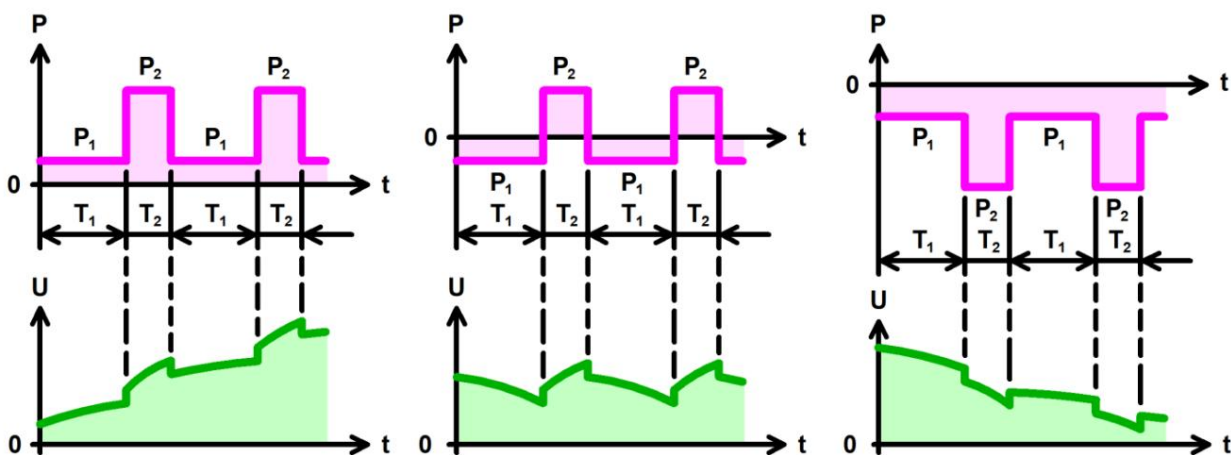


Figure 29 - Power Pulse Mode

The hardware current subrange for each power pulse is selected automatically, based on the value of the specified power (in absolute value) and the specified minimum voltage limit on the power supply "Umin:" (see Figure 104).

For example, with a specified power of **P1** minus 17 W and a specified minimum voltage limit on the HIT of 10 V, the hardware will be automatically selected
current sub-range is 2 A, since during the execution of pulse **P1** the current through the power supply unit will not exceed $17 \text{ W} / 10 \text{ V} = 1.7 \text{ A}$. The hardware current sub-range for power pulse **P2** is **selected in a similar manner**. If the minimum voltage limitation on the power supply unit is not used (the minimum voltage limitation input field on the power supply unit is left empty when specifying the power supply unit testing program), the minimum voltage value from the working range of the Analyzer is used to calculate the maximum current through the power supply unit during the execution of the power pulse step and the selection of the hardware current sub-range – 0.1 V, which will almost always lead to the selection of the highest current sub-range – 10 A.

ATTENTION! For the correct selection of the appropriate current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umin:" (see Figure 104)!

The power pulse step has two configurable step end criteria (see Figure 30):

- The voltage on the heat sink reaches the specified step end voltage **UEND**;
- Expiration of the allotted time of the **TEND** step.

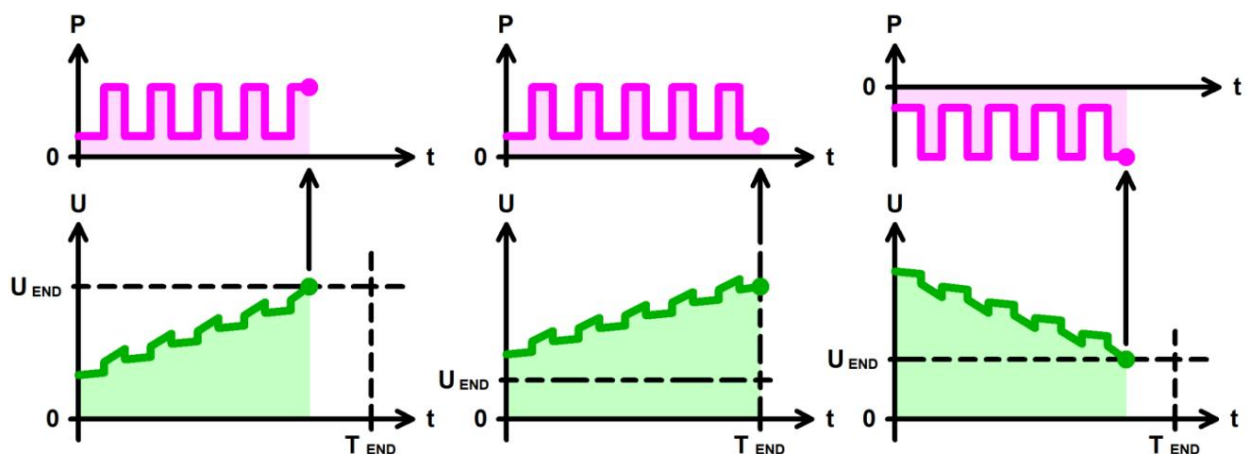


Figure 30 - Power pulse step termination criteria

The specified step end voltage **UEND** is selected by the User when forming the testing program for the power supply unit from the working range of the Analyzer (see Table 5). The power pulse step ends as soon as the current voltage on the power supply unit touches the specified step end voltage level **UEND**. The criterion for the step end upon reaching the specified voltage on the power supply unit can be

disabled, leaving the end voltage input field empty when specifying the test program (see subsection 19.2).

The maximum step duration (allocated step time) **TEND** is selected by the User from the permissible range of the device (see Table 5). The criterion for ending a step after a specified time has elapsed cannot be disabled, but it is possible to set a deliberately larger value for the maximum step time than expected during testing of the CIT, so that this criterion never triggers.

The end of the power pulse step occurs when any (at least one) of the specified termination criteria.

During the execution of the power pulse step, the Analyzer measures and records ESR of the tested CIT at each change of power (from **P1** to **P2** and back, from **P2** to **P1**). For more details, see subsection 13.3.

When setting up the test program (see subsection 19.2), the power pulse step setting takes up two fields (two lines). For this reason, the power pulse mode cannot be selected for the last preparation step (step No. 3), the last

step of the cyclic part (step No. 45) and the last step of testing completion (step No. 48).

In the measurement results files (see section 15) the power pulse step has the step marker **"IPP"**.

11.18 Resistance pulse mode

At the discharge resistance pulse step, the Analyzer supplies alternating resistance pulses of specified values **R1** and **R2 to the tested CIT**, whereby resistance **R1** is supplied for a specified time **T1**, and resistance **R2** is supplied for a specified time **T2** (see Figure 31).

The value of the discharge resistance of the chemical current source during the supply of each resistance pulse is maintained by the Analyzer channel at a level specified by the User and does not depend on the voltage on the chemical current source (within the operating range of voltage, current and power of the Analyzer channel).

The inclusion of the resistance pulse step in the HIT testing program is performed by selecting the **"Impulse R"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

Resistance **R1**, as well as resistance **R2**, can be chosen any values from the working range of the device (see table 5).

When the device channels operate in parallel (see Section 10), the working resistance range decreases proportionally to the number of channels operating in parallel in the group. In this case, the resistance of each pulse is multiplied equally between all channels in the group.

The specified duration **T1** and **T2** of resistance pulses can be selected from any of the device's operating range (see Table 5).

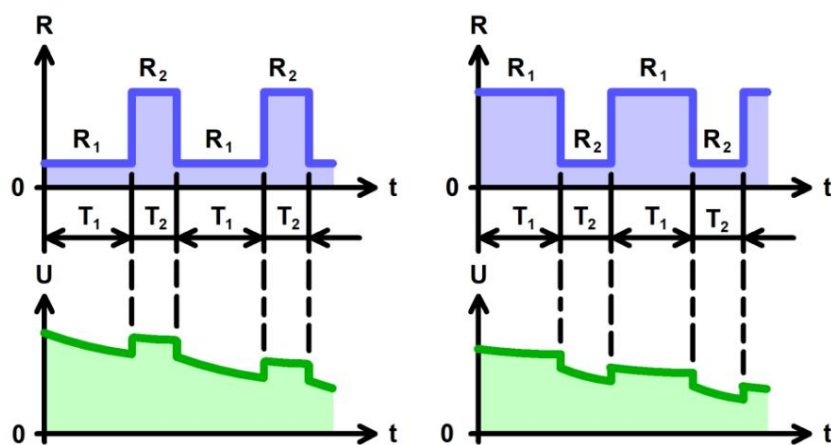


Figure 31 - Resistance pulse mode

The hardware current subrange for each pulse is selected by the device automatically based on the specified resistance and the specified maximum voltage limitation on the CIT "Umax:" (see Figure 104). If the maximum voltage limitation on the CIT is not used (the maximum voltage limitation input field on the CIT is left empty when specifying the CIT testing program), the maximum operating voltage of the device is used to calculate the maximum possible current through the CIT during the execution of the resistance pulse step and selection of the corresponding hardware current subrange (see Table 5).

ATTENTION! For the correct selection of the appropriate hardware current subrange by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "Umax:" (see Figure 104)!

The resistance pulse step has two adjustable step end criteria (see Figure 32):

- Reducing the voltage on the heat sink to the specified step end voltage **U_{END}**;
- Expiration of the allotted time of the **T_{END}** step.

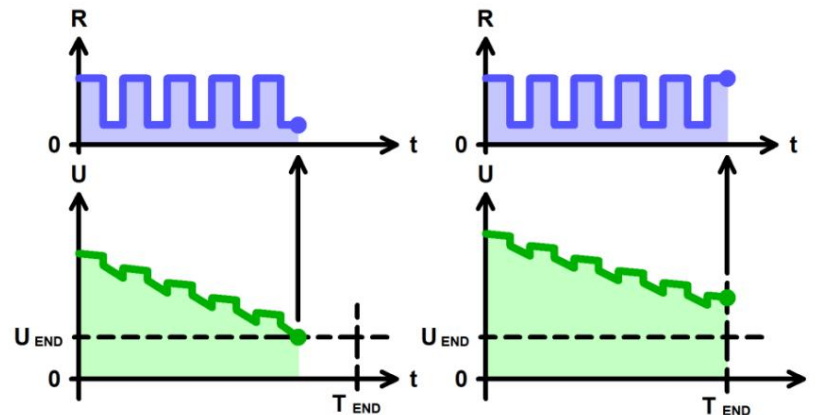


Figure 32 - Criteria for ending the resistance pulse step

The specified step end voltage **U_{END}** is selected by the User when forming the testing program for the CIT from the working range of the Analyzer. The criterion for ending the step upon reaching the specified voltage on the CIT can be

disabled by leaving the end voltage input field empty when setting the program testing (see subsection 19.2).

The maximum step duration (allocated step time) **T_{END}** is selected by the User from the permissible range of the device (see Table 5). End Criterion

It is not possible to disable the step after a specified time has elapsed, but it is possible to set a deliberately larger value for the step time limit than expected when testing of the HIT so that this criterion never works.

The end of the resistance pulse step occurs when any (at least one) of the specified end criteria is triggered.

During the resistance pulse step, the Analyzer measures and records the ESR of the tested CCT at each change of resistance (from **R₁** to **R₂** and back, from **R₂** to **R₁**). For more details, see subsection 13.3.

When setting the test program (see subsection 19.2), the resistance pulse step setting takes up two fields (two lines). For this reason, the resistance pulse mode cannot be selected for the last preparation step (step #3), the last step of the cyclic part (step #45), and the last test completion step (step #48).

In the measurement results files (see section 15) the resistance pulse step has the step marker **"IPR"**.

11.19 Working with tables

11.19.1 General instructions

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are capable of working with tables of voltage, current, power and resistance. Working with tables may be required if it is necessary to implement complex patterns (profiles) of change

voltage, current, power or resistance, which are difficult or impossible to set using standard testing modes (see subsections 11.3 – 11.18).

A typical example is the current and power profiles of energy storage devices of electric and hybrid vehicles during their movement in various driving cycles, for example, in the urban driving cycle, in the suburban driving cycle, NEDC, EPA, JC08, WLTC, NRTC profiles and any others. Working with tables of voltage, current, power and resistance will also be useful for simulating the operation of the tested HIS as part of

solar installations, wind turbines, as well as in any systems in which changes in current, power or load resistance on the heat exchanger occur in complex ways patterns.

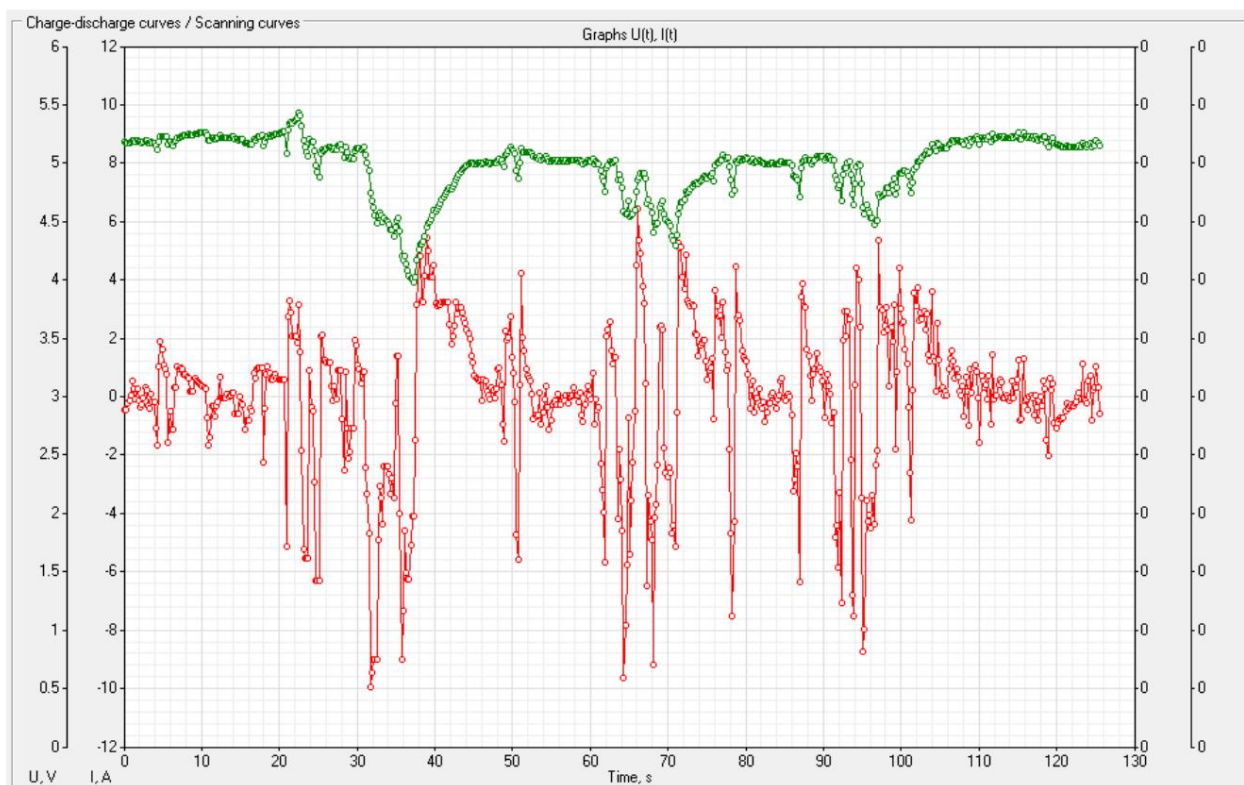


Figure 33 - Example of work according to the current table

ATTENTION! Working with tables is only possible under computer control! Even a short-term loss of connection between the Analyzer and the computer (Computer Interface, see section 19) during operation of the device according to the tables will lead to an emergency interruption of the testing program!

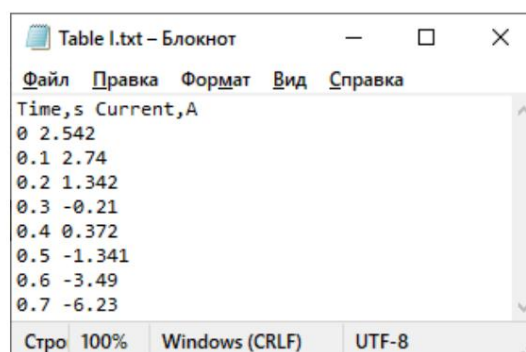
A table is a specially prepared text file, each line of which describes the required value of voltage, current, power or resistance at a given point in time.

The table file has special requirements. If the table file does not meet these requirements, it will not be possible to use it when specifying testing programs.

- The table file must contain at least three lines:
 - line #0 (start line) – header line (ignored during processing);
 - line No. 1 – the first line with tabular data;
 - line No. 2 (and subsequent lines) – lines with tabular data.
- Blank lines are not allowed either in the table body or at the end of the table.
- Rows with tabular data must contain two columns (two values):
 - the first column (the first value in the row) is the value of the moment in time, s;
 - the second column (the second value in the row) is the value of voltage, V, current, A, power, W, or resistance, Ohm.
- Columns (parameter values in file lines) must be separated special separator. The separator can be:
 - one space character;
 - one tab character;
 - one semicolon sign.
 Other options and other numbers of separators are not allowed.
- Rows with tabular data must have only one separator, only between columns (only between parameter values). Additional signs before the time value or after the voltage, current, power value or resistance is not allowed.
- The separator between the integer and fractional parts in the parameter values must be a period. Using a comma as a separator between the integer and fractional parts not allowed.
- The time column can start with any value (not necessarily zero). When executing the program according to the table, the step time is counted from the time value in line #1 (the time value from line #1 is equated to zero, and then the time is counted from it).
- The values in the time column must be ascending (the time value from each subsequent row of the table must be greater than the time value from the previous row).

When preparing tables in the MS Excel program, to save the file you should select the following formats:

- CSV (comma delimited) (*.csv);
- Text files (tab-delimited) (*.txt).



Time, s	Current, A
0	2.542
0.1	2.74
0.2	1.342
0.3	-0.21
0.4	0.372
0.5	-1.341
0.6	-3.49
0.7	-6.23

Figure 34 - Example of current table

Files with voltage, current, power and resistance tables should be selected when forming the HIT testing program on the "Tables" page of the Computer Interface (see subsection 19.3). In a specific HIT test, one (the same) voltage table, one current table, one power table and one resistance table can be used.

Work according to the voltage, current, power or resistance table can be selected for any step of the HIT testing program (see subsection 11.1). Program testing may contain several steps of work on the voltage table, several steps of work on the current table, several steps of work on the power table and several steps of work on the resistance table.

All steps of work according to the voltage table specified in the program will be performed according to the selected (same) voltage table. In other words, the selected voltage table will be used for all steps of work according to the voltage table specified in the testing program. The same applies to steps of work according to the current, power and resistance tables.

Work on tables is carried out as follows.

At the beginning of the execution of the step of work on the table, the Analyzer transmits to the computer the command to start work on the specified table. The computer starts counting the step time.

In accordance with the current time from the beginning of the step, the computer determines the required instantaneous value of the parameter (voltage, current, power or resistance) according to the specified table. If the time value falls between the rows of the specified table, the parameter value is determined by the linear approximation method using two adjacent parameter values from the table. The instantaneous value of the parameter found is transmitted to the Analyzer and accepted for execution.

When the entire specified table is completed, the computer sends a command to the Analyzer to complete work on the table and the device moves on to the next specified step of the testing program.

The execution of the step of work according to the table can be completed when the voltage on the CIT reaches the specified limits of the step end voltage. In this case, the device completes the execution of the step independently, without waiting for the command to complete the work according to the table from the computer, and proceeds to the execution of the next step of the CIT testing program.

The quality of the Analyzer's processing of a given table depends on the refresh rate of the parameter value (voltage, current, power or resistance) during the execution of the given table. The design refresh rate of the parameter value when working with tables is at least 10 Hz. The actual refresh rate depends on the quality of the connection between the device and the computer, on the workload and speed of the computer itself and on the workload of the computer network to which the Analyzer and the computer are connected, and may be lower than 10 Hz.

ATTENTION! To work with tables, it is recommended to use a direct connection between the Analyzer and the computer (bypassing the computer network, see Section 8), or use a limited local computer network without Internet access, close all programs on the computer except for the Computer Interface of the device (see Section 19), turn off Wi-Fi on the computer (if available), and temporarily disable Windows updates.

Any delays and gaps in the data exchange between the Analyzer and the computer will have a direct impact on the quality of the specified table execution. Interruption of communication between the device and the computer for more than 1 s will result in an emergency stop of the HIT testing!

11.19.2 Working with the voltage table

At the step of working according to the voltage table, the Analyzer ensures stabilization of the voltage at the terminals of the power supply with an instantaneous value selected depending on the current step time according to the specified voltage table (see Figure 35).

In this case, the device channel operates as a source or as a consumer of energy with a stabilized voltage at the output (input) equal to the current instantaneous table voltage. If the current voltage on the HIT differs from the instantaneous required table voltage, the device will give the HIT a charge current or take from the HIT a discharge current of such a level that the voltage on the HIT becomes equal to the instantaneous required one.

The Analyzer's capabilities for maintaining a given voltage on the HIT are limited by the maximum current for the selected hardware current sub-range and the maximum power of the device channel (see Table 5).

The step of working according to the voltage table is included in the testing program of the HIT by selecting the "**Table U**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2). The file with the voltage table must be pre-selected and loaded on the "Tables" page of the Computer Interface of the device (see subsection 19.3).

When selecting and loading a voltage table, the time and voltage values from the selected table can be adjusted (scaled) by the specified coefficients (multipliers) directly on the "Tables" page of the Computer Interface. When using a negative coefficient (multiplier) for voltage values, you can change the sign of the voltage from the table to the opposite. Thus, it is possible to use tables with any initial time and voltage values, since these values can be easily adjusted (adjusted) to the parameters of the tested CCS and the parameters of the device.

The corrected and prepared for execution voltage values when working according to the voltage table must be positive and fall within the working range of the Analyzer (see Table 5).

The adjusted duration of the voltage table should also fall within within the operating range of the device (see Table 5).

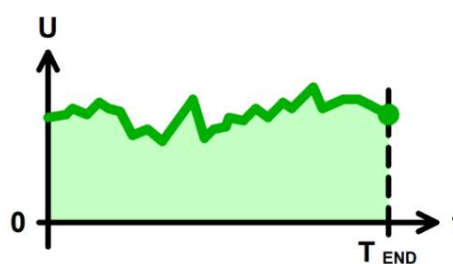


Figure 35 - Working with the voltage table

When the device channels operate in parallel (see section 10), the group's capabilities maximum current and power are expanded proportionally to the number of channels in the group. In this case, the current between parallel channels in the group is distributed randomly within the capabilities of the Analyzer channels for current and power.

The hardware subrange of the current in the operating mode according to the voltage table cannot be pre-determined by the device and must be set manually by the User when creating the testing program for the HIT (see subsection 19.2).

The end of the step of work according to the voltage table occurs after the expiration of duration of the **TEND table**.

In the measurement results files (see section 15), the step of working according to the current table has the step marker "**TBU**".

11.19.3 Working with the current table

At the current table step, the Analyzer ensures that a stabilized current flows through the tested CIT with an instantaneous value selected depending on the current step time according to the specified current table. The instantaneous current value can be either positive (charge current) or negative (discharge current) (see Figure 36).

The current value is maintained by the Analyzer in accordance with the current instantaneous value according to the table and does not depend on the voltage on the HIT (within the operating range of voltage and power for the Analyzer channel).

The step of working according to the current table is included in the testing program of the HIT by selecting the **"Table I"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2). The file with the current table must be pre-selected and loaded on the "Tables" page of the Computer Interface of the device (see subsection 19.3).

When selecting and loading a current table, the time and current values from the selected table can be adjusted (scaled) by the specified coefficients (multipliers) directly on the "Tables" page of the Computer Interface. When using a negative coefficient (multiplier) for current values, you can change the sign of the current from the table to the opposite (change the charge current to the discharge current, and vice versa). Thus, it is possible to use tables with any initial time and current values, since these values can be easily adjusted (adjusted) to the parameters of the tested CCS and the parameters of the device channel.

Corrected and prepared for execution current values during operation according to the current table, they must fall within the operating range of the Analyzer (see table 5).

The adjusted current table duration must also fall within the operating range of the device (see Table 5).

When the device channels operate in parallel (see Section 10), each channel in the group will process the specified table current value. Thus, the specified table current values are multiplied by the number of channels in the group. For example, if the instantaneous table current value is 5 A, and there are 4 channels operating in the group, then the resulting instantaneous current through the CCS will be 20 A.

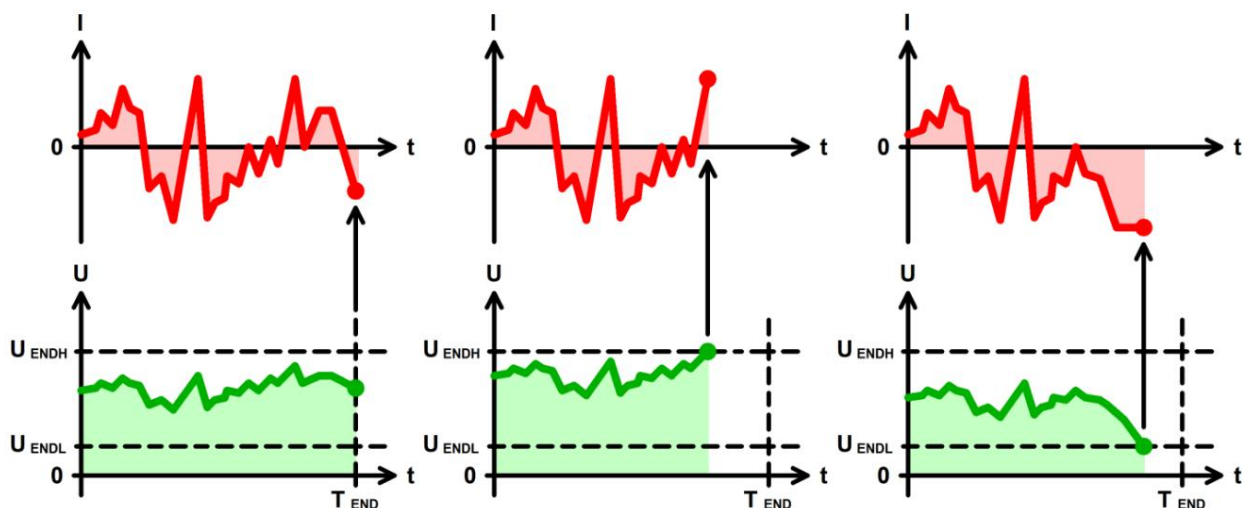


Figure 36 - Working with the current table

The current table work step has two configurable step end criteria (see Figure 36):

- Increasing the voltage on the heat exchanger to the specified upper voltage end of step **UENDH**;
- Reducing the voltage on the power supply to a specified lower voltage end of step **UENDL**.

In addition, the step of working on the current table will be automatically completed when the execution of the entire table is completed (after the duration of the **TEND table has expired**).

The specified lower (**UENDL**) and upper (**UENDH**) step end voltages are selected by the User when forming the testing program from the working range of the Analyzer (see Table 5).

Criterion for ending a step upon reaching a specified lower voltage on the heat exchanger can be disabled by leaving the lower end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

Criterion for ending a step upon reaching a specified upper voltage on the heat exchanger can be disabled by leaving the upper end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The selection of the hardware current sub-range is performed by the device automatically based on the maximum absolute value of the corrected current found in the selected current table file.

The end of a step of work according to the current table occurs when any (at least one) of the specified end criteria is triggered, or when the duration of the entire table has expired.

In the measurement results files (see section 15), the step of working according to the current table has the step marker **"TBI"**.

11.19.4 Working with the power table

At the step of working according to the power table, the Analyzer ensures the supply of the specified power to the tested CCS with an instantaneous value selected depending on the current time of the step according to the specified power table. The instantaneous power value can be either positive (charge power) or negative (discharge power) (see Figure 37).

The power value is maintained by the Analyzer in accordance with the current instantaneous table value and does not depend on the voltage on the power supply (within the operating range of the voltage and current of the device channel, see Table 5).

The step of working according to the power table is included in the testing program of the HIT by selecting the **"Table P"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2). The file with the power table must be pre-selected and loaded on the "Tables" page of the Computer Interface of the device (see subsection 19.3).

When selecting and loading a power table, the time and power values from the selected table can be adjusted (scaled) by the specified coefficients (multipliers) directly on the "Tables" page of the Computer Interface. When using a negative coefficient (multiplier) for power values, you can change the sign of the power from the table to the opposite (change the charge power to the discharge power, and vice versa). Thus, it is possible to use tables with any initial time and power values, since these values can be easily adjusted (adjusted) to the parameters of the tested CCS and the parameters of the device channel.

The corrected and prepared for execution power values when working according to the power table must fall within the working range of the Analyzer (see Table 5).

The adjusted power table duration should also fall within within the operating range of the device (see Table 5).

When the device channels operate in parallel (see Section 10), each channel in the group will operate at a specified table power value. Thus, the specified table power values are multiplied by the number of channels in the group. For example, if the instantaneous table power value is 25 W, and there are 4 channels operating in the group, the resulting instantaneous power on the CIT will be 100 W.

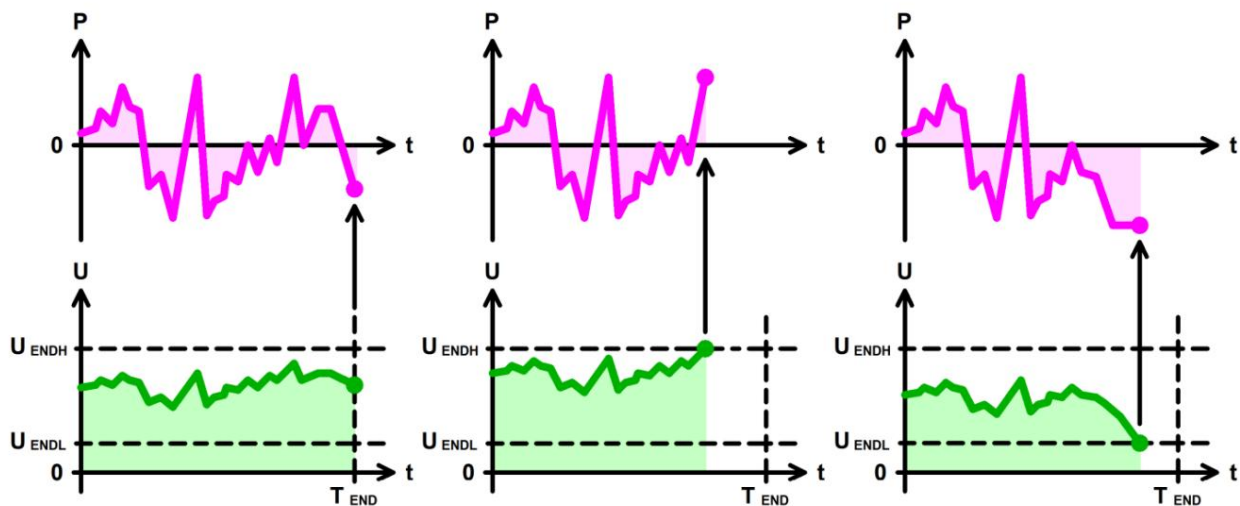


Figure 37 - Working with the power table

The power table work step has two configurable step end criteria (see Figure 37):

- Increasing the voltage on the heat exchanger to the specified upper voltage end of step **U_ENDH**;
- Reducing the voltage on the power supply to a specified lower voltage end of step **U_ENDL**.

In addition, the step of working on the power table will be automatically completed. upon completion of the execution of the entire table (upon expiration of the **T_END** table duration).

The specified lower (**U_ENDL**) and upper (**U_ENDH**) step end voltages are selected by the User when forming the testing program from the working range of the Analyzer (see Table 5).

Criterion for ending a step upon reaching a specified lower voltage on the heat exchanger can be disabled by leaving the lower end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

Criterion for ending a step upon reaching a specified upper voltage on the heat exchanger can be disabled by leaving the upper end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The selection of the hardware current subrange is performed by the device automatically based on the maximum absolute value of the corrected power found in the selected power table file, as well as on the larger of the specified values of the lower end voltage of the step **U_ENDL** and the minimum voltage limitation on the HIT "Umin:" (see Figure 104). For example, with the maximum absolute value of the power prepared for execution according to the specified power table, amounting to 18 W, the specified lower end voltage of the step of 10 V and the specified minimum voltage limitation on the HIT of 8 V, the hardware current subrange of -2 A will be automatically selected, since during the execution of the step the current through the HIT

will not exceed $18 \text{ W} / 10 \text{ V} = 1.8 \text{ A}$. If the criterion for ending the step by reducing the voltage on the heat sink to the lower end-of-step voltage **UENDL** is not used (field input of the lower voltage of the end of the step is left empty when specifying the testing program of the HIT) and the limitation of the minimum voltage on the HIT is also not used (the input field of the limitation of the minimum voltage on the HIT is also left empty when specifying the testing program), to calculate the maximum current through the HIT during the execution of the power table and the selection of the hardware sub-range of the current, the minimum voltage value from the working range of the Analyzer is used - 0.1 V, which will almost always lead to the selection of the highest hardware sub-range of the current - 10 A.

ATTENTION! To ensure that the device correctly selects the appropriate voltage sub-range and increases the accuracy of measurements, it is recommended to enter adequate values in the input fields of the restriction parameters, in particular "Umin:" (see Figure 104)!

The end of a step of work according to the power table occurs when any (at least one) of the specified end criteria is triggered, or when the duration of the entire table has expired.

In the measurement results files (see section 15) the step of working according to the power table has a step marker **"TBP"**.

11.19.5 Working with the resistance table

At the resistance table step, the Analyzer simulates a resistive load with an instantaneous value of electrical resistance selected depending on the current step time according to the specified resistance table (see Figure 38).

The resistance value is maintained by the Analyzer in accordance with the current instantaneous value according to the table and does not depend on the voltage on the HIT (within the operating range of voltage, current and power for the Analyzer channel).

The step of working according to the resistance table is included in the testing program of the HIT by selecting the **"Table R"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2). The file with the resistance table must be pre-selected and loaded on the "Tables" page of the Computer Interface of the device (see subsection 19.3).

When selecting and loading a resistance table, the time and resistance values from the selected table can be adjusted (scaled) by the specified coefficients (multipliers) directly on the "Tables" page of the Computer Interface. When using a negative coefficient (multiplier) for resistance values, you can change the sign of the resistance from the table to the opposite. Thus, it is possible to use tables with any initial time and resistance values, since these values can be easily adjusted (adjusted) to the parameters of the tested CCT and the parameters of the device.

The corrected and prepared resistance values for execution when working according to the resistance table must be positive and fall within the working range of the Analyzer (see Table 5).

The adjusted duration of the resistance table should also fall within the operating range of the device (see Table 5).

When the device channels operate in parallel (see Section 10), each channel in the group will process the specified table resistance value. Thus, the specified table resistance values are divided by the number of channels in the group. For example, if the instantaneous table resistance value is 1 Ohm,

and there are 4 channels operating in the group, then the resulting instantaneous load resistance on the heat sink will be 0.25 Ohm.

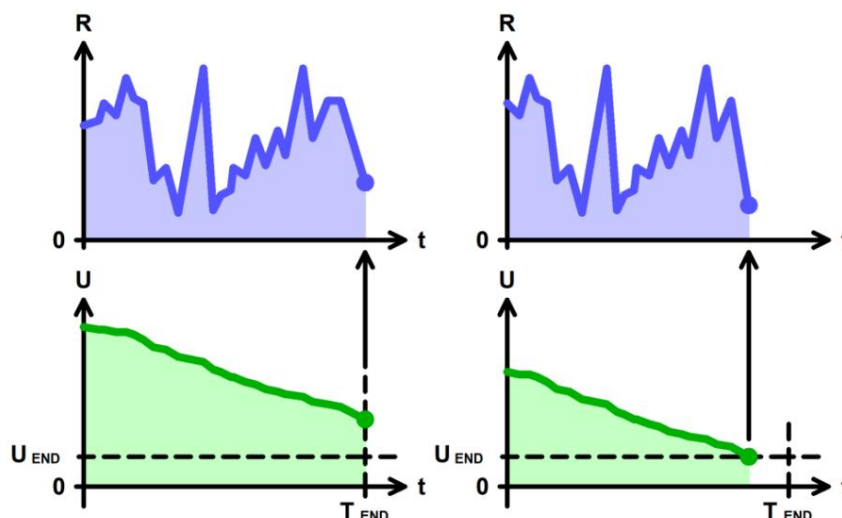


Figure 38 - Working with the resistance table

The step of working according to the resistance table has a single adjustable criterion for the end of the step – a decrease in the voltage on the heat exchanger to the specified final voltage of the step **U_{END}** (see Figure 38).

In addition, the step of working on the resistance table will be automatically completed upon completion of the execution of the entire table (upon expiration of the **T_{END}** table duration).

The specified step end voltage **U_{END}** is selected by the User when forming the testing program from the working range of the Analyzer (see Table 5). The criterion for ending the step upon reaching the specified voltage on the CIT

can be disabled by leaving the lower end voltage input field empty when specifying the HIT testing program (see subsection 19.2).

The hardware subrange of the current is selected by the device automatically based on the minimum value of the corrected resistance found in the selected resistance table file, as well as on the specified limitation of the maximum voltage on the HIT "U_{max}:" (see Figure 104). If the limitation of the maximum voltage on the HIT is not used (the input field for the maximum voltage limitation on the HIT is left empty when specifying the HIT testing program), the maximum operating voltage of the device is used to calculate the maximum current through the HIT during the step of working with the resistance table and selecting the hardware subrange (see Table 5).

ATTENTION! For the correct selection of the appropriate voltage and current sub-range by the device and to increase the accuracy of measurements, it is recommended to enter adequate values in the input fields of the limitation parameters, in particular "U_{max}:" (see Figure 104)!

In the measurement results files (see section 15) the step of working according to the table The resistance has a **"TBR" step marker**.

11.20 Voltage recorder

At the voltage recorder step, the Analyzer interrupts the current supply through the HIT, continuing to measure and record the voltage at the terminals of the heat exchanger.

Unlike the Relaxation mode (see subsection 11.10), during the recorder step the tested chemical and electrical equipment is completely disconnected from the internal power circuits of the Analyzer channel, and only the voltmeter V, which has a very high input resistance (see Section 6), remains connected to the chemical and electrical equipment terminals. Due to this, in the voltage recorder mode at the chemical and electrical equipment terminals, there are practically no additional current leaks from the Analyzer, and the change in voltage at the chemical and electrical equipment terminals is determined exclusively by the internal processes occurring in the chemical and electrical equipment itself. The disadvantages of the voltage recorder mode, in contrast to the Relaxation mode, include the additional time required to disconnect the tested chemical and electrical equipment from the internal power circuits of the device (no more than 1 s), which will be added to the total chemical and electrical equipment testing time, but will not be taken into account in the measurement results.

The inclusion of the voltage recorder step in the HIT testing program is performed by selecting the "**Logger U**" mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The voltage recorder step has two adjustable step end criteria (see Figure 39):

- The voltage on the heat sink reaches the specified step end voltage **U_{END}**;
- Expiration of the allotted time of the **T_{END}** step.

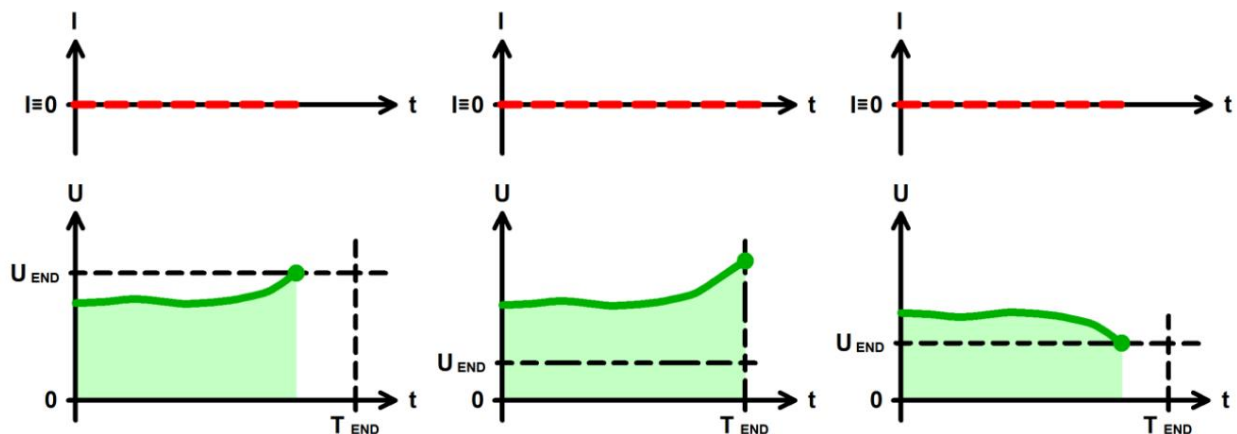


Figure 39 - Power pulse step termination criteria

The specified step end voltage **U_{END}** is selected by the User when forming the testing program for the CIT from the working range of the Analyzer (see Table 5). The end of the voltage recorder step occurs as soon as the current voltage on the CIT touches the specified step end voltage level **U_{END}**. The criterion for the end of the step upon reaching the specified voltage on the CIT can be

disabled, leaving the end voltage input field empty when specifying the test program (see subsection 19.2).

The maximum step duration (allocated step time) **T_{END}** is selected by the User from the permissible range of the device (see Table 5). The criterion for ending a step after a specified time has elapsed cannot be disabled, but it is possible to set a deliberately larger value for the maximum step time than expected during testing of the CIT, so that this criterion never triggers.

The end of the voltage recorder step occurs when any (at least one) of the specified end criteria is triggered.

In the measurement result files (see section 15) the voltage recorder step has the step marker "**LGU**".

11.21 Break

In pause mode, the Analyzer interrupts the current supply through the HIT. During the pause, the HIT parameters are not recorded and are not saved in the measurement results files (see Section 15).

The pause mode can be used as a timer for delayed start of the testing program, or for "cutting out" a specified time from the measurement results. For example, the dark time of day when testing solar batteries.

During the pause step, the tested HIT is completely disconnected from the internal power circuits of the Analyzer channel, and only the voltmeter V, which has a very high input resistance (see Section 6), remains connected to the HIT terminals. Due to this, in the pause mode, additional current leaks from the Analyzer are practically absent.

The pause step is included in the HIT testing program by selecting the **"Pause"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The pause step has a single step end criterion – the expiration of the specified pause time. The specified pause time must be selected by the User from the device's operating range (see Table 5).

11.22 MPRT mode

In the MPPT (Maximum Power Point Tracking) mode, the Analyzer creates an electrical load on the tested chemical power source with such conductivity as to ensure the extraction of the maximum possible electrical power from the tested chemical power source.

The MPPT mode can be used for testing solar panels, fuel cells, micro wind generators, and other energy sources.

The voltage on the tested CCS must fall within the operating range of the device (see Table 5). The maximum power taken from the CCS is limited by the maximum discharge power that the device channel can provide, as well as the maximum current for the hardware current subrange selected when setting the MPPT step.

In the MPPT mode, the Analyzer uses an algorithm for searching for the maximum power point, the operating principle of which is based on the continuous change in the conductivity of the **G** channel of the device at a rate specified by the User, with continuous tracking of the instantaneous power consumed from the CHIT **P** (see Figure 40).

After the MPPT step is launched, the device channel begins to increase its conductivity. As the conductivity increases (and the current taken from the CCS increases), the power consumed from the CCS will also increase. The increase in the power taken from the CCS is limited by the CCS capabilities, and at a certain conductivity value, the maximum electric power taken from the CCS is reached (see Figure 40). A further increase in conductivity leads to a drop in the power taken from the CCS, due to a significant drop in voltage on the CCS. When the device registers a decrease in power by a specified value **dP** (relative to the previously found maximum), the direction of the channel conductivity change changes to the opposite - the device begins to decrease conductivity, which leads to a gradual decrease in the current consumed from the CCS and an increase in voltage on the CCS. When the direction of conductivity change changes, the search for the maximum power value begins anew. Having again overcome the power maximum and again registered a decrease in power by the value **dP**, the device channel again changes the direction of conductivity change to the opposite. Then this process is continuously repeated. As a result, the conductivity of the device channel will fluctuate around the maximum power point for its continuous tracking and following the maximum power point when the characteristics of the CIT change.

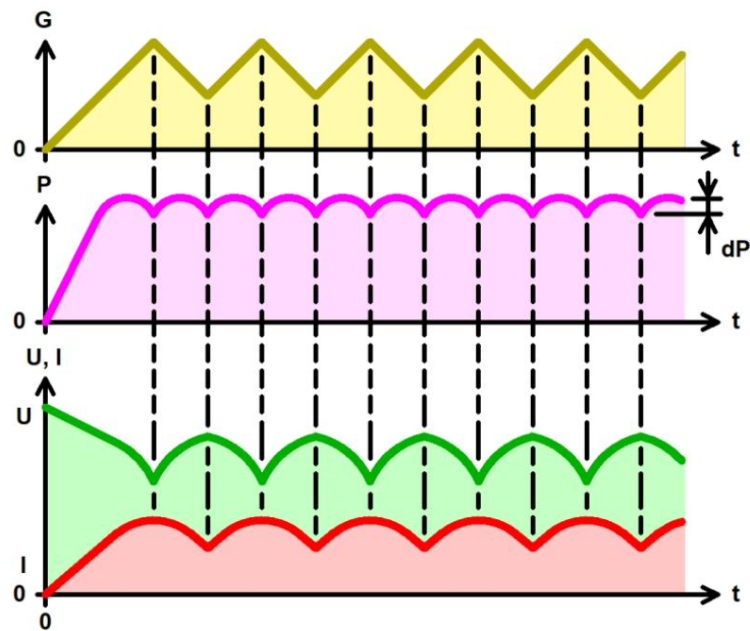


Figure 40 - Operation of the device in MPPT mode

The MPPT step is included in the HIT testing program by selecting the **"MPPT"** mode from the drop-down list of operating modes for a specific step (see subsection 19.2).

The hardware subrange of the current in the MPPT mode cannot be pre-determined by the device and must be set manually by the User when creating the testing program for the HIT (see subsection 19.2).

The given rate of change of conductivity, as well as the given change power dP , must fall within the operating range of the device (see Table 5).

When the device channels operate in parallel (see Section 10), the current and power consumed from the CIT are divided equally between all channels in the group. In this case, the group's capabilities for the current and power taken from the CIT increase proportionally to the number of channels in the group.

The MPPT step has a single criterion for step termination – the expiration of a specified time. The specified time of the MPPT mode must be selected by the User from the device's operating range.

By selecting smaller values of the power change dP , it is possible to achieve a better approximation to the maximum power point of the CCS, and obtain maximum energy from the CCS. By increasing the specified rate of conductivity change, it is possible to achieve a faster response to the change in the properties of the CCS, the maximum power point will be tracked more efficiently. However, too small a specified value of the power change dP , as well as too high a rate of conductivity change, will lead to instability in the search for the maximum power point, and in the worst case - to the impossibility of finding this point at all. Therefore, the values of the rate of conductivity change and the value of the power change dP should be selected experimentally for specific CCS and specific test conditions.

For example, for a solar panel with a nominal voltage of 12 V and a maximum power of 100 W, a good starting point for finding the optimal values for the given parameters would be a conductivity change rate of 50 mS/s and a power change value of 0.5 W.

In the measurement results files (see section 15) the MPPT step has the step marker **"MPT"**.

12 DEFINITION OF HIT PARAMETERS

The ASK75.10.20.2 and ASK75.10.20.12 analyzers allow you to determine:

- Capacity of the heat-transfer device by charge, Q , A h;
- Capacity of the heat storage device by energy, E , W h;
- Electrical capacity (supercapacitors), C , F;
- Charge storage efficiency (by A h), **EF_q**, %;
- Energy storage efficiency (Wh) **EF_e**, %;
- Equivalent series resistance of the heat sink (ESR), mOhm;
- Average leakage current through the HIT for each charge-discharge cycle, **Ilk**, A.

The primary parameters that the Analyzers measure are:

- Current through the heat exchanger, A;
- Voltage at the terminals of the HIT, V;
- Temperature of the heat exchanger, °C;
- Time, sec.

The remaining parameters of the HIT are calculated and are determined by formulas, given in this and subsequent sections.

Positive values of current through the CIT correspond to the process of charging the CIT, Negative current values correspond to the process of discharging the chemical energy source.

The primary parameters are recorded by the device with a given time step, configured by the User when setting the testing program for the CCS. As a result, at each step of the testing program (see Section 11), the device saves a certain set of points* . Each point contains the instantaneous value of the voltage on the CCS, the instantaneous value of the current through the CCS, the instantaneous value of the temperature of the CCS, as well as the time from the beginning of the current step of the testing program for the CCS, at which these instantaneous values of voltage, current and temperature were recorded.

For further description of the calculation formulas, we will designate the total number of points registered by the device at a specific step of the HIT testing program with the letter "k".

Table 7 - Set of primary parameters for each step of testing the HIT

Parameter	Parameter value									
Point number*	0	1	2	3 ...		i-1	i	...	k	
Time, s	0	t1	t2	t3 ...	ti-1		ti ...	tk		
Voltage, V	equal to U1	U1	U2	U3 ...	Ui-1	Ui ...	Uk			
Current, A	equals I1	I1	I2	I3 ...	Ii-1		Ii ...	I		
Temperature, °C		T1	T2	T3 ...	Ti-1		Ti ...	Tk		

* see the list of accepted abbreviations and designations.

For each point of the primary parameters registered by the Analyzer during the testing process, the charge portion dQ , A h, and the energy portion dE , W h, received by the CCS during the time elapsed between the registration of the current (i) and previous ($i - 1$) points are calculated:

$$dQ_i = \frac{I_i + I_{i-1}}{2} \cdot (t_i - t_{i-1}) \quad (1)$$

$$dE_i = \frac{I_i \cdot U_i + I_{i-1} \cdot U_{i-1}}{2} \cdot (t_i - t_{i-1}) \quad (2)$$

where: dQ_i is the portion of charge received by the tested CCS at point number i , A h;

dE_i - portion of energy received by the tested chemical supply system at point i , W h; I_i - current through the chemical supply system at point i , A; U_i

- voltage across the chemical supply system at point i , V;

I_{i-1} - current through the chemical supply system at point

$i - 1$, A; U_{i-1} - voltage across the chemical supply system at point

$i - 1$, V; t_i - time from the start of the step at point i , h; t_{i-1} -

time from the start of the step at point $i - 1$, h.

Positive values of charge portions dQ_i correspond to the charge transferred on the HIT during the charging process, and will be further designated as " dQ^+ ".

Negative values of charge portions dQ_i correspond to the charge returned from the HIT during the discharge process, and will henceforth be designated as " dQ^- ".

Positive values of energy portions dE_i correspond to the energy transferred to the CIT during the charging process and will be further designated as " dE^+ ".

Negative values of the energy portions dE_i correspond to the energy returned from the HIT during the discharge process, and will henceforth be designated as " dE^- ".

The total charge received or returned by the tested CCT at a specific step of the testing program is calculated as the integral of the current over time for a specific step of the CCT testing program, according to formulas (1) and (3):

$$Q_s = \sum_{i=1}^{i=k} dQ_i \quad (3)$$

where: Q_s is the charge received by the CIT in a specific step of the testing program, A h;

i is the number of the next point; k

is the total number of points registered by the Analyzer in this step; dQ_i is the portion of charge received by the tested CCS at point number i , A h.

Total energy received or returned by the tested HIT at a specific step testing programs, is calculated as the integral of the power over time for a specific step of the testing program of the heat and power equipment, according to formulas (2) and (4):

$$E_s = \sum_{i=1}^{i=k} dE_i \quad (4)$$

where: E_s is the energy obtained by the chemical energy source in a specific step of the testing program, W h;

i is the number of the next point; k

is the total number of points registered by the Analyzer in this step;

dE_i is the portion of energy received by the tested heat source at point number i , W h.

Values of charge Q_s and energy E_s , received or returned by the HIT at a specific step of the testing program (see Section 11), are included in the summary measurement results files (see Subsection 15.3), in the lines describing the HIT parameters for a specific completed step of the testing program.

To calculate the characteristics of the tested CIT within the full cycle
The following parameters are entered for charge-discharge:

- Q_{CLK}^+ - total charge transferred to the heat-saving battery within a given cycle, A h;
- Q_{CLK}^- - total charge returned from the heat source within a given cycle, A h;
- E_{CLK}^+ - total energy transferred to the heat source within a given cycle, W h;
- E_{CLK}^- - total energy returned from the heat source within a given cycle, W h.

These parameters are reset at the beginning of the next charge-discharge cycle of the HIT (see Section 11) and are calculated (summed up) using formulas (5), (6), (7) and (8) throughout the entire cycle, including all steps of the test program performed within a specific charge-discharge cycle.

The total charge transferred to the tested CCT within a specific cycle of the testing program is calculated as the sum of all positive portions of charge transferred to the tested CCT during the entire cycle:

$$Q_{CLK}^+ = \sum_{CLK} dQ^+ \quad (5)$$

where: Q_{CLK}^+ - total charge received by the heat source within a given cycle, A h;
 dQ^+ - the next portion of charge transferred to the tested HIT, A h.

The total charge returned from the tested CIT within a specific cycle of the testing program is calculated as the sum of all negative portions of charge returned from the CIT for the entire cycle:

$$Q_{CLK}^- = \sum_{CLK} dQ^- \quad (6)$$

where: Q_{CLK}^- - total charge returned from the heat source within a given cycle, A h;
 dQ^- - the next portion of charge returned from the tested HIT, A h.

The total energy transferred to the tested chemical equipment within a specific cycle of the testing program is calculated as the sum of all positive portions of energy transferred to the tested chemical equipment for the entire cycle, according to formula (7).

$$E_{CLK}^+ = \sum_{CLK} dE^+ \quad (7)$$

where: E_{CLK}^+ - total energy received by the heat source within a given cycle, W h;
 dE^+ - the next portion of energy transferred to the tested power supply unit, W h.

The total energy returned from the tested CIT within a specific cycle of the testing program is calculated as the sum of all negative portions of energy returned from the tested CIT for the entire cycle:

$$E_{CLK}^{-} = \sum_{CLK} dE^{-} \quad (8)$$

where: E_{CLK}^{-} - total energy returned from the heat source within a given cycle, W h;

dE^{-} - the next portion of energy returned from the tested heat source, W h.

The total capacity of the HIT Q , A h, is equal to the total charge Q_{CLK} returned from the HIT within the entire cycle:

$$Q = Q_{CLK}^{-} \quad (9)$$

where: Q is the total capacity of the CIT per cycle, with given testing parameters, A h;

Q_{CLK}^{-} - total charge returned from the heat source within a given cycle, A h.

The result of calculating the total capacity of the heat supply system Q is included in the "GNRL" lines of the summary file of results by steps or by cycles of the heat supply system testing program (see subsection 15.3), and is also displayed on the analysis graphs (see subsection 19.6).

The total energy capacity of the HIT E , W h, is equal to the total charge of the E_{CLK} returned from the HIT within the entire cycle:

$$E = E_{CLK}^{-} \quad (10)$$

where: E is the total energy capacity of the chemical energy storage device per cycle, under the specified testing conditions, W h;

E_{CLK}^{-} - total energy returned from the heat source within a given cycle, W h.

The result of calculating the total energy capacity of the HIT E is included in the "GNRL" lines of the summary file of the results of the testing program (see subsection 15.3), and is also displayed on the analysis graphs (see subsection 19.6).

In case of using a complex testing program of the CCS, consisting of a sequence of *alternating* charge and discharge modes, the result of determining the full capacity of the CCS Q and the full energy capacity of the CCS E may exceed the charge and energy actually accumulated in the CCS. In this regard, to measure the real capacity of the CCS, A h, and the real energy capacity of the CCS, W h, simple testing programs should be used, consisting of only *one* sequence of charge and discharge.

The efficiency of the CCS by charge EFq is calculated as the ratio of the total charge returned by the CCS within a specific charge-discharge cycle to the total charge received by the CCS during this cycle:

$$EFq = \frac{Q_{CLK}^{-}}{Q_{CLK}^{+}} \cdot 100 \quad (11)$$

where: EFq - efficiency of the heat source by charge (efficiency of charge storage), %;

Q_{CLK}^{-} - total charge returned from the heat source within a given cycle, A h;

Q_{CLK}^{+} - total charge transferred to the HIT within a given cycle, A h.

The energy efficiency of the power supply EFe is calculated as the ratio of the total energy returned by the power supply within a specific charge-discharge cycle to the total energy received by the power supply during this cycle:

$$EFe = \frac{E_{CLK}^{-}}{E_{CLK}^{+}} \cdot 100 \quad (12)$$

where: EFe - energy efficiency of the heat source (energy storage efficiency), %;

E_{CLK}^{-} - total energy returned from the heat source within a given cycle, W h;

Q_{CLK}^+ - total energy transferred to the heat source within a given cycle, W h.

The average leakage current through the CCS I_{lk} during the charge-discharge cycle is calculated as the difference between the received and returned charge (i.e. the charge lost in the CCS), divided by the total duration of the entire charge-discharge cycle (the sum of the durations of all steps included in the given charge-discharge cycle):

$$I_{lk} = \frac{Q_{CLK}^+ - |Q_{CLK}^-|}{\sum t_k} \quad (13)$$

where: I_{lk} is the average leakage current through the CIT per charge-discharge cycle, A;

Q_{CLK}^+ - total charge transferred to the heat-saving battery within a given cycle, A h;

Q_{CLK}^- - total charge returned from the heat source within a given cycle, A h;

t_k - duration of the next step in the charge-discharge cycle of the chemical power station, h.

The average leakage current through the CIT I_{lk} is of an estimated nature, and at values of the charge efficiency (η_{eq}) close to 100%, it may differ from the actual value of the leakage current through the CIT by ten or more times.

Measurement of the equivalent series internal resistance (ESR) of the heat exchanger using the ASC75.10.20.2 and ASC75.10.20.12 analyzers is carried out in three in various ways.

The method for measuring the equivalent series resistance (ESR) of the power supply, in Ohm, used by the ASK75.10.20.2 and ASK75.10.20.12 Analyzers, is presented in Section 13.

The algorithm for determining the electrical capacitance of the heat exchanger C, F, is described in section 14.

The summary file of measurement results by steps or by cycles (see subsection 15.3) contains a number of parameters averaged within a specific step or cycle of the HIT testing program.

In particular, such parameters include the temperature of the CCS, the electrical capacitance of the CCS, the equivalent series resistance (ESR) measured by the method of periodically interrupting the charge (discharge) current or in the modes of voltage, current, power or resistance pulses (see subsections 13.2 and 13.3), as well as

equivalent series resistance of the chemical energy storage device, measured at the beginning of the charge and at the beginning of the discharge (see subsection 13.4).

Averaging of values for a specific parameter is carried out within a specific step or cycle only for parameter values that differ from zero. Zero

The parameter values mean that its measurement was not performed and are not included in the calculation of the average value for this parameter.

13 ESR MEASUREMENT HIT

13.1 General instructions

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are capable of measuring the equivalent series resistance (ESR) of the HIT. ESR measurement can be carried out in three different ways.

All three methods of measuring ESR are based on measuring the voltage jump at the terminals of the tested power supply dU , caused by a step change in the current dI , passing through the CIT, while the abrupt change in current can occur either intentionally, by adding special pauses to the current, or as a result of a natural change in current during the execution of a given CIT testing program.

The equivalent series resistance of the heat sink is determined by formula (14):

$$ESR = \frac{dU}{dI} \quad (14)$$

where: **ESR** – equivalent series resistance of the heat source, Ohm;

dI – disturbance (drop) of current through the heat exchanger, A;

dU – voltage surge at the terminals of the heat exchanger, V, caused by current drop **dI**.

Each of the methods has its own advantages and disadvantages, and can be used depending on the specific testing conditions and specific parameters of the studied chemical experiment.

13.2 Measurement of ESR of a heat source using the periodic current interruption method

The ESR measurement of a chemical power supply by the method of periodic interruption of current works as follows. During the charge or discharge modes of the tested chemical power supply, after a specified period of time **PESR**, adjustable within wide limits, the current through the chemical power supply is interrupted. (switches off) for a period of time **DESER**, also adjustable within wide limits, then the voltage surge at the terminals of the heat supply **dU** and the value of the interrupted current **dI** is determined the next instantaneous ESR value:

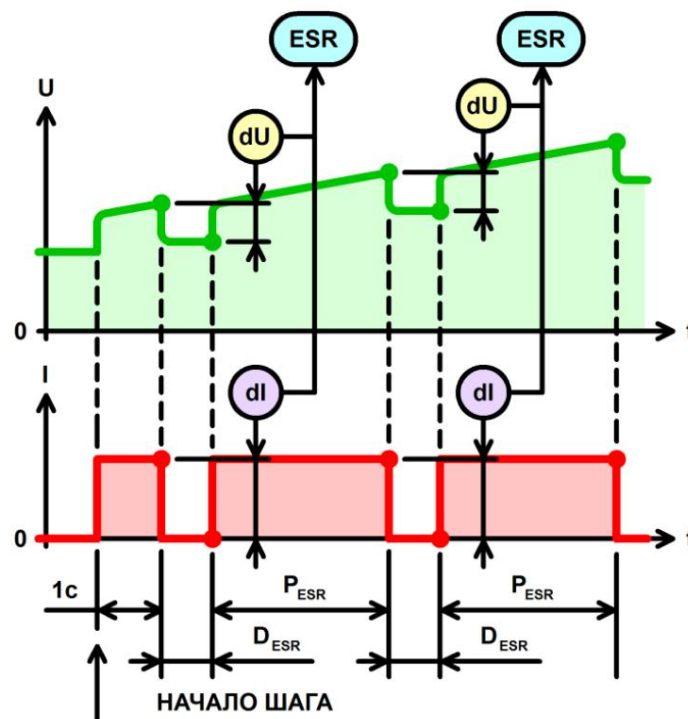


Figure 41 - Periodic measurement of the ESR of the HIT during the charging (discharging) process

Advantages of the method of measuring the ESR of a heat source using the periodic current interruption method:

- It is the most accurate way to measure the ESR of a heat source, implemented using the ASK75.10.20.2 and ASK75.10.20.12 Analyzers;
- Allows obtaining multiple instantaneous ESR values directly during the charging or discharging process, which makes it possible to construct the dependence of ESR on the charging (discharging) time, as well as on the degree of charge or on the depth of discharge of the studied chemical power source;
- When selecting the **DESR** pause duration = "1000Hz" (0.5 ms), the ESR measurement results correspond to the ESR measurement results obtained using the generally accepted method for measuring the ESR of a power supply with alternating current frequency of 1000 Hz.

Flaws:

- As a result of adding pauses in the current through the tested CIT, the average value of the current through the CIT decreases, as a consequence, the charging or discharging time of the CIT – increases;
- The method does not work in all HIT testing modes.

Enable or disable this ESR measurement method, ESR measurement period

PESR, as well as the duration of the ESR **DESR** measurement, are configured in the Computer Interface when preparing the HIT testing program (see subsection 19.2).

The measurement of the ESR of the heat sink in this way can be performed only at the steps:

- Constant current charging (CHCC);
- Constant Charge Power (CHCP);
- Direct Current Discharge (DCHCC);
- Discharge constant power (DCHCP);
- Discharge to constant resistance (DCHCR);
- Current sweeps;
- Power sweeps;
- Resistance sweeps.

The first ESR measurement is always taken 1 s after the start of the step.

Considering that this method implies interruption of the charging (discharging) current for some time, the average value of the current through the CCS will be less than the specified one. For example, if **PESR** = **DESR** = 1 s, and the specified charging (discharging) current is 10 A, the average charging (discharging) current will be only 5 A. In this case, the charging (discharging) rate of the CCS will be halved.

During the pause for ESR (**DESR**) measurement, the saving of instantaneous values of the parameters of the tested CCS in the primary data files is blocked (see subsection 15.2 and subsection 19.2), as well as the output of parameters to the charge-discharge graphs is blocked (see subsection 19.4). Instantaneous values of the parameters of the tested CCS are saved in the primary data files and output to the charge-discharge graphs only during the **PESR** time (between ESR measurements).

With this method of storing measurement results, to take into account the absence of current through the CIT during the **DESR** time, when processing the measurement results obtained from the device, a reduction coefficient is introduced:

$$K_{ESR} = \frac{P_{ESR}}{P_{ESR} + D_{ESR}} \quad (15)$$

where: **KESR** is the reduction factor for current;

PESR – period between pauses in current for ESR measurement, s;

DESR – duration of pause in current for ESR measurement, s.

The results of measuring the instantaneous current through the CIT, saved in the primary data files, are multiplied by this reduction factor. Thus,

The measurement results files do not contain the true instantaneous current through the ESR, but the corrected value of the current through the ESR, taking into account pauses for measuring the ESR.

For example, if **PESR** = **DESR** = 1 s, **KESR** = 1 / (1 + 1) = 0.5. In this case, if the instantaneous current measurement result obtained during the **PESR** time (i.e. between ESR measurements) is 10 A, the instantaneous current value equal to 10 · 0.5 = 5 A will be saved in the primary data file. In the future, when processing the obtained primary measurement results, this will allow us to obtain correct results for calculating the capacity, A h, energy capacity, W h, electrical capacity, F, and other necessary parameters of the tested power supply.

For the period between ESR **PESR** measurements the following values can be selected:

- "Not meas." – ESR measurement will not be performed at a specific step;
- "1s" (1 c)
- "2s" (2 c)
- "5s" (5 c)
- "10s" (10 c)
- "20s" (20 c)
- "1min" (1 min)
- "2min" (2 min)
- "5min" (5 min)
- "10min" (10 minutes)
- "20min" (20 min)
- "1h" (1 ħ)
- "2h" (2 ħ)
- "5h" (5 ħ)
- "10h" (10 ħ)
- "1d" (1 ħħ)

For the duration of the current pauses for ESR **DESR** measurement the following can be selected values:

- | | | | |
|------------------|-------------------|------------------|---------------------|
| • "1min" (1 min) | • "2s" (2 c) | • "5Hz" (0,1 c) | • "100Hz" (5 ħc) |
| • "20s" (20 c) | • "1s" (1 c) | • "10Hz" (50 ħc) | • "250Hz" (2 ħc) |
| • "10s" (10 c) | • "1Hz" (0,5 c) | • "25Hz" (20 ħc) | • "500Hz" (1 ħc) |
| • "5s" (5 c) | • "2.5Hz" (0,2 c) | • "50Hz" (10 ħc) | • "1000Hz" (0,5 ħc) |

When the Analyzer channels operate in parallel (see section 10), the value of the pause duration in the current for measuring ESR **DESR** is limited to a minimum value of 0.1 s (the equivalent frequency of alternating current is no more than 2.5 Hz).

The highest accuracy of ESR measurement by the method of interrupting the current through the CIT is ensured due to the absence of transient processes associated with operation charge and discharge regulators included in the Analyzer channels (see Figure 7 and Figure 8), since during the pause the current through the CIT is completely switched off and immediately takes a zero value. As a result, for all specified **DESR** values, the ESR measurement results are repeatable and most accurate.

Selecting specific values for the period between ESR **PESR** measurements and duration pauses in current for measuring ESR **DESR** depends on the specific tasks during testing of the chemical equipment.

In general, the ESR measurement period **PESR** should be chosen so that, on average, 25 to 100 ESR measurements are obtained during a given step of the test program. For example, if the discharge step of a given battery under given test conditions lasts approximately 1 hour, a good choice for **PESR** would be 1 min. This will allow approximately 60 ESR measurements to be obtained during this step.

The duration of the ESR **DESR** measurement determines the greater or lesser contribution to the results of the ESR measurement of the polarization component of the equivalent series resistance of the CCS. With longer **DESR** durations, the contribution of the polarization component will be greater, and, as a consequence, the value of the result

ESR measurements will generally be higher, and vice versa, for shorter **DESR** durations the contribution of the polarization component will be smaller and, as a consequence, the value of the ESR measurement result will, in general, be lower.

The results of ESR measurement using the periodic current interruption method are into the results files in the columns "ESR,Ohm" and "ESRa,Ohm" (see section 15).

In addition, the ESR measurement results obtained by this method can be displayed on the charts on the "Charts" page (see subsection 19.4).

If the CHCC, CHCP, DCHCC, DCHCP, DCHCR, current, power or resistance sweep steps are used in the CHC test program (see Section 11 and Subsection 19.2), but the CHC ESR measurement function by the current chopping method is disabled for these steps, zero values will be stored in the "ESR,Ohm" and "ESRa,Ohm" columns.

In general, the ESR measurement will be more accurate the larger the absolute value.
the value of the interrupted current, and the greater the measured value of the ESR of the heat exchanger itself.

13.3 Measuring ESR of a power supply in pulse modes

In this method, the ESR of the CCS is determined by the voltage surges on the CCS dU , caused by the surges dI in the current through the CCS as a result of the change in voltage, current, power or resistance pulses when performing steps of voltage, current, power or resistance pulses (see subsections 11.15, 11.16, 11.17 and 11.18).

The method of measuring ESR is shown in Figure 42 using the current pulse mode as an example:

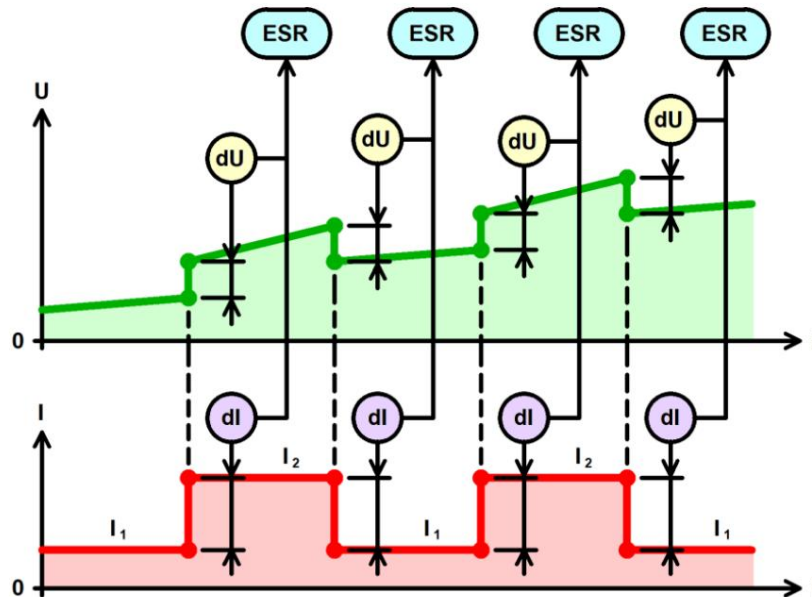


Figure 42 - Measurement of ESR of the heat sink in voltage pulse modes, current, power or resistance

Advantages of the method of measuring the ESR of a heat sink in pulse modes:

- Allows obtaining multiple instantaneous ESR values directly during the execution of a step of voltage, current, power or resistance pulses, which makes it possible to construct the dependence of ESR on time, as well as on the degree of charge or on the depth of discharge of the studied chemical power source;
- Does not reduce the average value of current through the HIT, does not increase the charging time or discharge and does not require additional device resources.

Flaws:

- Works only in voltage, current, power or pulse modes resistance;
- In some cases, the results of ESR measurements may be distorted under the influence of transient processes of current stabilization through the CIT (see below for more details).

When measuring ESR in the voltage, current, power or resistance pulse modes, it should be taken into account that the change in current through the CCS when changing pulses does not occur instantly, but takes, on average, time **Tst**, from 1 to 10 ms, depending on the specific testing conditions and the specified parameter values (see Figure 43). In addition, there is a certain delay **Tdl** in the measurement and processing of the measurement results by the device. All this leads to a delay in obtaining the second voltage value at the CCS terminals when determining the voltage jump dU by a total time **(Tst + Tdl)** from 2 to 20 ms. As a consequence, the obtained result of measuring the voltage jump dUm is subject to **Err** distortions, since

the voltage on the tested CIT changes not only due to the voltage jump dU_i on the ESR itself, but also due to the fact that the tested CCS manages to partially charge or discharge during the measurement delay time ($T_{st} + T_{dl}$). Thus, the results of ESR measurements are overestimated or underestimated relative to the true ESR value of the CCS. This negative effect will be more pronounced the smaller the capacity of the CCS being tested, the lower its ESR value, and the faster the voltage changes on it during charging or discharging (which occurs, for example, when testing supercapacitors). In this case, to obtain the most accurate ESR values, it is recommended to use the ESR measurement method of periodic current interruption (see subsection 13.2).

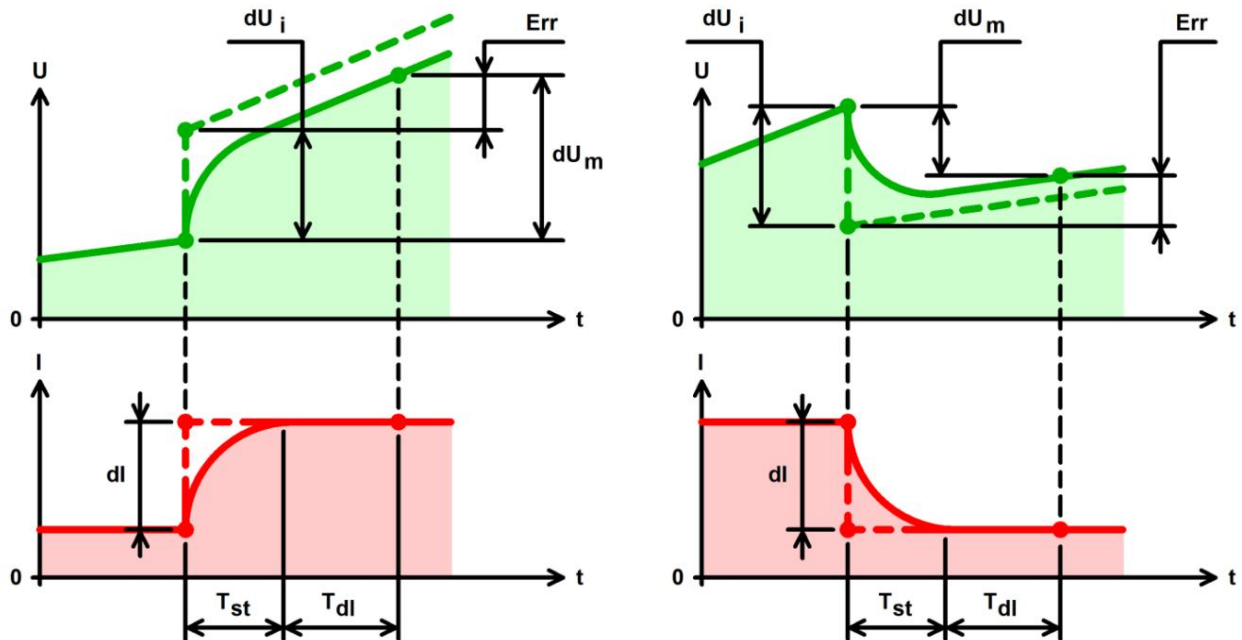


Figure 43 - Causes of errors when measuring ESR

here: dU_i – ideal value of voltage surge on ESR of the heat sink;

dU_m – measured value of voltage surge on ESR of the heat source;

Err – error in measuring the voltage surge on the ESR of the heat source;

dl – current surge causing a surge in voltage at the terminals of the heat exchanger;

T_{st} – time to establish the set value of current through the heat exchanger;

T_{dl} – delay of measurements and processing of measurement results.

ESR measurement results in current, power or resistance pulse modes are included in the results files in the columns "ESR,Ohm" and "ESRa,Ohm" (see section 15).

In addition, the ESR measurement results obtained by this method can be displayed on the charts on the "Charts" page (see subsection 19.4).

In general, the ESR measurement will be more accurate the greater the absolute value of the change in current through the CCS, the lower the rate of change in voltage on the CCS during charging and discharging (the greater its capacity), and the greater the measured ESR value of the CCS itself.

13.4 Measuring the ESR of the HIT when changing program steps

In this method, the ESR of the CCS is determined by the voltage jump on the CCS at the beginning of the next step of the CCS testing program. If the current through the CCS at the beginning of a new step has become greater than it was at the end of the previous step of the program, the current jump is positive, and the result of the ESR measurement at the beginning of this step refers to the beginning of the charge (**ESRc** – ESRcharge). If the current through the CCS at the beginning of a new step has become less than it was at the end of the previous step of the program, the current jump is negative, and the result

ESR measurements at the beginning of this step refer to the beginning of the discharge (**ESRd** – **ESRdischarge**). Both types of these values are included in the summary file of measurement results, but for a specific step of the program only one of them will be significant, the second will be zero (see subsection 15.3).

If several charge (discharge) steps are used in a cycle, the "**GNRL**" lines of the summary file for program cycles contain the averaged results of **ESRc** and **ESRd** measurements for all charge steps and all discharge steps in a given cycle.

This method of measuring ESR always works and cannot be switched off when preparation of a HIT testing program.

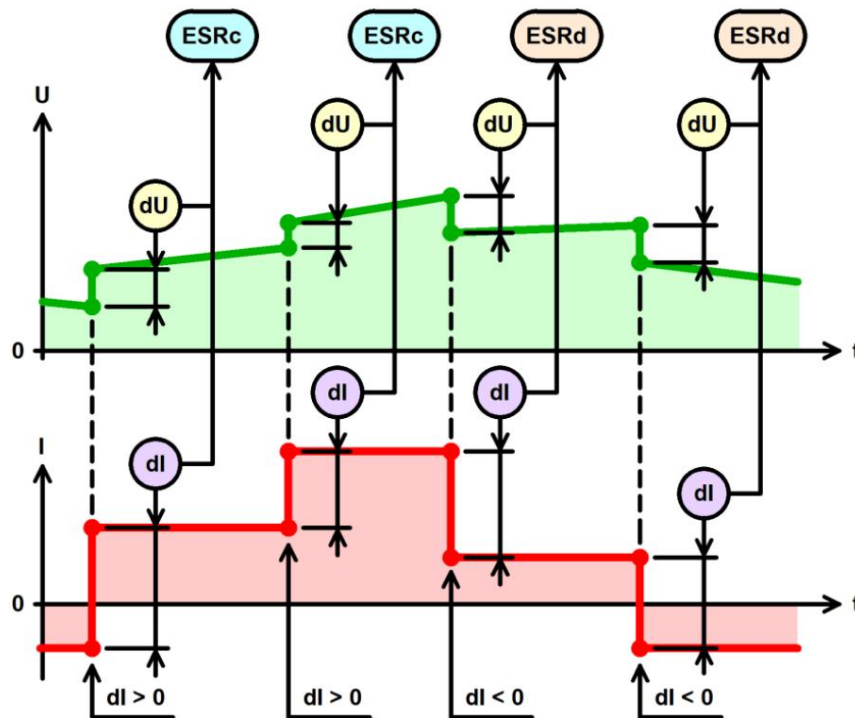


Figure 44 - Measuring the ESR of the HIT when changing the steps of the HIT testing program

Advantages of the method of measuring the ESR of the heat sink when changing program steps:

- Does not reduce the average value of current through the HIT, does not increase the charging time or discharge and does not require additional device resources.

Flaws:

- Allows you to obtain only one ESR measurement result at the beginning of the next step of the HIT testing program;
- Does not work for all types of steps of the HIT testing program;
- In some cases, ESR measurement results may be distorted by transients and measurement delays (see subsection 13.3 and Figure 43).

The method does not work on the constant voltage charge (CHCV), constant voltage charge (DCHCV) and voltage sweep steps. For these steps, the **ESRc** and **ESRd** measurement results will always be zero.

The method has similar features and limitations as the method for measuring the ESR of a chemical power supply in the modes of voltage, current, power or discharge resistance pulses (see subsection 13.3 and Figure 43).

In general, the ESR measurement will be more accurate the greater the absolute value of the change in current through the CIT at the beginning of the next step of the program, the lower the rate of change of voltage on the CIT during charging and discharging (the greater its capacity), and the greater the measured value of the CIT ESR itself.

14 DETERMINATION OF THE ELECTRIC CAPACITY OF THE HIT

The ASK75.10.20.2 and ASK75.10.20.12 analyzers allow measuring the electrical capacity of the HIT in Farads. Measuring the electrical capacity is necessary when testing supercapacitors (ionistors).

The electrical capacity is determined only at the step of discharge of the CCS with a specified constant current (DCHCC), which must be included in the testing program (see Section 11). If the step of discharge with a constant current (DCHCC) is not included in the testing program, the result of measuring the electrical capacity will be zero.

The calculation of the electrical capacity of the heat source is carried out using two points on the graph changes in voltage during the discharge of the chemical energy source with a given constant current.

The first point is taken at a level of 10% of the total change in voltage on the CIT during the DCHCC step, minus the initial voltage jump on the CIT ESR (see section 13).

The second point is at the level of 90% of the total change in voltage on the power supply, also minus the initial voltage surge on the ESR of the HIT.

The change in voltage on the heat exchanger ΔU is determined by these two points. and the control discharge time Δt during which this change in voltage ΔU occurred:

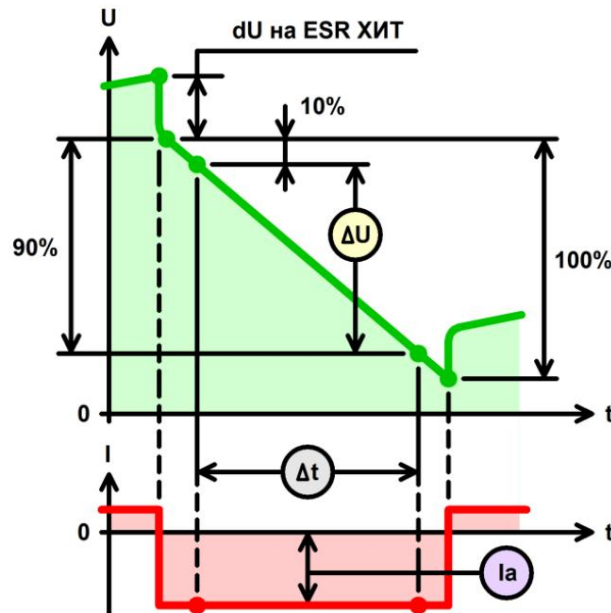


Figure 45 - Determination of the electrical capacity of the heat exchanger

The calculation of electrical capacity is carried out according to formula (16):

$$C = I_a \cdot \frac{\Delta t}{\Delta U} \quad (16)$$

where: **C** is the electrical capacity of the heat source, F;

I_a is the average discharge current of the chemical energy storage device during the control discharge time Δt , A;

Δt - control discharge time, s;

ΔU is the change in voltage on the CCS during the control discharge time Δt , V.

If several DC discharge steps (DCHCC) are used in a cycle, the "GNRL" lines of the cycle summary file (see Section 15.3) contain the averaged result of the measurement of the electrical capacitance of the power supply voltage source over all DCHCC steps in a given cycle.

To correctly determine the electrical capacity of supercapacitors (ionistors), it is recommended to select such a frequency of data recording in the measurement results files (see subsection 19.2) that at the step (steps) of discharge with direct current

(DCHCC) the device recorded at least twenty points* of measurement of the CHIT parameters.

* see the list of accepted abbreviations and designations.

15 MEASUREMENT RESULTS FILES

15.1 General instructions

As a result of testing the HIT using the ASK75.10.20.2 Analyzers and ASK75.10.20.12, the User receives two different types of output files.

All measurement result files created by the device are in text format (*.txt) and can be opened for viewing and processing in any text editor (for example, Notepad in the Windows environment), or in spreadsheet and data processing programs (for example, MS Excel, see section 18).

The results files of both types consist of a header and a table with the measurement results obtained during the testing of the HIT. The columns in the results files are separated from each other by spaces.

The description of the parameters included in the measurement results files is presented in subsections 15.2 and 15.3. The methods for determining the parameters included in the measurement results files are presented in sections 12, 13 and 14.

Both file types have a similar header containing:

- The type of Analyzer used to obtain this file;
- IP address of the Analyzer (see Sections 9 and Subsection 19.7);
- The channel number of the device on which the test was performed, or a list of channel numbers in the case of their parallel operation;
- The name of the HIT entered by the User when starting the test (see subsection 19.2);
- Description of the specified test preparation steps (if the program specifies preparation steps, see section 11);
- Description of the specified steps of the cyclic part of the testing program (if the program has a cyclic part);
- Specified number of repetitions of the cyclic part of the testing program;
- Description of the specified testing completion steps (if completion steps are specified in the program);
- Specified limitations of the parameters of the HIT (see subsection 19.2 and section 22);
- The specified period of recording data in primary data files (see subsection 19.2);
- Test start date and time (for test result summary files only, see subsection 15.3);
- Header of parameters in the measurement results table.

The description of each specified step of the test program in the header of the results files consists of the number of this step in the specified test program, a description of the step type, the operating parameters of the step, the specified criteria for completing the step, and the parameters for measuring the ESR of the CCS using the periodic current interruption method (if the ESR measurement by this method is available for the selected step type).

In addition, in the body of the files itself, in the table with the test results, there is a column containing the number and marker of the step type for which the results are presented in a specific line of the measurement results file.

Correspondence of step types, their text description in the file header and their markers in the lines of the result files is presented in Table 8.

In case of manual interruption of the process by the "Stop" button for the corresponding channel of the device (see subsection 19.1), a final record of the interruption of the testing program by the User is added to the primary data file for the interrupted cycle and to the summary measurement results file, and the date and time of such interruption are also noted.

interruptions.

Table 8 - Markers and headings of the steps of the HIT testing program

Marker steps	Description type of step	Type of step of the testing program of HIT	Note
SSS	Charge CC Direct	current charge see subsection 11.3	
USSR	Charge CP Constant	power charge see subsection 11.4	
CCV	Charge CV Constant	voltage charging see subsection 11.5	
DCC Discharge	Direct	Current Discharge see subsection 11.6	
DCP Discharge	CP Discharge	constant power see subsection 11.7	
DCR Discharge	CR Discharge	to constant resistance see subsection 11.8	
DCV Discharge	CV Constant	Voltage Discharge see subsection 11.9	
RLX	Rest	Relaxation	see subsection 11.10
SNU Scanning	U Voltage	Scan	see subsection 11.11
SNI	Scanning I	Current scan	see subsection 11.12
SNP Scanning	P Power	Scan	see subsection 11.13
SNR Scanning	R Resistance	Scanning	see subsection 11.14
CUP	Impulse U Voltage	pulse mode	see subsection 11.15
IPI	Impulse I Current	pulse mode	see subsection 11.16
IPP	Impulse P Power	pulse mode	see subsection 11.17
IPR	Impulse R Resistance	pulse mode	see subsection 11.18
TBU	Table U	Working on the voltage table Working	see subsection 11.19.2
TBI	Table I	on the current table Working	see subsection 11.19.3
TBP	Table γ	on the power table Working on the	see subsection 11.19.4
TBR	Table R	resistance table	see subsection 11.19.5
LGU	Logger U Voltage	recorder	see subsection 11.20
PMT	MPPT	MPRT mode	see subsection 11.22

In the event of emergency situations during testing, the testing process of the HIT is automatically interrupted, and a final line is entered into the primary data files (see subsection 15.2), in which the step marker is replaced by the marker of the emergency that occurred.

The description of emergency markers is given in Table 9.

A detailed description of possible emergency situations and the reasons for their occurrence set out in section 22.

Table 9 - Emergency condition markers

Marker accidents	Description of the accident	Note
OAL Crash HIT		see section 22
OUB Overvoltage at channel power terminals		
OUS Overvoltage at potential terminals of the channel		
RPB	Reverse polarity on the channel power terminals	
RPS	Reverse polarity on potential terminals	
ULL	Minimum voltage limitation on the heat exchanger	
UHL Maximum Voltage Limitation on the Heat Supply		
ICL	Limiting the maximum charging current of the HIT	
IDL	Limiting the maximum discharge current of the HIT	
TLL	Minimum temperature limitation of the HIT	
THL	Limiting the maximum temperature of the HIT	
PCA Lost connection with PC when working with tables		
OUP Overvoltage on channel power supply		
PSA	Instrument Channel Power Supply Failure	
OLS	Overload of channel power transistors	
IVA	Failure of the power inverter of the device channel	
WAS	Device channel overcurrent	
OLP	Device channel power overload	
OTL	Overheating of the load resistor R (see Figure 7 and 8)	
FINISHED	Overheating of the power inverter of the device channel	
IAL	Lost connection with the device channel	
SHE	Device channel initialization failure	
MAL General channel device fault		

The following subsections provide a description of the parameters included in the files.
measurement results.

15.2 Primary data files

Primary data files (RAW) contain instantaneous values of voltage on the CIT, V, current through the CIT, A, and temperature of the CIT, °C, with a specified recording step parameters in time.

Also, the primary data files contain instantaneous ESR values of the CIT, mOhm, measured by the method of periodically interrupting the current through the CIT or obtained in voltage, current, power or discharge resistance pulse modes (see section 13).

In addition, the primary data files contain the values of the received (delivered) charge, A h, and energy, W h, cumulatively from the start of the next step of the specified testing program of the HIT (see Figure 46).

Lii-50E-00000005.txt – Блокнот

Файл Правка Формат Вид Справка

Analyzer: ACK75.10.20.2 IP: 192.168.100.120 Channel: 1
 Object: Lii-50E
 Program:
 4 Charge CC 7A to 4.2V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
 5 Charge CV 4.2V to 1A or 1d.
 6 Discharge CC 10A to 2.5V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
 Preset number of cycles: 10000.
 Limitation Umin: 2.4V; Umax: 4.3V; ICmax: not set; IDmax: not set; t°min: 10°C; t°max: 80°C.
 Data recording period: 5s

Cycle	Step	Time,s	U,V	I,A	T,°C	ESR,mR	Q,Ah	E,Wh
5	4CCC	0.050000	2.8512	6.9866	54.7	.00000	.00010	.00028
5	4CCC	5.050000	3.0444	6.9990	54.6	18.022	.00981	.02891
5	4CCC	10.050000	3.1196	6.9990	54.7	17.859	.01953	.05887
5	4CCC	15.050000	3.1704	6.9994	54.7	17.760	.02925	.08944
5	4CCC	20.050000	3.2098	6.9993	54.9	17.695	.03897	.12045
5	4CCC	25.050000	3.2421	6.9990	54.9	17.596	.04869	.15181
5	4CCC	30.050000	3.2698	6.9994	54.9	17.596	.05841	.18346
5	4CCC	35.050000	3.2941	6.9992	55.0	17.531	.06814	.21537
5	4CCC	40.050000	3.3157	6.9992	55.0	17.498	.07786	.24749
5	4CCC	45.050000	3.3352	6.9992	55.0	17.498	.08758	.27982

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 46 - Primary data file

The following notations are used in the primary data file above:

- **Charge CC** - charging with a specified constant current;
- **Charge CV** - additional charge at a given constant voltage;
- **Discharge CC** - discharge with a given constant current;
- **4, 5, 6** - numbers of the testing program steps (see section 11, 19.2);
- **Umin, Umax, ICHmax, IDCHmax, Tmin, Tmax** – specified voltage limits, current and temperature for the tested chemical energy source (see subsection 19.2 and section 22);
- **Cycle** - the program cycle number. **3** – this entire file is for cycle No. 3;
- **Step** - marker of the step of the HIT testing program.
4CCC – the number of the test program step (4) and the marker of the test mode at this step (**CCC**) (see Table 8);
- **Time** - time from the start of the next step in the selected format (see section 17);
- **U,V** - instantaneous voltage on the heat exchanger, V;
- **I,A** - instantaneous value of current through the CIT, A, adjusted by the coefficient KESR (see subsection 13.2). Positive values correspond to the process of charging the CCS, negative values correspond to the process of discharging;
- **T, °C** – instantaneous temperature of the heat exchanger, °C. Measured only when connected to the channel of the external temperature sensor HIT. If the temperature sensor is not connected, the temperature measurement result will be zero;
- **ESR,mR** - instantaneous value of ESR, mOhm, measured by the method of periodically interrupting the current through the CCS (see subsection 13.2) or in the modes of current, power or resistance pulses (see subsection 13.3);
- **Q,Ah** - received (given) charge, A h, cumulative total from the beginning of the step;
- **E,Wh** - received (given) energy, W h, cumulatively from the beginning of the step.

Based on the primary data files, charge-discharge curves of the CCS can be constructed for each completed cycle of the testing program, the dependence of the energy given on the CCS voltage, the dependence of some CCS parameters on others can be identified, special graphs can be constructed when studying the CCS using the Cyclic VoltAmperometry (CVA) method, etc.

Primary data files are an inevitable result of testing the HIT. For each cycle of the testing program, the device will create its own primary data file.

Thus, let us assume that when preparing to launch the HIT testing, the User entered the name of the file for saving the measurement results **"TEST.txt"** in the results file selection window (see subsection 19.2).

In this case, each time the next primary data file is saved, the program cycle number for which the data is saved in this file will be automatically added to the file name specified by the User, in our example:

```
("TEST-00000000.txt")
"TEST-00000001.txt"
"TEST-00000002.txt"
"TEST-00000003.txt"
"TEST-00000004.txt"
"TEST-00000005.txt"
.....
"TEST-[N].txt"
("TEST-[N + 1].txt")
```

A file with the number "-00000000" is created for the measurement results at steps preparation, if any, specified in the test program (see subsection 11.1).

If the test program specifies completion steps, and the specified number of cycles is N , the measurement results for the completion steps will be saved to a file with the number $N + 1$. So, if for our example the specified number of cycles is **100** cycles, then the file containing the measurement results for the test completion steps (if such are specified in the program) will have the name "TEST-00000101.txt".

All files will be saved in the folder selected by the User, specified when starting the testing (see subsection 19.2) and after the completion of the specified test program, their number will correspond to the number of completed cycles of the HIT testing program, including the file of the results of the execution of the preparation steps and the file of the results of the execution of the completion steps of the testing program. The number of primary data files can be significantly reduced if the mode of thinning the measurement results files is configured when specifying the testing program (see section 16).

ATTENTION! To avoid malfunctions and confusion in the measurement results files, each time a test is launched, a separate folder must be created on the User's computer to store its results!

15.3 Summary measurement results files

The second type of measurement results file is a summary file of results by steps or by cycles and contains summary (generalized, processed) data for each step or cycle of the completed testing program of the investigated chemical equipment.

The creation of a file of this type can be allowed or prohibited in the settings of the HIT testing program. The file itself, depending on the settings, can contain summary data for each completed *step* or for each completed *cycle* HIT testing programs (see subsection 19.2).

The ending "-CLK" is automatically added to the file name of the summary results of measurements by steps or cycles. For the example considered in subsection 15.2, the name of this file will be **"TEST-CLK.txt"**.

Lii-50E-CLK.txt – Блокнот

Файл Правка Формат Вид Справка

Analyzer: ACK75.10.20.2 IP: 192.168.100.120 Channel: 1
Object: Lii-50E
Program:
4 Charge CC 7A to 4.2V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
5 Charge CV 4.2V to 1A or 1d.
6 Discharge CC 10A to 2.5V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
Preset number of cycles: 10000.
Limitation Umin: 2.4V; Umax: 4.3V; ICmax: not set; IDmax: not set; t°min: 10°C; t°max: 80°C.
Data recording period: 5s
Testing started: 19.04.2023 2:54:53

Cycle	Step	Drt,s	Ue,V	Ie,A	T,°C	ESR,mR	Q,Ah	E,Wh	C,F	ERC,mR	ERd,mR	Ilk,A	EFq,%	EFe,%
1	4CCC	678.3100	4.1984	7.0051	43.1	14.260	1.3188	5.3522	.000000	.00000	.00000			
1	5CCV	1059.070	4.1983	.99612	38.4	.00000	.90872	3.8152	.000000	.00000	.00000			
1	6DCC	1621.080	2.5005	-9.999	43.5	15.107	-4.503	-15.41	12628.2	.00000	16.673			
1	GNRL	3358.460	2.5005	-9.999	41.7	14.684	-4.503	-15.41	12628.2	.00000	16.673	-2.4393	202.2	168.1
2	4CCC	1846.110	4.1984	7.0005	47.4	15.235	3.5893	13.735	.000000	20.002	.00000			
2	5CCV	1070.450	4.1983	.99539	38.1	.00000	.91488	3.8410	.000000	.00000	.00000			
2	6DCC	1618.480	2.5006	-10.00	43.1	15.273	-4.496	-15.37	12753.3	.00000	16.928			
2	GNRL	4535.040	2.5006	-10.00	42.9	15.254	-4.496	-15.37	12753.3	20.002	16.928	.006551	99.8	87.5
3	4CCC	1841.990	4.1984	6.9994	46.9	15.329	3.5813	13.711	.000000	20.264	.00000			
3	5CCV	1075.180	4.1983	.99940	37.9	.00000	.91726	3.8510	.000000	.00000	.00000			
3	6DCC	1616.970	2.5005	-10.00	43.0	15.316	-4.492	-15.35	12663.7	.00000	17.007			
3	GNRL	4534.140	2.5005	-10.00	42.6	15.322	-4.492	-15.35	12663.7	20.264	17.007	.005426	99.8	87.4
4	4CCC	1834.100	4.1984	7.0037	46.8	15.578	3.5659	13.659	.000000	20.458	.00000			
4	5CCV	1089.280	4.1984	.99989	37.8	.00000	.92691	3.8915	.000000	.00000	.00000			
4	6DCC	1614.640	2.5006	-10.00	43.0	15.445	-4.485	-15.32	12768.0	.00000	17.225			
4	GNRL	4538.020	2.5006	-10.00	42.5	15.512	-4.485	-15.32	12768.0	20.458	17.225	.006035	99.8	87.3

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 47 - Summary measurement results file in the "For each step" option
(Every step)

*Lii-50E-CLK.txt – Блокнот

Файл Правка Формат Вид Справка

Analyzer: ACK75.10.20.2 IP: 192.168.100.120 Channel: 1
Object: Lii-50E
Program:
4 Charge CC 7A to 4.2V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
5 Charge CV 4.2V to 1A or 1d.
6 Discharge CC 10A to 2.5V or 1d. Period ESR: 5s. Duration ESR: 1000Hz.
Preset number of cycles: 10000.
Limitation Umin: 2.4V; Umax: 4.3V; ICmax: not set; IDmax: not set; t°min: 10°C; t°max: 80°C.
Data recording period: 5s
Testing started: 19.04.2023 2:54:53

Cycle	Step	Drt,s	Ue,V	Ie,A	T,°C	ESR,mR	Q,Ah	E,Wh	C,F	ERC,mR	ERd,mR	Ilk,A	EFq,%	EFe,%
1	GNRL	3358.460	2.5005	-9.999	41.7	14.684	-4.503	-15.41	12628.2	.00000	16.673	-2.4393	202.2	168.1
2	GNRL	4535.040	2.5006	-10.00	42.9	15.254	-4.496	-15.37	12753.3	20.002	16.928	.006551	99.8	87.5
3	GNRL	4534.140	2.5005	-10.00	42.6	15.322	-4.492	-15.35	12663.7	20.264	17.007	.005426	99.8	87.4
4	GNRL	4538.020	2.5006	-10.00	42.5	15.512	-4.485	-15.32	12768.0	20.458	17.225	.006035	99.8	87.3
5	GNRL	4534.750	2.5005	-10.00	43.0	15.789	-4.484	-15.30	12687.1	20.637	17.228	.006013	99.8	87.1

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 48 - Summary measurement results file in the "For each cycle only" option
(Only by cycles)

The summary results file can be saved by the Analyzer in two variants: "For each step" or "For each cycle only". The choice of saving option is specified by the User when creating the testing program before starting the test (see subsection 19.2).

When selecting the "For each step" option, the summary results for each completed step of the testing program are entered into the file lines.

completion of the next cycle, based on the results of all the steps performed, a line with summary data for the entire cycle is calculated and formed. This line has marker 79

"GNRL" and contains the final data collected and averaged over all the steps performed, as well as a number of additional parameters (see Figure 47).

If you select the "For each cycle only" option, the results for each step are not saved, and only the "GNRL" lines are included in the summary results file (see Figure 48).

The summary results file contains the following parameters:

- The total duration of a step or cycle in the selected time format (see section 17);
- Final voltage (extreme voltage value) of a step or cycle, V;
- Final current (extreme current value) of a step or cycle, A, corrected by the KESR coefficient (see subsection 13.2);
- Averaged per step or per cycle temperature of the chemical power supply, °C; ¹
- Averaged per step or per cycle result of the ESR measurement, obtained by the method of periodically interrupting the current through the chemical power supply or in the modes of voltage, current, power or resistance pulses, mOhm (see subsections 13.2 and 13.3);
- Charge (capacity) received (given) by the heat source per step or per cycle, A h;
- Energy received (given) by the heat source per step or per cycle, W h;
- Electric capacity HIT, F. Makes sense only for supercapacitors (ionistors) and is determined *only* at the DC discharge step (DCHCC) (see section 14) (for summary results per step), or the average result of determining the electrical capacity for all DCHCC steps in a given cycle (for "GNRL" lines); ²
- ESR of the CCD, measured by the voltage jump at the beginning of the charge (see subsection 13.4) (for summary results per step), or the average result of ESR measurement for all charge steps in a given cycle (for "GNRL" lines), mOhm;
- ESR of the HIT, measured by the voltage jump at the beginning of the discharge (see subsection 13.4) (for summary results per step), or the average result of the ESR measurement for all discharge steps in a given cycle (for the "GNRL" lines), mOhm;
- Average leakage current through the power supply unit per cycle, A (estimated value); ³
- Power supply unit efficiency by charge (by A h), %;
- Efficiency of the heat source by energy (in W h), %.

In the above result files (see Figures 47 and 48) the following notations are used:

- **Charge CC** - charging with a specified constant current;
- **Charge CV** - additional charge at a given constant voltage;
- **Discharge CC** - discharge with a given constant current;
- 4, 5, 6 – numbers of the testing program steps (see section 11, 19.2);
- **Umin, Umax, ICHmax, IDCHmax, Tmin, Tmax** – specified voltage, current and temperature limits for the tested power supply unit (see subsection 19.2 and section 22);
- **Cycle** - the cycle number of the testing program;
- **Step** - marker of the step of the HIT testing program.
4CCC, 5CCV, 6DyC – test program step number (4, 5, 6) and test mode marker (CyC, CCV, DCC) (see Table 8);
- **GNRL** - full cycle marker (GENERAL - full, generalized);
- **Drt** - the duration of a step or cycle in the selected time format.
For time format options, see section 17;
- **Ue,V** - voltage on the heat supply voltage, V, at the end of a step or cycle (Uend);

- **le,A** - current through the CIT, A, at the end of a step or cycle (Iend),
adjusted for the KESR coefficient (see subsection 13.2). Positive values correspond to charging, negative values to discharging;
- **ESR,mR** - average per step or per cycle ESR value, mOhm, measured by the method of periodic interruption of current through the CIT or in the modes of voltage, current, power or resistance pulses (see subsections 13.2 and 13.3);
- **Q,Ah** – received (given) charge (capacity), A h, per step or per cycle;
- **E,Wh** – received (given) energy, W h, per step or per cycle;
- **C,F** - electrical capacity of the CCS, F. It makes sense only for supercapacitors (ionistors) and is determined only at the steps of discharge with direct current (**DCHCC**) (see section 14); • **ERc,mR** - ¹
ESR of the CCS, measured by the voltage jump at the beginning of the charge (see subsection 13.4) (for summary results per step), or the averaged result of ESR measurement for all charging steps in a given cycle (for "**GNRL**" lines), mOhm;
- **ERd,mR** – ESR of the CIT, measured by the voltage jump at the beginning of the discharge (see subsection 13.4) (for summary results per step), or the averaged result of the ESR measurement for all discharge steps in a given cycle (for the "**GNRL**" lines), mOhm;
- **Ilk,A** - leakage current through the CIT, A, averaged within the cycle. Indirectly reflects self-discharge rate of the heat exchanger; ²
- **EFq,%** - Efficiency Factor of the chemical power supply, calculated by charge (by A h). Indirectly reflects the leakage current and self-discharge rate of the chemical power supply;
- **EFe,%** - Efficiency factor of the heat source, calculated by energy (W h).
Reflects energy losses in the chemical energy storage device during cyclic charge-discharge.

Notes:

1. Measuring the HIT temperature is possible only when an external HIT temperature sensor is connected to the device channel; if the HIT temperature sensor is not connected, the HIT temperature measurement result will be zero;
2. The electric capacity of supercapacitors (ionistors) can be determined *only* in the discharge mode with a given constant current (DCHCC); if this step is not in the testing program, the result of measuring the electric capacity will be zero;
3. The parameter is purely evaluative. For more details, see Section 12.

16 THINNING MEASUREMENT RESULTS FILES

The ASK75.10.20.2 and ASK75.10.20.12 analyzers are capable of testing any type of HCS, including supercapacitors (ionistors). During resource tests of the latter, the number of charge-discharge cycles can reach 1,000,000 or more. In this case, the volume of data recorded by the device and the number of files of measurement results created (one primary data file for each charge-discharge cycle, plus one general file, see Section 15) becomes excessively large, which can significantly complicate or make impossible subsequent processing and analysis of measurement results.

Considering that the parameters of supercapacitors change insignificantly from charge-discharge cycle to cycle, there is no point in saving the measurement results for each charge-discharge cycle. When conducting resource tests of supercapacitors, it is more reasonable to save the measurement results only for, for example, every hundredth charge-discharge cycle. In this case, the number of primary data files is sharply reduced, as well as

The number of lines in the test summary file is sharply reduced (see subsection 15.3). As a result, subsequent processing and analysis of the measurement results files, as well as their copying, moving and deleting, if necessary, is significantly simplified.

In order to thin out the measurement results, the User can select the required thinning coefficient for the measurement results before starting the HIT testing program (see Section 11). The thinning coefficient is selected on the "Tasks" page of the Computer Interface (see Subsection 19.2), from the corresponding drop-down list (see Figure 49):

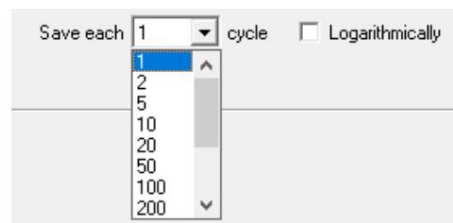


Figure 49 - Setting up the thinning of measurement results

By selecting the appropriate data decimation factor, you can specify the device on the need to save the measurement results only for each

- 1;
 - 2;
 - 5;
 - 10;
 - 20;
 - 50;
 - 100;
 - 200;
 - 500;
 - 1 000;
 - 2 000;
 - 5 000;
 - 10 000
- charge-discharge cycle.

Thus, when selecting the thinning coefficient of 100, the measurement results will be saved in the files for cycle #0 (preparation steps, if they are in the testing program, see Section 11), cycle #1, then cycle #100, #200, #300, and so on. The measurement results for the last charge-discharge cycle specified in the HIT testing program (for example, this is cycle #5385) and for the subsequent cycle containing the testing program completion steps, if they are present (in our example, this is cycle #5386) will also be saved.

Usually, the greatest changes in the parameters of the CIT occur during the first charge-discharge cycles. In order to record the changes in the parameters of the CIT during the first charge-discharge cycles in the most detail, and at the same time use the ability to thin out data with a given thinning coefficient, in the software

The software includes a checkbox "Logarithmically" (see Figure 49). Setting this checkbox activates saving the measurement results with a non-uniform thinning coefficient.

Thus, if the thinning factor of 100 is selected and the "Logarithmically" checkbox is checked, cycle #0 (if any) will be saved in the measurement results files, followed by ten consecutive charge-discharge cycles (these are cycles #1, #2, ..., #9 and #10), then the next ten charge-discharge cycles will be saved with a thinning factor of two (these are cycles #12, #14, ..., #28 and #30), then the next ten cycles will be saved with a thinning factor of five (these are cycles #35, #40, ..., #75 and #80) and so on, until the specified thinning factor of one hundred is reached. From this moment on, every hundredth charge-discharge cycle will be saved in the measurement results files, as if the "Logarithmically" checkbox was not checked.

installed.

Setting the "Logarithmically" checkbox is strongly recommended for new HCSs that have not yet been used. For such HCSs, the probability of significant changes in parameters is highest during the first charge-discharge cycles.

An example of graphs for analyzing changes in the parameters of the CIT (see subsection 19.6) at the beginning of testing the CIT with a specified data thinning coefficient of ten and the "Logarithmically" checkbox checked is shown in Figure 50. The points on the graphs correspond to the charge-discharge cycles for which the measurement results were saved in the results files:

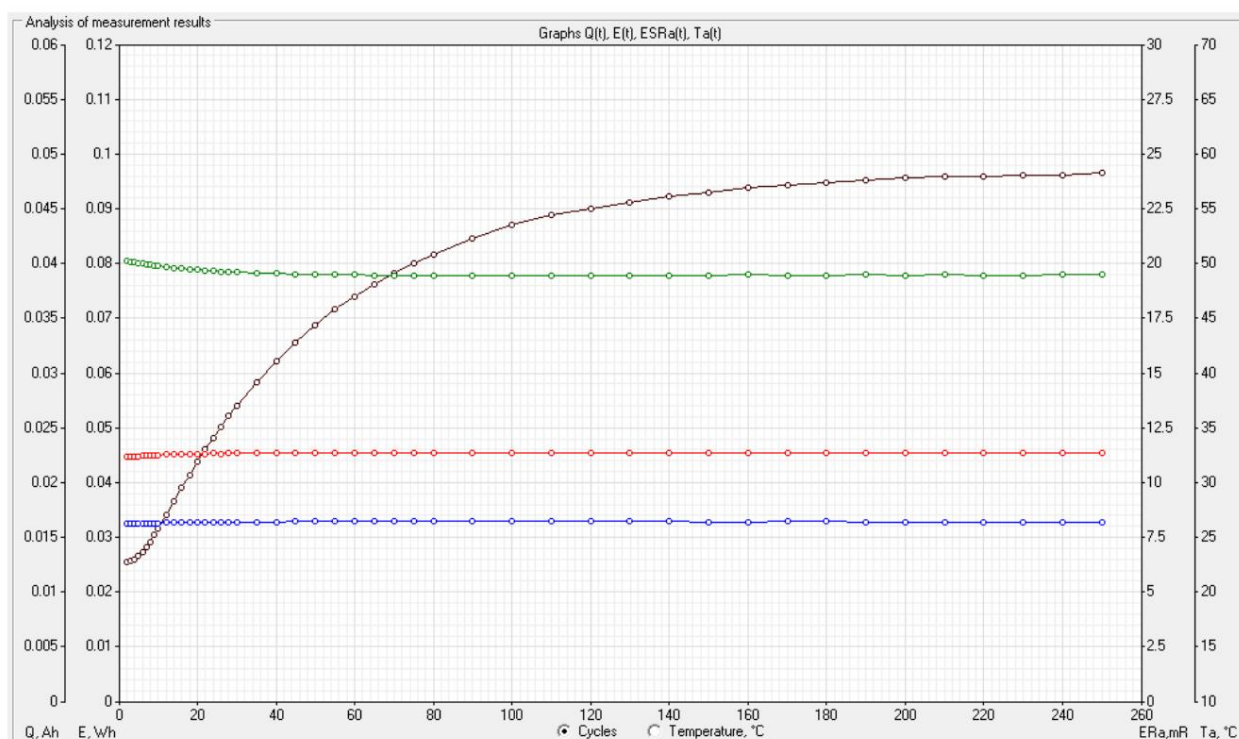


Figure 50 - Graphs of the analysis of the parameters of the heat exchanger with the thinning coefficient data - ten and the checkbox "Logarithmically" is checked

When selecting the thinning coefficient - one (no thinning), the installation The "Logarithmically" checkbox will not affect the way the measurement results are saved.

The selected decimation factor directly affects only the saving of data to measurement result files executed by the software on the PC. The decimation factor does not affect the use of the internal memory of the device and cannot be used to increase the battery life of the device (see Section 21).

17 TIME FORMAT IN MEASUREMENT RESULTS FILES

Before starting the testing of the HIT using the ASK75.10.20.2 and ASK75.10.20.12 Analyzers, the User can select various time format options in which the time values will be saved in the measurement results files:

- seconds;
- minutes;
- times;
- day;
- "HH:MM:SS".

The time format is selected on the "Tasks" page of the Computer Interface (see subsection 19.2), from the "Time format" drop-down list:

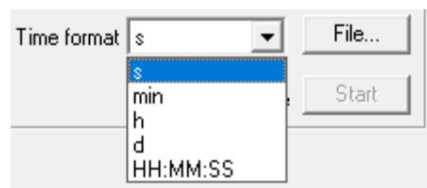


Figure 51 - Selecting the time format before starting the HIT testing

When selecting the time format "s" - seconds, the instantaneous time from the start of the step in the primary data files (see subsection 15.2), as well as the duration of the steps and cycles performed in the file of the summary results of measurements will be expressed in seconds. The time column header takes the form "Time,s" for the primary data files and "Drt,s" for the file of the summary results of testing:

Cycle	Step	Time,s	U,V	I,A	T, °C	ESR,mR	Q,Ah	E,Wh
10	4CCC	0.050000	1.9649	10.002	25.7	.00000	.00014	.00027
10	4CCC	0.150000	2.0575	9.9983	25.7	.00000	.00042	.00083
10	4CCC	0.250000	2.1181	9.9926	25.7	.00000	.00069	.00141
10	4CCC	0.350000	2.1712	9.9924	25.7	.00000	.00097	.00201
10	4CCC	0.450000	2.2215	9.9919	25.7	.00000	.00125	.00262

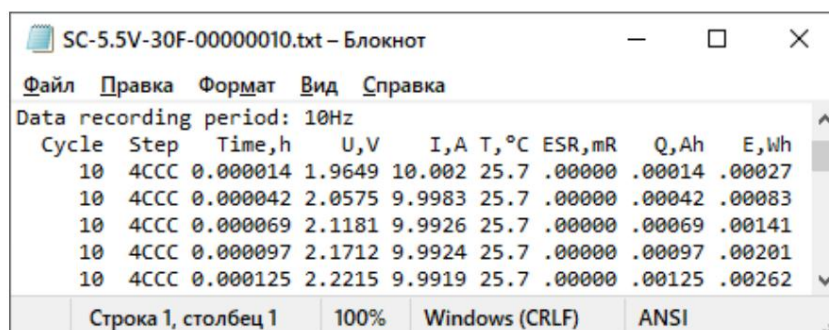
Figure 52 - Time format - seconds

When selecting the time format "min" - minutes, the time in the measurement results files will be expressed in minutes. The time column header takes the form "Time,m" for the primary data files and "Drt,m" for the summary test results file:

Cycle	Step	Time,m	U,V	I,A	T, °C	ESR,mR	Q,Ah	E,Wh
10	4CCC	0.000833	1.9649	10.002	25.7	.00000	.00014	.00027
10	4CCC	0.002500	2.0575	9.9983	25.7	.00000	.00042	.00083
10	4CCC	0.004167	2.1181	9.9926	25.7	.00000	.00069	.00141
10	4CCC	0.005833	2.1712	9.9924	25.7	.00000	.00097	.00201
10	4CCC	0.007500	2.2215	9.9919	25.7	.00000	.00125	.00262

Figure 53 - Time format - minutes

When selecting the time format "h" - hours, the time in the measurement results files will be expressed in hours. The time column header takes the form "Time,h" for files primary data and "Drt,h" for the test summary results file:



SC-5.5V-30F-00000010.txt - Блокнот

Файл Правка Формат Вид Справка

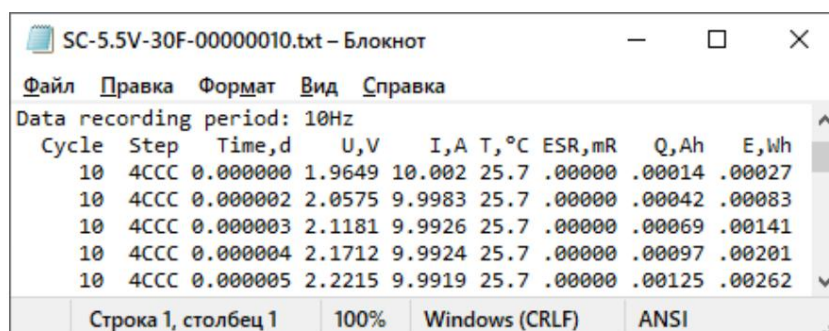
Data recording period: 10Hz

Cycle	Step	Time,h	U,V	I,A	T,°C	ESR,mR	Q,Ah	E,Wh
10	4CCC	0.000014	1.9649	10.002	25.7	.00000	.00014	.00027
10	4CCC	0.000042	2.0575	9.9983	25.7	.00000	.00042	.00083
10	4CCC	0.000069	2.1181	9.9926	25.7	.00000	.00069	.00141
10	4CCC	0.000097	2.1712	9.9924	25.7	.00000	.00097	.00201
10	4CCC	0.000125	2.2215	9.9919	25.7	.00000	.00125	.00262

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 54 - Time format - hours

When selecting the time format "d" - day, the time in the measurement results files will be expressed in days. The time column header takes the form "Time,d" for files primary data and "Drt,d" for the test summary results file:



SC-5.5V-30F-00000010.txt - Блокнот

Файл Правка Формат Вид Справка

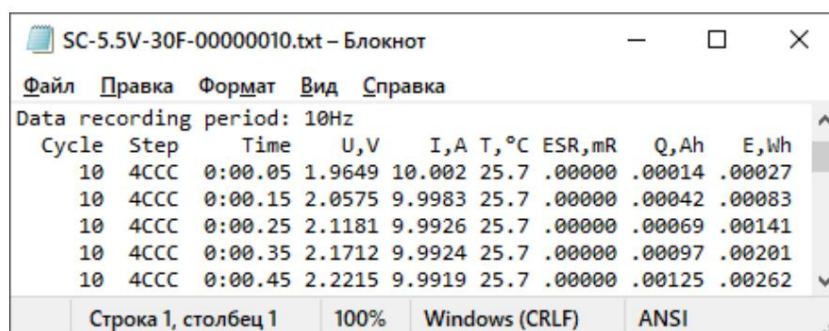
Data recording period: 10Hz

Cycle	Step	Time,d	U,V	I,A	T,°C	ESR,mR	Q,Ah	E,Wh
10	4CCC	0.000000	1.9649	10.002	25.7	.00000	.00014	.00027
10	4CCC	0.000002	2.0575	9.9983	25.7	.00000	.00042	.00083
10	4CCC	0.000003	2.1181	9.9926	25.7	.00000	.00069	.00141
10	4CCC	0.000004	2.1712	9.9924	25.7	.00000	.00097	.00201
10	4CCC	0.000005	2.2215	9.9919	25.7	.00000	.00125	.00262

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 55 - Time format - day

Finally, when selecting the time format "HH:MM:SS", the time in the measurement results files will be displayed as hours, minutes and seconds. The time column header will be "Time" for the primary data files and "Drt" for the summary file. test results:



SC-5.5V-30F-00000010.txt - Блокнот

Файл Правка Формат Вид Справка

Data recording period: 10Hz

Cycle	Step	Time	U,V	I,A	T,°C	ESR,mR	Q,Ah	E,Wh
10	4CCC	0:00.05	1.9649	10.002	25.7	.00000	.00014	.00027
10	4CCC	0:00.15	2.0575	9.9983	25.7	.00000	.00042	.00083
10	4CCC	0:00.25	2.1181	9.9926	25.7	.00000	.00069	.00141
10	4CCC	0:00.35	2.1712	9.9924	25.7	.00000	.00097	.00201
10	4CCC	0:00.45	2.2215	9.9919	25.7	.00000	.00125	.00262

Строка 1, столбец 1 100% Windows (CRLF) ANSI

Figure 56 - Time format - "HH:MM:SS"

To improve the accuracy of time recording, depending on the current values time, the time recording will be done in one of three subformats:

- MM:SS.SS
- HH:MM:SS
- CCCC:MM

The time format "HH:MM:SS" is convenient for human perception, but is difficult for subsequent processing of measurement results in MS Excel or similar programs, if necessary.

18 WORKING WITH RESULT FILES IN MS Excel

Text files of measurement results obtained using the ASK75.10.20.2 and ASK75.10.20.12 Analyzers can be easily opened in MS Excel for subsequent processing and editing if necessary.

To open result files in MS Excel, you must:

- Launch the MS Excel program (blank page);
- Select the "File" tab in the menu at the top left, then the "Open" item;
- In the window that opens, select the option "All files" from the list of file types (see figure 57);
- Select the path to the desired results file and click the "Open" button;
- In the opened window "Text Wizard (import) - step 1 of 3" select the option "with delimiters", make sure that the drop-down list "File format" is set to "1251: Cyrillic (Windows)", and click the "Next" button (see Figure 58);
- In the next window "Text Wizard (import) - step 2 of 3" check the box "space" and press the "Done" button (see Figure 59);
- The measurement results are loaded onto the MS Excel page and become ready for further work (see Figure 60).

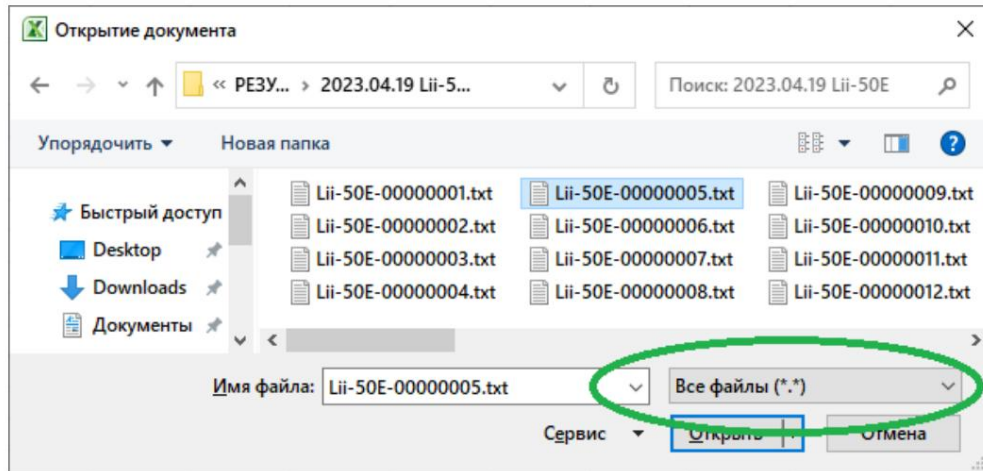


Figure 57 - Selecting a file with measurement results

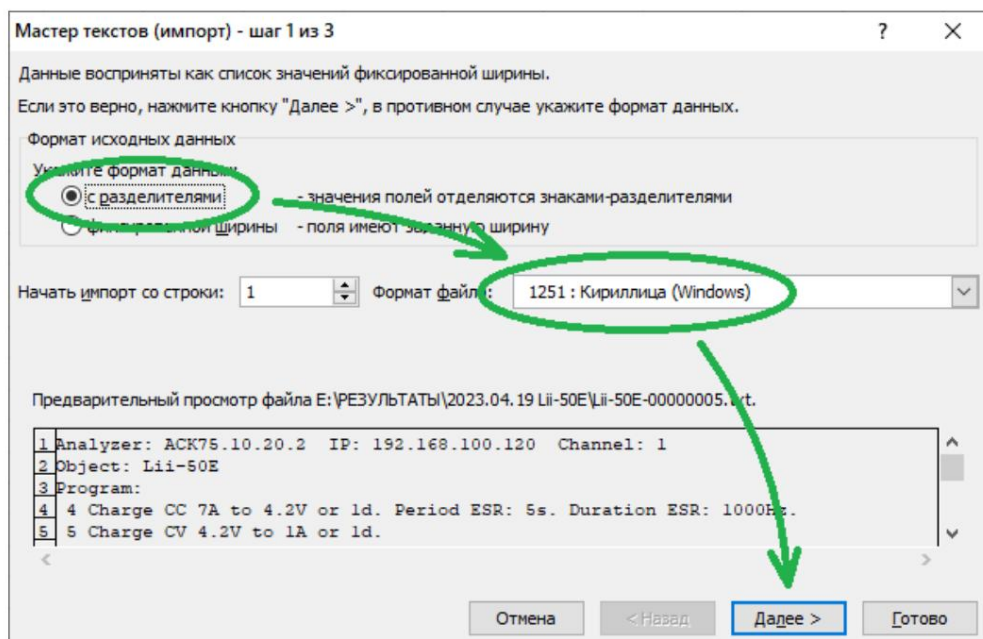


Figure 58 - Measurement results files contain separators

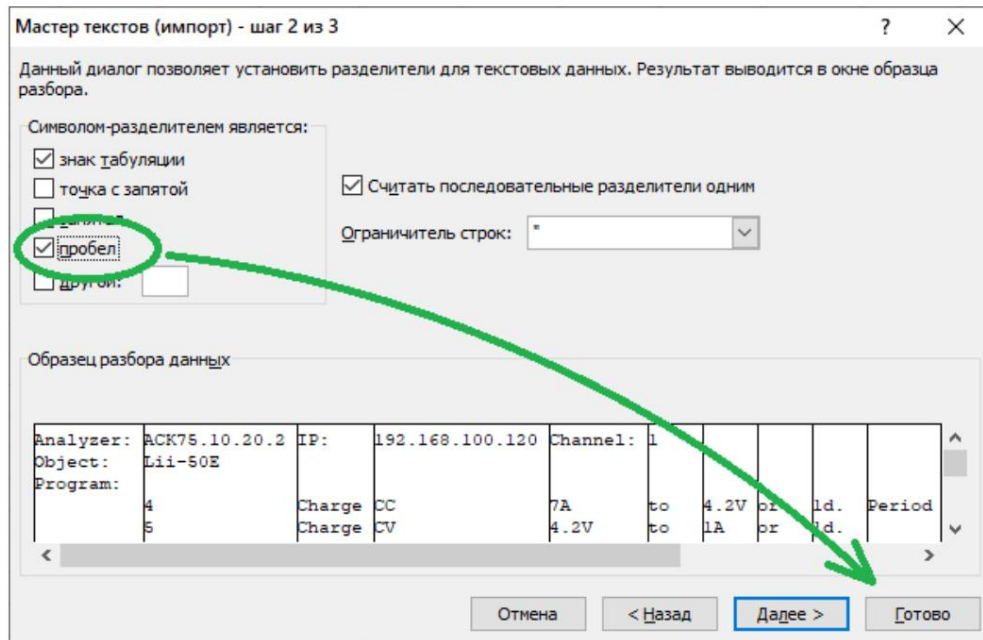


Figure 59 - Columns in the results files are separated by spaces

	A	B	C	D	E	F	G	H	I	J	K
1	Analyzer:	ACK75.10.	IP:	192.168.100.120	Channel:	1					
2	Object:	Lii-50E									
3	Program:										
4		4	Charge	CC	7A	to	4.2V	or	1d.	Period	ESR:
5		5	Charge	CV	4.2V	to	1A	or	1d.		
6		6	Discharge	CC	10A	to	2.5V	or	1d.	Period	ESR:
7	Preset	number	of	cycles:	10000						
8	Limitation	Umin:	2.4V;	Umax:	4.3V;	ICmax:	not	set;	IDmax:	not	set;
9	Data	recording	period:	5s							
10		Cycle	Step	Time,s	U,V	I,A	T,°C	ESR,mR	Q,Ah	E,Wh	
11		5	4CCC	0.05	2.8512	6.9866	54.7	0	0.0001	0.00028	
12		5	4CCC	5.05	3.0444	6.999	54.6	18.022	0.00981	0.02891	
13		5	4CCC	10.05	3.1196	6.999	54.7	17.859	0.01953	0.05887	
14		5	4CCC	15.05	3.1704	6.9994	54.7	17.76	0.02925	0.08944	
15		5	4CCC	20.05	3.2098	6.9993	54.9	17.695	0.03897	0.12045	
16		5	4CCC	25.05	3.2421	6.999	54.9	17.596	0.04869	0.15181	
17		5	4CCC	30.05	3.2698	6.9994	54.9	17.596	0.05841	0.18346	
18		5	4CCC	35.05	3.2941	6.9992	55	17.531	0.06814	0.21537	
19		5	4CCC	40.05	3.3157	6.9992	55	17.498	0.07786	0.24749	

Figure 60 - Measurement results file opened in MS Excel
and available for further processing

When planning subsequent processing of measurement results files in the MS Excel program, when specifying the HIT testing program, the format for saving time units should be selected as "s", "min", "h" or "d" (see Section 17).

Using the wide functionality of the MS Excel program, additional parameters can be calculated from the measurement results files, various graphs and histograms can be constructed, visual reports on the testing of the HIT can be created, and so on.

19 COMPUTER INTERFACE

19.1 General instructions

To control the device, create and launch the HIT testing program, as well as to monitor the progress of testing, obtain and process measurement results, the application software installed on the User's computer is used – the **Ym128 Computer Interface**.

Information on installing and configuring the connection between the device and the Computer The interface is described in Section 9 and Subsection 19.7.

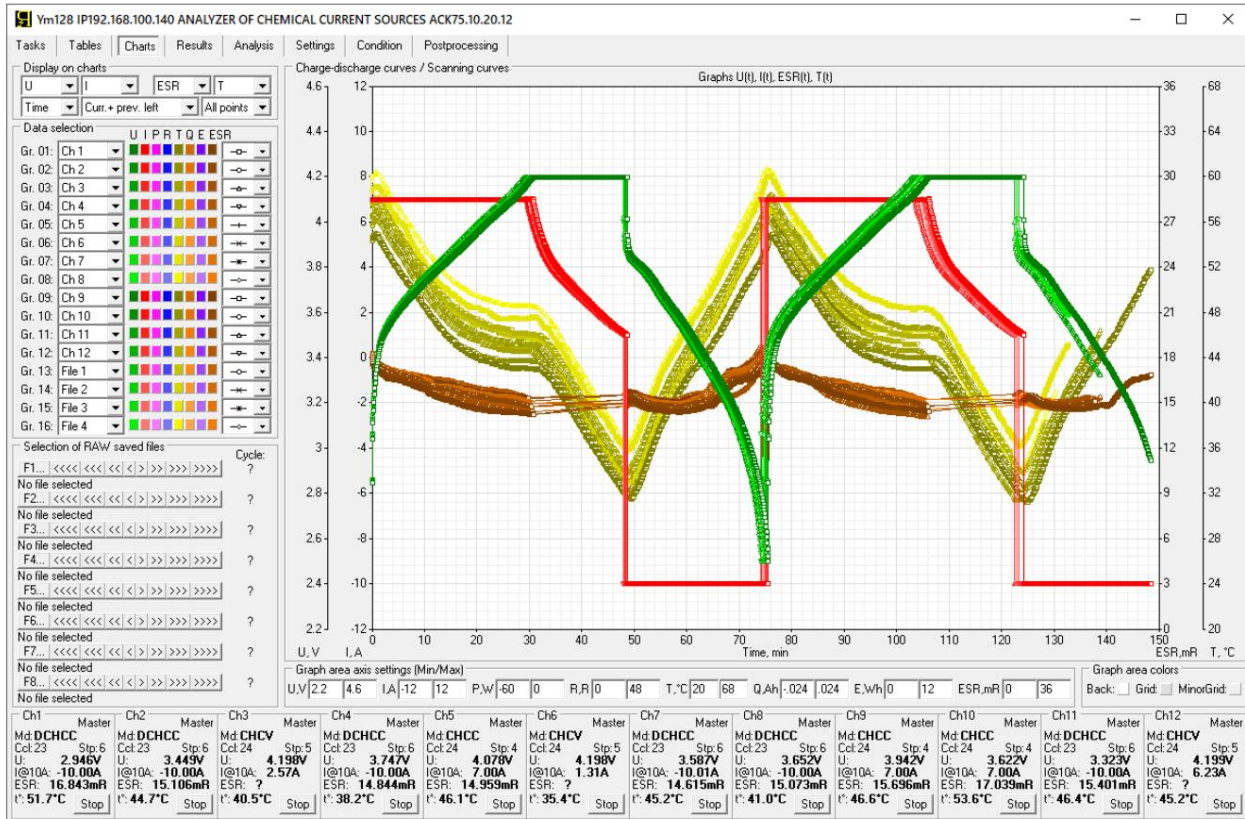


Figure 61 - External appearance of the Ym128 Computer Interface

The Ym128 Computer Interface allows you to:

- Create the necessary HIT testing program for each channel;
- Set the boundary conditions for testing the HIT;
- Set parameters for saving data in measurement results files;
- Launch the prepared HIT testing program for execution;
- Monitor the operation of the device in real time;
- Display measurement results in various forms;
- Plot graphs of the dependencies of various parameters of the heat exchanger on time, temperature, the number of charge-discharge cycles passed and from each other;
- Interrupt the HIT test at any time on any of the device channels;
- Monitor the condition of the device;
- Determine the reason for abnormal termination of testing;
- View and display on graphs the results of measurements obtained earlier, in the past, and saved on the User's computer;
- Perform verification and calibration of the device.

The Ym128 Computer Interface is universal and is suitable for working with both the ASK75.10.20.2 Analyzers and the ASK75.10.20.12 Analyzers, as well as with other devices manufactured by YAROSTANMASH LLC.

Before using the Computer Interface and connecting to an instrument, a specific instrument type must be selected from the drop-down list in the "Instrument(s) Type" area of the "Settings" page, see subsection 19.7.

Once a connection with the Analyzer is established, the device type and its IP address are displayed in the title bar of the Computer Interface (see Figure 62).

The Computer Interface contains the pages:

- Tasks
- Tables
- Charts
- Results
- Analysis
- Settings
- Condition
- Postprocessing

The selection of pages is carried out using the corresponding buttons at the top left:



Figure 62 - Computer Interface Page Selection Buttons

All pages have a common lower area, reflecting the status and main parameters of the Analyzer channels, and also providing the ability to interrupt the HIT testing process for any channel from any page of the Computer Interface.

Depending on the type of device, the area will have two or twelve identical channel status panels (see Figure 63 and Figure 64):

Ch1 Master Md: DCHCC Col: 23 Stp: 6 U: 2.946V I@10A: -10.00A ESR: 16.843mR t: 51.7°C Stop	Ch2 Master Md: DCHCC Col: 23 Stp: 6 U: 3.449V I@10A: -10.00A ESR: 15.106mR t: 44.7°C Stop	Ch3 Master Md: CHCV Col: 24 Stp: 5 U: 4.198V I@10A: 2.57A ESR: ? t: 40.5°C Stop	Ch4 Master Md: DCHCC Col: 23 Stp: 6 U: 3.747V I@10A: -10.00A ESR: 14.844mR t: 38.2°C Stop	Ch5 Master Md: CHCC Col: 24 Stp: 4 U: 4.078V I@10A: 7.00A ESR: 14.953mR t: 46.1°C Stop	Ch6 Master Md: CHCV Col: 24 Stp: 5 U: 4.198V I@10A: 1.31A ESR: ? t: 35.4°C Stop	Ch7 Master Md: DCHCC Col: 23 Stp: 6 U: 3.587V I@10A: -10.01A ESR: 14.615mR t: 45.2°C Stop	Ch8 Master Md: DCHCC Col: 23 Stp: 6 U: 3.652V I@10A: -10.00A ESR: 15.073mR t: 41.0°C Stop	Ch9 Master Md: CHCC Col: 24 Stp: 4 U: 3.942V I@10A: 7.00A ESR: 15.636mR t: 46.6°C Stop	Ch10 Master Md: CHCC Col: 24 Stp: 4 U: 3.622V I@10A: 7.00A ESR: 17.039mR t: 53.6°C Stop	Ch11 Master Md: DCHCC Col: 23 Stp: 6 U: 3.323V I@10A: -10.00A ESR: 15.401mR t: 46.4°C Stop	Ch12 Master Md: CHCV Col: 24 Stp: 5 U: 4.199V I@10A: 6.23A ESR: ? t: 45.2°C Stop
---	---	---	---	--	---	---	---	--	---	--	--

Figure 63 - Channel status area for the ASC75.10.20.12 Analyzer

Ch1 Master Md: DCHCC Col: 23 Stp: 6 U: 2.946V I@10A: -10.00A ESR: 16.843mR t: 51.7°C Stop	Ch2 Master Md: DCHCC Col: 23 Stp: 6 U: 3.449V I@10A: -10.00A ESR: 15.106mR t: 44.7°C Stop
---	---

Figure 64 - Channel status area for the ASC75.10.20.2 Analyzer

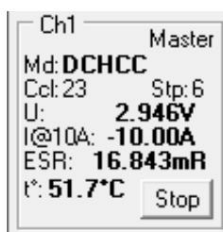


Figure 65 - Analyzer Channel Status Panel

Each device channel status panel (see Figure 65) contains:

- Channel status indicator – “Master” or “Slave”, see section 10 and subsection 19.2);
- Channel operation mode marker at the current step of the HCS testing program; • Indicator of the number of the HCS testing program cycle being executed (“Ccl:”); • Indicator of the number of the step being executed of the specified HCS testing program (“Stp:”); • Current HCS voltage value, V; • Indicator of the current hardware current subrange; • Current value of current through the HCS, A or mA; • Current HCS ESR measurement result, mOhm, obtained by the periodic current interruption method (see subsection 13.2) or in the voltage, current, power or resistance pulse mode (see subsection 13.3); • Current HCS temperature, °C (only when the temperature sensor is connected); • Button for interrupting HCS testing or exiting the channel emergency state.

The channel mode marker can take the following values:

- **"INIT"** – Channel initialization; • **"IDLE"** – Channel idle mode (the channel is ready for operation); • **"TST CMPLT"** – The HIT test is complete; • **"CNNECTING"** – Connecting the HIT to the internal circuits of the Analyzer channel; • **"DSCNNCTNG"** – Disconnecting the HIT from the internal circuits of the channel; • **"ESR MES"** – Current pause mode during HIT ESR measurement (see subsection 13.2); • **"CHCC"** – Charging the HIT with a specified constant current; • **"CHCP"** – Charging the HIT with a specified constant power; • **"CHCV"** – Recharging the HIT at a specified constant voltage; • **"DCHCC"** – Discharging the HIT with a specified constant current; • **"DCHCP"** – Discharging the HIT with a specified constant power; • **"DCHCR"** – Discharge of the chemical supply voltage to a given constant resistance; • **"DCHCV"** – Additional discharge of the chemical supply voltage at a given constant voltage; • **"REST"** – Relaxation mode; • **"SCANING U"** – Voltage scan on the chemical supply voltage; • **"SCANING I"** – Current scan through the chemical supply voltage; • **"SCANING P"** – Power scan on the chemical supply voltage; • **"SCANING R"** – Resistance scan of the chemical supply voltage discharge; • **"IMP U"** – Voltage pulse mode on the chemical supply voltage; • **"IMP I"** – Current pulse mode through the chemical supply voltage; • **"IMP P"** – Power pulse mode on the chemical supply voltage; • **"IMP R"** – Resistance pulse mode of the chemical supply voltage discharge; • **"TBL U"** – Operation according to the voltage table; • **"TBL I"** – Operation according to the current table; • **"TBL P"** – Working according to the power table; • **"TBL R"** – Working according to the resistance table; • **"LGG U"** – Voltage recorder mode at the terminals of the hot current source; • **"NO FW"** – There is no firmware in the microcontroller of the LZR (see section 23); • **"UPDATE"** – The update mode of the firmware of the LZR (see section 23); • **"OBJFLT ST"** – Failure to stabilize the set parameter on the hot current source; • **"OBJFLT BS"** – Violation of the connection circuits of the hot current source (poor contact); • **"OVER U B+"** – Overvoltage at the power terminals of the channel; • **"OVER U S+"** – Overvoltage at the potential terminals of the channel; • **"RVRS B+"** – Reverse polarity of voltage at the power terminals of the channel; • **"RVRS S+"** – Reverse polarity at the potential terminals of the channel; • **"U MIN LMT"** – The minimum voltage limitation on the heat supply voltage has been triggered; • **"U MAX L"**

- **"IC MX LMT"** – Maximum charging current limitation has been triggered;
- **"ID MX LMT"** – The maximum discharge current limitation has been triggered;
- **"T MIN LMT"** – The minimum temperature limitation of the heating element has been triggered;
- **"T MAX LMT"** – The maximum temperature limitation of the heat source has been triggered;
- **"PC FAULT"** – Loss of connection with the PC when working with tables;
- **"OVER U PS"** – Overvoltage on the power source of the device channel;
- **"PS FAULT"** – Failure of the power supply of the device channel;
- **"SW OVRLD"** – Overload of power transistors of the device channel;
- **"INV FAULT"** – Failure of the power inverter of the device channel;
- **"OVER I"** – Channel overload by current;
- **"OVER P"** – Channel power overload;
- **"OVER T LD"** – Overheating of the load resistor **R** of the device channel (see Figures 7 and 8);
- **"OVR T INV"** – Overheating of the power inverter of the device channel;
- **"CNNCT ERR"** – Loss of communication between the MZR and the MKU (see section 6).
- **"INIT FLT"** – Initialization failure of the internal circuits of the device channel;
- **"GNRL FLT"** – General channel fault of the device;
- **"UPDTE FLT"** – Error updating the device channel firmware;

The hardware current subrange indicator is located directly before the current value (see Figure 65). For example, the text "I@10A: **-5.49A**" means that the hardware current subrange of 10 A is currently operating, and the current value through the HIT itself is minus 5.49 A.

The "Stop" button is used to interrupt the testing process of the HIT for a specific channel of the device, or to reset the emergency state (acknowledge the error message) (see section 22).

The Computer Interface has many fields for entering the specified parameters of the HIT testing and displaying the measurement results. The values in the fields must be entered from the range of acceptable values for a specific configurable parameter.

If you enter incorrect values, the corresponding input field will be highlighted in red (see Figure 66):

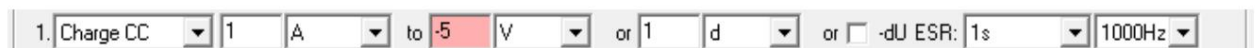


Figure 66 - Incorrect values are highlighted in red

A number of parameters allow the absence of entered values. In this case, control of the corresponding parameter is disabled. However, to attract the User's attention, the empty window is highlighted in yellow (see Figure 67):

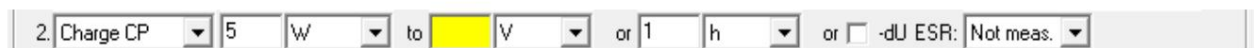


Figure 67 - Some fields can be left blank

Finally, a number of parameters are of particular importance when forming a testing program. To remind the User, such parameters are always highlighted in yellow, even if the parameter value has been entered and is correct (see Figure 68):

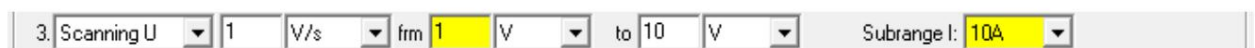


Figure 68 - Some parameters are of particular importance

A detailed description of the Ym128 Computer Interface is provided in subsections 19.2 – 19.9 and in section 21.

19.2 TASKS page

The Ym128 Computer Interface Tasks page is designed to HIT testing program settings:

Ym128 IP192.168.100.120 ANALYZER OF CHEMICAL CURRENT SOURCES ACK75.10.20.12

Tasks | Tables | Charts | Results | Analysis | Settings | Condition | Postprocessing

Channel select for setting: ☒ Ch1 ☐ Ch2 ☐ Ch3 ☐ Ch4 ☐ Ch5 ☐ Ch6 ☐ Ch7 ☐ Ch8 ☐ Ch9 ☐ Ch10 ☐ Ch11 ☐ Ch12

Channel mode: ☒ Master ☐ || to the previous

Preparation

1. Charge CC: 1 A to 5 V or 1 d or -dU ESR: 1s 1000Hz

2. Charge CP: 50 W to 10 V or 2 h or -dU ESR: 5s 10Hz

3. Charge CV: 10 V to 1 A or 10 min

Program

4. Discharge CC: 7 A to 12.5 V or 100 s

5. Discharge CP: 70 W to 10 V or 1 d

6. Discharge CR: 6.4 R to 2.5 V or 10 h

7. Discharge CV: 2.5 V to 0.125 A or 5 d

8. Rest: 8 min

9. Scanning U: 50 mV/s from 2.5 V to 10 V

10. Scanning I: 1 A/s from 5 A to -10 A

11. Scanning P: 3 W/s from -75 W to 50 W

12. Scanning R: 0.1 R/s from 1 R to 100 R

13. Impulses U: 5 V on 30 s and 12 V

14. on 1 s to 5 min

15. Impulses I: -0.25 A on 1 min and -10 A

16. on 0.1 s to 3.3 V or 10 d

17. Impulses P: 15 W on 25 s and -60 W

18. on 5 s to 13.3 V or 1 h

19. Impulses R: 0.5 R on 4 s and 17 R

20. on 2 min to 2.1 V or 60 min

21. Table U

22. Table I

23. Table P

24. Table R

25. Logger U

1000 cycles Program: As previous CH... Load... Save...

Completion

46. Pause: 1 h

47. MPPT: 50 mSm/s dP 0.5 W on 24 h

48. Not

Limits: Empty = switched off

Umin: 2.4 V Umax: 4.3 V Imin: 8 A Imax: 9900 mA tmin: 10 °C tmax: 80 °C

Settings for saving results

RAW file: 100Hz Object name: Sample No.1

☒ CLK file ☐ For each cycle only ☐ Save each 1 cycle ☐ Logarithmically Time format: HH.MM.SS File...

RAW file: E:\RESULTS\TEST1\Smpl.no.1-00000000.txt

CLK file: E:\RESULTS\TEST1\Smpl.no.1-CLK.txt

Start

Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch10	Ch11	Ch12
Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE	Md: IDLE
Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0	Cct: 0
U: 3.85V	U: 3.735V	U: 4.025V	U: 3.635V	U: 4.018V	U: 4.046V	U: 3.676V	U: 3.837V	U: 3.984V	U: 3.847V	U: 3.736V	U: 4.057V
I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA	I@0.1A: 0.0mA
ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?	ESR: ?
t: 23.5°C	t: 23.8°C	t: 22.9°C	t: 23.0°C	t: 23.4°C	t: 23.3°C	t: 23.8°C	t: 22.9°C	t: 23.0°C	t: 23.4°C	t: 23.2°C	t: 23.5°C

Figure 69 - Ym128 Computer Interface Tasks Page

The device channel for setting up the testing program is selected in the "Channel select for setting" area. The number of channels offered for selection will depend on the specific device type (ACS75.10.20.2 or ACS75.10.20.12) and is 2 or 12 channels, respectively (see Figures 70 and 71).

If it is necessary to increase the testing current of the HIT above 10 A, the device channels can be combined into groups for parallel operation. In this case, for each channel in the group, you must specify the operating mode - "Master" or "|| to the previous". The channel operating mode is configured in the "Channel mode" area. The first channel in the group must be configured as "Master", the others must be configured as "|| to the previous". Additional information on parallel channel operation is provided in Section 10.

Ym128 IP192.168.100.140 ANALYZER OF CHEMICAL CURRENT SOURCES ACK75.10.20.12

Tasks | Tables | Charts | Results | Analysis | Settings | Condition | Postprocessing

Channel select for setting: ☐ Ch1 ☒ Ch2 ☐ Ch3 ☐ Ch4 ☐ Ch5 ☐ Ch6 ☐ Ch7 ☐ Ch8 ☐ Ch9 ☐ Ch10 ☐ Ch11 ☐ Ch12

Channel mode: ☒ Master ☐ || to the previous

Figure 70 - Selecting channels for setting up the HIT testing program and setting the mode of the selected channel for the ASC75 Analyzer.10.20.12

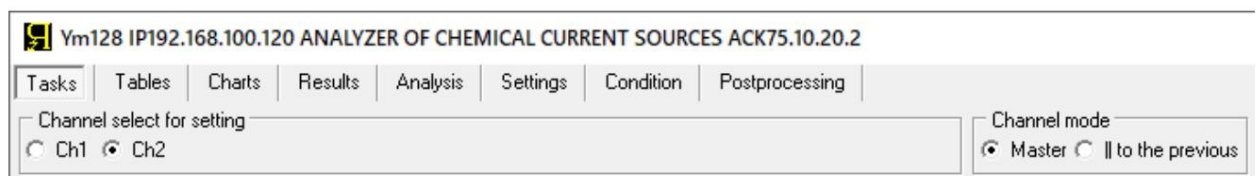


Figure 71 - Selecting channels for setting up the HIT testing program and setting up the mode of the selected channel for the ASK75.10.20.2 Analyzer

Setting up a HIT testing program for a specific channel is possible if:

- The connection between the device and the Computer Interface is established;
- The selected channel is in the **"IDLE"** mode (ready for operation);
- The channel operating mode is set to "Master" (see section 10);
- The channels prepared for parallel operation with the selected channel are also in the **"IDLE"** mode (also ready for work);
- All data from the internal memory of the device is transferred to the Computer Interface and the internal memory of the device is empty (see section 21).

In all other cases, the test program setup elements for the selected channel will not be shown or blocked.

The "Tasks" page allows you to configure the required HIT testing program (see Section 11, Figure 12), which, in general, may contain:

- up to 3 test preparation steps, performed once;
- up to 42 steps of the main cyclic part of the program, performed a specified number of times (cycles);
- up to 3 test completion steps, performed once after completion of the main cyclic part of the program.

At each step (with some exceptions, see section 11) the device can perform:

- Chemical battery charging with a given constant current (CHCC);
- Chemical battery charging with a given constant power (CHCP);
- Chemical battery additional charging at a given constant voltage (CHCV);
- Chemical battery discharge with a given constant current (DCHCC);
- Chemical battery discharge with a given constant power (DCHCP);
- Chemical battery discharge at a given constant resistance (DCHCR);
- Chemical battery additional discharge at a given constant voltage (DCHCV);
- Relaxation (pause);
- Voltage sweep (U) across the chemical battery in time;
- Current sweep (I) through the chemical battery in time;
- Power sweep (P) across the chemical battery in time;
- Chemical battery discharge resistance sweep (R) in time;
- Voltage pulse mode (U);
- Current pulse mode (I);
- Power pulse mode (P);
- Discharge resistance pulse mode of the chemical power plant (R);
- Work according to the voltage table (U);
- Work according to the current table (I);
- Work according to the power table (P);
- Work according to the resistance table (R);
- Recording the voltage at the device terminals

The selection and configuration of preparation steps is carried out in the "Preparation" area (see Figure 72).

The selection and configuration of the steps of the cyclic part is carried out in the "Program" area.

Selecting and configuring completion steps is done in the "Completion" area.

The screenshot displays the HIT testing program configuration interface, organized into three main sections: Preparation, Program, and Completion.

Preparation Section:

- 1. Charge CC: 1 A to 5 V or 1 d or ☐ -dU ESR: 1s 1000Hz
- 2. Charge CP: 50 W to 10 V or 2 h or ☒ -dU ESR: 5s 10Hz
- 3. Charge CV: 10 V to 1 A or 10 min

Program Section:

- 4. Discharge CC: 7 A to 12.5 V or 100 s
- 5. Discharge CP: 70 W to 10 V or 1 d
- 6. Discharge CR: 6.4 R to 2.5 V or 10 h
- 7. Discharge CV: 2.5 V to 0.125 A or 5 d
- 8. Rest: 8 min
- 9. Scanning U: 50 mV/s from 2.5 V to 10 V
- 10. Scanning I: 1 A/s from 5 A to -10 A
- 11. Scanning P: 3 W/s from -75 W to 50 W
- 12. Scanning R: 0.1 R/s from 1 R to 100 R
- 13. Impulses U: 5 V on 30 s and 12 V
- 14. on 1 s to 5 min
- 15. Impulses I: -0.25 A on 1 min and -10 A
- 16. on 0.1 s to 3.3 V or 10 d
- 17. Impulses P: 15 W on 25 s and -60 W
- 18. on 5 s to 13.3 V or 1 h
- 19. Impulses R: 0.5 R on 4 s and 17 R
- 20. on 2 min to 2.1 V or 60 min
- 21. Table U
- 22. Table I: between 5.1 V and 18.3 V
- 23. Table P: between 2 V and 8.5 V
- 24. Table R: to 3.5 V
- 25. Logger U: to 7.5 V or 5 h

ESR settings on the right:

- ESR: 1min 1s
- ESR: Not meas.
- ESR: 1s 1000Hz
- Subrange I: 10A
- ESR: 1s 250Hz
- ESR: Not meas.
- ESR: 10s 50Hz
- Subrange I: 2A
- Subrange I: 10A

Buttons at the bottom: 1000 cycles, Program: As previous CH..., Load..., Save...

Completion Section:

- 46. Pause: 1 h
- 47. MPPT: 50 mSm/s dP 0.5 W on 24 h
- 48. Not

Subrange I: 10A

Figure 72 - Elements for setting up the HIT testing program

The "Program" area expands and contracts as the Computer Interface on the User's computer screen is resized, and also has a scroll bar on the right side to view and adjust all 42 steps of the program's cyclical portion.

To set up a given HIT testing program, each step must be set up separately.

To the left of each line of the parameter settings for each step, the numbers of the steps in the testing program are indicated. These numbers are then entered into the measurement results files (see Section 15) for unambiguous comparison of the obtained measurement results and the specified testing program.

The operating mode of the device channel at each step is selected from the drop-down list at the beginning of the line describing the parameters for this step.

When selecting an item, this step will be skipped during program execution, and the device channel will immediately proceed to the next step.

Setting up a specific step consists of selecting its type and entering it into the settings windows. correct values of the parameters that determine the operation of the channel at this step.

The entered parameter values must fall within the acceptable range of the channel. or groups of channels prepared for parallel operation (see Section 10). If the entered value of any of the parameters is not within the permissible range, units of measurement are not selected for this parameter, or the Computer Interface cannot recognize the entered text, the window with the incorrect parameter will be highlighted in red (see Figure 66), and the launch of the prepared HIT testing program

it will become impossible to execute.

To include *the constant current charge (CHCC) step in the testing program*, you must select the mode from the drop-down list for a specific step.

"Charge CC" (see Figure 73):



Figure 73 - Elements of setting the step of charging the CHIT with a given constant current (CHCC)

In the CHCC charging mode, the device channel ensures the flow of a stabilized positive current (charging current) of a given value through the tested chemical energy source.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.3.

For the CHCC mode, the following are configured separately:

- Specified charging current, A, mA or μ A;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day;
- Enabling or disabling the "-dU" charge end criterion;
- **PESR** period between ESR measurements by the current interruption method (see subsection 13.2);
- Duration of **DESR** current pauses for ESR measurement.

The input field for the specified step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow.

The "-dU" criterion is enabled or disabled by checking or unchecking the box with the same name. The value of the



negative voltage increment at which the device will stop the charging step is configured in the "-dU" area on the "Tasks" page of the Computer Interface, for all charging steps at once (see Figure 74):

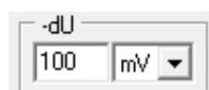


Figure 74 - Setting the value of the end of charge criterion "-dU"

During the CHCC step, the device can measure the ESR of the CHCC using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **P_{ESR}** and the duration of pauses in current **D_{ESR}** are set by selecting the required values from the corresponding drop-down lists (see Figure 75):

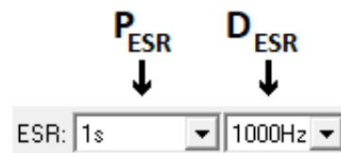


Figure 75 - Setting up the ESR measurement of the HIT by the method of periodic interruption of current (see subsection 13.2)

The ranges of permissible values for the ESR measurement period **P_{ESR}** and the duration of current pauses **D_{ESR}** are presented in subsection 13.2.

In the measurement results files (see section 15) the DC charging step has the step marker "**CCC**".

To include the *constant constant power charge step (CHCP)* in the testing program, you must select the "**Charge CP**" mode from the drop-down list for a specific step (see Figure 76):



Figure 76 - HIT charge step adjustment elements specified constant power (CHCP)

In the CHCP charging mode, the device ensures that a positive current (charging current) of such a value flows through the tested CHCP that the absolute value of the electrical power on the CHCP corresponds to the specified value.
power value.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.4.

For CHCP mode, the following are configured separately:

- Specified charging power, W, mW or μ W;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day;
- Enabling or disabling the "-dU" charge end criterion;
- **P_{ESR}** period between ESR measurements by the current interruption method (see subsection 13.2);
- Duration of **D_{ESR}** current pauses for ESR measurement.

The input field for the specified step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified voltage will be disabled, and the step itself
The end voltage input field will be highlighted in yellow to remind the User to disable the criterion.

The "-dU" criterion is enabled or disabled by checking or unchecking the box with the same name.



The amount of negative voltage increment at which the device will stop the charging step is configured in the "-dU" area on the "Tasks" page of the Computer Interface, for all charging steps at once (see Figure 74).

During the CHCP step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in current **DESR** are set by selecting the required values from the corresponding drop-down lists (see Figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15) the constant power charging step has the step marker **"CCP"**.

To include *the step of recharging at a given constant voltage (CHCV)* in the program, you must select the mode from the drop-down list for a specific step **"Charge CV"** (see Figure 77):

Figure 77 - Elements of setting up the additional charge of the HIT at a given constant current voltage (CHCV)

In the CHIT recharge mode at a given constant voltage (CHCV), the device channel is a source of stabilized voltage.

A detailed description of this device channel operating mode with explanatory drawings, a description of the step end criteria, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.5.

ATTENTION! The recharge voltage should be set equal to the voltage completion of the previous step!

To remind the User to enter correct values voltage, the window for entering the recharging voltage is always highlighted in yellow.

For additional charging of the HIT at a given constant voltage (CHCV) separately are configured:

- Specified recharge voltage, V or mV;
- End-of-step current, A, mA or μ A;
- Maximum step duration, s, min, h, or days.

The input field for the specified step end current can be left empty. In this case, the criterion for the step end upon reaching the specified current will be disabled, and the input field for the end current itself will be highlighted in yellow to remind the User about the criterion being disabled.

In the measurement results files (see section 15), the recharging step at constant voltage has a step marker **"CCV"**.

To include *the discharge step with a given constant current (DCHCC)* in the testing program, it is necessary to select the **"Discharge CC"** mode from the drop-down list for a specific step (see Figure 78):

Figure 78 - Elements of setting the step of discharge of the CHIT with a given constant current (DCHCC)

In the DCCC discharge mode, the device ensures that a stabilized negative current (discharge current) of a specified value flows through the tested DCCC.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.6.

For DCHCC mode, the following are configured separately:

- Specified discharge current, A, mA or μ A;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day;
- **PESR** period between ESR measurements by the current interruption method (see subsection 13.2);
- Duration of **DESR** current pauses for ESR measurement.

The input field for the specified step end voltage can be left empty. In this case, the criterion for the step end upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow to remind the User about the criterion being disabled.

During the DCHCC step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in current **DESR** are set by selecting the required values from the corresponding drop-down lists (see Figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15) the DC discharge step has the step marker "**DCC**".

To include *the discharge step with a specified constant power (DCHCP)* in the testing program, it is necessary to select the "**Discharge CP**" mode from the drop-down list for a specific step (see Figure 79):



Figure 79 - Elements of setting the step of the discharge of the HIT specified constant power (DCHCP)

In the DCCC discharge mode, the device ensures that a negative current (discharge current) flows through the tested DCCC such that the absolute value of the electrical power on the DCCC corresponds to the specified value.

power value.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.7.

For DCHCP mode, the following are configured separately:

- Specified discharge power, W, mW or μ W;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day;
- **PESR** period between ESR measurements by the current interruption method (see subsection 13.2);
- Duration of **DESR** current pauses for ESR measurement.

The input field for the specified step end voltage can be left empty. In this case, the criterion for the step end upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow to remind the User about the criterion being disabled.

During the DCHCP step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in current **DESR** are set by selecting the required values from the corresponding drop-down lists (see Figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15) the constant power discharge step has the step marker "**DCP**".

To include *the discharge step at a given constant resistance (DCHCR)* in the testing program, it is necessary to select the "**Discharge CR**" mode from the drop-down list for a specific step (see Figure 80):



Figure 80 - Elements of setting the step of the discharge of the HIT to a given constant resistance (DCHCR)

In the discharge mode of the HIT at a given constant resistance (DCHCR), the device simulates a resistive load with a given value of electrical resistance, ensuring the flow of a negative current (discharge current) through the tested CIT of such a value that the input resistance of the device corresponds to the given value

resistance value.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.8.

For DCHCR mode, the following are configured separately:

- Specified discharge resistance, Ohm, kOhm or MOhm;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day;
- **PESR** period between ESR measurements by the current interruption method (see subsection 13.2);
- Duration of **DESR** current pauses for ESR measurement.

The input field for the specified step end voltage can be left empty. In this case, the criterion for the step end upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow to remind the User about the criterion being disabled.

During the DCHCR step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in current **DESR** are set by selecting the required values from the corresponding drop-down lists (see Figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15) the discharge step is at a given constant The resistance has a step marker "**DCR**".

To include *the DCHCV step in the program*, you must select the mode from the drop-down list for a specific step.

"Discharge CV" (see Figure 81):

Figure 81 - Elements of setting the additional discharge of the HIT at a given direct current voltage (DCHCV)

In the mode of additional discharge of the HIT at a given constant voltage (DCHCV), the channel the device is a consumer of stabilized voltage.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.9.

ATTENTION! The pre-discharge voltage should be set equal to the voltage completion of the previous step!

To remind the User to enter correct values pre-discharge voltage, the pre-discharge voltage input window is always highlighted in yellow.

For the mode of additional discharge of the HIT at a given constant voltage (DCHCV) separately are configured:

- Specified pre-discharge voltage, V or mV;
- End-of-step current, A, mA or μ A;
- Maximum step duration, s, min, h, or days.

The input field for the specified step end current can be left empty. In this case, the criterion for the step end upon reaching the specified current will be disabled, and the input field for the end current itself will be highlighted in yellow to remind the User about the criterion being disabled.

In the measurement results files (see section 15), the step of additional discharge at a constant voltage has a step marker **"DCV"**.

To include a relaxation step in the HIT testing program, it is necessary From the drop-down list for a specific step, select the **"Rest"** mode (see Figure 82):

Figure 82 - Relaxation Step Setting Elements

At the relaxation step, the device interrupts the current supply through the CIT for a specified time, continuing to measure and record the voltage on the heat exchanger.

A detailed description of this operating mode is given in subsection 11.10.

For the relaxation mode, only the step duration is set: sec, min, h or day.

In the measurement result files (see section 15) the relaxation (pause) step has the step marker **"RLX"**.

To include *the voltage sweep step on the HIT* in the HIT testing program, it is necessary to select the **"Scanning U"** mode from the drop-down list for a specific step (see Figure 83):

Figure 83 - Elements for setting the voltage sweep step on the HIT

In the voltage sweep mode on the CIT, the device will linearly change (sweep) the voltage on the CIT from the initial sweep voltage to the final sweep voltage, with a specified rate of change (sweep) of voltage.

A detailed description of this operating mode of the device with explanatory drawings, a description of the criteria for ending a step, as well as a description of the ranges of acceptable values of adjustable parameters, is given in subsection 11.11.

ATTENTION! The initial sweep voltage should be set equal to tension at the end of the previous step!

To remind the User of the need to enter correct initial sweep voltage values, the initial sweep voltage input window is always is highlighted in yellow.

The hardware current subrange in the voltage sweep mode cannot be pre-determined by the device and must be set manually by the User from the "Subrange I:" drop-down list. To remind the User of the need to manually select the hardware current subrange, the drop-down list for selecting the current subrange is highlighted in yellow (see Figure 83).

For the voltage sweep mode on the HIT, the following are configured separately:

- Voltage sweep rate, V/s, mV/s, μ V/s or nV/s;
- Initial sweep voltage, V or mV;
- Final sweep voltage, V or mV;
- Required current sub-range.

Sequential sweeps of two voltages combination corresponding initial and final sweep voltages allows the investigation of the CCS by the Cyclic VoltAmperometry (CVA) method (see Figure 84 and Figure 85).

Figure 84 - Example of a program for testing the CVT using the CVA method

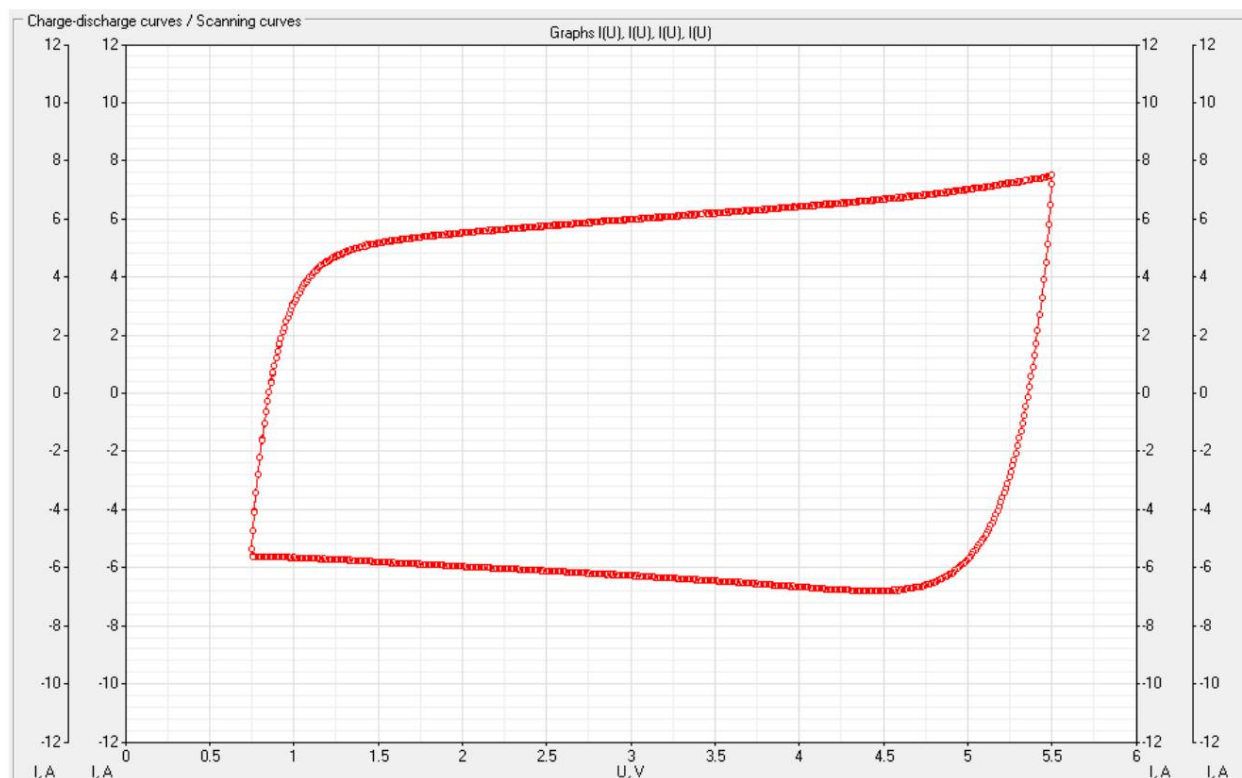


Figure 85 - Result of the HIT testing program execution
by the CVA method (see Figure 84)

In the measurement results files (see section 15), the voltage sweep step on the HIT has the step marker **"SCU"**.

To include *the current scanning step through the HIT* in the HIT testing program, it is necessary to select the **"Scanning I"** mode from the drop-down list for a specific step (see Figure 86):



Figure 86 - Current sweep step adjustment elements through the HIT

In the current sweep mode through the CIT, the device will linearly change (sweep) the current through the CIT from the initial sweep current to the final sweep current, with a given rate of current change.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.12.

For the current sweep mode through the HIT, the following are configured separately:

- Current sweep rate, A/s, mA/s, μ A/s or nA/s;
- Initial sweep current, A, mA or μ A;
- Final sweep current, A, mA or μ A.

During the current sweep step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in the current **DESR** are set by selecting the required values from the corresponding drop-down lists (see Figure 75).

Ranges of acceptable values for the ESR **PESR** measurement period and duration pauses in **the DESR** current are presented in subsection 13.2.

In the measurement result files (see section 15) the current sweep step has the step marker **"SCI"**.

To include the *power scanning step on the HIT* in the testing program, it is necessary to select the "**Scanning P**" mode from the drop-down list for a specific step (see Figure 87):



Figure 87 - Power sweep step adjustment elements on the HIT

In the power sweep mode on the CIT, the device will linearly change (sweep) the power on the CIT from the initial sweep power to the final sweep power, with a specified rate of power change.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.13.

For the power sweep mode on the HIT, the following are configured separately:

- Power scan rate, W/s, mW/s, μ W/s or nW/s;
- Initial sweep power, W, mW or μ W;
- Final sweep power, W, mW or μ W.

During the power sweep step, the device can measure the ESR of the power supply using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in current **DESR** are set by selecting the required values from the corresponding drop-down lists (see figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15) the power sweep step has the step marker "**SCP**".

To include the HIT *discharge resistance sweep step* in the HIT testing program, it is necessary to select the mode from the drop-down list for a specific step "**Scanning R**" (see Figure 88):



Figure 88 - Elements of setting the step of the discharge resistance sweep of the HIT

In the discharge resistance sweep mode, the device will linearly change (sweep) the discharge resistance of the HIT from the initial sweep resistance to the final sweep resistance, with a specified resistance change rate.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.14.

For the discharge resistance sweep mode of the HIT, the following are configured separately:

- Resistance sweep speed, mOhm/s, Ohm/s or kOhm/s;
- Initial sweep resistance, Ohm, mOhm or Mohm;
- Final sweep resistance, Ohms, mOhms or Mohms.

During the discharge resistance sweep step, the device can measure the ESR of the CCS using the periodic current interruption method (see subsection 13.2). The period between ESR measurements **PESR** and the duration of pauses in the current **DESR** are adjusted by selecting the required values from the corresponding drop-down lists (see figure 75).

The ranges of permissible values for the ESR measurement period **PESR** and the duration of current pauses **DESR** are presented in subsection 13.2.

In the measurement results files (see section 15), the discharge resistance sweep step has the "**SCR**" step marker.

To include *the voltage pulse step* in the testing program , you must select the "**Impulse U**" mode from the drop-down list (see Figure 89):

Figure 89 - Voltage pulse step adjustment elements

In the voltage pulse mode, the device supplies alternating voltage pulses of specified values **U1** and **U2** to the tested CIT , whereby voltage **U1** is supplied for a specified time **T1**, and voltage **U2** is supplied for a specified time **T2**.

A detailed description of this operating mode of the device with explanatory drawings, a description of the criteria for ending a step, as well as the ranges of acceptable values of adjustable parameters, is given in subsection 11.15.

The hardware current subrange in the voltage pulse mode cannot be pre-determined by the device and must be set manually by the User from the "Subrange I:" drop-down list. To remind the User of the need to manually select the hardware current subrange, the drop-down list for selecting the current subrange is highlighted in yellow (see Figure 89).

For the voltage pulse mode, the following are configured separately:

- Set voltage value **U1**, V or mV;
- Specified duration **T1** of voltage pulses **U1**, s, min, h or days;
- Set voltage value **U2**, V, or mV;
- Specified duration **T2** of voltage pulses **U2**, s, min, h or days;
- Step end voltage, V or mV;
- Step duration, s, min, h or day;
- Required current sub-range.

In the measurement results files, the voltage pulse step has the step marker "**IPU**".

To include the current pulse step in the testing program , you must select the "**Impulse I**" mode from the drop-down list (see Figure 90):

Figure 90 - Current pulse step adjustment elements

In the current pulse mode, the device supplies alternating current pulses of specified values **I1** and **I2 through the tested CIT**, with current **I1** supplied for a specified time **T1**, and current **I2** supplied for a specified time **T2**.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.16.

For the current pulse mode, the following are configured separately:

- Set value of current **I1**, A, mA or μ A;
- Specified duration **T1** of current pulses **I1**, s, min, h or days;
- Set value of current **I2**, A, mA or μ A;
- Specified duration **T2** of current pulses **I2**, s, min, h or days;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day.

The input field for the specified step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow.

In the measurement results files, the current pulse step has the step marker "**IPI**".

To include the power pulse step in the testing program, you must select the "**Impulse P**" mode from the drop-down list (see Figure 91):

17.	Impulses P	15	W	on	25	s	and	-60	W	
18.		on	5	s	to	13.3	V	or	1	h

Figure 91 - Power Pulse Step Setting Elements

In the power pulse mode, the device supplies alternating power pulses of specified values **P1** and **P2 to the tested CCT**, whereby power **P1** is supplied for a specified time **T1**, and power **P2** is supplied for a specified time **T2**.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.17.

For the power pulse mode, the following are configured separately:

- Set power value **P1**, W, mW or μ W;
- Specified duration **T1** of power pulses **P1**, s, min, h or days;
- Set power value **P2**, W, mW or μ W;
- Specified duration **T2** of power pulses **P2**, s, min, h or days;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day.

The input field for the specified step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow.

In the measurement results files (see section 15) the power pulse step has the step marker "**IPP**".

To include *the resistance pulse step in the testing program*, you need to select the "**Impulse R**" mode from the drop-down list (see Figure 92):

19.	Impulses R	0.5	R	on	4	s	and	17	R	
20.		on	2	min	to	2.1	V	or	60	min

Figure 92 - Elements for setting the resistance pulse step

In the resistance pulse mode, the device supplies alternating discharge resistance pulses of specified values **R1** and **R2 to the tested chemical energy source**, with resistance **R1** supplied for a specified time **T1**, and resistance **R2** supplied for a specified time **T2**.

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.18.

For the resistance pulse mode, the following are configured separately:

- Specified resistance value **R1**, Ohm, kOhm or MOhm;
- Specified duration **T1** of resistance pulses **R1**, s, min, h or day;
- Specified resistance value **R2**, Ohm, kOhm or MOhm;
- Specified duration **T2** of resistance pulses **R2**, s, min, h or days;
- Step end voltage, V or mV;
- Maximum step duration, s, min, h or day.

The input field for the specified step end voltage can be left empty. In this case, the criterion for the step end upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow to remind the User about the criterion being disabled.

In the measurement results files (see section 15) the resistance pulse step has the step marker "IPR".

To include in the testing program *the step of working according to the voltage table*, you need to select the "Table U" mode from the drop-down list (see Figure 93):



Figure 93 - Current table step adjustment elements

In the voltage table mode, the device supplies the tested CIT with a stabilized voltage with an instantaneous value selected depending on the current step time according to the specified voltage table.

ATTENTION! Operation according to the voltage table is possible only under computer control! Even a short-term loss of connection between the Analyzer and the computer during the operation of the device according to the voltage table will lead to an emergency interruption of the HIT testing program!

Detailed description of this operating mode of the device with explanatory drawings provided in subsection 11.19.2.

The hardware current subrange in the voltage pulse mode cannot be pre-determined by the device and must be set manually by the User from the "Subrange I:" drop-down list. To remind the User of the need to manually select the hardware current subrange, the drop-down list for selecting the current subrange is highlighted in yellow (see Figure 93).

In the measurement results files (see section 15) the step of working according to the voltage table has a step marker "TBU".

To include in the testing program *the step of working according to the current table*, you need to select the "Table I" mode from the drop-down list (see Figure 94):



Figure 94 - Current table step adjustment elements

In the current table mode, the device ensures that a stabilized current flows through the tested CIT with an instantaneous value selected depending on the current step time according to the specified current table.

ATTENTION! Operation according to the current table is possible only under computer control! Even a short-term loss of connection between the Analyzer and the computer during operation of the device according to the current table will result in an emergency interruption of the HIT testing program!

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.19.3.

For the operating mode according to the current table, the following are configured separately:

- Lower step end voltage, V or mV;
- Upper step end voltage, V or mV.

The input field for the specified lower step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified lower voltage will be disabled, and the input field for the lower end voltage itself will be highlighted in yellow to remind the User about disabling the criterion.

The input field for the specified upper step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified upper voltage will be disabled, and the input field for the upper end voltage will be highlighted in yellow to remind the User that the criterion has been disabled.

In the measurement results files (see section 15), the step of working according to the current table has the step marker **"TBI"**.

To include in the testing program *the step of working according to the power table*, you need to select the **"Table P"** mode from the drop-down list (see Figure 95):



Figure 95 - Elements for setting the step of work according to the power table

In the power table mode, the device supplies the tested CIT with a specified power with an instantaneous value selected depending on the current step time according to the specified power table.

ATTENTION! Operation according to the power table is possible only under computer control! Even a short-term loss of connection between the Analyzer and the computer during operation of the device according to the power table will result in an emergency interruption of the HIT testing program!

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.19.4.

For the operating mode according to the power table, the following are configured separately:

- Lower step end voltage, V or mV;
- Upper step end voltage, V or mV.

The input field for the specified lower step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified lower voltage will be disabled, and the input field for the lower end voltage itself will be highlighted in yellow to remind the User about disabling the criterion.

The input field for the specified upper step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified upper voltage will be disabled, and the input field for the upper end voltage will be highlighted in yellow to remind the User that the criterion has been disabled.

In the measurement results files (see section 15), the step of working according to the power table has the step marker **"TBP"**.

To include in the testing program *the step of working according to the resistance table*, you need to select the **"Table R"** mode from the drop-down list (see Figure 96):



Figure 96 - Elements for setting the step of work according to the resistance table

In the resistance table mode, the device simulates a resistive load with an instantaneous value of electrical resistance selected depending on the current step time according to the specified resistance table.

ATTENTION! Work according to the resistance table is possible only under computer control! Even a short-term loss of connection between the Analyzer and the computer during the device's work according to the resistance table will lead to an emergency interruption of the HIT testing program!

A detailed description of this device operating mode with explanatory drawings, a description of the criteria for step completion, ranges of acceptable values of adjustable parameters, as well as conditions for selecting the hardware current subrange, are given in subsection 11.19.5.

For the resistance table mode, only the lower one is adjusted.
step end voltage, V or mV.

The input field for the specified lower step end voltage can be left empty. In this case, the criterion for ending the step upon reaching the specified lower voltage will be disabled, and the input field for the lower end voltage itself will be highlighted in yellow to remind the User about disabling the criterion.

In the measurement results files (see section 15) the step of working according to the table The resistance has a **"TBR" step marker**.

To include *the voltage recorder step at the terminals of the HIT* in the testing program, it is necessary to select the mode from the drop-down list for a specific step **"Logger U"** (see Figure 97):



Figure 97 - Elements of the setting of the voltage recorder mode at the terminals of the HIT

In the voltage recorder mode at the terminals of the HIT, the device interrupts the current supply through the HIT, continuing to measure and record the voltage on the HIT.

A detailed description of this operating mode and its differences from the "Relaxation" mode given in subsection 11.20.

For the voltage recorder mode, the following are configured separately at the terminals of the heat exchanger:

- Step end voltage, V or mV;
- Maximum step duration, s, min, h, or days.

The input field for the specified step end voltage can be left empty. In this case, the criterion for the step end upon reaching the specified voltage will be disabled, and the input field for the end voltage itself will be highlighted in yellow to remind the User about the criterion being disabled.

In the measurement result files (see section 15) the voltage recorder step has the step marker **"LGU"**.

To include a pause step in the HIT testing program, it is necessary
From the drop-down list for a specific step, select the **"Pause"** mode (see Figure 98):

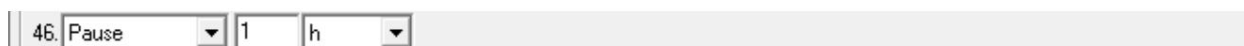


Figure 98 - Pause Step Setting Elements

During the pause step, the device interrupts the current supply through the HIT for a specified time. During the pause, the HIT parameters are not recorded and are not saved in the measurement results files - the pause time is "cut" from the measurement results.

A detailed description of this operating mode is given in subsection 11.21.

For the pause mode, only the step duration is configured: s, min, h or day.

To include *the MPPT step* in the HIT testing program, it is necessary from the drop-down list for a specific step, select the **"MPRT"** mode (see Figure 99):



Figure 99 - MPPT step setting elements

At the MPPT (Maximum Power Point Tracking) step, the Analyzer creates an electrical load on the tested power supply with such conductivity as to ensure the extraction of the maximum possible electrical power from the tested power supply.

The hardware current subrange in the MPPT mode cannot be pre-determined by the device and must be set manually by the User from the "Subrange I:" drop-down list. To remind the User of the need to manually select the hardware current subrange, the drop-down list for selecting the current subrange is highlighted in yellow (see Figure 99).

A detailed description of this operating mode of the device with explanatory drawings, a description of the criteria for ending a step, as well as the ranges of acceptable values of adjustable parameters, is given in subsection 11.22.

For the MPPT mode, the following are configured separately:

- Specified rate of change of conductivity, S/s, mS/s, μ S/s or nS/s;
- A specified reduction in the power taken from the power supply, leading to a change in the direction of searching for the maximum power point, W, mW or μ W;
- Step duration, s, min, h, or day;
- Required current sub-range.

In the measurement results files (see section 15) the MPPT step has the step marker **"MPT"**.

The required number of charge-discharge cycles of the chemical power supply (cycles), which the Analyzer will perform in the main cyclic part of the chemical power supply testing program, is specified in the corresponding field of the "Program" area. The permissible value of the specified number of charge-discharge cycles is from 1 to 9,999,998 (see Figure 100).

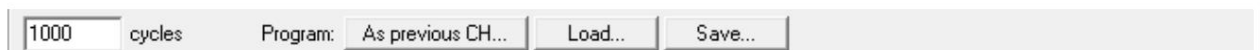


Figure 100 - Window for setting the number of charge-discharge cycles of the tested chemical energy source

The "As previous CH..." button (see Figure 100) is designed to copy the specified test program of the previous channel to the test program setup elements on the selected channel. Thus, the test program specified, for example, for channel #1, can be easily copied to the other channels of the device. For channel #1, the "As previous CH..." button is always unavailable (channel #1 does not have a previous channel).

When pressing the "As previous CH..." button, the User will be shown a dialog box (see Figure 101):

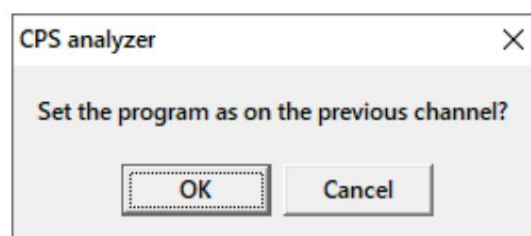


Figure 101 - Confirmation window for the HIT testing program task as on the previous channel of the device

The test program for the selected device channel will be replaced by the specified test program for the previous channel only if the "OK" button is pressed.

The "Save..." button (see Figure 100) allows you to save the prepared HIT testing program in a special testing program file. The values of all settings and the values of all parameters entered in the "Preparation", "Program", "Completion", "Limits. Empty = switched off", "-dU" and "Settings for saving results" areas are saved in this file. Clicking on the button calls up a dialog box for selecting a file to save the specified HIT testing program. The dialog box for selecting a file to save the specified HIT testing program is shown in Figure 102.

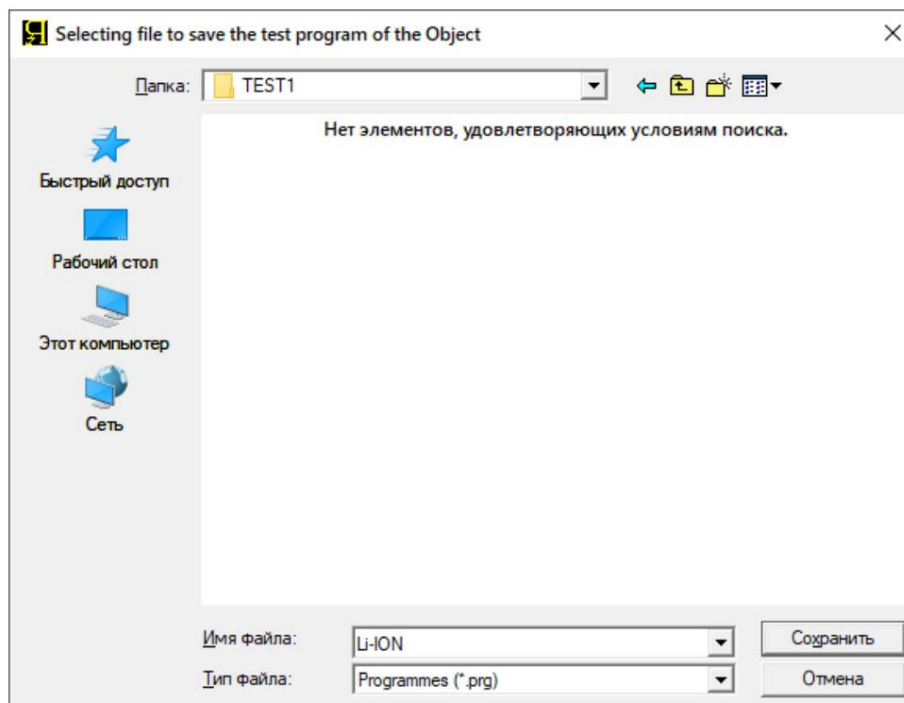


Figure 102 - Selecting a file to save the specified HIT testing program

The "Load..." button (see Figure 100) allows you to load a previously saved HIT testing program into the corresponding parameter settings windows. Clicking on the button calls up a standard dialog box for selecting a file to load the specified HIT testing program. The dialog box for selecting a file to load the specified HIT testing program is shown in Figure 103.

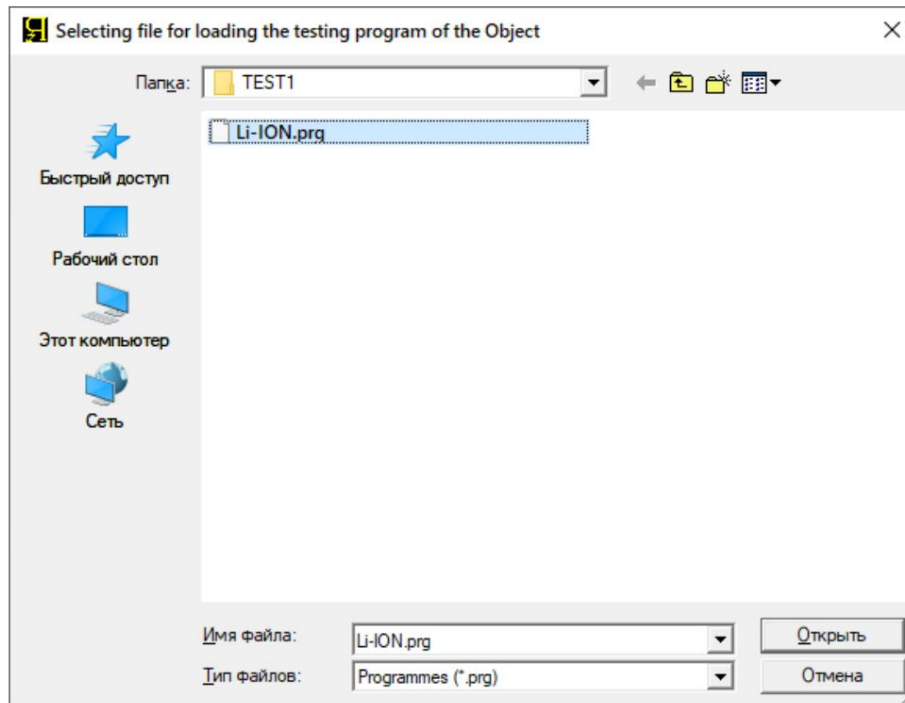


Figure 103 - Selecting a file for loading the HIT testing program

The use of the function of saving and loading a specified HIT testing program can be convenient in the event of the need to frequently change various HIT testing programs from a certain set of programs executed on one device or on one channel of the device.

In a number of modes, such as current sweep, it is impossible to predict in advance the maximum voltage on the CCS, just as in the voltage sweep mode, it is impossible to predict in advance the maximum current through the CCS.

To avoid possible overcharging, overdischarging, overcooling, overheating, as well as overloading of the HIT by charge and discharge current, the "Limits. Empty = switched off" area is used:

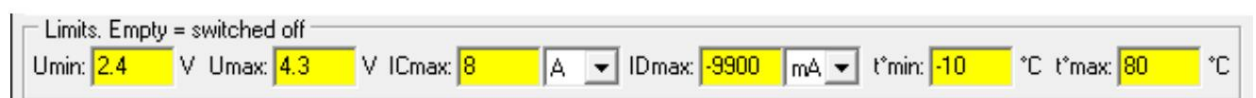


Figure 104 - Specified limits of the main parameters for the tested HIT

In the "Restrictions" area the following are specified:

- Limitation of minimum voltage on the heat supply unit "Umin:", V;
- Limitation of maximum voltage on the heat supply unit "Umax:", V;
- Limitation of the maximum charging current of the HIT, "ICmax:", A;
- Limitation of the maximum discharge current of the HIT, "IDmax:", A;
- Limitation of the minimum temperature of the heat exchanger, "t°min:", °C;
- Limitation of the maximum temperature of the heat exchanger, "t°max:", °C.

If during the execution of the testing program the voltage on the CCS decreases to the specified minimum voltage limit "Umin:", the execution of the CCS testing program will be interrupted, and the device will go into the emergency state **U MIN LMT**. Further discharge of the CCS will be stopped. In the primary data file (see subsection 15.2) the last entry will be made, in which instead of the step marker the emergency marker **"ULL"** will be recorded. The specified minimum voltage limit

on the CIT must fall within the operating range of the device (see Table 5), or remain empty. If there is no entered value (the window for entering the limitation parameter value is left empty), the minimum voltage limitation on the CIT will not operate. The parameter is important, so to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see Figure 104).

If during the execution of the testing program the voltage on the CCS increases to the specified maximum voltage limit "Umax:", the execution of the CCS testing program will be interrupted, and the device will go into the emergency state **"U MAX LMT"**. Further charging of the CCS will be stopped. In the primary data file (see subsection 15.2) the last entry will be made, in which instead of the step marker the emergency marker **"UHL" will be recorded**. The specified maximum voltage limit on the CCS must fall within the operating voltage range of the device (see table 5), or remain empty. If the value is not entered (the window for entering the limitation parameter value is left empty), the limitation on the maximum voltage on the CCS will not be triggered. The parameter is important, therefore, to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see figure 104).

If during the execution of the testing program the positive current through the CCS (charging current) increases to the specified charging current limit "ICmax:", the CCS testing program will be interrupted and the device will go into the emergency state **"IC MX LMT"**. Further charging of the CCS will be stopped. The last entry will be made in the primary data file (see subsection 15.2), in which the emergency marker **"ICL" will be recorded instead of the step marker**. The specified limitation of the maximum CCS charging current must fall within the operating range of the device (see table 5), or remain empty. If the value is not entered (the window for entering the limitation parameter value is left empty), the limitation of the maximum charging current through the CCS will not be triggered. The parameter is important, therefore, to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see figure 104).

If during the execution of the testing program the absolute value of the negative current through the CCS (discharge current) increases to the specified discharge current limit "IDmax:", the execution of the CCS testing program will be interrupted, and the device will go into the emergency state **"ID MX LMT"**. Further discharge of the CCS will be stopped. In the primary data file (see subsection 15.2) the last entry will be made, in which the emergency marker **"IDL" will be recorded instead of the step marker**. The specified limitation of the maximum discharge current of the CCS must fall within the operating current range of a specific type of device, or remain empty. If the value is not entered (the window for entering the limitation parameter value is left empty), the limitation on the maximum discharge current through the CCS will not be triggered. The parameter is important, therefore, to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see Figure 104).

In addition to the protective functions, the parameters "Umin:", "Umax:", "ICmax:" and "IDmax:" are used to select the required hardware current subrange. This becomes necessary for those modes of device operation (steps of the HCS testing program) for which the required current subrange cannot be determined explicitly based on the specified HCS testing parameters. For more details, see the description of specific HCS testing modes (section 11).

ATTENTION! For the correct selection of the appropriate sub-ranges by the device voltage and current and improve the accuracy of measurements, it is strongly recommended to enter reasonable values in the input fields of the limitation parameters "Umin:", "Umax:", "ICmax:" ÿ "IDmax:"!

For example, when testing a certain battery with the following characteristics:

- Maximum charging voltage 4.2 V;
- Minimum discharge voltage 3.0 V;
- Maximum charge and discharge current 1.5 A.

It is reasonable to enter the following into the input fields for the HIT testing parameter restrictions:

- $U_{min} = 2.95 \text{ V}$;
- $U_{max} = 4.25 \text{ V}$;
- $I_{Cmax} = 1.6 \text{ A}$;
- $I_{Dmax} = 1.6 \text{ A}$.

In this case, the device for testing this battery will use the hardware subrange of 2 A current. The accuracy of reproduction and measurement of the current through the HIT will be maximum.

If during the execution of the testing program the CCS temperature drops to the specified minimum temperature limit " $t^{\circ}min$:", the execution of the CCS testing program will be interrupted and the device will go into the emergency state **"T MIN LMT"**. Further work with the CCS will be stopped. In the primary data file (see subsection 15.2) the last entry will be made, in which the emergency marker **"TLL" will be recorded instead of the step marker**. The specified minimum temperature limit of the CCS must fall within the operating range of the device (see table 5), or remain empty. If the value is not entered (the window for entering the limitation parameter value is left empty), the limitation by the minimum temperature of the CCS will not be triggered. The parameter is important, therefore, to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see figure 104).

If during the execution of the testing program the CCS temperature increases to the specified maximum temperature limit " $t^{\circ}max$:", the execution of the CCS testing program will be interrupted and the device will go into the emergency state **"T MAX LMT"**. Further work with the CCS will be stopped. The last entry will be made in the primary data file (see subsection 15.2), in which the emergency marker **"TJMX" will be recorded instead of the step marker**. The specified maximum temperature limit of the CCS must fall within the operating range of the device (see table 5), or remain empty. If the value is not entered (the window for entering the limitation parameter value is left empty), the limitation by the maximum temperature of the CCS will not be triggered. The parameter is important, therefore, to remind the User, the window for entering the limitation parameter is always highlighted in yellow (see figure 104).

"Settings for saving results" area

is designed to configure the period of data recording in primary data files, the method of recording data in results files, the time format in results files, entering the name of the tested HIT, and selecting the path to the measurement results files (see Figure 105).

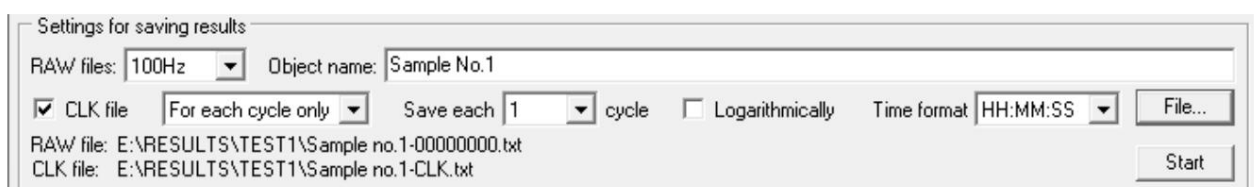


Figure 105 - Measurement results saving settings area

Detailed information about the types and structure of measurement result files, as well as information on the order of their naming and storage is presented in section 15.

The period for saving measurement results in primary data files is selected from the drop-down list "RAW files:" (raw data files) (see Figure 105).

The period for recording measurement results can take the following values:

- "100Hz" (100 Hz); • "2Hz" (2 Hz); • "10s" (10 s); • "5min" (5 min);
- "25Hz" (25 Hz); • "1s" (1 s); • "20s" (20 s); • "10min" (10 minutes);
- "10Hz" (10 Hz); • "2s" (2 s); • "1min" (1 min); • "20min" (20 min);
- "5Hz" (5 s); • "5s" (5 s); • "2min" (2 min); • "1h" (1 h).

The event of triggering the criterion for the end of the current step of the testing program, as well as the event of changing the next pulse in the modes of voltage, current, power or resistance pulses, has a higher priority for recording the results

measurements in the primary data file than the specified data saving method. For example, if the data recording period of "10min" is selected, the next line (point*) will be written to the primary data files once every 10 minutes of the HIT testing process. However, if the criterion for the end of the current step or pulse change in pulse modes is triggered before the moment of recording the next point (for example, 5 minutes after recording the previous point), the current HIT parameters will still be written to the primary data file and thus the last (extreme) point in this program step or in this pulse will be formed. Immediately after this, a transition to the next step (or cycle) of HIT testing, or to the next pulse, will be made, at the start of which the first data point for the new step or pulse will be formed, and the data recording period count will start over. Thus, the first and last points for a specific step or a specific pulse are always saved to the primary data file exactly at the moment of the specified event, without reference to the selected data recording method.

The choice of the period for storing measurement results in primary data files depends on the expected speed of processes during testing of a particular CHCC and the required detail of recording measurement results. In most cases, a good choice of data recording period will be one that stores from 100 to 500 points at each CHCC test step (e.g., CHCC or DCHCC)*

measurements of the operating parameters of the chemical equipment (lines in the primary data file).

For example, for a certain battery, the expected duration of a charge or discharge step is 1 hour. In this case, a reasonable choice for the data storage period would be 10 s. In this case, about 400 points will be saved during the charge or discharge step, which is more than enough to calculate the required characteristics of the tested CCT with maximum accuracy.

Shorter periods of recording primary data, in this case, will lead to an unjustified increase in the volume of measurement results and to a decrease in the performance of the Computer Interface when processing the received data.

* see the list of accepted abbreviations and designations.

ATTENTION! Do not select short measurement result storage periods (1 sec or less) for long-term HIT testing processes (lasting 10 minutes or more). This will result in an unjustified increase in the volume of measurement results, as well as a reduction in the device's autonomous operation time (see Section 21).

The "Object name" field is required to enter the name of the tested HIT, which will be included in the measurement results files (see Figure 106):



Figure 106 - Input field for the name of the tested HIT

The entered HIT name will be saved in all measurement results files for this test and will allow you to easily identify the results obtained in the future (see Figure 107).

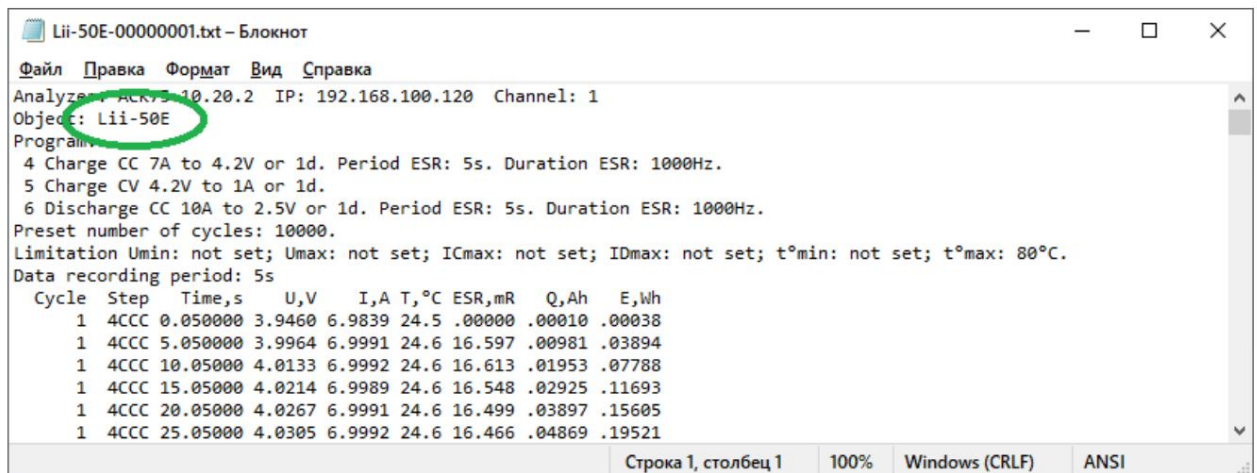


Figure 107 - The name of the HIT entered in the "Object name" field (see Figure 106) is included in the measurement results files

ATTENTION! It is strongly recommended that in the "Object name" field enter a reasonable name for the tested HIT. This will allow you to correctly identify files with HIT test results in the future.

The "CLK file" checkbox (see Figure 105) allows you to enable or disable the creation of a summary measurement results file. The drop-down list located next to this checkbox allows you to select the summary results file type - for each completed step or for each completed cycle of the HIT testing program. Detailed information on summary measurement results files is presented in subsection 15.3.

Drop-down list "Save each" and checkbox "Logarithmically" (see Figure 105) are used to configure the decimation of measurement results. Decimation of measurement results allows to significantly reduce the volume of measurement result files created by the device in case of the need to perform a significant number of charge-discharge cycles of the tested CCS (for example, when testing supercapacitors). Detailed information on decimation of measurement results is presented in Section 16.

The "Time format" drop-down list (see Figure 105) allows you to select one of the available formats for saving time units in measurement results files. Detailed information about the various time formats is provided in Section 17.

The "File..." button (see Figure 105) opens a dialog box for selecting the location to save the measurement results and entering the base name of the result files. Detailed information about the measurement result files is provided in Section 15.

During the operation of the device, the test cycle numbers (when creating and saving primary data files) or a label will be added to the base file name.

"-CLK" for the summary measurement results file. For example, if you select the base file name **"TEST.txt" in the dialog box**, multiple result files will be created, one for each program cycle, and one common one with summary measurement results (if the "CLK file" (File by cycles) checkbox is checked (see Figure 105). The names

The created files will be as follows:

"TEST-00000000.txt" – for the primary data file for preparation steps (if any);

"TEST-00000001.txt" – for the primary data file for cycle No. 1;

"TEST-00000002.txt" – for the primary data file for cycle No. 2;

"TEST-00000003.txt" – for the primary data file for cycle No. 3;

"TEST-CLK.txt" – for the file of summary measurement results (if specified).

All files will be saved in the folder selected by the User and after completion of the specified test program, their number will correspond to the number of completed cycles of the HIT testing program, including the file of results of the execution of preparation steps and the file of results of the execution of completion steps of the testing program, as well as the file of summary measurement results.

ATTENTION! To avoid malfunctions and confusion in the measurement results files, each time a test is launched, a separate folder must be created on the User's computer to store its results!

The "Start" button (see Figure 105) is designed to start the prepared execution testing programs.

The "Start" button will be *inactive* and the HIT test will not be able to be started if:

- There is no established connection between the device and the Computer Interface;
- The selected device channel is in a state other than **"IDLE"** (the channel is not ready for operation);
- Channels prepared for parallel operation with the selected channel are in a state other than **"IDLE"** (channels are not ready for operation);
- No or incorrect values have been entered for one or more parameters of the specified HIT testing program (windows with incorrect values are highlighted in red). The window with the incorrectly entered parameter value may be located in the invisible part of the "Program" area on the "Tasks" page of the Computer Interface. Use the scroll bar on the right side of the "Program" area to view all entered parameter values for all possible 42 steps of the cyclic part of the specified HIT testing program;

- The specified testing program does not contain any steps (for all steps

the item is selected from the drop-down );

lists • The name of the tested HIT is not entered;

- The path to the measurement results files is not selected.

After all the above conditions are met, run the prepared test

The "Start" button will be allowed.

19.3 Page "TABLES"

The "Tables" page of the Ym128 Computer Interface is designed to select and prepare tables of voltage, current, power and resistance for the Analyzer operating modes according to the specified tables of voltage, current, power and resistance (see Figure 108):

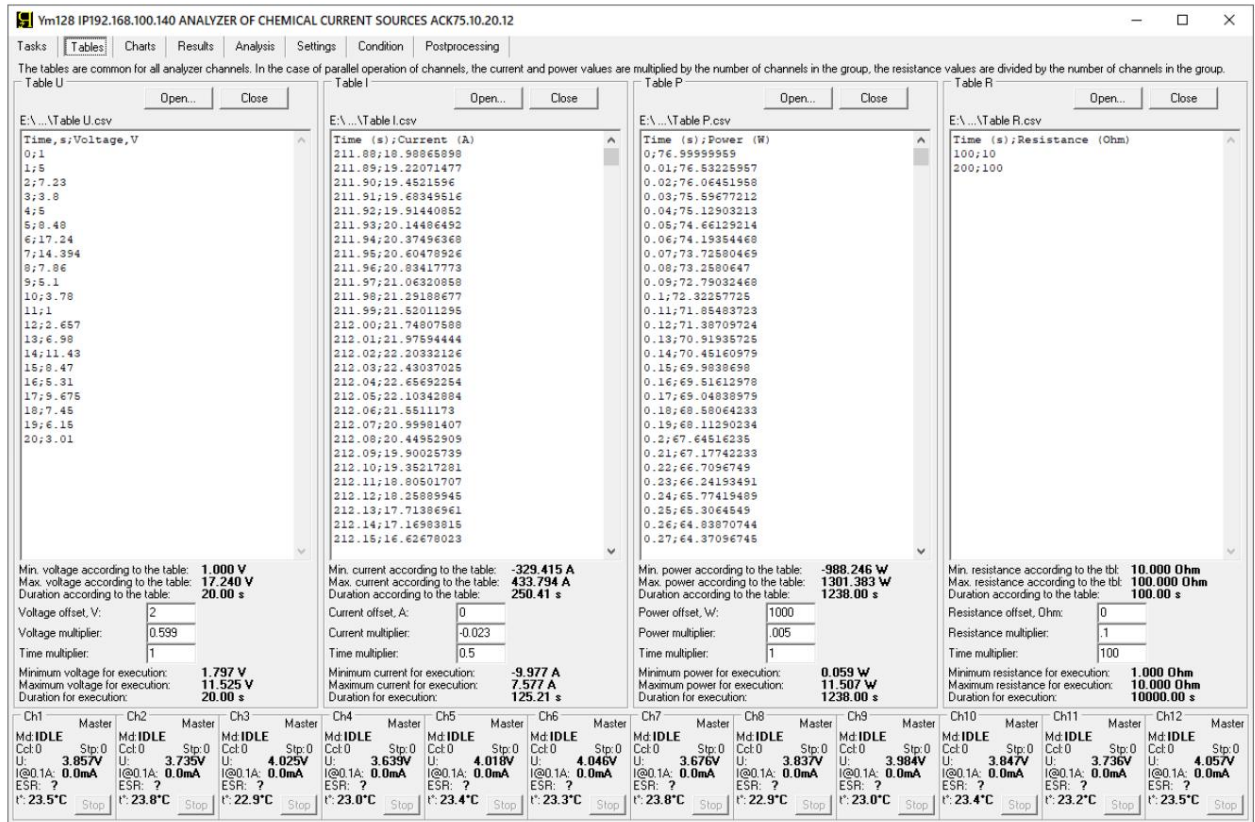


Figure 108 - Ym128 Computer Interface "Tables" Page

The "Tables" page contains four similar areas: "Table U" (Voltage Table), "Table I" (Current Table), "Table P" (Power Table), and "Table R" (Resistance Table).

The "Table U" area is used to select and load a voltage table that can be used when testing the HIT if the testing program includes steps for working with the voltage table (see subsection 11.19.2).

The "Table I" area is used to select and load a current table that can be used when testing the CCT if the testing program includes steps for working with the current table (see subsection 11.19.3).

The "Table P" area is used to select and load a power table that can be used when testing the CIT if the testing program includes steps for working with the power table (see subsection 11.19.4).

The "Table R" area is used to select and load a resistance table that can be used when testing the CIT if the test program includes steps for working with the resistance table (see subsection 11.19.5).

The selected table files are common to all device channels. When the device channels operate in parallel (see Section 10), the specified table values of current and power are multiplied by the number of channels in a specific group. For example, if the instantaneous current value according to the table is 8 A and in a specific group there are

three channels, the actual current through the analyzer will be $8 \text{ A} \cdot 3 \text{ channels} = 24 \text{ A}$. The tabular resistance values, on the contrary, are divided by the number of channels in a specific group. For example, if the instantaneous resistance value according to the table is 1 Ohm and two channels are operating in a specific group, the actual load resistance on the analyzer will be $1 \text{ Ohm} / 2 \text{ channels} = 0.5 \text{ Ohm}$. To remind the User about multiplying or dividing the specified tabular values during parallel operation of the device channels, the corresponding phrase is given at the top of the "Tables" page (see Figure 108): "The tables are common for all analyzer channels. In the case of parallel operation of channels, the current and power values are multiplied by the number of channels in the group, the resistance values are divided by the number of channels in the group."

The "Open..." buttons are used to select and download the required file with a table. Clicking on the corresponding "Open..." button opens a dialog box for selecting a file with a table. The Computer Interface allows you to load table files in two formats – text (*.txt) and files with delimiters (*.csv) (see Figure 109).

The structure and content of files with tables are subject to special requirements. the requirements set out in subsection 11.19.1.

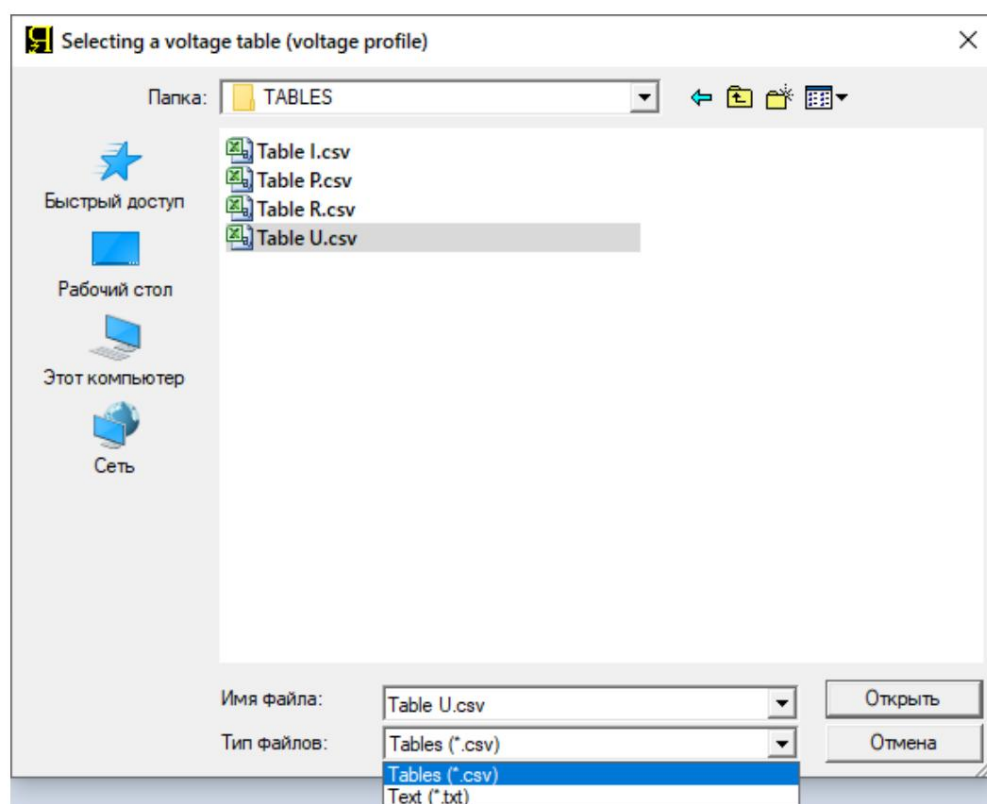


Figure 109 - File selection window with table

The selected file with the table is loaded into the window of the corresponding table (see Figure 108) and undergoes an automatic verification procedure. If any errors are detected, errors The user will be shown a window with a message about the type and location. If an error is detected, the selected file will be automatically closed and work on the corresponding table will be blocked.

As a result of automatic checking of the file with the table, the following are found and highlighted:
the following values:

- The minimum value of voltage, current, power or resistance, it can also be the maximum value of current or discharge power (with a minus sign), V, A, W or Ohm;
- Maximum value of voltage, current, power or resistance, V, A, W or Ohm;
- Total duration of the table in time, sec.

The total duration of the table is defined as the difference between the time value from the last row of the table file and the time from the first row of the table file. The time value from the first row of the table does not necessarily have to be zero and can have any value. The time during the execution of the table is counted from the time value from the first row (see subsection 11.19.1).

The minimum and maximum voltage values found ("Min. voltage according to the table:", "Max. voltage according to the table:"), "Min. current according to the table:", "Max. current according to the table:", "Min. power according to the table:", "Max. power according to the table:" or resistance ("Min. resistance according to the table:", "Max. resistance according to the table:"), and the calculated duration tables ("Duration according to the table:") in accordance with the original data in the selected table files, are displayed in the corresponding fields under the table windows (see figure 110):

Min. voltage according to the table:	1.000 V	Min. current according to the table:	-329.415 A
Max. voltage according to the table:	17.240 V	Max. current according to the table:	433.794 A
Duration according to the table:	20.00 s	Duration according to the table:	250.41 s

Figure 110 - Results of analysis of selected files with voltage and current tables
(for the power table and the resistance table - similarly)

The values of voltage, current, power and resistance specified in the table files can be shifted by the required amount, individual for each table (voltage shift, current shift, power shift and resistance shift).

After applying the shift, the shifted table values can be scaled (increased or decreased) by the required multiplier, which is set individually for each table (voltage multiplier, current multiplier, power multiplier and resistance multiplier).

The offsets of the voltage, current, power and resistance values are specified in the corresponding windows "Voltage offset, V:", "Current offset, A:", "Power offset, W:" and "Resistance offset, Ohm" (see Figure 111).

The multipliers for the shifted values for each table are specified in the corresponding windows "Voltage multiplier:", "Current multiplier:" (Current multiplier), "Power multiplier:" (Power multiplier) and "Resistance multiplier:" (Resistance multiplier) (see Figure 111).

Using the required values of shifts and multipliers, it is possible to adjust the initial tabular values of voltage, current, power and resistance from the selected files to the parameters of the tested CIT or to the parameters of the Analyzer.

Using multipliers with a minus sign, you can change the sign of the parameter. For example, use the discharge current values specified in the table as the charging current of the HIT, and vice versa, use the charge current values specified in the original table as the discharge current of the HIT.

The duration of the selected tables can be scaled (multiplied) by a time multiplier specific to each table. Time multipliers for each table are specified in the corresponding "Time multiplier:" windows (see Figure 111). Time multipliers can have any positive value, both greater than one (stretching the table in time) and less than one (compressing the table in time).

Voltage offset, V:	2	Current offset, A:	0
Voltage multiplier:	0.599	Current multiplier:	-0.023
Time multiplier:	1	Time multiplier:	0.5

Figure 111 – Parameters for setting up the voltage table and current table
(for the power table and the resistance table - similarly)

Maximum and minimum voltage values ("Minimum voltage for execution:", "Maximum voltage for execution:"), $\ddot{y}\ddot{y}\ddot{y}$ ("Minimum current for execution:", "Maximum current for execution:"), $\ddot{y}\ddot{y}\ddot{y}\ddot{y}\ddot{y}\ddot{y}$ ("Minimum power for execution:", "Maximum power for execution:") and resistance ("Minimum resistance for execution:", "Maximum resistance for execution:"), adjusted by the specified coefficients and prepared for execution by the device, are displayed in the corresponding fields under the windows for entering the shift and scale coefficients (see Figure 112). The final execution durations of the corresponding tables are also displayed there ("Duration for execution:").

Minimum voltage for execution:	1.797 V	Minimum current for execution:	-9.977 A
Maximum voltage for execution:	11.525 V	Maximum current for execution:	7.577 A
Duration for execution:	20.00 s	Duration for execution:	125.21 s

Figure 112 - The final minimum and maximum values of voltage and current, as well as the final durations of the voltage and current tables that the device will execute (for the power and resistance table - similarly)

The adjusted values of voltage, current, power, resistance and duration must fall within the working range of the device (see Table 5). Parameter values that are outside the working range of the device are highlighted **in red**.

color, and the inclusion of such a table in the HIT testing program becomes

impossible.

The "Close" buttons (see Figure 108) are used to discard the selected voltage, current, power or resistance table and clear the corresponding table window.

The "Open..." and "Close" buttons (see Figure 108), as well as the windows for entering the shift and multipliers for a specific table (see Figure 111), will be available for work (active) only if the connection between the Computer Interface and the Analyzer is established (see Section 9 and Subsection 19.1), and only if the specific table is not currently used for work on any of the instrument channels. In other words, if there is no channel for which the HIT testing program contains work according to a specific table (voltage, current, power or resistance), and which is not currently in the "IDLE" mode .

In all other cases, the "Open..." and "Close" buttons, as well as
The windows for entering shifts and multipliers for the corresponding table will be blocked.

19.4 Page "GRAPHICS"

The "Charts" page is required to display the charge-discharge curves of the devices tested on the HIT device and allows you to monitor the progress of the testing process in real time (see Figure 113):

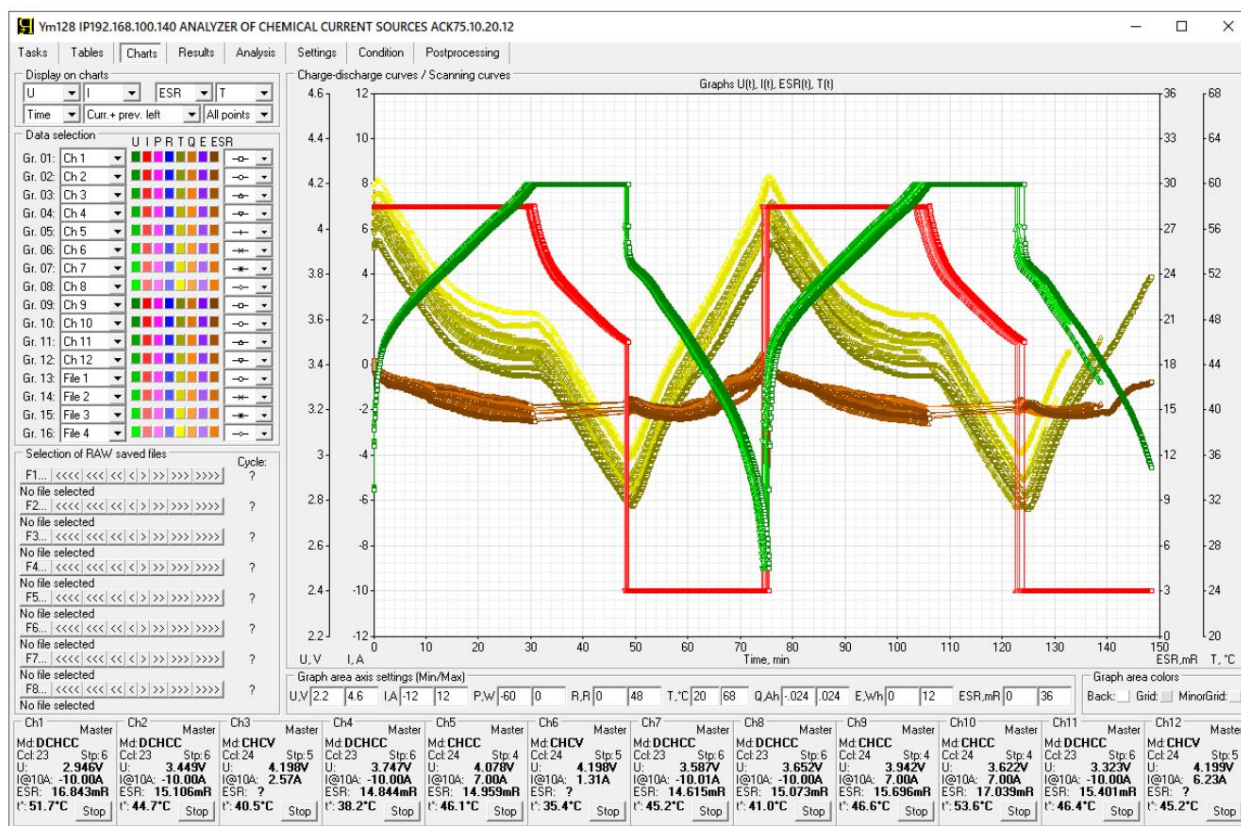


Figure 113 - Ym128 Computer Interface "Charts" Page

The main component of the "Charts" page is the plot area. charge-discharge graphs of the tested chemical energy sources.

In addition, the "Charts" page contains a "Selection of RAW saved" panel. files" (Selecting saved primary data files), which allows you to select, open and browse primary data files previously saved on the User's computer (historical files). Eight historical files can be selected, opened and viewed independently of each other at the same time. Data from the selected historical files can be displayed in the graph plotting area, in addition to the charge-discharge graphs displayed in real time during the testing of the HCS. This feature allows, for example, to visually compare the charge-discharge curves of the tested HCS and some reference HCS, or even to compare the charge-discharge curves of the same HCS, but in different testing cycles during the testing.

For more information on viewing previously saved primary data files and summary measurement results files, see Section 20.

The graph plotting area can simultaneously display 16 data groups, and for each group, the User can select measurement results from the following sources:

- "Not" – No (the group is not displayed on the charts);
- "File 1" – File with historical primary data No. 1;
- "File 2" – File with historical primary data No. 2;
- "File 3" – File with historical primary data No. 3;

- "File 4" – File with historical primary data ŷ4; • "File 5" – File with historical primary data ŷ5; • "File 6" – File with historical primary data ŷ6; • "File 7" – File with historical primary data ŷ7; • "File 8" – File with historical primary data ŷ8; • "Ch 1" – Current data for channel ŷ1; • "Ch 2" – Current data for channel ŷ2; • "Ch 3" – Current data for channel ŷ3; • "Ch 4" – Current data for channel ŷ4; • "Ch 5" – Current data for channel ŷ5; • "Ch 6" – Current data for channel #6; • "Ch 7" – Current data for channel #7; • "Ch 8" – Current data for channel #8; • "Ch 9" – Current data for channel #9; • "Ch 10" – Current data for channel #10; • "Ch 11" – Current data for channel #11; • "Ch 12" – Current data for channel #12.

Depending on the selected device type (ASK75.10.20.2 or ASK75.10.20.12), the ability to select channels No. 3 – No. 12 may not be available. The desired data source for a specific group of graphs is selected using the corresponding drop-down list in the "Data selection" area (see Figure 114):

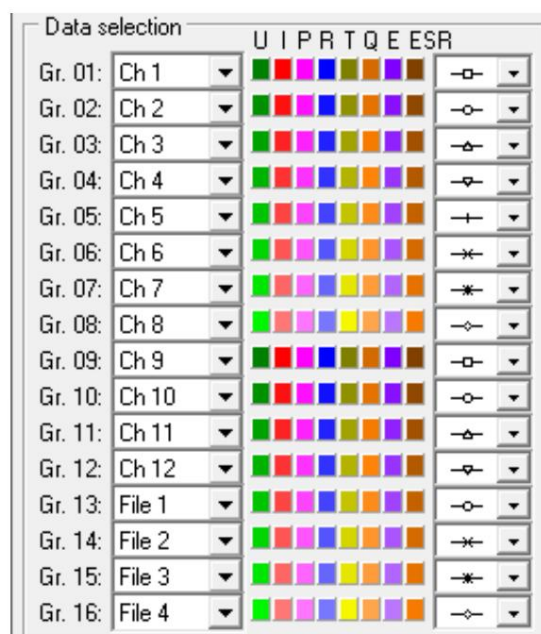


Figure 114 – Data selection and display settings panel

The following can be output to the area of plotting charge-discharge graphs HIT testing parameters:

- Voltage on the chemical power supply, U, V; • Current through the chemical power supply, I, A; • Power on the chemical power supply, P, W (Power P = Voltage U Current I); • Discharge resistance of the chemical power supply, R, Ohm (Resistance R = Voltage U / Current I); • Temperature of the chemical power supply, T, °C; • Charge received by the chemical power supply from the beginning of the next cycle Q, A h;

- The energy received by the heat source from the beginning of the next cycle E, W h;
- **ESR** HIT (see section 13.2, 13.3), mOhm.

Of these eight parameters, up to four parameters can be output to the plotting area at the same time in any combination, depending on time or depending on each other.

The power on the heat exchanger is displayed on the graphs with the same sign as the current through the heat exchanger.

The discharge resistance of the HIT **R** is displayed on the graphs only for those points* for which the voltage on the CIT is positive and the current through the CIT is negative (i.e. the CIT discharge occurs). For the remaining points, the CIT discharge resistance is not calculated and is not displayed on the graphs.

The ESR of the HIT is displayed on the graphs only for those program steps where the ESR is measured using the periodic current interruption method (see subsection 13.2), or the ESR of the HIT is measured when changing voltage, current, power or resistance pulses at the corresponding steps (see subsection 13.3). For all other steps of the testing program, the ESR values are not displayed on the graphs.

For each parameter of each data group, you can set an individual graph color. To select the color of the graph of a specific parameter of a specific data group, you need to click on the corresponding color rectangle in the "Data" panel

selection" (see Figure 114). This opens a color selection dialog box that allows you to set a color for a specific graph (see Figure 115):

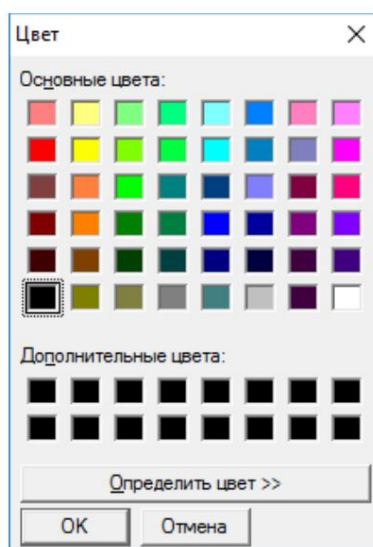


Figure 115 - Window for selecting the color of the parameter graphs (U, I, P, R, T, Q, E, ESR)

For each data group, you can set up the type of points for which graphs are plotted. The type of points is selected from the drop-down lists on the right side of the "Data selection" panel (see Figure 116):

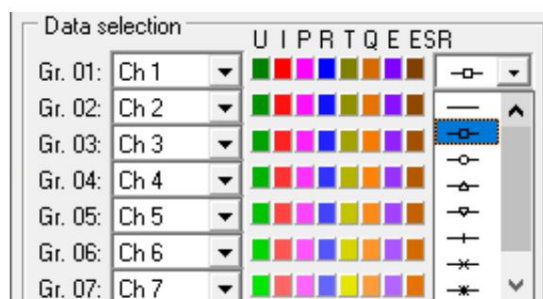


Figure 116 - Setting up the view of points for charge-discharge graph groups

* see the list of accepted abbreviations and designations.

The chart plotting area contains four vertical axes and one horizontal axis. The data for each axis is selected from the corresponding drop-down lists in the "Display on charts" area in the upper left corner of the "Charts" page. The relationship of a specific drop-down list to a specific axis in the chart plotting area is shown in Figure 117.

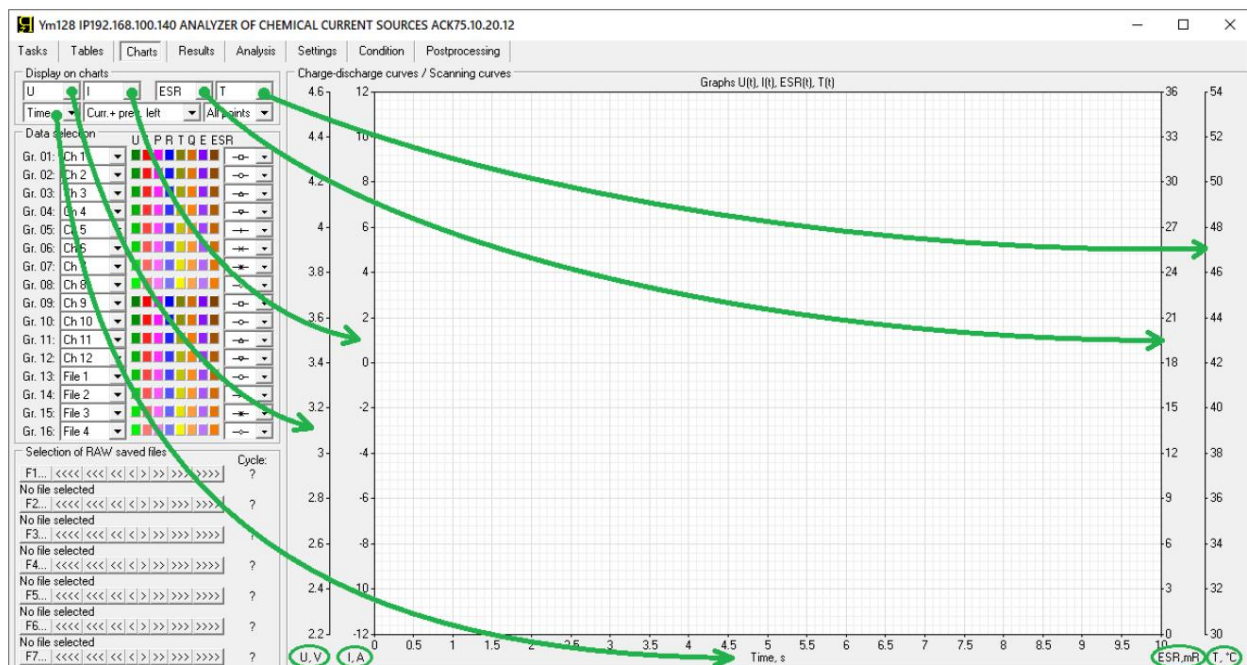


Figure 117 - Selecting parameters for the axes of the plotting area

For each vertical axis of the plotting area from the corresponding
From the drop-down list you can select one of the parameters:

- "U" – voltage on the CCS; •
- "I" – current through the CCS; • "P" – power on the CCS; • "R" – discharge resistance of the CCS; • "T" – temperature of the CCS; • "Q" – charge received by the CCS from the beginning of the current cycle; • "E" – energy received by the CCS from the beginning of the current cycle; • "ESR" – equivalent series resistance of the CCS; • "-I" – current through the CCS with an inverted sign (convenient for the MPPT mode); • "-P" – power on the CCS with an inverted sign (convenient for the MPPT mode); • "-Q" – charge given up by the CCS from the beginning of the current cycle; • "-E" – charge given up by the CCS from the beginning of the current cycle; • or

do not select any ("No"). For the lower horizontal axis of the plotting area from the corresponding
From the drop-down list you can select:

- "Time" – time; • "U" – voltage on the HCS; • "I" – current through the HCS; • "P" – power on the HCS; • "R" – discharge resistance of the HCS; • "T" – temperature of the HCS; • "Q" – charge received by the HCS from the beginning of the current cycle; • "E" – energy received by the HCS from the beginning of the current cycle; • "-Q" – charge returned by the HCS during the discharge process;

- "-E" – energy returned by the heat source during the discharge process;
- "ESR" – equivalent series resistance of the heat source;
- "-I" – current through the HIT with an inverted sign (convenient for testing the SB);
- "-P" – power at the heat exchanger with an inverted sign.

The selected combination of parameters is displayed in the plot area header. graphs directly above the graphs (see Figure 117).

The minimum and maximum of the graph area scales are configured separately for each parameter (U, I, P, R, T, Q, E, ESR). The scales for the parameters are configured in the "Graph area axis settings (Min/Max)" area (see Figure 118):

Graph area axis settings (Min/Max)																							
U,V	2.2	4.6	I,A	-12	12	P,W	-60	0	R,R	0	48	T,°C	30	54	Q,Ah	-0.024	0.024	E,Wh	0	12	ESR,mR	0	36

Figure 118 - Setting up parameter scales (U, I, P, R, T, Q, E, ESR)

The parameter scales in the plotting area are divided into 12 main divisions. To correctly display the plotting area grid, it is recommended to enter such minimum and maximum values for a specific parameter that the difference between the entered maximum and minimum is well divisible by 12 without the occurrence of irrational numbers.

The time scale is always adjusted automatically based on the data group. with maximum process duration.

For convenience of further consideration, the graphs in the figures below are presented for only one channel of the Analyzer. The data output is made from historical files of results for a test performed earlier, in the past, and the results for which were saved on the User's computer (see Section 20).

Data on charge-discharge graphs can be displayed in the following modes:

- Only the currently running cycle or data for the cycle from the selected primary data history file ("Current cycle", see Figure 119);
- The currently running cycle (or data for the cycle from the selected primary data history file), as well as the previous completed cycle (or data from the history file for the previous cycle, relative to the cycle in the selected history file). In this case, the graphs for the previous cycle are located to the left of the current cycle (as it happened in the course of time) ("Curr. + prev.

left", see Figure 119). This display mode is convenient for monitoring the testing process in real time. In this case, one complete previous cycle and an incomplete current cycle are displayed in the graph area. When the current cycle is completed, it becomes the previous one, and the previous previous cycle is removed from the graph area. Thus, the graphs are dynamically updated, but one complete completed cycle of the testing program and a partially executed cycle can always be observed on the graphs.

- The currently running cycle (or data for the cycle from the selected primary data history file), as well as the previous completed cycle (or data from the history file for the previous cycle, relative to the cycle in the selected history file). In this case, the graphs for the previous cycle are located behind, under the graphs for the current cycle, so that the graphs for the currently running cycle are superimposed on the graphs for the previous completed cycle ("Curr. + prev. bottom", see Figure 119). This display mode is convenient for monitoring the behavior of the tested chemical supply system with rapidly changing characteristics. In this mode, it is easy to see how much the charge-discharge graphs of the chemical supply system have changed in the current cycle relative to the charge-discharge graphs of the same chemical supply system, but in the previous cycle.

The selection of the data display mode for the current and previous cycles is carried out using the drop-down list located at the bottom in the middle in the "Display on charts" area (see Figure 119).

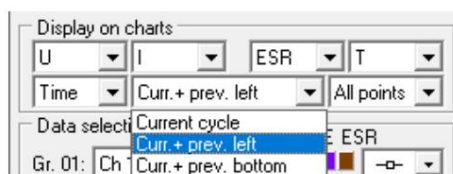


Figure 119 - Setting the data display mode for the current and previous

test program cycles

Examples of displaying charge-discharge graphs in various options for setting the data display mode for the current and previous cycles are shown in Figures 120–122.

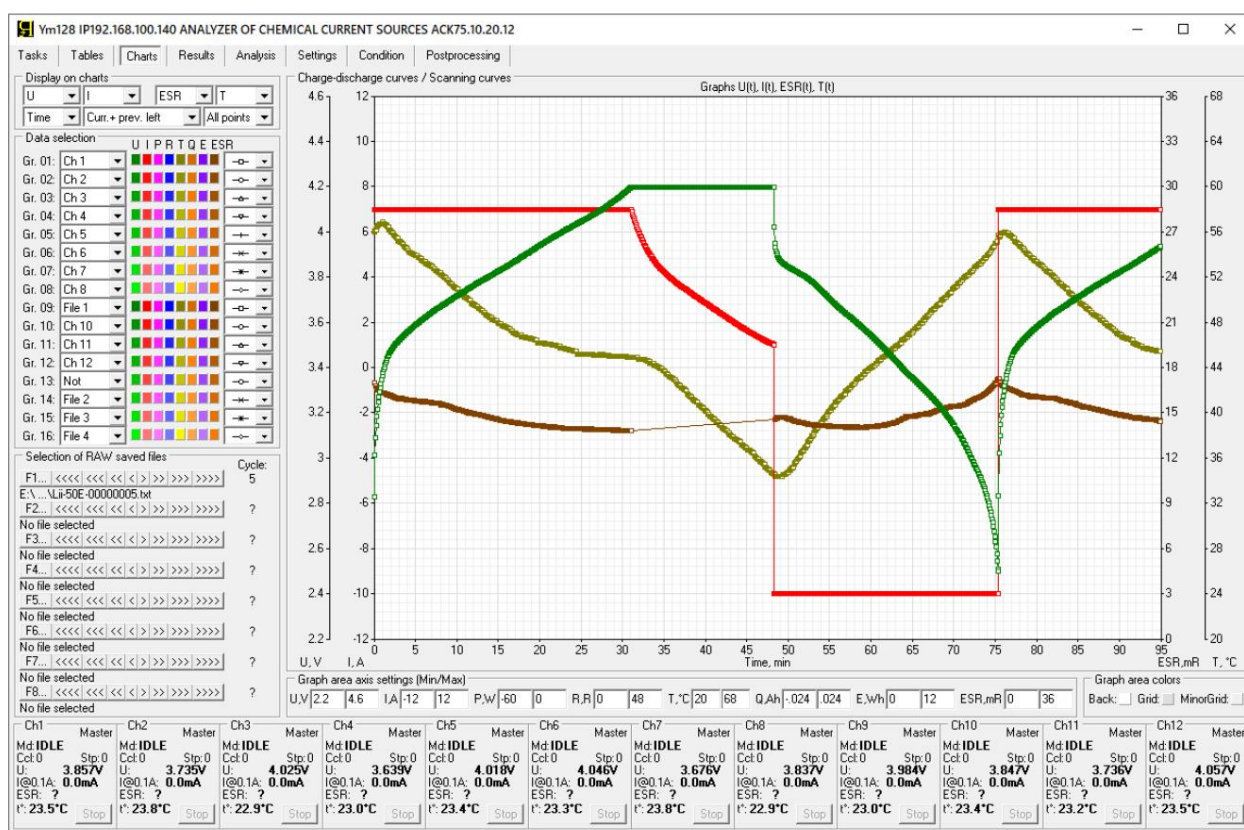


Figure 120 - Output of data on graphs for the currently running cycle,

and also for the previous completed cycle. The graphs for the previous cycle are displayed to the left of the graphs of the cycle being executed, in accordance with the passage of time ("Curr. + prev. left", see Figure 119).

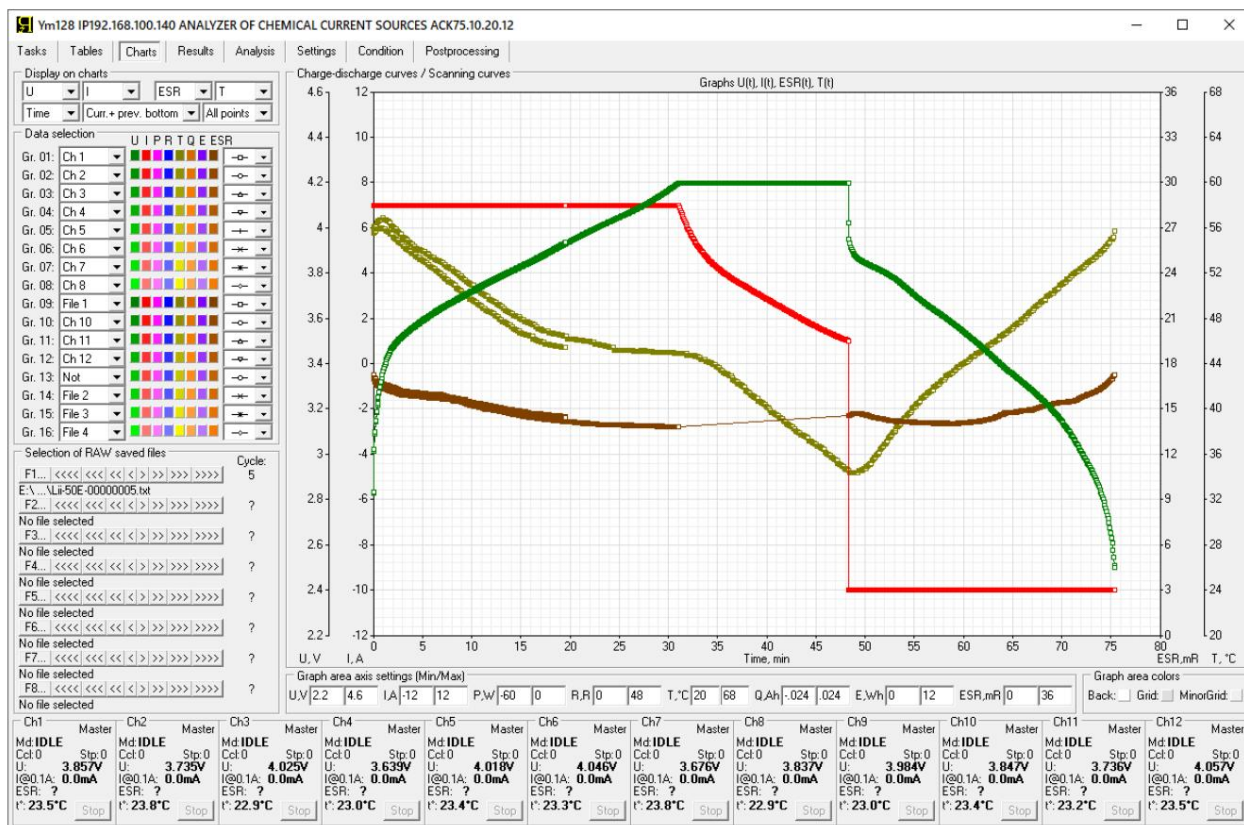


Figure 121 - Output of data on graphs for the currently running cycle, and also for the previous completed cycle. The graphs for the previous completed cycle are displayed behind the graphs for the currently running cycle and are placed under the graphs for the currently running cycle. ("Curr. + prev. bottom", see Figure 119).

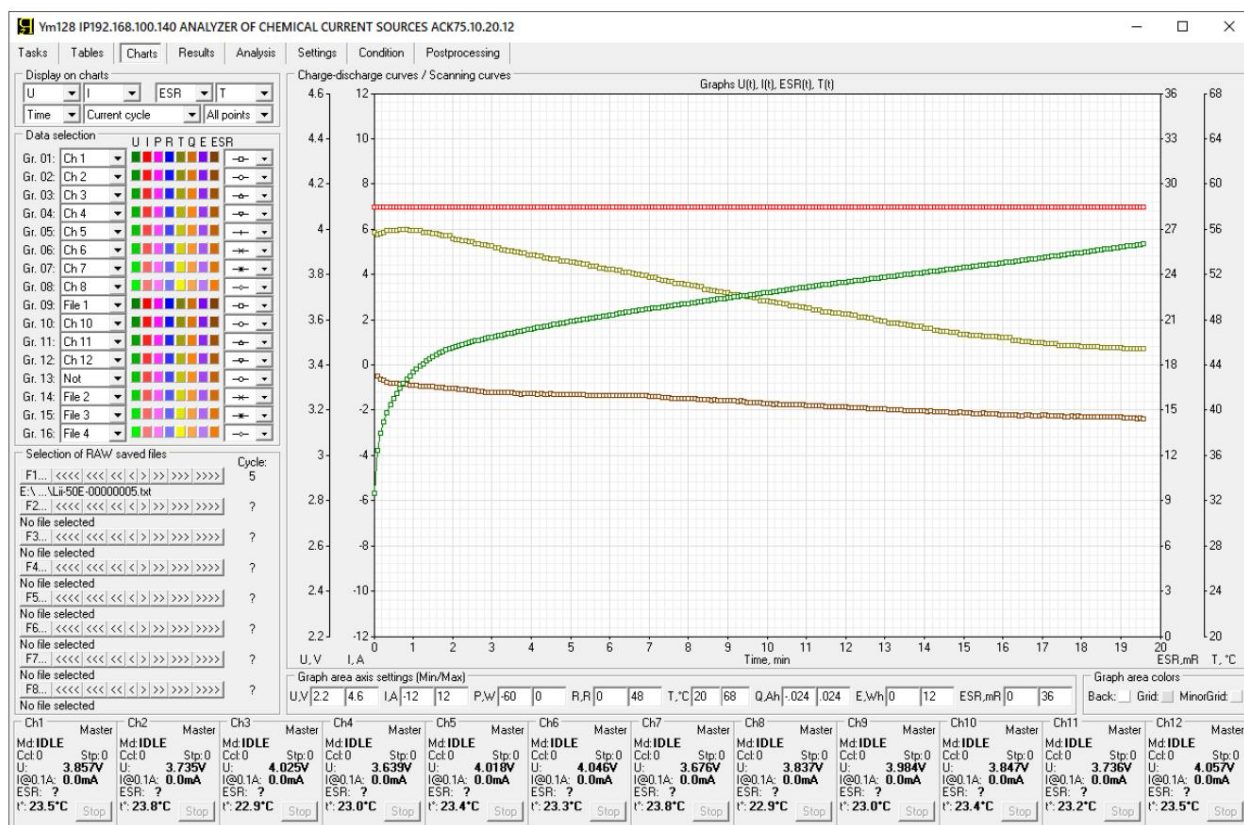


Figure 122 - Displaying data on graphs only for the currently running cycle ("Current cycle", see Figure 119)

The "Graph area colors" panel, located in the lower right corner of the "Charts" page, contains controls for setting the background and grid colors of the graph area (see Figure 123):

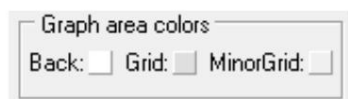


Figure 123 - "Graph area colors" panel

The background color and grid color of the area for plotting charge-discharge graphs are set by clicking on the corresponding colored square (see Figure 123). This opens a color selection dialog box (see Figure 115), which allows you to set the desired color.

The "Back:" square sets the background color of the chart plotting area.

The "Grid:" square specifies the color of the main grid lines of the plotting area.

The "MinorGrid:" square specifies the color of the minor grid lines.
areas of graph construction.

When working with large volumes of measurement results, the performance of the "Charts" page of the Computer Interface may be slow. Considerable acceleration of processing of large volumes of data can be achieved by using thinning of the data displayed on the charge-discharge graphs. The setting of thinning of the data displayed on the graphs is specified using the drop-down list located at the bottom right in the "Display on charts" area (see Figure 124):

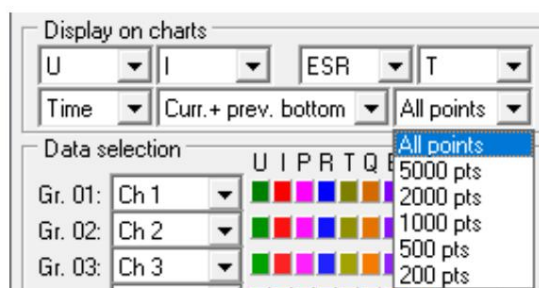


Figure 124 - Selecting the limitation of the number of points, displayed on charge-discharge graphs

When the "All points" option is selected (see Figure 124), the graphs of the selected parameter combination are displayed using all available points* from the current real-time test results or from the selected historical files.

When you select "5000 pts", "2000 pts", "1000 pts", "500 pts" or "200 pts", the number of points displayed on the graphs will be reduced (thinned out) so that the total number of points that will be used to plot graphs for a particular group of data does not exceed the specified limit.

An example of limiting the number of points on which charge-discharge graphs are plotted to 200 points is shown in Figure 125 (the output of graphs without limiting the number of points for the same case is shown in Figure 126).

* see the list of accepted abbreviations and designations.

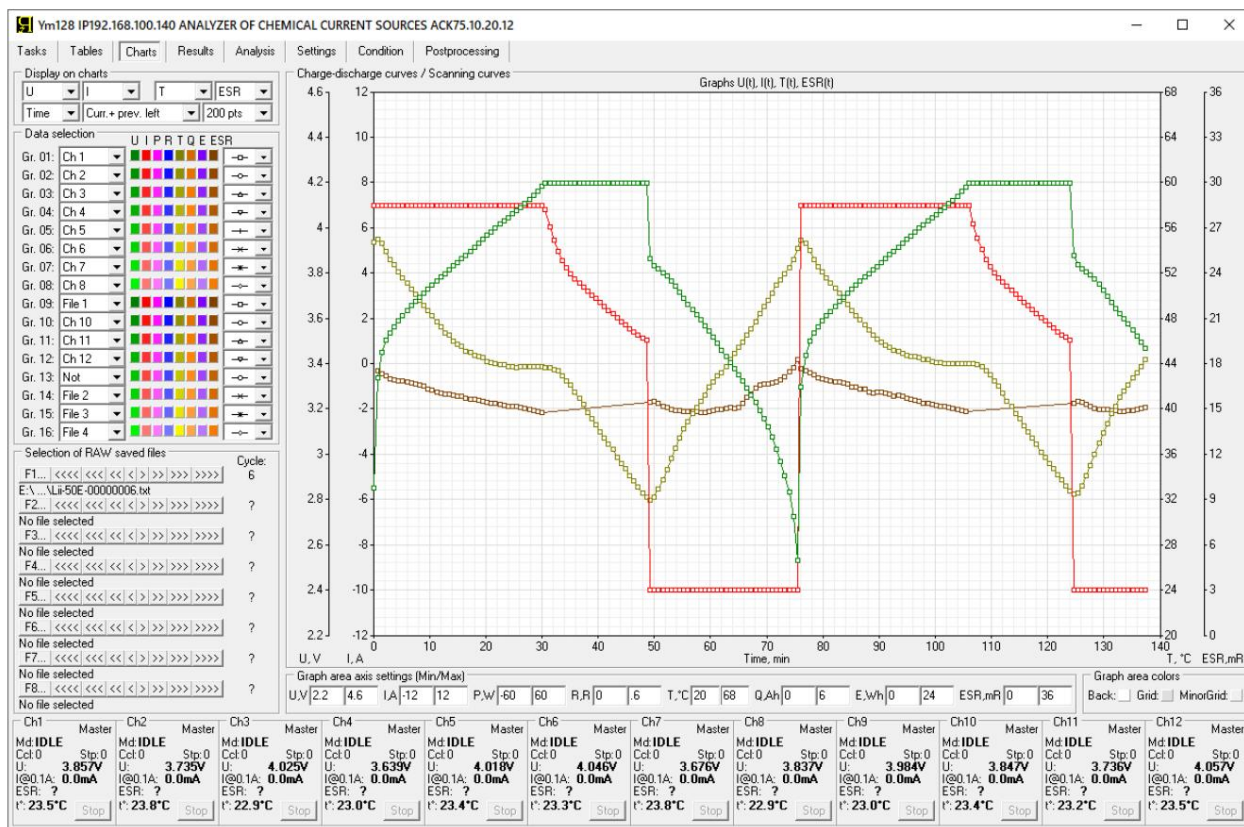


Figure 125 - No more than 200 points are displayed on the charge-discharge graphs for each data group

Below, in figures 126 – 132, charge-discharge graphs are shown for some combinations of HIT parameters.

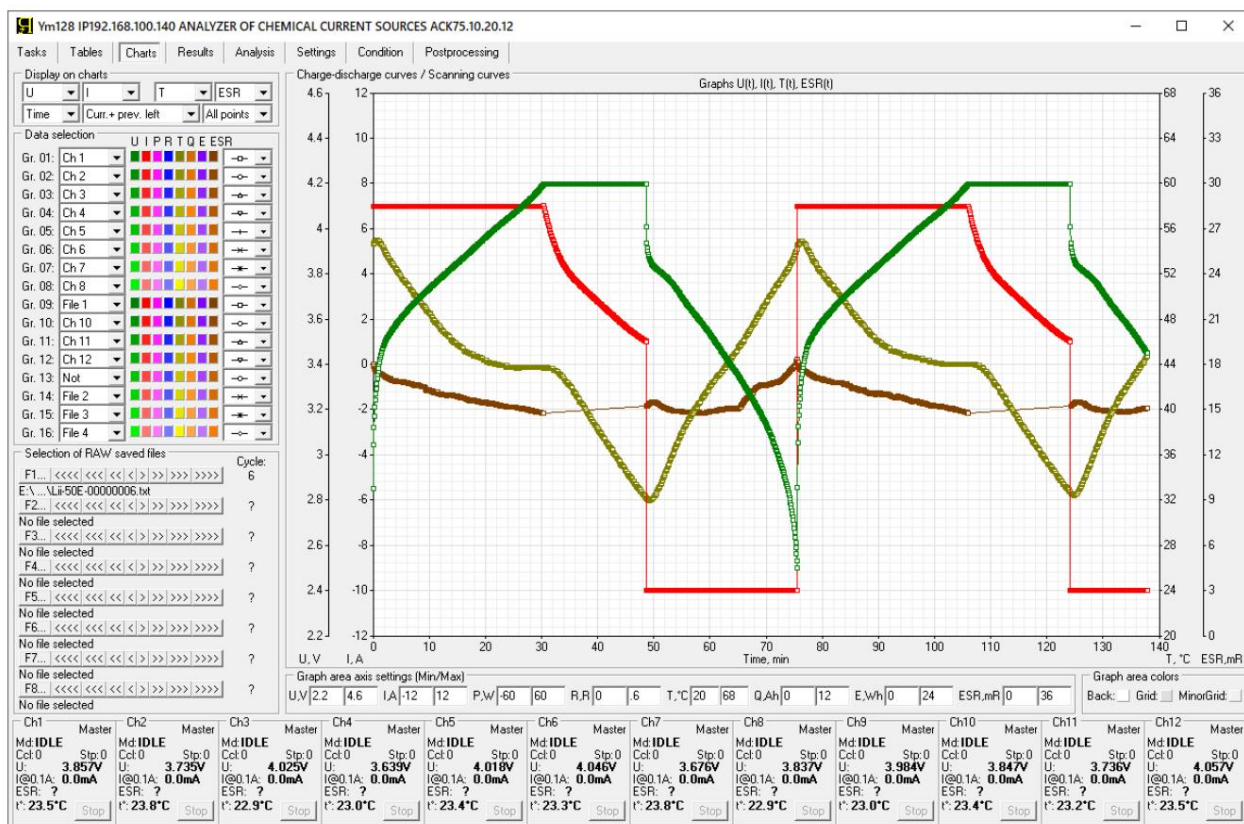


Figure 126 - The graphs show a combination of parameters U(t), I(t), T(t), ESR(t)

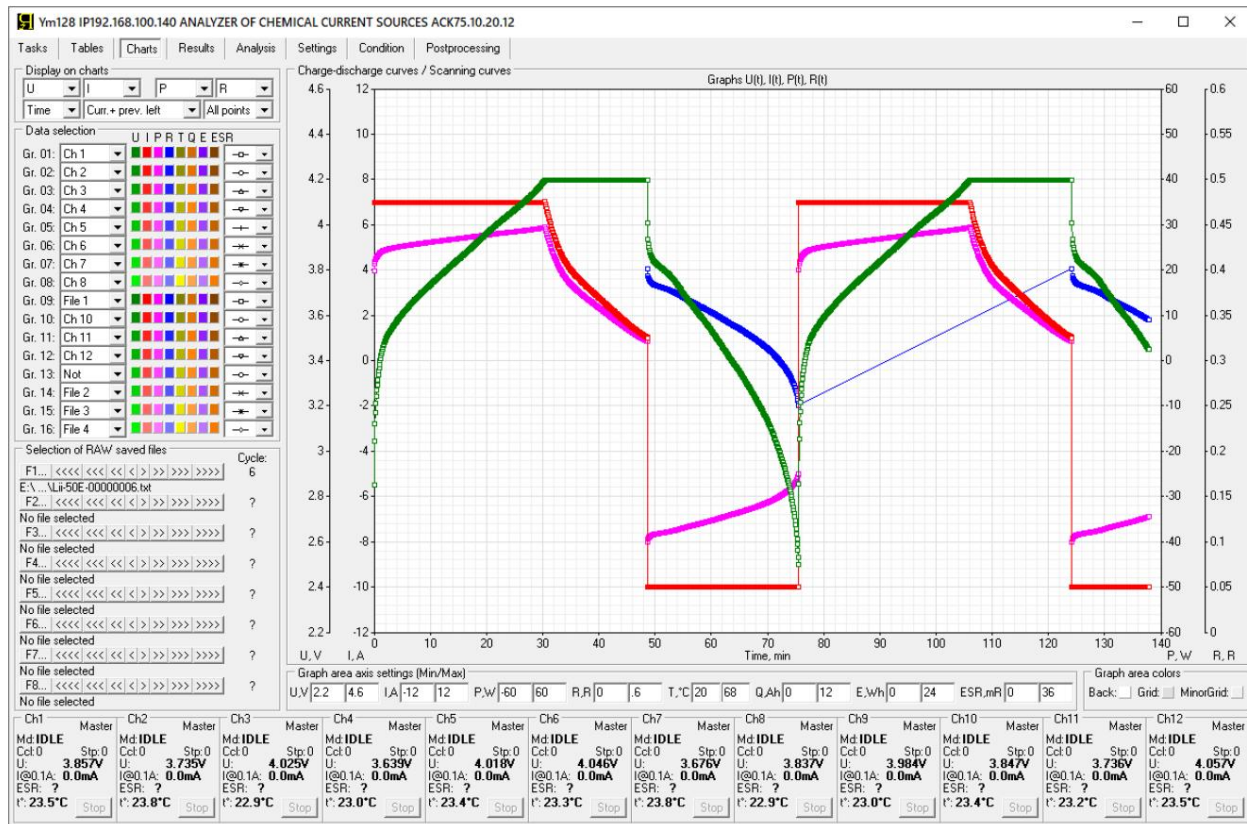


Figure 127 - The graphs show a combination of parameters U(t), I(t), P(t), R(t)

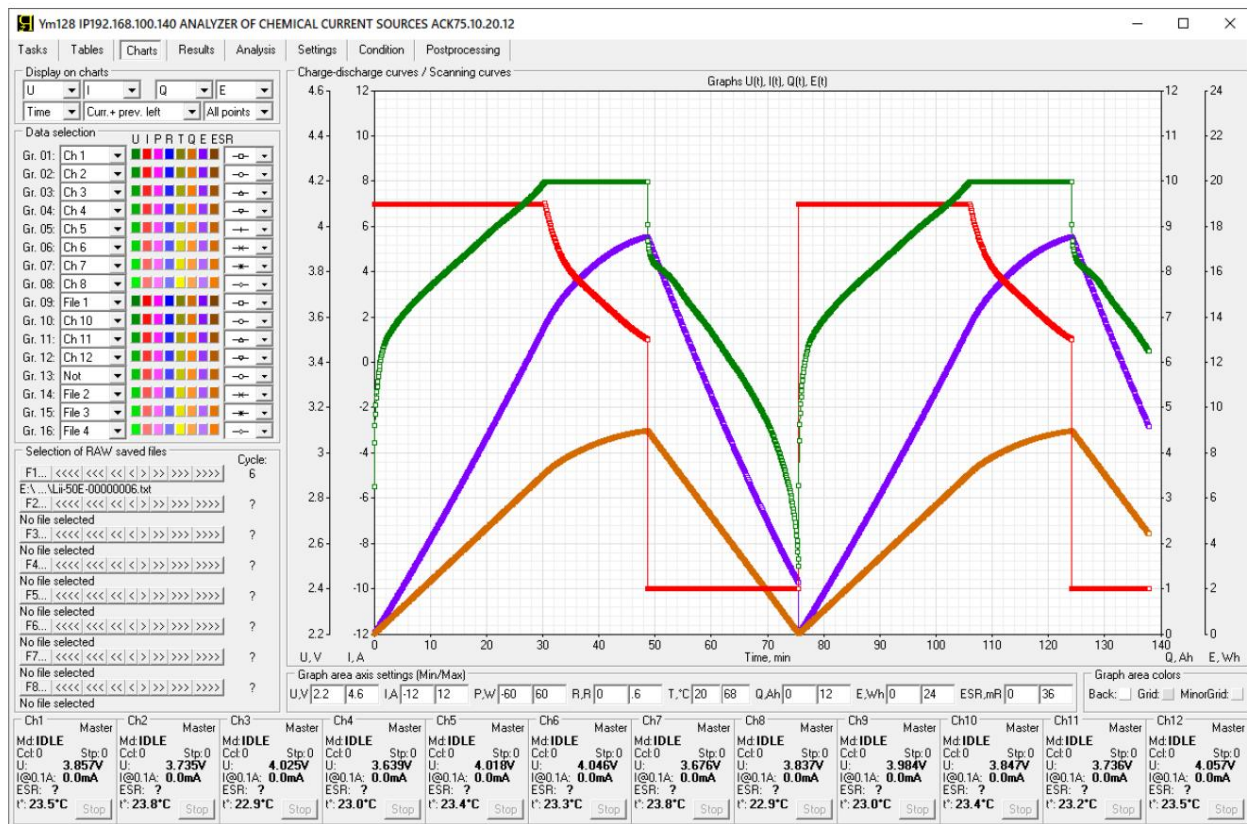


Figure 128 - The combination of parameters U(t), I(t), Q(t), E(t) is displayed on the graph.

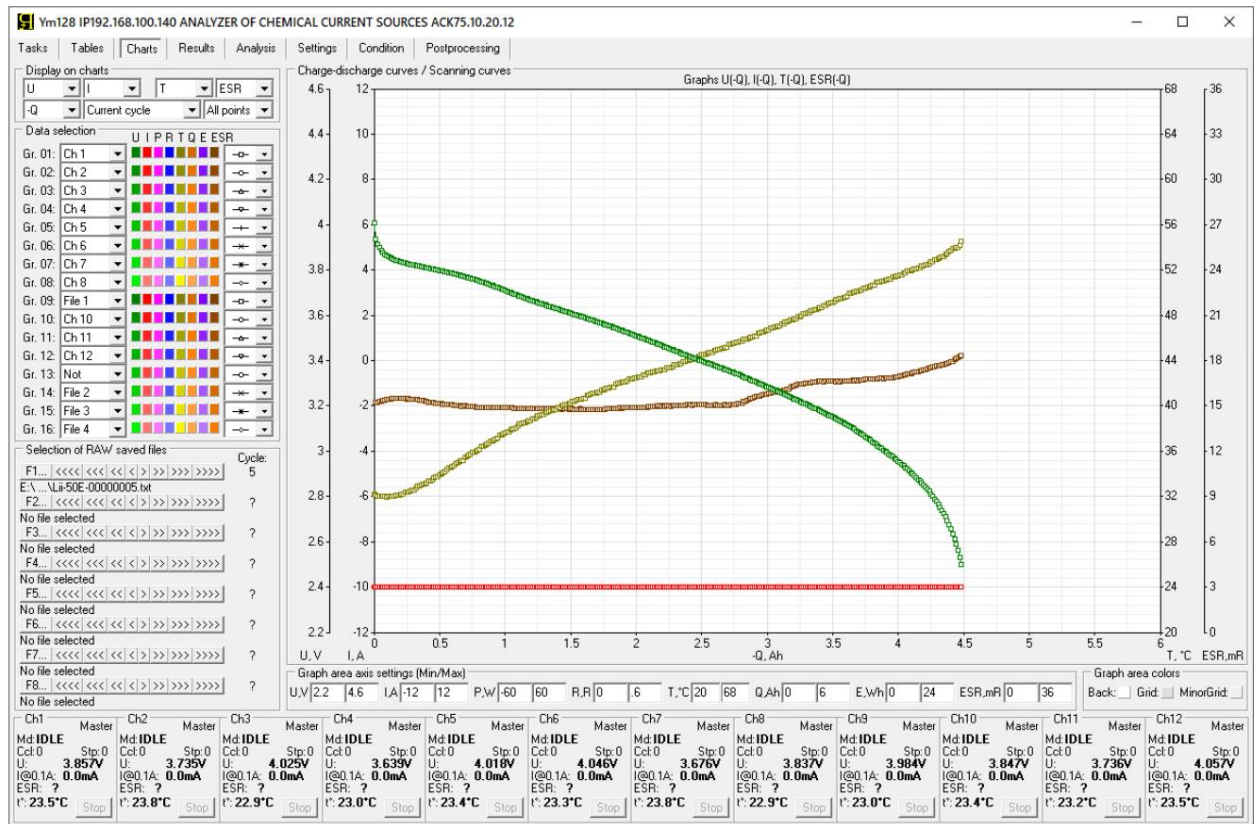


Figure 129 - The graphs show the combination of parameters U(-Q), I(-Q), T(-Q), ESR(-Q)

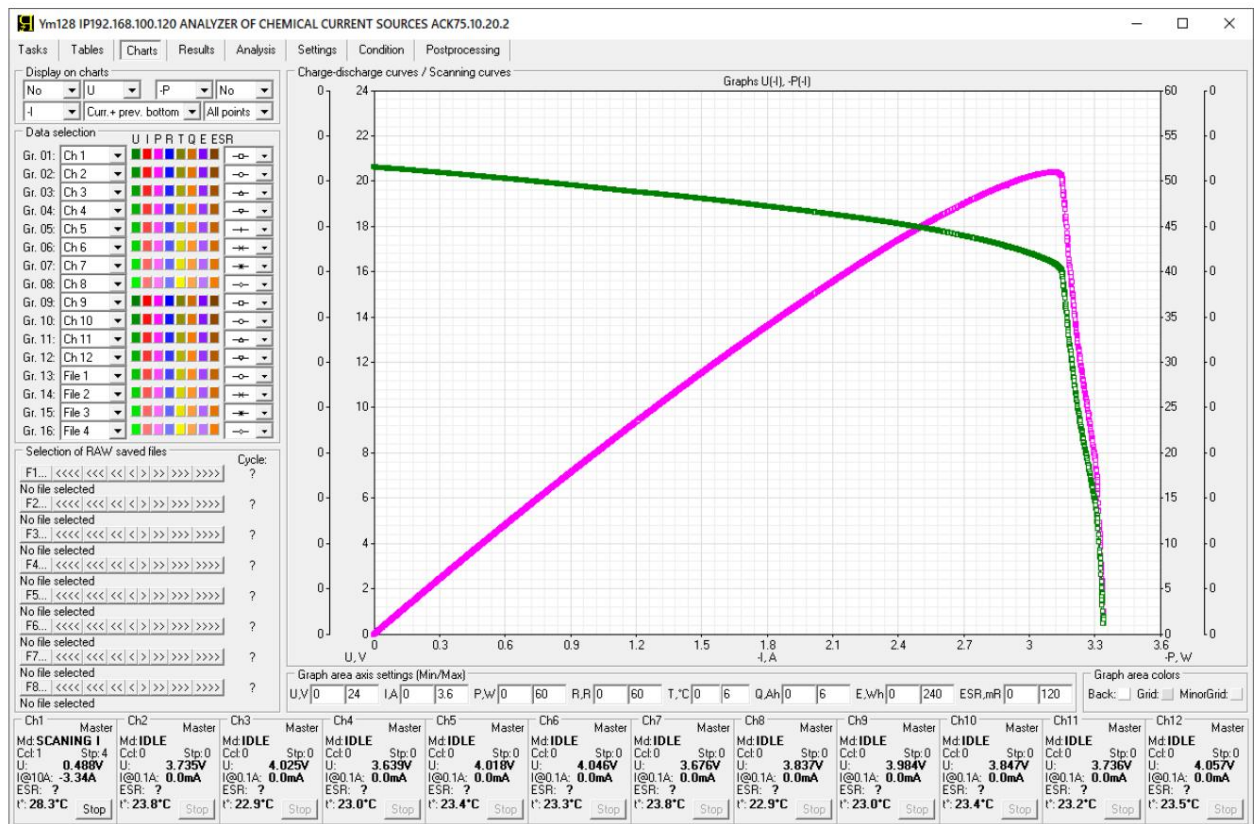


Figure 130 - The graphs show a combination of parameters U(-I), -P(-I) (Volt-Amp and Watt-Amp characteristics of a solar battery)

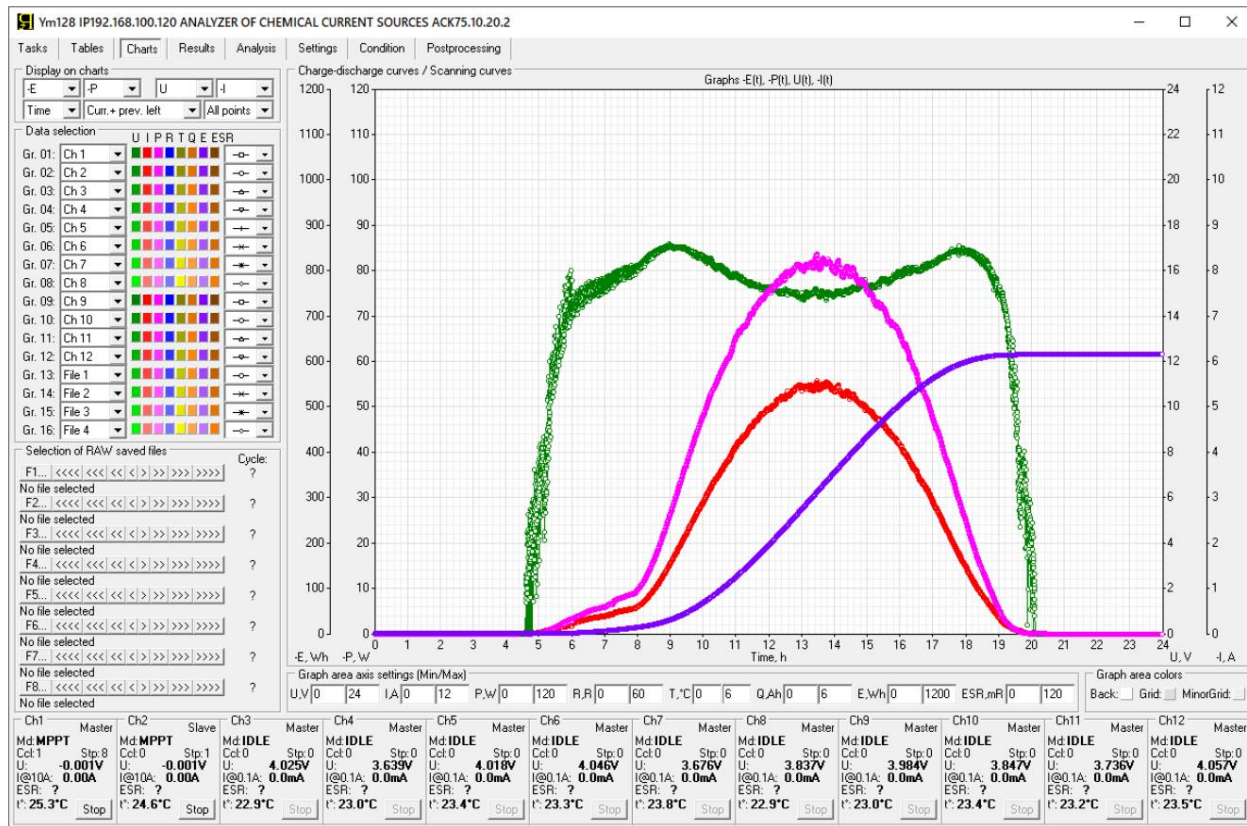


Figure 131 - The combination of parameters $-P(t)$, $U(t)$, $-I(t)$, $-E(t)$ is displayed on the graph.
(daily output of a solar battery rated at 12 V, 150 W)

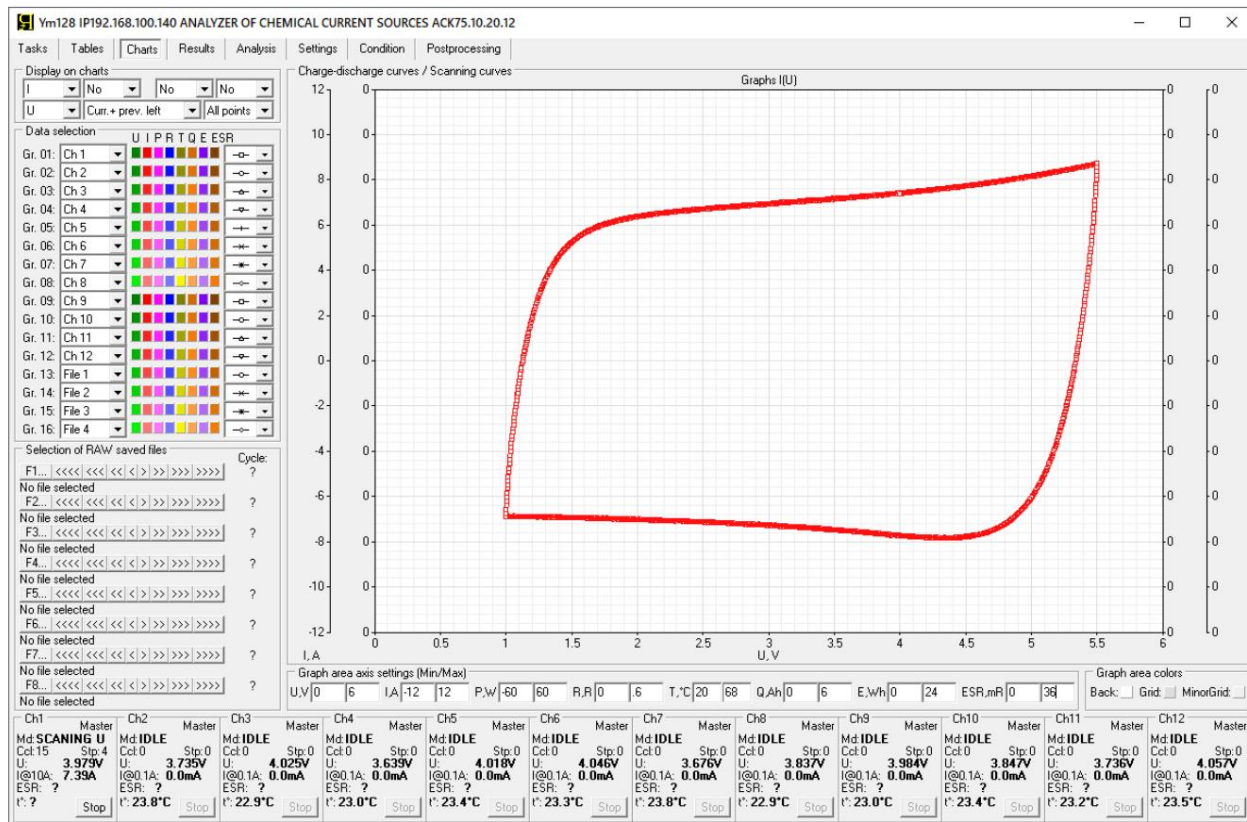


Figure 132 - The combination of parameters $I(U)$ is displayed on the graphs

To save the area of charge-discharge graphs as a picture and use in various reports, any area of the graphs (as well as any area of the computer screen) can be copied to the clipboard by pressing the Win + Shift + S keys simultaneously, and then selecting the required area of the computer screen using the mouse.

19.5 Page "RESULTS"

The "Results" page of the Ym128 Computer Interface is designed to display the current state of the arrays of accumulated measurement results for each channel of the device (see Figure 133):

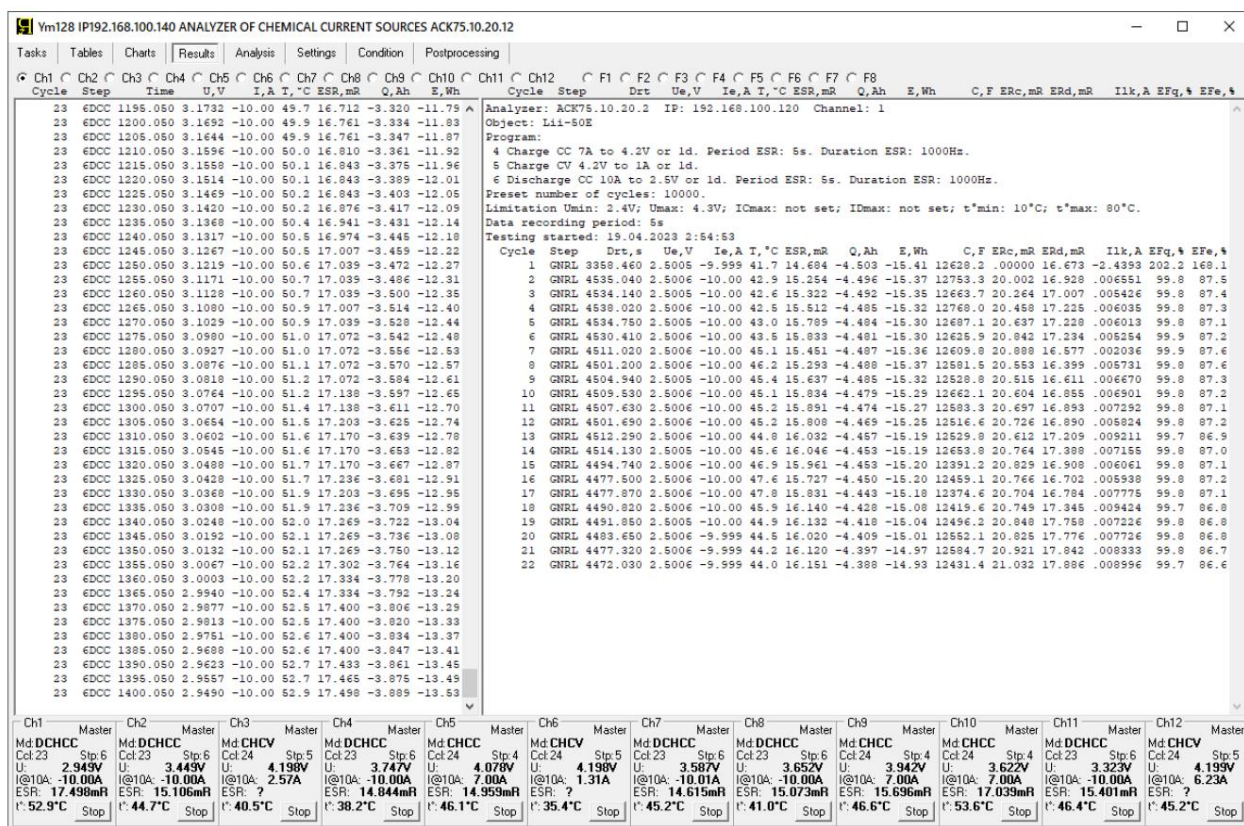


Figure 133 - Ym128 Computer Interface Results Page

The "Results" page displays the current state of the measurement results files, the final form of which will be available to the User after the completion of the HIT testing (see section 15).

The Results page displays both types of result windows. The window on the left is for displaying the primary data, and the window on the right is for displaying the summary results of the measurements.

The results windows can display measurement results for one of the device channels or from one of eight results files previously saved on the User's computer (from historical files) and selected by the User for viewing (see Section 20).

The choice of data source to view is made by the switches at the top of the page. The number of channels offered for selection will depend on the type device (ACS75.10.20.2 or ACS75.10.20.12) and consists of two or twelve channels, respectively (see Figures 134 and 135):

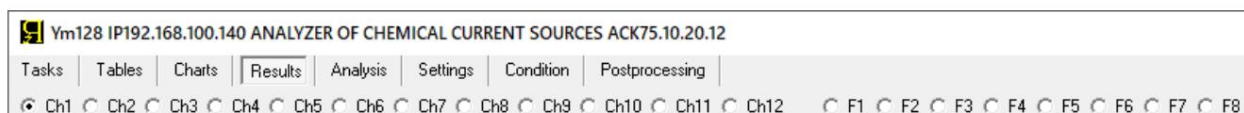


Figure 134 - Data source selection switches for output in the results windows of the "Results" page for the ACS75.10.20.12 Analyzer

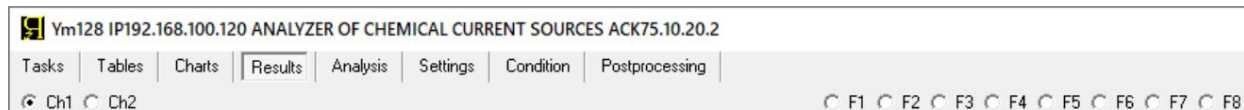


Figure 135 - Data source selection switches for output in the results windows of the "Results" page for the ASC75.10.20.2 Analyzer

Selecting, opening, and browsing historical raw data files is done in the "Selection of RAW saved files" area on the "Charts" page. Selecting and opening historical summary result files is done in the "Selection of CLK saved files" area on the "Analysis" page.

For more information on viewing previously saved primary data files and summary measurement results files, see Section 20.

19.6 "ANALYSIS" page

The "Analysis" page of the Ym128 Computer Interface contains an area for constructing graphs for analyzing the parameters of the tested chemical energy sources and allows one to observe changes in the characteristics of the chemical energy sources depending on the number of charge-discharge cycles completed or from the temperature of the heat exchanger (see Figure 136):

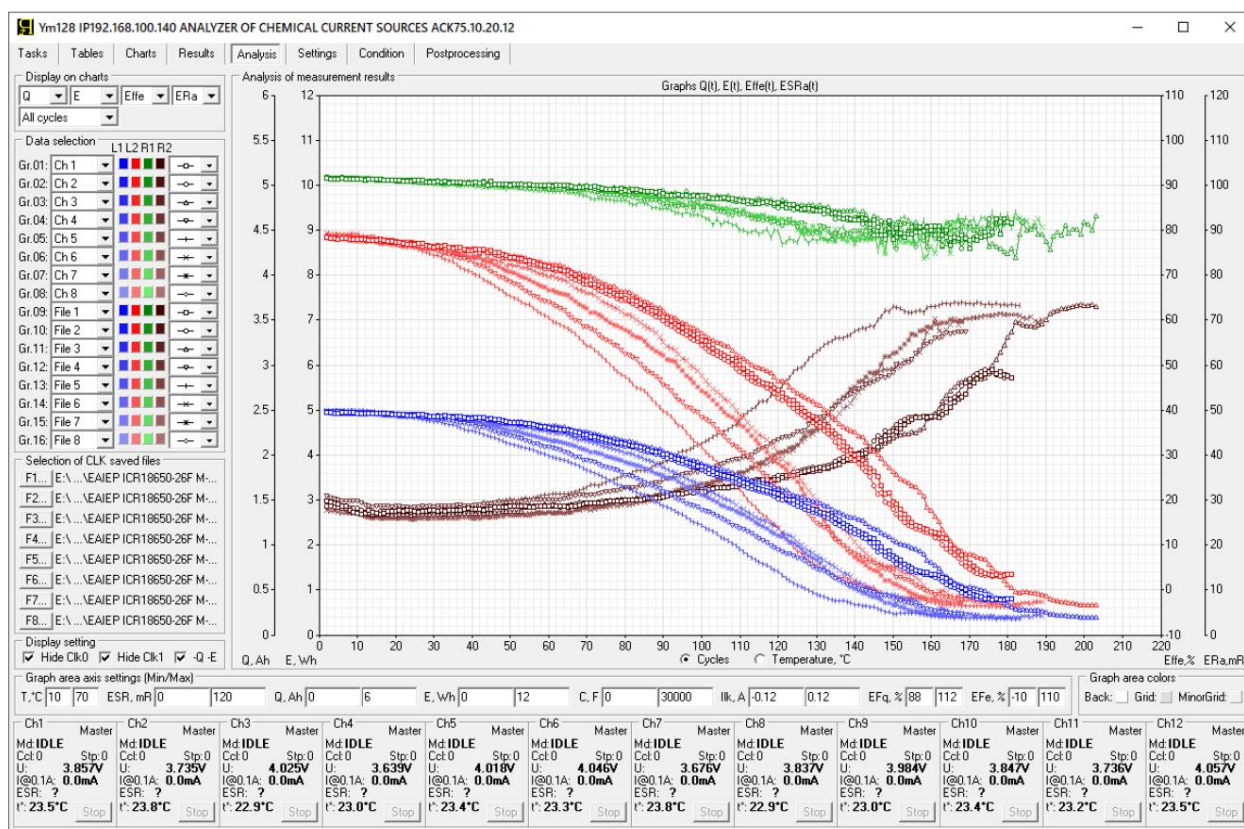


Figure 136 - Ym128 Computer Interface Analysis Page

The main component of the "Analysis" page is the area for constructing graphs for analyzing changes in the characteristics of the tested HITs.

In addition, the "Analysis" page contains the "Selection of CLK saved files" panel, which allows you to select, open, and view the CLK saved files previously saved on the User's computer (historical files).

eight historical files can be selected, opened and viewed independently of each other. Data from the selected historical files can be displayed in the graph plotting area, in addition to graphs of the analysis of changes in the characteristics of the HIT displayed in real time during testing. This feature allows, for example, to visually compare the change in the characteristics of the tested HIT and some reference HIT.

For more information on viewing previously saved primary data files and summary measurement results files, see Section 20.

In the area of plotting analysis graphs, 16 groups of data can be simultaneously displayed, and for each group, at the User's choice, measurement results from the following data sources can be displayed:

- "Not" – No (the group is not displayed on the charts);
- "File 1" – File with historical summary results of measurements No. 1;
- "File 2" – File with historical summary results of measurements No. 2;
- "File 3" – File with historical summary results of measurements No. 3;
- "File 4" – File with historical summary results of measurements No. 4;
- "File 5" – File with historical summary results of measurements No. 5;
- "File 6" – File with historical summary results of measurements No. 6;
- "File 7" – File with historical summary results of measurements No. 7;
- "File 8" – File with historical summary results of measurements No. 8;
- "Ch 1" – Current data for channel No. 1;
- "Ch 2" – Current data for channel No. 2;
- "Ch 3" – Current data for channel No. 3;
- "Ch 4" – Current data for channel No. 4;
- "Ch 5" – Current data for channel No. 5;
- "Ch 6" – Current data for channel No. 6;
- "Ch 7" – Current data for channel No. 7;
- "Ch 8" – Current data for channel No. 8;
- "Ch 9" – Current data for channel No. 9;
- "Ch 10" – Current data for channel No. 10;
- "Ch 11" – Current data for channel No. 11;
- "Ch 12" – Current data for channel #12.

Depending on the selected device type (ACS75.10.20.2 or ACS75.10.20.12), the ability to select channels No. 3 – No. 12 may not be available.

The selection of the desired data source for a specific group of graphs is carried out using the corresponding drop-down list in the "Data selection" area (see Figure 137).

The parameters determined by the ASK75.10.20.2 and ASK75.10.20.12 Analyzers during the testing of the HIT and located in the files of summary measurement results in lines with the "GNRL" marker (in lines with general parameters for the cycle) (see subsection 15.3) can be displayed in the graph plotting area, namely:

- "Ta" - Average temperature of the heat exchanger over the cycle, °C;
- "ESRa" - Averaged over the cycle value of the ESR of the heat source, measured by the method periodic interruption of current through the CIT or in the modes of voltage, current, power or resistance pulses, mOhm (see subsection 13.2 and subsection 13.3);
- "Q" - Charge (capacity) given by the heat source during discharge, A h;
- "E" - Energy released by the heat source during discharge, W h;
- "C" - Electrical capacity of the heat source, F (determined only at the DCHCC steps);

- "ESRc" - ESR of the HIT, measured by the voltage jump at the beginning of the charging steps, mOhm (see subsection 13.4);
- "ESRd" - ESR of the HIT, measured by the voltage jump at the beginning of the discharge steps, mOhm (see subsection 13.4);
- "Ilk" - Average leakage current through the heat-transfer device per cycle, A (see section 12);
- "EFq" - efficiency of the heat source by charge (by A h), %;
- "EFe" - heat source efficiency by energy (by W h), %.

Of these ten parameters, up to four parameters can be simultaneously output to the graph plotting area in any combination, depending on the number of completed cycles of the testing program or depending on the average temperature of the heat exchanger per cycle.

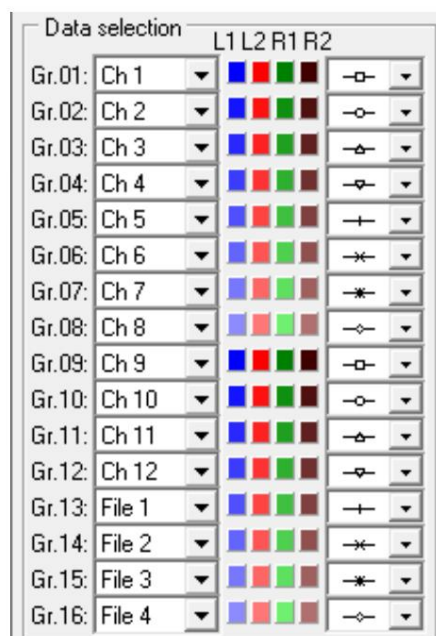


Figure 137 - Data selection and display settings panel

The chart plotting area contains four vertical axes and one horizontal axis. The data for each vertical axis is selected from the top drop-down lists located in the "Display on charts" area in the upper left corner of the "Analysis" page (see Figure 138).

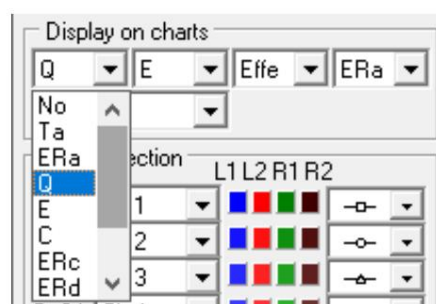


Figure 138 - Selecting parameters for display on the vertical axes of graphs

The left drop-down list allows you to select the parameter to display on the first left vertical axis of the plotting area, the second drop-down list for the second left vertical axis, the third drop-down list for the first right vertical axis, and the rightmost drop-down list for the second right vertical axis (see Figure 139). If you select "No", data for a specific vertical axis of the plotting area will not be displayed.

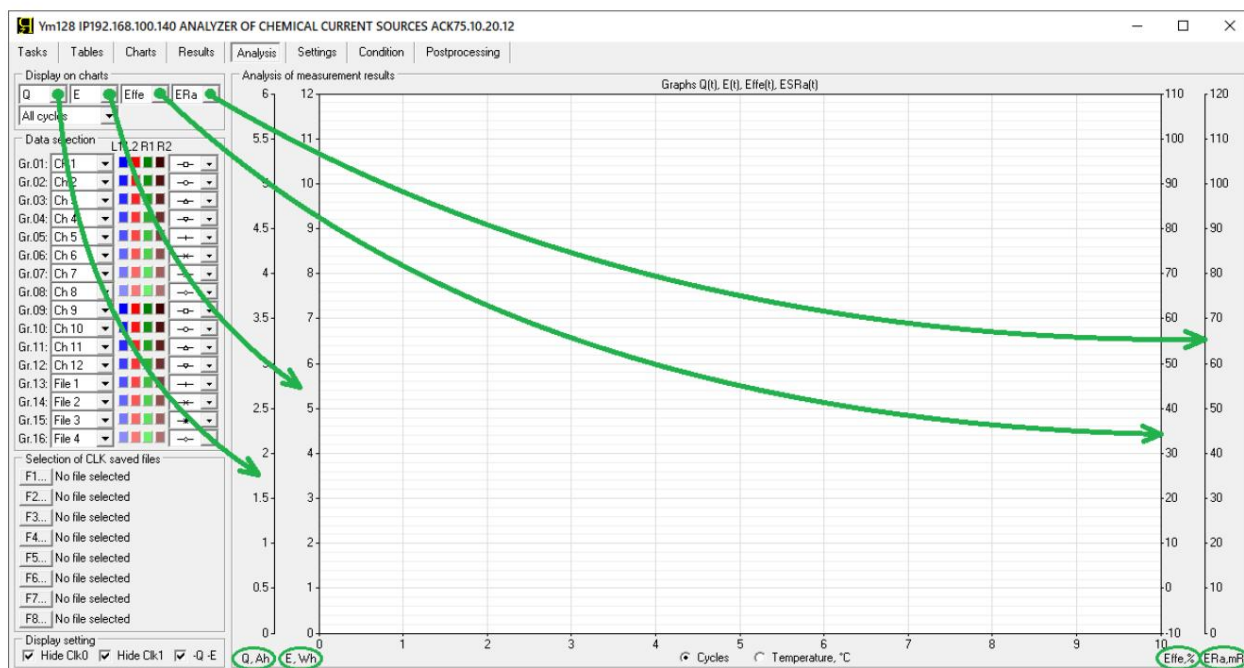


Figure 139 - Selecting parameters for the vertical axes of the analysis graphs

For each data group, individual graph colors can be selected for each vertical axis. The color for a specific data group and a specific vertical axis is selected by clicking on the corresponding color rectangle (see Figure 137). "L1" is the color of the parameter graphs displayed along the first left vertical axis, "L2" is the color of the parameter graphs displayed along the second left vertical axis, "R1" is the color of the parameter graphs displayed along the first right vertical axis, and "R2" is the color of the parameter graphs displayed along the second right vertical axis. Clicking on the corresponding color rectangle opens a color selection dialog box (see Figure 115), which allows you to set the required color.

Additionally, for each data group, you can set up the type of points for which graphs are built. The type of points is selected from the drop-down lists on the right side of the "Data selection" panel (see Figure 140):

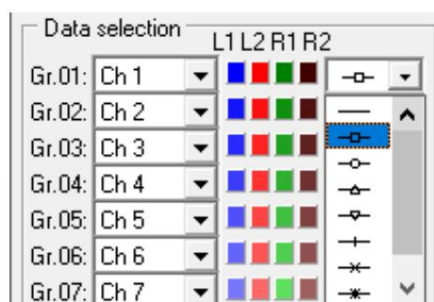


Figure 140 - Setting up the view of points for groups of analysis graphs

Parameters in the area of plotting analysis graphs can be displayed depending on the number of completed cycles of the testing program or depending on the average temperature of the chemical heat exchanger per cycle. The selection is made using switches located directly under the area of plotting graphs (see Figure 139).

Selecting the "Cycles" switch allows you to display the HIT parameters depending on the number of cycles passed in the testing program. On the lower axis

the number of cycles passed will be plotted. The axis is scaled automatically, according to the data group with the maximum number of cycles passed.

Selecting the "Temperature, °C" switch allows you to display the CCS parameters depending on their average temperature. The average CCS temperature for each completed cycle of the testing program will be plotted along the lower axis. This display method is extremely convenient when conducting climatic tests of CCS, in which the CCS characteristics are determined depending on their temperature.

Using the lower drop-down list in the "Display on charts" area (see Figure 141), you can set a limit on the number of points for which the analysis charts are built. Limiting the number of points can be useful when working with large volumes of measurement results and significantly speeds up the operation of the Computer Interface.

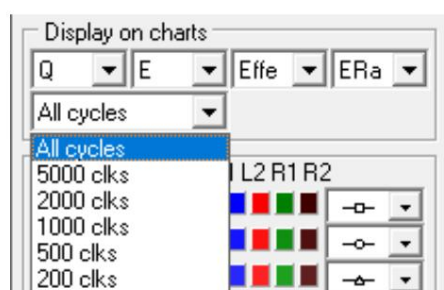


Figure 141 - Selecting a limit on the number of points displayed on the analysis graphs

When the "All cycles" option is selected (see Figure 141), the graphs of the selected parameter combination are displayed using all available points (lines with the "GNRL" marker) from the current real-time summary files or from the selected historical summary files.

When you select "5000 clks", "2000 clks", "1000 clks", "500 clks" or "200 clks", the number of points plotted on the graphs will be reduced (thinned out) so that the total number of points plotted on the graphs for a particular data group does not exceed the specified limit.

The minimum and maximum scale for each parameter is configured separately in the "Graph area axis settings (Min/Max)" area (see Figure 142):



Figure 142 - Setting up parameter scales displayed on analysis graphs

Depending on the specified testing program and the initial state of the tested CCTs, the measurement results for cycle #0 (test preparation steps, see Section 11) and for cycle #1 may differ significantly from the measurement results in subsequent CCT testing cycles. The "Hide Clk0" and "Hide Clk1" checkboxes in the "Display setting" area (see Figure 143) allow you to prohibit the display of measurement results on the graphs obtained for cycle #0 and cycle #1, respectively.

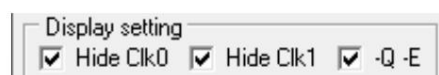


Figure 143 - "Display setting" area of analysis graphs

In general, the charge Q and energy E displayed on the analysis graphs have negative values, since the "GNRL" lines of the summary measurement results files contain the values of charge Q and energy E returned by the tested CCS at discharge steps. Working with negative values of charge Q and energy E is inconvenient for visual perception of changes in the characteristics of the tested CCS. Checking the "-Q -E" box (see Figure 143) inverts the sign of charge Q and energy E displayed on the analysis graphs, allowing the graphs of changes in charge Q and energy E returned by the tested CCS to be transferred to the positive region, which makes the graphs of these parameters more visual.

The "Graph area colors" panel, located in the lower right corner of the "Analysis" page, contains elements for setting the background color and grid of the analysis graph area (see Figure 144):

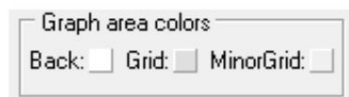


Figure 144 - "Graph area colors" panel

Setting the background color and grid color of the plotting area of the analysis graphs is done by clicking on the corresponding color square (see Figure 144). This opens a color selection dialog box (see Figure 115), which allows you to set the desired color.

The "Back:" square sets the background color of the chart plotting area.

The "Grid:" square specifies the color of the main grid lines of the plotting area.

The "MinorGrid:" square specifies the color of the minor grid lines.
areas of graph construction.

To save the area of the analysis graphs as a picture and use in various reports, any area of the graphs (as well as any area of the computer screen) can be copied to the clipboard by pressing the Win + Shift + S keys simultaneously, and then selecting the required area of the computer screen with the mouse. The part of the screen copied to the clipboard can then be pasted as a picture into the program "Paint", "Microsoft Word", "Microsoft Excel" and many other programs, as well as pasted into the text of an e-mail, into chats of various messengers, and so on.

19.7 "SETTINGS" page

The Ym128 Computer Interface "Settings" page is designed to configure the connection between the selected Analyzer and the Computer Interface, viewing the status of the internal systems of the device, viewing the version number of the Computer Interface, as well as performing factory calibration and testing of the device:

Figure 145 - Ym128 Computer Interface "Settings" Page

Before you can start working with the Analyzer, you must select the type of instrument you are using from the drop-down list in the "Instrument(s) Type" area located in the upper left corner of the "Settings" page (see Figure 146):

Figure 146 - Selecting the type of Analyzer used

CAUTION! Do not change the device type during the HIT test! This will lead to malfunction of the device and loss of test results!

The connection to the device is configured in the "TCP" area (see Figure 147). Here, in the "Analyzer IP" window, you must specify the IP address indicated on the front panel of the device (position 14, Figure 1 and Figure 4), and check the "Connect automatically" box.

"Computer IP" - the IP address of the User's computer. In general, to establish a connection between the Computer Interface and the Analyzer, it is necessary

the condition is that the first three digits in the IP address of the device and the computer match (that is, in this example, 192.168.100). For more information, see Section 9.

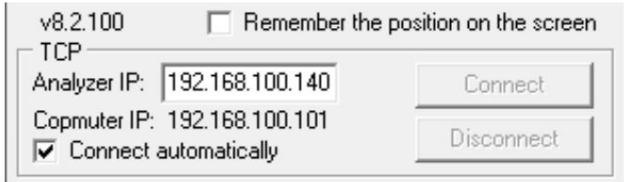


Figure 147 - Setting up connection to the Analyzer

The Computer Interface version number is displayed directly above the "TCP" area (in Figure 147 it is "v8.2.100").

Checking the "Remember the position on the screen" box (see Figure 147) before closing the Computer Interface allows you to remember its position on the User's computer monitor screen. The next time you launch the Computer Interface, it will be opened in the same place on the monitor screen.

The "Analyzer memory" area (see Figure 148) contains information about the state of the Analyzer's internal memory, used to save measurement results during autonomous operation. For more details, see Section 21.

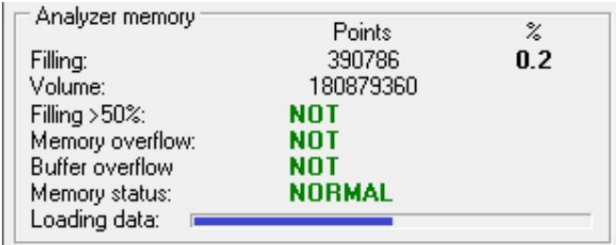


Figure 148 - Parameters of the Analyzer's internal memory state

The "Additionally" area (see Figure 149) reflects the additional operating parameters of the device.

The "MCD connection quality" values reflect the quality of communication on the internal data bus between the control and monitoring module (CMM) and the charge-discharge modules (CDM) (see Figure 7 and Figure 8). The values of 1200:1200 for the ASK75.10.20.12 Analyzer and 200:200 for the ASK75.10.20.2 indicate perfect communication quality. Other values indicate delays or malfunctions in the operation of the device hardware.

When the connection between the Analyzer and the Computer Interface is established, the "Analyzer clock" value should increase continuously. This parameter can be used as an indicator of the establishment of a connection between the Analyzer and the Computer Interface when the device is connected.

The parameter "Data arrival period, ms:" reflects the time interval between adjacent data packets arriving at the computer from the Analyzer via the Ethernet network. The parameter value "100.0" indicates good connection quality. Other values indicate delays in the Ethernet network or low performance of the User's computer.



Figure 149 - Auxiliary parameters of the device operation

The parameter "Analyzer operating time, h:" (see Figure 149) reflects the total operating time of the Analyzer since production. This parameter is included in the warranty conditions for the device.

The "Update..." button is used to update the firmware Analyzer. For more information on updating the device firmware, see Section 23.

The "Analyzer verification / calibration" area (see Figure 145) is intended for calibration and verification of the device under factory conditions.

19.8 Page "STATUS"

The "Condition" page of the Ym128 Computer Interface is designed to monitor the operation of the Analyzer's internal systems, as well as to identify the causes of emergency modes of the device's channels:

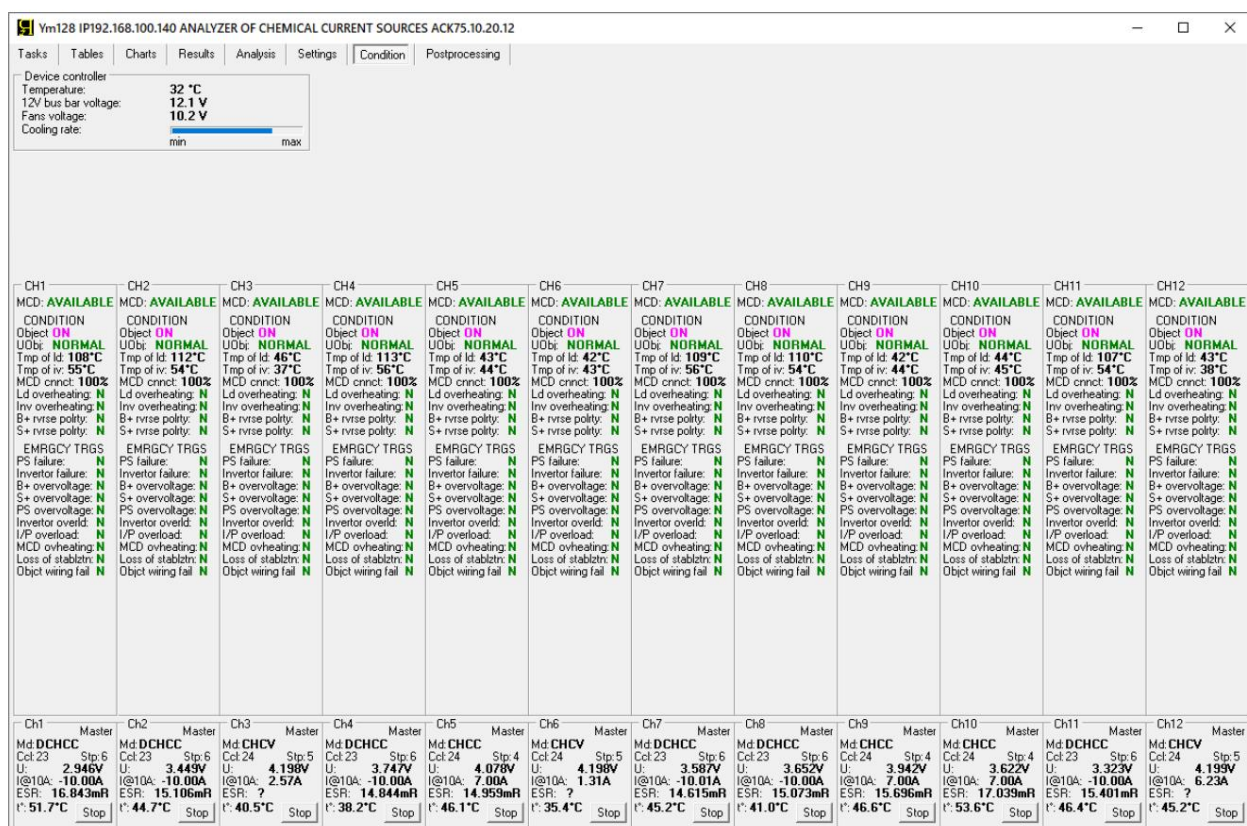


Figure 150 - Ym128 Computer Interface Condition Page

The "Device controller" area displays the current state control and management module (CMM) of the Analyzer (see Figure 7 and Figure 8):

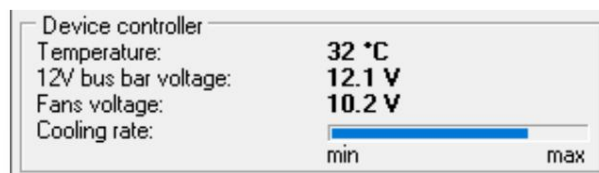


Figure 151 - Parameters of the control and management module of the device

The "Temperature" parameter reflects the temperature of the control module board and device control (MCC) (see Figure 7 and Figure 8).

The "12V bus bar voltage" parameter reflects the voltage on the power supply (PS) that supplies power to the control module.

and device control (MCC) (see Figure 7 and Figure 8).

The ASK75.10.20.2 and ASK75.10.20.12 analyzers have an adaptive cooling system, the performance of which is adjusted automatically depending on the temperature of the device components. The cooling system intensity is controlled by changing the supply voltage of the cooling fans. The "Fans voltage" parameter reflects the current supply voltage of the device cooling fans. The "Cooling rate:" parameter reflects the current capacity of the Analyzer cooling system in a clear graphical form.

The middle part of the "Condition" page contains uniform areas reflecting some internal parameters and emergency triggers of the device channels. The number of these areas is determined by the type of the device used - two for the ASK75.10.20.2 Analyzer and twelve for the ASK75.10.20.12.

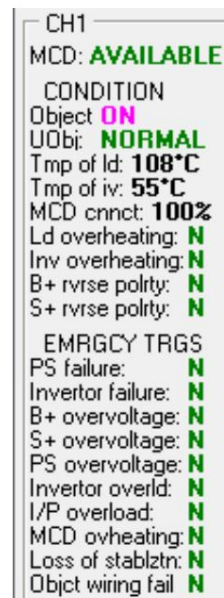


Figure 152 - Device channel status parameters

The "MCD:" indicator (yyy) reflects the state of data exchange between a specific channel of the device (charge-discharge module - yy) and the control and monitoring module (yy) (see Figure 7 and Figure 8). The indicator can take two values:

"AVAILABLE" (AVAILABLE) – data exchange between a specific MZR and MCU is in normal mode, and **"INACCESS"** (INACCESS) – data exchange between the MZR and MCU is disrupted. The MZR does not respond to requests from the MCU, work with this channel is impossible.

The "CONDITION" parameter block contains the current parameter values.
channel.

The "Object" indicator reflects the state of the connection of the HIT with the internal circuit of the device channel. The **"ON"** value means that the tested HIT is connected to the power circuit of the channel and is ready for charging or discharging. The **"OFF"** value means that the tested HIT is disconnected from the power part of the channel, and the device can only monitor the voltage at the HIT terminals and its temperature. The connection and disconnection of the tested HIT to the power circuit of the device channel occurs automatically, in accordance with the specified testing program.

The **"NORMAL"** state of the "UObj:" indicator means that the voltage on the UOC falls within the operating range of the device, and the device can fully provide the declared UOC testing parameters. When the voltage goes beyond the operating range of the device, the indicator value changes to **"OORANGE"** (OUT OF RANGE). This indicator state does not prohibit the operation of the device channel, but it should be borne in mind that outside the operating range of the voltage, the actual UOC testing parameters may not correspond to the specified ones (for example, when

(if the voltage on the HIT is less than 1.2 V, the device channel will not be able to provide a full discharge current of 10 A).

The parameter "Tmp of Id:" reflects the current temperature of the load resistor **R** (See Figure 7 and Figure 8). The temperature of the load resistor increases significantly in the discharge modes of the tested battery. The maximum temperature of the load resistor, which causes the overheating protection to operate, is 180 °C.

The "Tmp of iv:" parameter reflects the current temperature of the power inverter of the device channel - the charge converter and the discharge converter (see Figure 7 and Figure 8). The maximum inverter temperature that triggers the overheating protection is 80 °C.

The "MCD cnct:" parameter reflects the quality of communication between a specific device channel (charge-discharge module - MCD) and the control and monitoring module (CM) (see Figure 7 and Figure 8). The value "100%" means that all data packets transmitted from the MCD reach and are correctly processed by the corresponding MCD - the quality of communication is impeccable. Values less than 100% indicate slowdowns or malfunctions of the device.

The parameter "Ld overheating:" is an indicator of overheating of the load resistor **R** (see Figures 7 and 8). The value "**N**" indicates that there is no overheating and the load resistor is operating normally, the value "**Y**" indicates that the resistor is overheating.

The parameter "Inv overheating:" is an indicator of overheating of the power inverter. device channel - charge converter and discharge converter (see Figure 7 and Figure 8). The value "**N**" indicates the absence of overheating and normal operation of the channel, the value "**Y**" indicates overheating.

The parameter "B+ rvrse polrty:" is an indicator of the reverse polarity of the voltage at the power terminals of the device channel (see positions 2 and 6, Figure 1 and Figure 4). The value "**N**" indicates the absence of reverse polarity at the power terminals of the channel, the value "**Y**" indicates the presence of reverse polarity of the voltage.

The parameter "S+ rvrse polrty:" is an indicator of the reverse polarity of the voltage at the potential terminals of the device channel (see positions 3 and 5, Figure 1 and Figure 4). The value "**N**" indicates the absence of reverse polarity at the power terminals of the channel, the value "**Y**" indicates the presence of reverse polarity of the voltage.

The "EMRGY TRGS" parameter block reflects the state of emergency triggers of a specific channel of the device. For more information on emergency triggers, emergency protections and emergency states of device channels, see Section 22.

19.9 "POSTPROCESSING" page

Ym128 Computer Interface Postprocessing Page is designed to correct measurement results files if necessary.

The main purpose of this page is to merge two separate result files (both primary data files, see subsection 15.2, and summary measurement results files, see subsection 15.3) obtained for the same tested CCS, in the event of a power failure during testing.

Description of working with the "Postprocessing" page is beyond the scope of this Manual and can be found on the Internet at <https://battery-analyzers.ru/>.

20 VIEWING PREVIOUSLY RECORDED MEASUREMENT RESULTS

The Ym128 Computer Interface (see Section 19) allows not only to control Analyzers of various types, as well as to receive and process measurement results in real time, but also to view measurement results recorded earlier, in the past, and saved on the User's computer (historical result files).

Both historical raw data files and historical summary measurement result files can be loaded and viewed (see Section 15). The Ym128 Computer Interface allows you to simultaneously select, open and scroll up to eight historical raw data files and up to eight historical summary measurement result files.

The measurement results from selected and open historical files can be displayed in the measurement results windows on the Results page (see subsection 19.5), as well as in the charge-discharge graphs on the Charts page (for primary data historical files, see subsection 19.4) and in the analysis graphs on the Analysis page (for summary measurement results historical files, see subsection 19.6).

The output of data from historical files to graphs on the "Charts" and "Analysis" pages can be implemented in addition to the graphs of measurement results displayed in real time during the testing of the chemical and electrical equipment. This feature allows, for example, to visually compare the graphs of the change in the parameters of the chemical and electrical equipment being tested with the graphs for a certain reference chemical and electrical equipment, or, for example, to compare the charge-discharge curves even for the same chemical and electrical equipment, but in different testing cycles during the testing.

Viewing historical files of measurement results is possible, including without connection to the device, in "Offline" mode.

To view historical files of *primary data* and display measurement results on charge-discharge graphs (see subsection 19.4), use the "Selection of RAW saved files" area, located on the "Charts" page of the Computer Interface (see Figure 153):

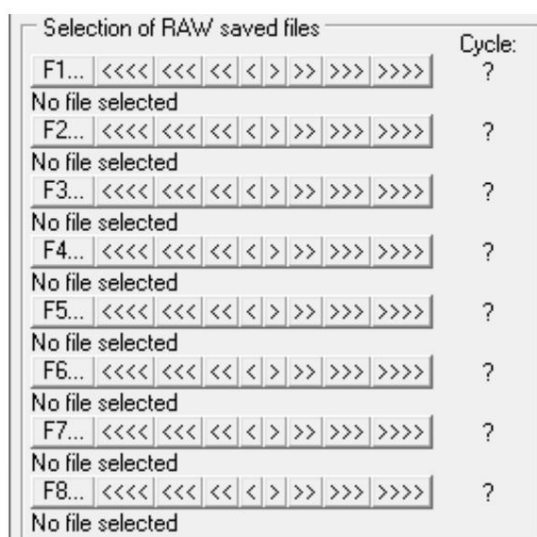


Figure 153 - View control area
historical files of primary data on the "Charts" page

The area contains eight identical fields intended for selecting, opening and scrolling through historical files of primary data.

Pressing the corresponding button "F1..." – "F8..." (for example, let's consider pressing the button "F3...", see Figure 154) leads to calling up the dialog box for selecting the historical file of primary data:

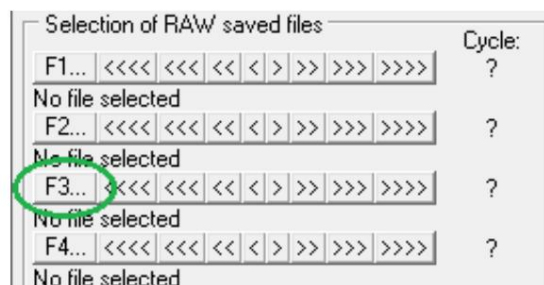


Figure 154 - Button for selecting a historical file for the data group "F3..."

In the window that opens, you should select the primary data file for the cycle with the desired cycle number:

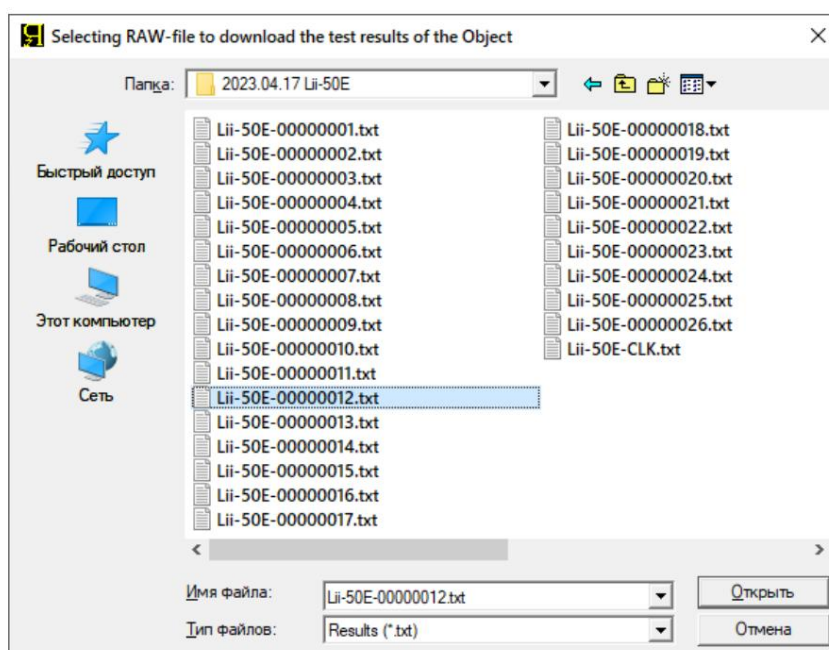


Figure 155 - Selecting Historical Primary Data Files

The data from the selected historical file is loaded into the internal memory of the Computer Interface in the selected data group (in our case – F3), at the same time, under the file selection button used, the path to the selected historical file of primary data is displayed, and to the right of the group of buttons for selecting and scrolling historical files – the cycle number, the data for which was obtained from the open historical file (see Figure 156):

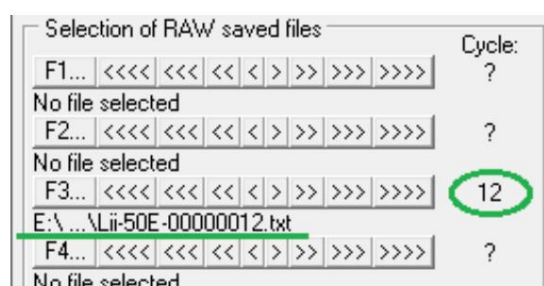


Figure 156 - Path to the selected history file
and the cycle number obtained from it

You can view the downloaded historical measurement results on the "Results" page by setting the switch to the required data group (in our example, this is the "F3" group) (see Figure 157):

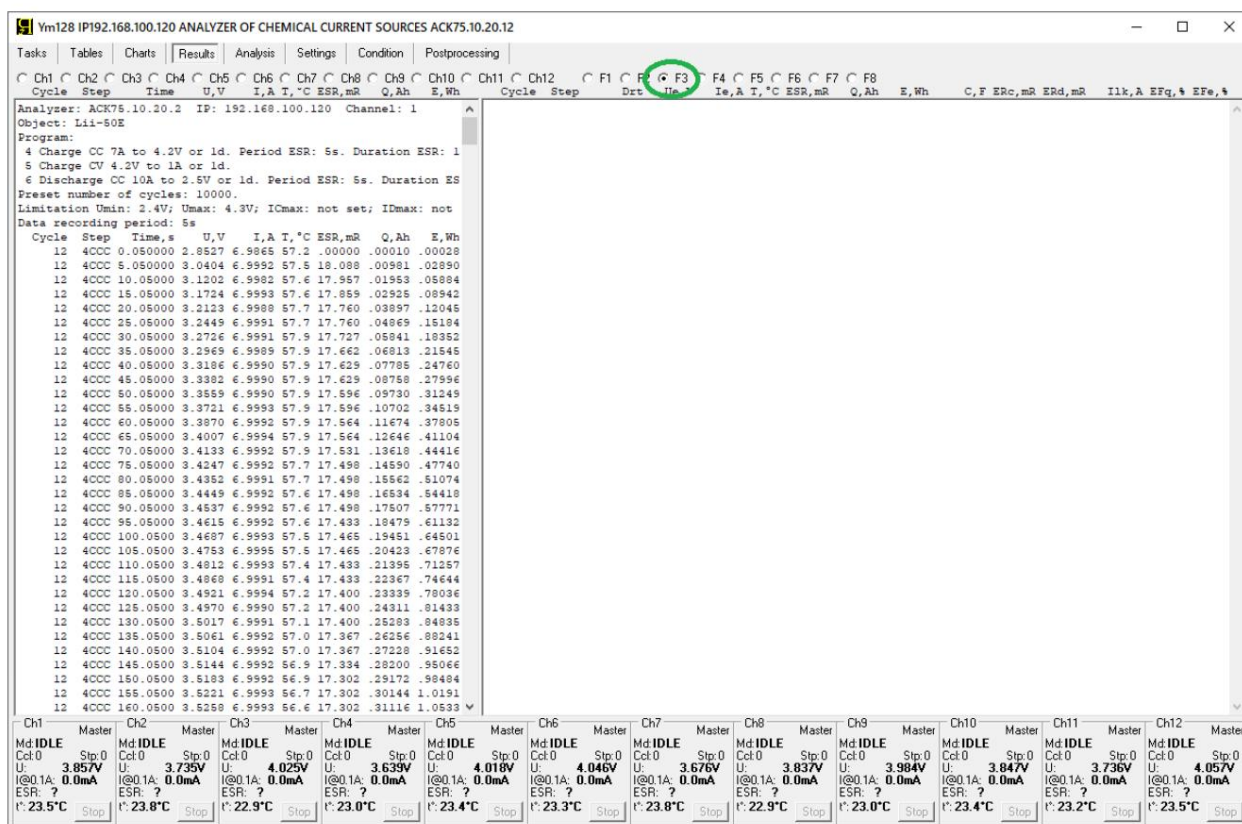


Figure 157 - Data from the downloaded history file can be viewed by setting the switch to the required data source

The data from the open file can be displayed on charge-discharge graphs. To do this, select the required data source for one of the graph groups (in our case, "File 3") (see Figure 158).

At the same time, when opening and loading the historical file for the selected cycle number, an attempt is made to open the historical file for the previous cycle. This is necessary to display the charge-discharge graphs in the graph output option for both the current cycle and the previous cycle in various combinations (see subsection 19.4, figures 120, 121 and 122). If the historical file for the previous cycle is not found, the data from it will not be output to the charge-discharge graphs (for example, when saving the results of measurements with thinning, the file for the previous cycle may not be saved when performing the HIT testing, see section 16).

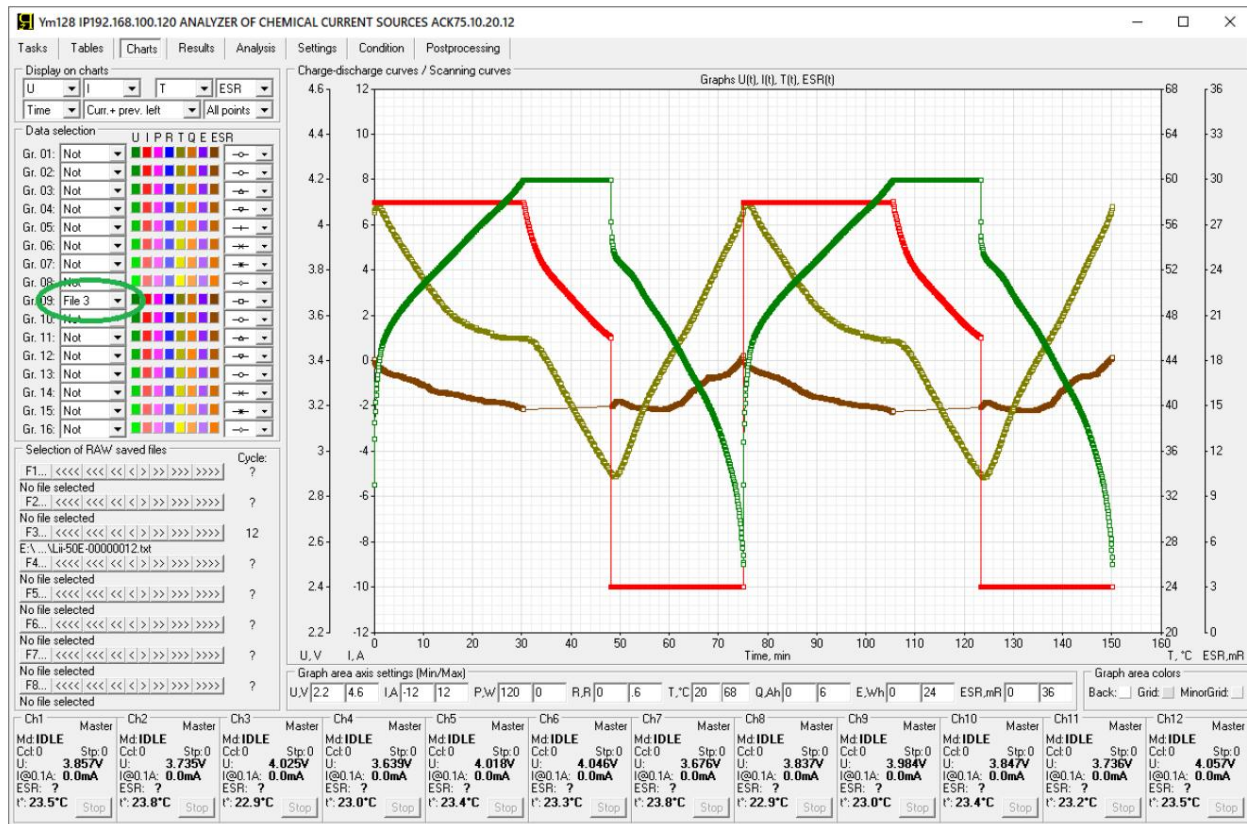


Figure 158 - Data from the downloaded history file can be output in the required format to the charge-discharge graphs by selecting the appropriate data source for one of the graph groups

The cycle number of the executed test program for the selected and loaded historical primary data file can be changed (scrolled) using special buttons for each specific group in the area

"Selection of RAW saved files":

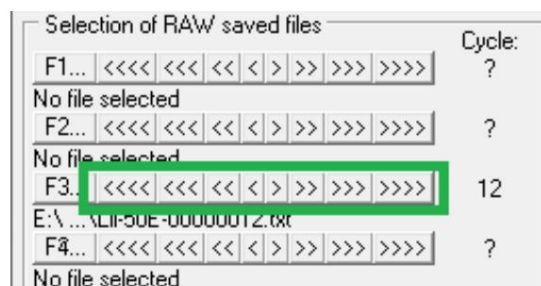


Figure 159 - Cycle scrolling buttons for viewing

The purpose of the cycle scrolling buttons is as follows:

- The ">" button allows you to select the next cycle to view; • The ">>" button allows you to select the cycle with the number "+10" from the current one; • The ">>>" button allows you to select the cycle with the number "+100" from the current one; • The ">>>>" button allows you to select the cycle with the number "+1000" from the current one; • The "<" button allows you to select the previous cycle to view; • The "<<" button allows you to select the cycle with the number "-10" from the current one; • The "<<<" button allows you to select the cycle with the number "-100" from the current one; • The "<<<<" button allows you to select the cycle with the number "-1000" from the current one.

When the required button is pressed, the Computer Interface will try to find and load the historical file of primary data for the cycle with the required number. The search is performed within the folder from which the initial file was selected by the User when the corresponding file open button was pressed (in our case it was the "F3..." button).

If you try to select a file for a cycle number for which a file does not exist, the historical raw data file for the next available cycle or for the cycle with the highest or lowest number that could be found in the current measurement results file folder will be opened and displayed.

Data from selected historical files can be output to the area of plotting charge-discharge graphs, in addition to the graphs displayed in real time during the testing of the CCS. This feature allows, for example, to visually compare the charge-discharge curves of the tested CCS and some reference CCS, or even to compare the charge-discharge curves of the same CCS, but in different testing cycles during the testing.

To view historical files of *summary measurement results* (see section 15) and displaying the measurement results on the analysis graphs (see subsection 19.6), the "Selection of CLK saved files" area, located on the "Analysis" page of the Computer Interface, is intended:

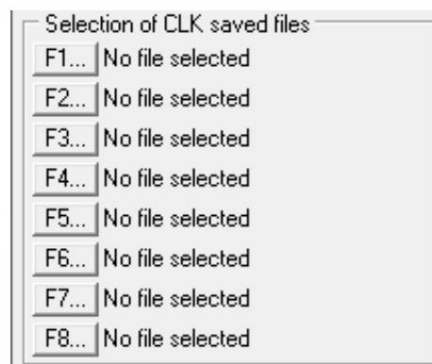


Figure 160 - View control area
historical files of summary measurement results on the "Analysis" page

Pressing the corresponding button "F1..." – "F8..." (for example, let's consider pressing the button "F5...", see Figure 161) calls up the dialog box for selecting a historical file of summary results.

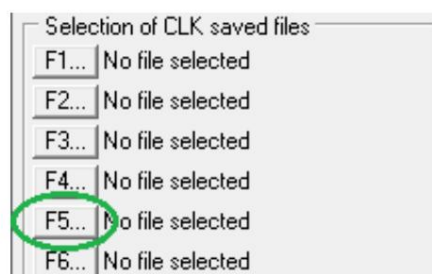


Figure 161 - Selecting a historical summary results file
for data group "F5"

In the window that opens, you should select the file with summary measurement results (see Figure 162).

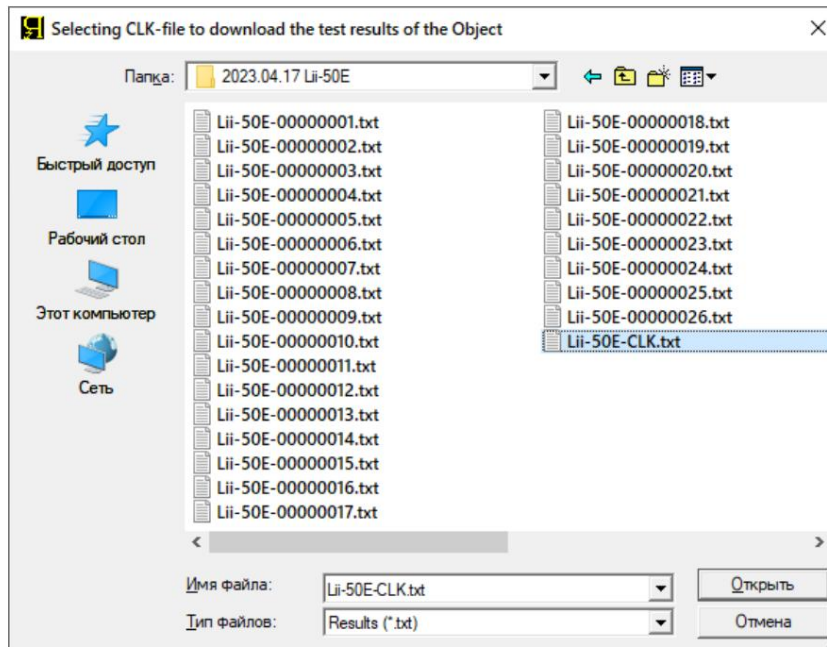


Figure 162 - Selecting historical files of summary measurement results

The data from the selected historical file is loaded into the internal memory of the Computer Interface in the selected data group (in our case – F5), while the path to the selected historical file of summary measurement results is displayed to the right of the file selection button used (see Figure 163):

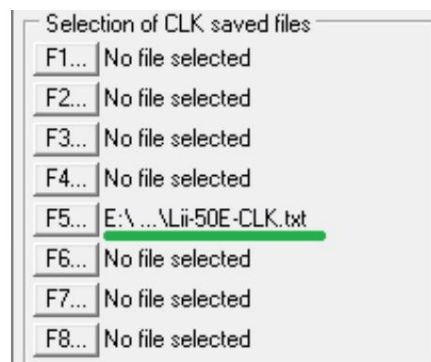


Figure 163 - Path to the selected historical summary results file

You can view the downloaded historical measurement results on the "Results" page by setting the switch to the required data group (in our example, this is the "F5" group) (see Figure 164).

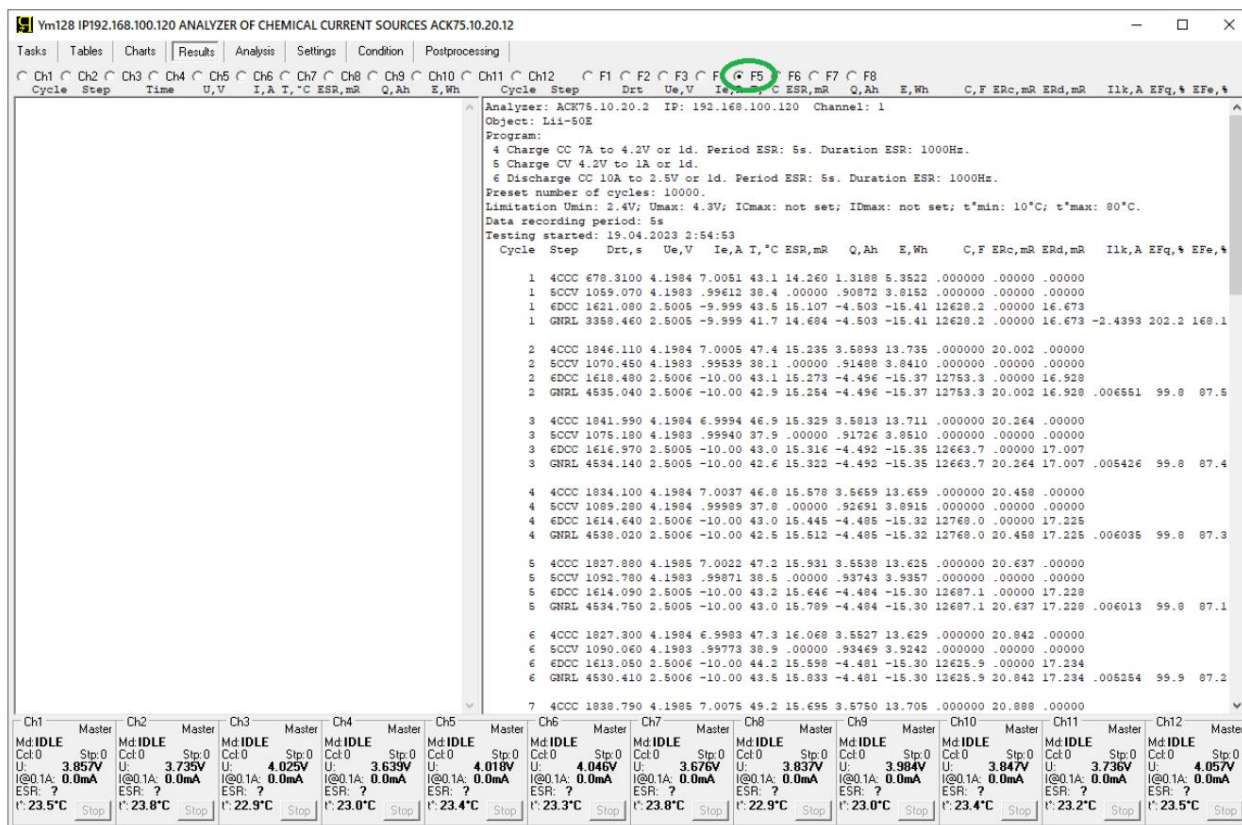


Figure 164 - Data from the downloaded history file can be viewed by setting the switch to the required data source

Data from an open file can be displayed on analysis graphs. To do this, select the required data source for one of the groups of analysis graphs (in our case, this is "File 5") (see Figure 165).

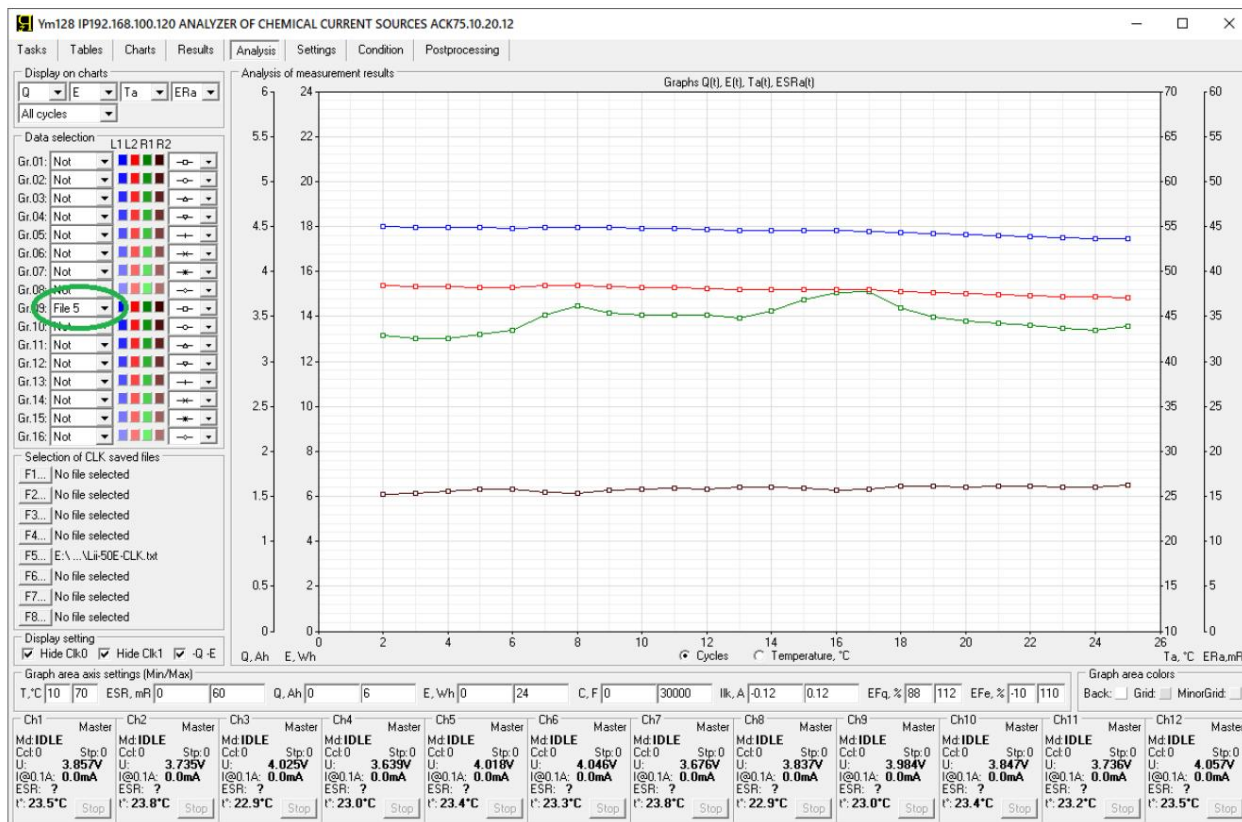


Figure 165 - Data from the downloaded history file can be output to the analysis graphs by selecting the appropriate source for one of the graph groups

21 OPERATION OF THE DEVICE IN AUTONOMOUS MODE

The device is controlled using the Computer Interface (see Section 19). The Computer Interface is required to create and launch the HIT testing program (see Section 11), as well as to monitor the progress of testing, obtaining and processing measurement results.

However, after the HIT testing program has been launched, the Computer Interface may be closed and the User's computer may be turned off. In this case, the device will continue to operate in autonomous mode according to the testing program specified by the User.

ATTENTION! Operation of the device according to the tables of voltage, current, power and resistance (see subsection 11.19) is possible only with an established connection between the device and the Computer Interface. Autonomous operation in the modes of operation according to tables is impossible!

Measurement results in offline mode are saved to a Micro SD memory card (position 11, figures 1 and 4, also see figures 7 and 8). The memory card included with the device has a capacity of at least 4 GB, which allows storing at least 1,800,000,000 measurement results (points*). The next time the User's computer is turned on and the Computer Interface is launched, the data accumulated in the memory card will be transferred to the computer and saved in the measurement results files (see section 15). After this, the memory card is cleared without interrupting the testing of the HIT and becomes ready for the next period of autonomous operation.

If necessary, the memory card can be removed from the device and replaced with another one by the User. The measurement results are stored on the memory card in "raw" binary form and cannot be read by standard Windows operating system tools. In addition, to ensure uniform wear of the memory card cells, data recording on the card begins in a new randomly determined location each time the Analyzer is turned on.

The battery life of the device directly depends on the period of saving data in the files of primary measurement results, which is configured in the "Settings for saving results" area on the "Tasks" page of the Computer Interface (see Figure 166).

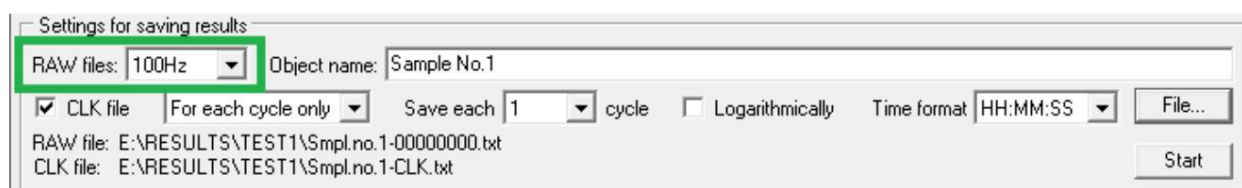


Figure 166 - Setting the measurement results storage period

With a shorter data storage period, the measurement results are saved more often, the memory card fills up faster, and the battery life is reduced. However, given the significant volume of the memory card, the battery life of the device in real operating conditions is practically unlimited. Even with the simultaneous operation of all twelve channels of the ASK75.10.20.12 Analyzer and the maximum data storage speed of 100 times per second, with a memory card capacity of 4 GB, the battery life of the Analyzer will be more than 17 days.

* see the list of accepted abbreviations and designations.

In case of very long operation of the device in autonomous mode with maximum data recording frequency, the memory card of the device may be full. In this case, new results will replace the oldest data, and some of the oldest measurement results will be irretrievably lost.

To indicate the status of the memory card, the "MEM" (MEMORY) indicator is located on the front panel of the device (position 9, Figure 1 and Figure 4). This indicator reflects the following states of the device's memory card:

- Not lit – the memory card is working properly and has enough free space;
- Blinking – the memory card is more than 50% full. It is necessary to turn on the computer and launch the Computer Interface to transfer the collected data to the computer, save the measurement results to files and free up the memory card;
- Constantly on – the memory card is faulty or full. Some data may be lost.

To monitor the memory card status using the Computer Interface, the "Settings" page (see subsection 19.7) provides an "Analyzer" area. memory" (Analyzer Memory):

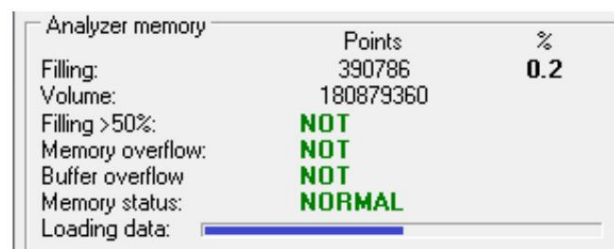


Figure 167 - Analyzer memory state parameters

The "Analyzer memory" area contains detailed information about the Analyzer memory status. When you next establish a connection with the computer, After autonomous operation, the data from the device memory is loaded into the Computer Interface, processed and saved in measurement result files. This process takes some time, its progress can be monitored by the "Filling:" values, expressed in the number of points* ("Points") and as a percentage of the installed memory card volume, which will decrease as the data is loaded onto the computer, as well as the linear scale "Loading data:".

In case of filling or overflow of the Analyzer memory, the indicators "Filling >50%", "Memory overflow:", "Buffer overflow:" instead of the values **"NOT"** , **"NOT"** , " NOT", will take the values **"YES"** , **"HAPPENED"** respectively.

(HAPPENED), **"HAPPENED"** (HAPPENED). When the memory card is full, some data will be irretrievably lost.

If the memory card is missing or fails, the "Memory status:" indicator will change from **"NORMAL"** to **"FAILURE"** .

If this message appears repeatedly, the memory card should be replaced with a new one.

* see the list of accepted abbreviations and designations.

22 DEVICE PROTECTIONS AND EMERGENCY MODES

The ASK75.10.20.2 and ASK75.10.20.12 analyzers have the following hardware and software protections:

- Protection against failure to stabilize a given parameter;
- Protection against violation of the integrity of the HIT connection circuits;
- Overvoltage protection on the channel power terminals;
- Overvoltage protection at the channel potential terminals;
- Protection against reverse polarity of voltage at the channel power terminals;
- Protection against reverse polarity of voltage on potential terminals;
- Protection to limit the minimum voltage on the HIT (overdischarge of the HIT);
- Protection to limit the maximum voltage on the HIT (overcharging of the HIT);
- Protection to limit the maximum charging current through the HIT;
- Protection to limit the maximum discharge current of the chemical energy storage device;
- Protection to limit the minimum temperature of the heat exchanger (overcooling of the heat exchanger);
- Protection to limit the maximum temperature of the heat source (heat source overheating);
- Protection against loss of connection between the device and the computer when working with tables;
- Overvoltage protection on the power supply **of the IP** (see Figures 7 and 8);
- Protection against failure of the power supply of **the IP** channel (see Figures 7 and 8);
- Overload protection of channel power transistors;
- Protection against failure of the channel power inverter;
- Channel overcurrent protection;
- Channel power overload protection;
- Overheating protection of the load resistor **R** (see Figures 7 and 8);
- Protection against overheating of the channel power inverter;
- Protection against loss of connection with the device channel;
- Protection against failure of initialization of internal channel systems;
- Protection against device channel failure.

Triggering of any of the protections leads to interruption of the testing process of the HIT on a specific channel of the device and the channel's transition to emergency mode.

The type of emergency mode is displayed in the status panel of the corresponding channel. at the bottom of the Computer Interface (see subsection 19.1), and the reason that caused the transition to the emergency state is displayed on the "Condition" page Computer Interface in the "EMRGY TRGS" section for a specific channel of the device (see Figure 168).

Emergency triggers are designed to record the cause of certain types of emergency conditions. For example, if overvoltage occurs simultaneously on both the power and measuring terminals of a channel, either overvoltage protection on the power terminals or overvoltage protection on the potential terminals of the channel (only for one overvoltage event) will be triggered. Find out

that the occurrence of the accident was preceded by overvoltage both on the power and potential terminals of the channel (both events), will be helped by emergency triggers, each of which is set independently of the other and maintains its state until the emergency condition is acknowledged.

Not every emergency event has a corresponding emergency trigger.

In addition, some types of alarm triggers are set when multiple types of alarm events occur (the same trigger is set when multiple alarm events occur).

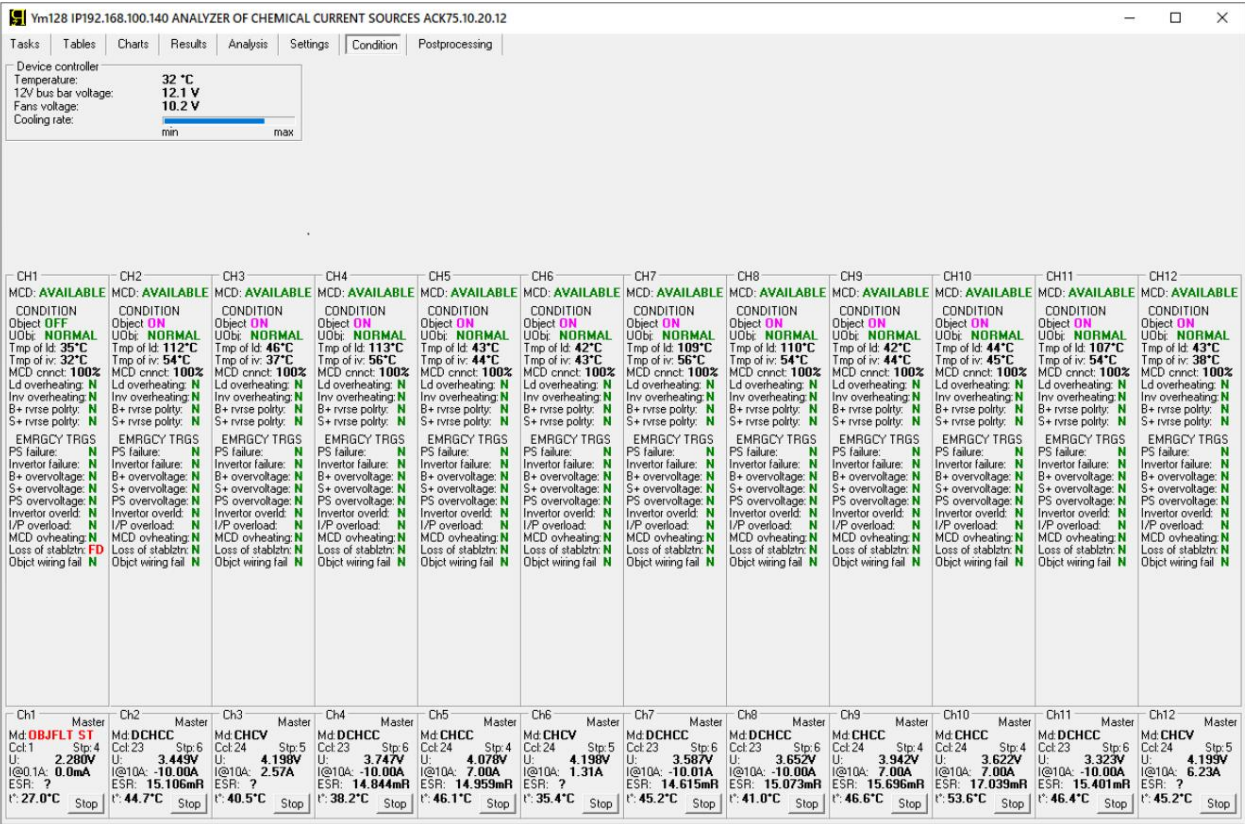


Figure 168 - Crash triggers help identify the cause of the crash

If an accident occurs during the testing of the HIT, an extreme entry will be made in the primary data file for the current cycle (see subsection 15.2), in which the accident marker will be recorded instead of the step marker.

Protection against failure of stabilization of a given parameter is triggered when a discrepancy is registered between the given and actual value of the stabilized parameter (voltage, current or power) by more than 5% of the maximum value of a specific parameter that is achievable in the selected hardware subrange of current. For the discharge modes of the HIT to a given resistance, the protection is triggered if the actual discharge resistance that ensures

the device channel differs by more than 25% from the specified resistance.

When the voltage on the HIT is less than 1.5 V, this protection is disabled for modes that cause negative current (discharge current) to pass through the HIT or negative power (discharge power) to be taken from the HIT.

The protection triggering indicates that it is impossible for one reason or another to maintain the specified value of the required parameter on the tested HIT. The reasons may be as follows:

- Incorrectly set parameters when creating a testing program for the HIT, an incorrectly selected testing mode that does not correspond to the type of the tested HIT or its initial state before starting to perform the specified step;
- The wrong sequence of steps of the HIT testing program is used;
- The voltage sweep speed in the voltage sweep mode at the terminals of the heat exchanger is set too high (see subsection 11.11);
- Equivalent series resistance (ESR) of the heat sink is too high for the given testing parameters;
- The integrity of the power or potential circuits of the HIT connection is compromised;
- Appliance defective.

The protection triggering results in the transition to the emergency mode **"OBJFLT ST"** (STABILIZATION FAILURE), while on the "Condition" page of the Computer Interface the emergency trigger *"Loss of stablzn:"* (Stabilization failure) is set to the state **"FD"** (Triggered) instead of the state **"N"** (No), and in the primary data file (see subsection 15.2) the last entry will be made, in which instead of the step marker the emergency marker **"OAL" will be recorded.**

If the HIT testing program is set correctly, and the HIT connection circuits to the device are in good working order and reliable, if this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Protection against violation of the integrity of the HIT connection circuits is triggered if a difference is registered between the voltage at the HIT terminals and the voltage at the power terminals of the device that exceeds 1 V, in other words, the voltage drop on the connecting wires of the HIT connection is more than 1 V.

The activation of the protection indicates poor contact in the HIT connection circuits, too long and too thin HIT connection power wires, if HIT connection power wires not included in the device delivery set are used (see item 2 of Table 1 and Table 2), or the device is faulty.

Triggering of the protection leads to a transition to the emergency mode **"OBJFLT BS"** (VIOLATION OF CONNECTION CHAINS HIT), while on the page "Condition" The Computer Interface emergency trigger *"Objct wiring fail:"* (Violation of the HIT connection circuits) is set to the **"FD"** (Fault) state instead of the **"N"** (No) state, and an extreme entry will be made in the primary data file (see subsection 15.2), in which the emergency marker **"OAL" will be recorded instead of the step marker.**

If the quality of the HIT connection circuits is not in doubt, in the event of a repeated occurrence of this accident, contact YAROSTANMASH LLC for consultation or repair.

Overvoltage protection on the channel power terminals is triggered when voltage at the power terminals (position 2 and 6, Figure 1 and Figure 4) over 22 V.

The activation of the protection causes the device channel to switch to emergency mode **"OVER U B+"** (OVERVOLTAGE AT POWER TERMINALS), while on the "Condition" page of the Computer Interface the emergency trigger *"B+ overvoltage:"* (Overvoltage at power terminals) is set to the **"FD"** (Faulted) state instead of the **"N"** (No) state, and in the primary data file (see subsection 15.2) an extreme entry will be made, in which the emergency marker **"OUB" will be recorded instead of the step marker.**

If this accident occurs repeatedly without any apparent reason, contact to OOO "YAROSTANMASH" for consultation or repair.

Overvoltage protection at the channel potential terminals is triggered when the voltage at the potential terminals (position 3 and 5, Figure 1 and Figure 4) exceeds 22 V.

The activation of the protection causes the device channel to switch to emergency mode **"OVER U S+"** (OVERVOLTAGE AT POTENTIAL TERMINALS), while on the "Condition" page of the Computer Interface the emergency trigger *"S+ overvoltage:"* (Overvoltage on potential terminals) instead of the state **"N"** (No) is set to the state **"FD"** (Tripped), and in the primary data file (see subsection 15.2) an extreme entry will be made, in which instead of the step marker the alarm marker **"OUS" will be recorded.**

If this accident occurs repeatedly without any apparent reason, contact to OOO "YAROSTANMASH" for consultation or repair.

Protection against reverse polarity of voltage at the channel power terminals is triggered when reverse polarity of voltage is registered at the power terminals (position 2 and 6, Figure 1 and Figure 4) (the "+" and "-" are reversed when connecting the HIT).

Reverse voltage polarity is registered when a negative voltage with an absolute value greater than 0.2 V is applied.

The protection is triggered at the moment of attempting to start the HIT testing and leads to a transition to the emergency mode **"RVRS B+"** (REVERSE POLARITY OF VOLTAGE AT POWER TERMINALS), while in the primary data file (see subsection 15.2) the last entry will be made, in which instead of the step marker the emergency marker **"RPB" will be recorded.**

If this accident occurs repeatedly without any apparent reason, contact to OOO "YAROSTANMASH" for consultation or repair.

Protection against reverse polarity of voltage at the potential terminals of the channel is triggered when reverse polarity of voltage is registered at the potential terminals (position 3 and 5, Figures 1 and 4) (the "+" and "-" are reversed when connecting the HIT).

Reverse voltage polarity is registered when a negative voltage with an absolute value greater than 0.2 V is applied.

The protection is triggered at the moment of attempting to start the HIT testing and leads to a transition to the emergency mode **"RVRS S+"** (REVERSE POLARITY OF VOLTAGE AT POTENTIAL TERMINALS), while in the primary data file (see subsection 15.2) the last entry will be made, in which instead of the step marker the emergency marker **"RPS" will be recorded.**

If this accident occurs repeatedly without any apparent reason, contact to OOO "YAROSTANMASH" for consultation or repair.

The protection for limiting the minimum voltage on the CCS (for overdischarge of the CCS) is triggered if the voltage on the CCS drops to or becomes less than the specified limitation of the minimum voltage on the CCS, entered in the "Umin:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"U MIN LMT"**, and in this case the last entry will be made in the primary data file for the current test cycle (see subsection 15.2), in which the emergency marker **"ULL"** will be recorded instead of the step marker .

The protection operation is disabled if the "Umin:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2).

If this protection is triggered without any apparent objective reasons, Contact YAROSTANMASH LLC for consultation or repair.

Protection for limiting the maximum voltage on the CIT (for overcharging the CIT) is triggered if the voltage on the CIT increases to or becomes greater than the specified limitation of the maximum voltage on the CIT, entered in the "Umax:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"U MAX LMT"**, and in this case the last entry will be made in the primary data file for the current testing cycle (see subsection 15.2), in which the emergency marker **"UHL"** will be recorded instead of the step marker .

The protection operation is disabled if the "Umax:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2).

If this protection is triggered without any apparent objective reasons, Contact YAROSTANMASH LLC for consultation or repair.

The protection for limiting the maximum current of the HIT charging is triggered if the HIT charging current reaches or exceeds the specified limitation of the maximum current of the HIT charging, entered in the "ICmax:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"IC MX LMT"**, and in this case the last entry will be made in the primary data file for the current test cycle (see subsection 15.2), in which the emergency marker **"ICL"** will be recorded instead of the step marker .

The protection operation is disabled if the "ICmax:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2).

If this protection is triggered without any apparent objective reasons,
Contact YAROSTANMASH LLC for consultation or repair.

The protection for limiting the maximum discharge current of the HIT is triggered if the discharge current of the HIT reaches or exceeds the specified limitation of the maximum discharge current of the HIT, entered in the "IDmax:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"ID MX LMT"**, and in this case the last entry will be made in the primary data file for the current test cycle (see subsection 15.2), in which the emergency marker **"IDL"** will be recorded instead of the step marker .

The protection function is disabled if the "IDmax:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2).

If this protection is triggered without any apparent objective reasons,
Contact YAROSTANMASH LLC for consultation or repair.

The protection for limiting the minimum temperature of the HIT (for overcooling of the HIT) is triggered if the HIT temperature drops to or becomes less than the specified limitation of the minimum temperature of the HIT, entered in the "Tmin:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"T MIN LMT"**, and in this case the last entry will be made in the primary data file for the current test cycle (see subsection 15.2), in which the emergency marker **"TLL"** will be recorded instead of the step marker .

The protection operation is disabled if the "Tmin:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2), or if the external temperature sensor of the HIT is not connected to the device channel.

If this protection is triggered without any apparent objective reasons,
Contact YAROSTANMASH LLC for consultation or repair.

The protection for limiting the maximum temperature of the HIT (for overheating of the HIT) is triggered if the HIT temperature increases to or becomes greater than the specified limitation of the maximum temperature of the HIT, entered in the "Tmax:" window on the "Tasks" page of the Computer Interface (see subsection 19.2).

The protection triggering results in a transition to the emergency mode **"T MAX LMT"**, and in this case the last entry will be made in the primary data file for the current test cycle (see subsection 15.2), in which the emergency marker **"THL"** will be recorded instead of the step marker .

The protection operation is disabled if the "Tmax:" window on the "Tasks" page of the Computer Interface is left empty (see subsection 19.2), or if the external temperature sensor of the HIT is not connected to the device channel.

If this protection is triggered without any apparent objective reasons,
Contact YAROSTANMASH LLC for consultation or repair.

Protection against loss of connection between the device and the computer when the device channel is operating according to tables is triggered if, during operation according to the voltage, current, power or resistance table (see subsection 11.19), the Analyzer registers a loss of connection with the Computer Interface for more than 1 s.

The activation of the protection indicates poor quality of the connection between the Analyzer and the computer, the load on the computer network to which the Analyzer and the computer are connected, insufficient computer performance, the computer being loaded with other tasks not related to the operation of the Analyzer (in particular, tasks for updating the Windows operating system), or a hardware malfunction (including the device, computer, network equipment, connecting cables, etc.).

The protection triggering leads to a transition to the emergency mode **"PC FAULT"**, and in this case the last entry will be made in the primary data file (see subsection 15.2), in which instead of the step marker the emergency marker **"TCA" will be recorded**.

Overvoltage protection on the channel power supply is triggered when the voltage on the MZR power supply increases above 110% of the nominal value.

The activation of the protection causes the device channel to switch to emergency mode **"OVER U PS"**, while on the "Condition" page of the Computer Interface, the emergency trigger **"PS overvoltage:"** (Overvoltage on the power supply) is set to the **"FD"** (Faulted) state instead of the **"N"** (No) state .

If an accident occurred during the testing of the HIT, an extreme entry will be made in the primary data file (see subsection 15.2), in which the accident marker **"OUP" will be recorded instead of the step marker**.

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

The device channel power supply failure protection is triggered when the power supply drops the voltage on the power source of the MZR is below 85% of the nominal value.

The activation of the protection causes the device channel to switch to emergency mode **"PS FAULT"**, while on the "Condition" page of the Computer Interface, the emergency trigger **"PS failure:"** (Power supply failure) is set to the **"FD"** (Fault) state instead of the **"N"** (No) state .

If an accident occurred during the testing of the HIT, an extreme entry will be made in the primary data file (see subsection 15.2), in which the accident marker **"PSA" will be recorded instead of the step marker**.

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

The protection against overload of the power transistors of the device channel is triggered when the instantaneous pulse current through any transistor of the power inverter of the channel is detected to exceed the permissible values.

The activation of the protection indicates a malfunction in the device and leads to a transition to the emergency mode **"SW OVRLD"**, while on the page "Condition" The Computer Interface alarm trigger **"Inverter overload:"** is set to the **"FD"** (Faulted) state instead of the **"N"** (No) state, and an extreme entry will be made in the primary data file (see subsection 15.2), in which the alarm marker **"OLS"** will be recorded instead of the step marker .

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Protection against failure of the power inverter channel of the device is triggered in case of malfunctions in the power inverter channel.

The protection triggering leads to the transition to the emergency mode **"INV FAULT"**, while on the page "Condition" (State) of the Computer Interface the emergency trigger *"Inverter failure:"* is set to **"FD"** (Failed) instead of **"N"** (No) and an extreme entry will be made in the primary data file (see subsection 15.2) in which the fault marker "IVA" will be recorded instead of the step marker .

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Channel overcurrent protection is triggered when an absolute current value is detected that exceeds the maximum current value for the current hardware current subrange by 12% or more.

The activation of the protection indicates a malfunction of the device and leads to a transition to the emergency mode **"OVER I"**, while on the page "Condition" The Computer Interface alarm trigger *"I/P overload:"* (Overcurrent or overpower) is set to the **"FD"** (Fault) state instead of the **"N"** (No) state .

If an accident occurred during the testing of the HIT, an extreme entry will be made in the primary data file (see subsection 15.2), in which the accident marker **"OLI" will be recorded instead of the step marker.**

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Channel overload protection by power is triggered when values are registered charge power or discharge power exceeding the corresponding maximum power values by 10% or more.

The activation of the protection indicates a malfunction of the device and leads to a transition to the emergency mode **"OVER P"**, while on the "Condition" page of the Computer Interface, the emergency trigger *"I/P overload:"* (Overload by current or by power) is set to the state **"FD"** (Triggered) instead of the state **"N"** (No) .

If an accident occurred during the testing of the HIT, an extreme entry will be made in the primary data file (see subsection 15.2), in which the accident marker **"OLP" will be recorded instead of the step marker.**

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Overheating protection of load resistor R (see Figure 7 and Figure 8)
Triggered when the load resistor temperature reaches or exceeds 180°C.

The activation of the protection indicates insufficient cooling of the load resistor **R** or a malfunction of the device and leads to a transition to emergency mode **"OVER T LD"**, in this case the last entry will be made in the primary data file (see subsection 15.2), in which the accident marker **"OTL" will be recorded instead of the step marker.**

If an accident occurs, make sure that there are no factors in the place where the device is installed that prevent its normal cooling, that the ventilation slots of the device are not clogged with dust or foreign objects, and that the temperature of the environment around the device meets the requirements specified in Section 4. If necessary, carry out unscheduled maintenance of the device in accordance with Section 25.

If the ambient temperature meets the requirements of Section 4, and there are no conditions that prevent normal cooling of the device, if this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation.

or repair.

Power inverter overheat protection is triggered when the inverter temperature (charge regulator or discharge regulator, see Figure 7 and 8) exceeds 80 °C.

The activation of the protection indicates insufficient cooling of the device or its malfunction and leads to a transition to the emergency mode **"OVR T INV"**, while in the primary data file (see subsection 15.2) an extreme entry will be made, in which instead of the step marker the emergency marker **"OTI" will be recorded**.

If an accident occurs, make sure that there are no factors in the place where the device is installed that prevent its normal cooling, that the ventilation slots of the device are not clogged with dust or foreign objects, and that the temperature of the environment around the device meets the requirements specified in Section 4. If necessary, carry out unscheduled maintenance of the device in accordance with Section 25.

If the ambient temperature meets the requirements of Section 4, and there are no conditions that prevent normal cooling of the device, if this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation.

or repair.

Protection against loss of connection with the device channel is triggered in the event of failure to establish connection between the control and management module of the MCU and a specific charge-discharge module of the MZR (see Figure 7 and Figure 8), and results in the channel switching to the emergency mode **"CNNCT ERR"**.

If an accident occurs during the testing of the HIT, an extreme entry will be made in the primary data file for the current cycle (see subsection 15.2) on a specific channel of the device, in which the accident marker **"IAL" will be recorded instead of the step marker**.

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Protection against failure of initialization of internal channel systems is triggered in the event of failure of the internal ADC microcircuit of the device channel, and leads to the channel switching to the emergency mode **"INIT FLT"**.

If an accident occurs during the testing of the HIT, an extreme entry will be made in the primary data file for the current cycle (see subsection 15.2) on a specific channel of the device, in which the accident marker **"ITA" will be recorded instead of the step marker**.

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

Device channel failure protection is triggered when there is a malfunction in the hardware of the device channel and causes the channel to switch to emergency mode **"MAL"**.

If an accident occurred during the testing of the HIT, an extreme entry will be made in the primary data file for the current cycle (see subsection 15.2) on a specific channel of the device, in which the accident marker **"MAL" will be recorded instead of the step marker**.

If this failure occurs repeatedly, contact YAROSTANMASH LLC for consultation or repair.

23 FIRMWARE UPDATE

In the course of improving its products, YAROSTANMASH LLC may produce firmware updates for their HIT Analyzers.

The current version of the firmware can be downloaded from the Internet on the website of YAROSTANMASH LLC <https://battery-analyzers.ru/> or request by email Yarst@mail.ru.

Starting the Analyzer firmware update is possible only if the connection between the device and the Computer Interface is established (see subsection 19.1), and only if all device channels are in the "IDLE" state.

Before starting the firmware update, make sure that the "Connect automatically" check box is checked in the "TCP" area of the Computer Interface "Settings" page.

To start the update, click the "Update..." button located on the "Settings" page of the Computer Interface in the "Additionally" area:

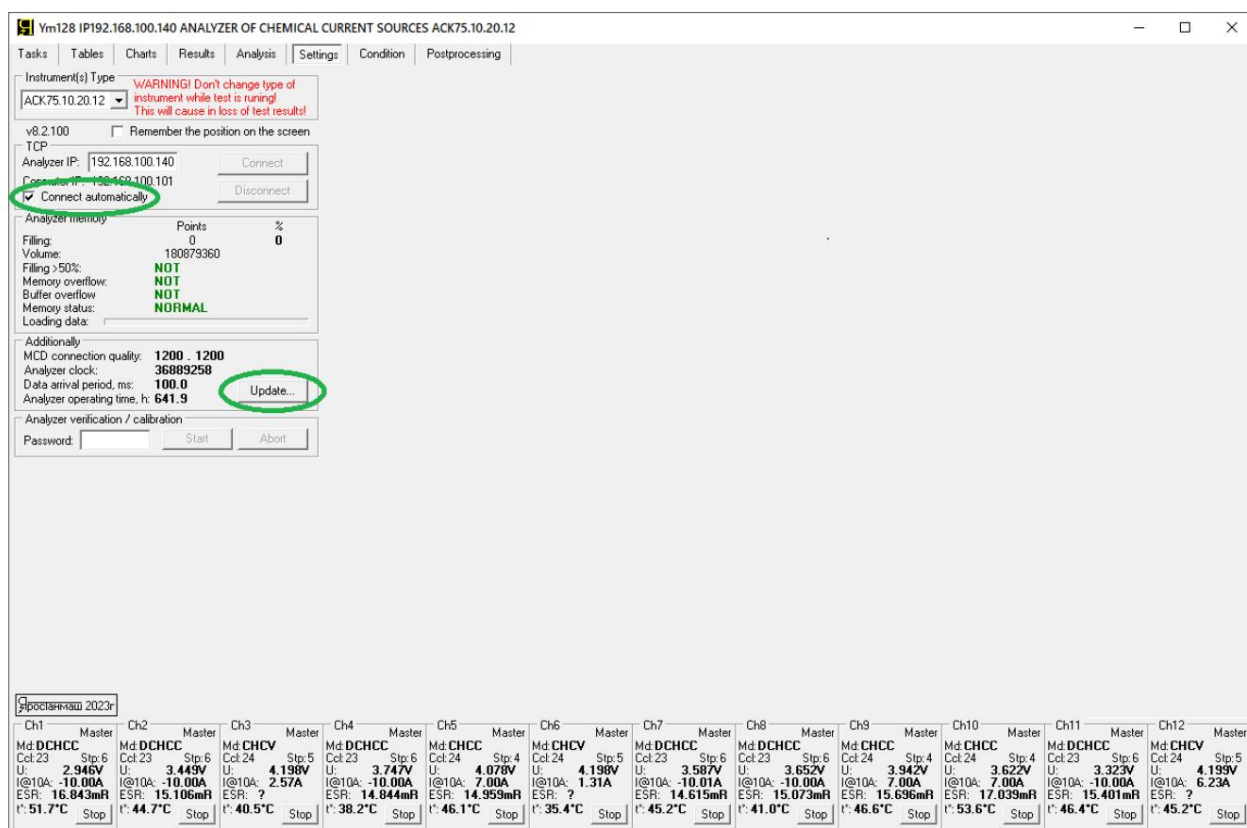


Figure 169 - Starting the firmware update

Clicking on the "Update..." button displays the additional "Instrument firmware update" area:

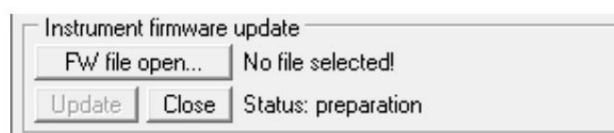


Figure 170 - "Instrument firmware update" area
(Updating the device firmware)

Here you should click on the "FW file open..." button, which will open the firmware file selection dialog box, in which you should select the firmware file to update the device (see Figure 171).

ATTENTION! The firmware file must be strictly intended for devices of the ASK75.10.20.x series!

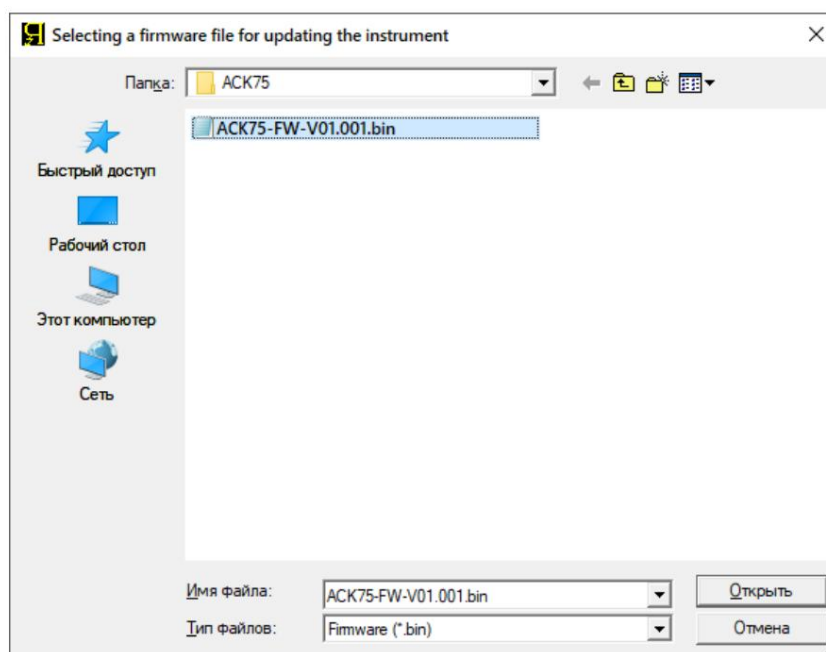


Figure 171 - Selecting a firmware file to update

If the selected firmware file is successfully opened and pre-analyzed, the path to the selected file will be shown to the right of the "FW file" button. open..." (Select firmware file), and the "Update" button will become active:

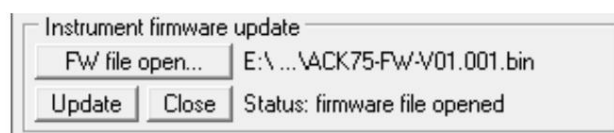


Figure 172 - Firmware update file opened successfully

To start the firmware update, click the "Update" button (see Figure 172). Updating the Analyzer firmware takes about 9 minutes. You can monitor the progress of the update in the status bar to the right of the "Close" button:

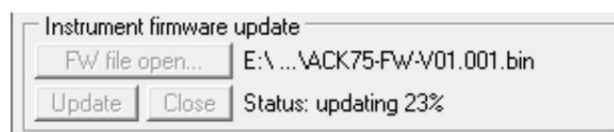


Figure 173 - Monitoring the firmware update progress

Upon successful completion, the status bar will read "Status: **update completed!**" (Status: Update completed!), and the "Close" button will become active. Pressing this button will return the device to normal operating mode.

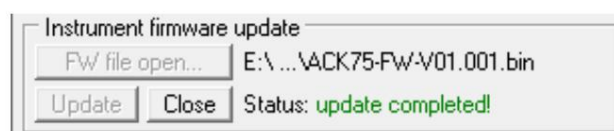


Figure 174 - Firmware update completed successfully

If any failures occur during the firmware update, the update procedure must be repeated.

24 INFORMATION ON THE CONTENT OF PRECIOUS METALS

Analyzers ASK75.10.20.2 and ASK75.10.20.12 do not contain precious metals.

25 MAINTENANCE AND REPAIR

Clean the device regularly from dust and dirt. Wipe the surface of the device with a rag dampened with water. Do not use solvents such as gasoline or acetone for wiping. This may damage the paintwork of the device.

To clean, disconnect the tested HIT from the device and disconnect the Analyzer from the 220 V power supply. Blow out the ventilation slots on the front, bottom, back and side panels of the device with a stream of compressed air.

In case of failure of the Analyzer, contact OOO "YAROSTANMASH" for consultation or repair. Attempting to repair the device yourself will result in loss of warranty and may cause electric shock and death.

26 TRANSPORTATION

The device in its original packaging can be transported by any type of ground, water or air transport subject to the following conditions:

- Ambient air temperature: from minus 30 to plus 50 °C;
- Relative air humidity: from 10 to 90% (without moisture condensation);
- Atmospheric pressure: from 27 to 107 kPa (from 200 to 800 mm Hg);
- Overloads and vibrations: no more than 0.5 g.

27 STORAGE

The device should be stored in its original packaging in dry, heated rooms with an ambient temperature of 5 to 40°C, relative air humidity of 10 to 90% (without moisture condensation), and atmospheric pressure of 84 to 107 kPa (630 to 800 mm Hg).

28 DISPOSAL

Do not dispose of the Analyzer with household waste. Parts and components of the device may pose a danger to the environment.

Dispose of the appliance as electrical and electronic equipment in accordance with current regulations and legislation.

Analyzers ASK75.10.20.2 and ASK75.10.20.12 do not contain precious metals.

29 CERTIFICATION INFORMATION

ЕВРАЗИЙСКИЙ ЭКОНОМИЧЕСКИЙ СОЮЗ ДЕКЛАРАЦИЯ О СООТВЕТСТВИИ



Заявитель Общество с ограниченной ответственностью "ЯРОСТАНМАШ"

Место нахождения и адрес места осуществления деятельности: Российская Федерация, Москва, 119435, переулок Малый Саввинский, дом 5, квартира 29, основной государственный регистрационный номер: 1097746667600, номер телефона: +79774875569, адрес электронной почты: Grigory-rylov@yandex.ru

в лице Генерального директора Рылова Григория Михайловича

заявляет, что анализаторы химических источников тока (Анализаторы ХИТ) АСК150.24.1750.1, АСК75.48.1750.1, АСК100.24.650.1, АСК50.48.650.1, АСК50.15.250.1, АСК25.27.250

изготовитель Общество с ограниченной ответственностью "ЯРОСТАНМАШ". Место нахождения и адрес места осуществления деятельности по изготовлению продукции: Российская Федерация, Москва, 119435, переулок Малый Саввинский, дом 5, квартира 29.

Продукция изготовлена в соответствии с ТУ 4032-001-63711239-2016 АНАЛИЗАТОРЫ ХИМИЧЕСКИХ ИСТОЧНИКОВ ТОКА (АНАЛИЗАТОРЫ ХИТ).

Код ТН ВЭД ЕАЭС 9027801700. Серийный выпуск

соответствует требованиям

ТР ТС 004/2011 "О безопасности низковольтного оборудования", утвержден Решением Комиссии Таможенного союза от 16 августа 2011 года № 768, ТР ТС 020/2011 "Электромагнитная совместимость технических средств", утвержден Решением Комиссии Таможенного союза от 09 декабря 2011 года № 879

Декларация о соответствии принята на основании

Протокола испытаний № 1037-12/20-05-ИМ от 15.12.2020 года, выданного Испытательной лабораторией Общества с ограниченной ответственностью «Центр испытаний и метрологии», аттестат аккредитации РОСС RU.31403.04ИВВ0.002.

Схема декларирования Id

Дополнительная информация

ГОСТ ИЕС 61010-1-2014 Безопасность электрических контрольно-измерительных приборов и лабораторного оборудования. Часть 1. Общие требования; ГОСТ Р МЭК 61326-1-2014 Оборудование электрическое для измерения, управления и лабораторного применения. Требования электромагнитной совместимости. Часть 1. Общие требования (Переиздание); ГОСТ 30804.3.2-2013 (ИЕС 61000-3-2:2009) Совместимость технических средств электромагнитная. Эмиссия гармонических составляющих тока техническими средствами с потребляемым током не более 16 А (в одной фазе). Нормы и методы испытаний (Переиздание); ГОСТ 30804.3.3-2013 Совместимость технических средств электромагнитная. Ограничение изменений напряжения, колебаний напряжения и фликера в низковольтных системах электроснабжения общего назначения. Технические средства с потребляемым током не более 16 А (в одной фазе), подключаемые к электрической сети при несоблюдении определенных условий подключения. Нормы и методы испытаний. Условия хранения продукции, срок хранения, срок службы указаны в прилагаемой к продукции товаросопроводительной и эксплуатационной документации.

Декларация о соответствии действительна с даты регистрации по 27.12.2025 включительно


(подпись)


Рылов Григорий Михайлович
(Ф.И.О. заявителя)

Регистрационный номер декларации о соответствии: ЕАЭС N RU Д-РУ.НХ37.В.17382/20

Дата регистрации декларации о соответствии: 28.12.2020

ДОБРОВОЛЬНАЯ СЕРТИФИКАЦИЯ ПРОДУКЦИИ

СЕРТИФИКАТ СООТВЕТСТВИЯ

№ РОСС RU.HB63.H00460/22

Срок действия с 31.01.2022

по 30.01.2025

№ 0070998

ОРГАН ПО СЕРТИФИКАЦИИ рег.№ RA.RU.11HB63, Орган по сертификации Общества с ограниченной ответственностью "НИЦ ТЕСТ", 123308, РОССИЯ, город Москва, улица 3-Я Хорошёвская, дом 2, строение 1, этаж 3, помещение 21, комната 2, офис 5, Тел: +7 9651234170, E-mail: ooo.nicetst@gmail.com

ПРОДУКЦИЯ ПРИБОРЫ ИЗМЕРИТЕЛЬНЫЕ: АНАЛИЗАТОРЫ
ХИМИЧЕСКИХ ИСТОЧНИКОВ ТОКА (АНАЛИЗАТОРЫ ХИТ)
АСК75.10.20.2, АСК75.10.20.12
Серийный выпуск

код ОК

Код ОК 034-2014
(КПЕС 2008)
26.51.53

СООТВЕТСТВУЕТ ТРЕБОВАНИЯМ НОРМАТИВНЫХ ДОКУМЕНТОВ
ТУ 4032-004-63711239-2022

код ТН ВЭД

9027890000

ИЗГОТОВИТЕЛЬ ОБЩЕСТВО С ОГРАНИЧЕННОЙ ОТВЕТСТВЕННОСТЬЮ
"ЯРОСТАНМАШ"

Место нахождения: 119435, Россия, город Москва, переулок Малый Саввинский, дом 5, квартира 29,
ИНН 7701854211

СЕРТИФИКАТ ВЫДАН ОБЩЕСТВО С ОГРАНИЧЕННОЙ ОТВЕТСТВЕННОСТЬЮ
"ЯРОСТАНМАШ"

Место нахождения: 119435, Россия, город Москва, переулок Малый Саввинский, дом 5, квартира 29
Телефон: +79774875569 Адрес электронной почты: Grigory-rylov@yandex.ru

НА ОСНОВАНИИ Протокола испытаний № 2022-NSP-01-0687 от 28.01.2022 года, выданного
Испытательной лабораторией Общества с ограниченной ответственностью
"НИИСЕРТ" (регистрационный номер аттестата аккредитации РОСС RU.32438.04PCT0.002)

ДОПОЛНИТЕЛЬНАЯ ИНФОРМАЦИЯ

Схема сертификации: 1с.



Руководитель органа

Эксперт

Подпись
Подпись

Н.Ю. Бизюкова
инициалы, фамилия

М. М. Доронин
инициалы, фамилия

Сертификат не применяется при обязательной сертификации